Kalai's 3^d conjecture in dimension five

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1 Problem and notation

Let P be a full-dimensional convex polytope in $\mathbb{R}^5$, centrally symmetric about the origin. A polytope is the convex hull of a finite set. For $0 \le i \le 4$, write $f_i(P)$ for its number of i-dimensional faces, and set

$$s(P) = 1 + f_0 + f_1 + f_2 + f_3 + f_4. \tag{1.1}$$

The empty face is excluded, and the whole polytope is included. An exposed face is the set of maximizers of a linear functional on P. Every face of a polytope is exposed; the zero functional exposes P itself.

Target 1.1 (Kalai's bound in dimension five). *Every full-dimensional, origin-symmetric convex polytope in $\mathbb{R}^5$ satisfies*

$$s(P) \ge 3^5 = 243. \tag{1.2}$$

The bound is sharp: each face of $[-1,1]^5$ chooses, in each coordinate, the negative endpoint, the positive endpoint, or the whole interval. These 3^5 choices include the whole cube.

This manuscript gives a mathematical proof through a fifteen-pair reduction, geometric endpoint arguments and exact finite tables. All real parameter cases, including the parallel and boundary cases in the three-full-pair branch, are included. The standard three-dimensional convex rigidity theorem is stated in Section 10; its dimensional extension and the stress-space constructions are proved there and in the following sections.

The argument reduces the face-count inequality to fifteen possible vertex/facet-count pairs. Euler and flag identities give the arithmetic reduction, a central cube-section argument handles small vertex counts, and facet geometry supplies the endpoint estimates. Sections 2–8 describe these reductions and selected endpoint arguments. Section 9 states the assembly principle. Sections 10–19 develop auxiliary arguments. Sections 20–29 supply product-facet proofs, exact endpoint certificates, two-facet stress localization, ridge capacities and the rank-three reduction. Sections 30–37 supply continuous cube-lift certificates, quadrilateral budgets and further complete cost tables. Sections 38–50 supply the remaining product-facet geometry, stronger weights, exact few-full reduction and one- and two-full-pair closures, and the five-point fibre exclusion. Sections 51–54 give exact balanced-section tables and complete the three-full-pair cases. Section

55 assembles every endpoint, and Section 56 gives the conclusion.

The target concerns Conjecture A in dimension five. It makes no simplicity, simpliciality, or vertex-count assumption. The all-dimensions conjecture and its unrestricted dimension-at-least-six variant are outside this target.

2 Euler, parity, and the incidence bound

Write $v = f_0$, $e = f_1$, $p = f_2$, $r = f_3$, and $h = f_4$. Put $t = e + r$ and $u = v + h$. Euler's relation on the boundary is

$$v - e + p - r + h = 2. \tag{2.1}$$

Lemma 2.1 (The counting target). *The number of nonempty faces is $3 + 2t$. Hence the desired inequality is equivalent to $t \geq 120$.*

Proof. Equation (2.1) gives $p = 2 + t - u$. Substitution into $1 + u + t + p$ gives $3 + 2t$. □

Central symmetry and full dimension imply that the origin lies in the interior. A proper exposed face has a strictly positive support value for its nonzero exposing functional. It cannot contain both a point and its antipode. Consequently $F \mapsto -F$ is a fixed-point-free involution on proper faces of each dimension.

$$v, e, p, r, h \text{ are even; } t \text{ is even.} \tag{2.2}$$

A flag is a chain of incident proper faces. Applying the same involution to every member of a flag also gives even flag counts. Let $I = f_{04}$ count vertex–facet incidences. In particular, I is even.

Lemma 2.2 (Antipodal incidence bound). *One has $2I \leq vh$.*

Proof. Each facet contains at most one vertex from each of the $v/2$ antipodal vertex pairs. Summing its vertex count over the h facets gives $I \leq vh/2$. □

If a counterexample exists, Lemma 2.1 and parity give $t \leq 118$. This two-unit gap will convert rational lower bounds just above 118 into the exact target 120.

3 Flag identities and nonnegative slacks

Let f_{ij} count pairs $F \subset G$ of dimensions i and j, and define triple flags similarly. Set $x = f_{02}$ and $y = f_{24}$. Euler's relation in face figures and intervals gives

$$f_{03} = I + x - 2e, \quad f_{14} = I + y - 2r, \tag{3.1}$$

$$\begin{aligned} f_{13} &= I + x + y - 2t, \\ f_{024} &= 2I + 2x + 2y - 4t. \end{aligned} \tag{3.2}$$

Define the following integer quantities:

$$\begin{aligned} A &= f_{024} - 3x - 3y + 9p, \\ B = 2f_{13} - 3f_{03}, &\quad C = 2f_{13} - 3f_{14}, \end{aligned} \tag{3.3}$$

$$\begin{aligned} G &= e + x - 3p - 5v + 10, \\ G^* &= r + y - 3p - 5h + 10. \end{aligned} \tag{3.4}$$

All five are nonnegative. The quantity A is the sum, over polygonal 2-faces Q, of the product of the vertex excess of Q and that of its polygonal face figure. The quantities B and C express minimum degree three in 3-faces and in the polar. The bounds G, $G^* \geq 0$ come from the centrally symmetric rigidity bound.

Lemma 3.1 (The two global identities). *The preceding quantities satisfy*

$$A + B + C = 9p - 6t, \tag{3.5}$$

$$G + G^* + A = 2I - 8u + 26. \tag{3.6}$$

Proof. Substitute (3.1)–(3.4), cancel terms, and use (2.1). Nonnegativity in (3.5), together with $p = 2 + t - u$, yields $3u \leq t + 6$. □

The rigidity bound can be expressed by triangulating the 2-faces without adding vertices. The resulting symmetric rigid framework has $e + x - 3p$ bars, and its rigidity implies $5v \leq e + x - 3p + 10$. This is precisely $G \geq 0$. Applying the same bound to the polar gives $G^* \geq 0$.

4 Small vertex counts and a cyclic orbit argument

Suppose $v \le 12$. Choose six vectors $v_1, \dots, v_6$ covering the vertices up to sign, padding unused positions with zero. The polar is defined by the six inequalities $|\langle y, v_i\rangle| \le 1$. The coordinate map $y \mapsto (\langle y, v_i\rangle)_i$ is injective and has a five-dimensional hyperplane as image in $\mathbb{R}^6$. Changing signs of coordinates makes its normal nonnegative. Thus its image is

$$K = [-1,1]^6 \cap \{z : \sum_i a_i z_i = 0\}, \quad a_i \ge 0. \tag{4.1}$$

Polarity and injective linear maps preserve the total face count. It remains to bound the number of cube faces whose relative interior meets the hyperplane. Such faces correspond bijectively to the faces of the section.

Lemma 4.1 (Central cube sections). *For $n \ge 1$ and nonnegative weights, at least 3^{n-1} of the cube's 3^n face patterns meet the central hyperplane in their relative interior.*

Proof. A pattern assigns to each coordinate one of $+$, $-$, or free. Let W be the total weight, and P, N, F the weights in the three states, so $P + N + F = W$. The minimum and maximum of the defining linear form on that face are

$$2P - W, \quad W - 2N. \tag{4.2}$$

A pattern lies weakly on the positive side without lying in the hyperplane exactly when $2P \ge W$ and $2N < W$. Cycle all coordinate states simultaneously by $+ \mapsto - \mapsto$ free $\mapsto +$. Each orbit has three elements. At most one element is on this positive side: two state weights at least $W/2$ would force equality and the third weight zero, and the strict inequality then excludes a second qualifying element. If $W = 0$, all patterns lie in the hyperplane.

Hence at most 3^{n-1} patterns are on the positive side. Antipodal symmetry gives the same bound on the negative side. All other patterns meet the hyperplane in their relative interior, so their number is at least $3^n - 2\cdot 3^{n-1} = 3^{n-1}$. □

Taking $n = 6$ gives 243. Applying the same argument to the polar also handles $h \le 12$. Parity therefore forces $v, h \ge 14$ in every counterexample.

5 The exact fifteen-pair reduction

Assume $s(P) < 243$. By Section 4, $v, h \geq 14$. Replace P by its polar if needed, so $v \leq h$. The total face count is preserved, and vertices and facets are exchanged.

Proposition 5.1 (Finite ledger). *The pair (v,h) belongs to the following fifteen-element list.*

Vertices v	Possible facets h
14	16, 18, 20, 22, 24, 26
16	16, 18, 20, 22, 24
18	18, 20, 22
20	20

Proof. Parity and (2.1) give $t \leq 118$. From Section 3, $3u \leq t + 6 \leq 124$. Since u is even,

$$v + h = u \leq 40. \tag{5.1}$$

Nonnegativity in (3.6) gives $2I \geq 8u - 26$. Since I is even, $2I$ is divisible by four, whereas $8u - 26$ is congruent to two modulo four. Therefore

$$2I \geq 8u - 24. \tag{5.2}$$

Combining this sharper integer bound with $2I \leq vh$ yields

$$(v - 8)(h - 8) \geq 40. \tag{5.3}$$

Because $v \leq h$ and their sum is at most 40, v can only be 14, 16, 18, or 20. Listing even h between v and $40 - v$ gives the table; (5.3) removes (14,14), for which the product is 36. Every other listed pair satisfies (5.3). □

This arithmetic reduction avoids searching through polytopes with arbitrary real coordinates. The geometric work is confined to fifteen vertex/facet-count pairs. It does not assert that the inequalities classify all realizable polytopes in those pairs, or that every row of the table is realizable.

6 Facet rows and their global sums

For a four-dimensional facet F, write its row as $R = (n, e_F, p_F, r_F; J)$, where J is its vertex–3-face incidence count. Summing rows over all facets gives

$$\begin{gathered}\sum_F n = I, \quad \sum_F e_F = f_{14}, \quad \sum_F p_F = y, \\ \sum_F r_F = 2r, \quad \sum_F J = 2f_{03}.\end{gathered} \tag{6.1}$$

Set $q = \#\{F : n = 7\}$. The following weights are used in the fourteen-vertex endpoints; the facet subscripts are suppressed inside a row.

$$\begin{aligned} L_1 &= 9p - 5e - n - J - 15, \\ L_2 &= 12n - 10e + 12p - 3J - 20 + 5[n = 7], \\ L_3 &= 16n + 2p + 31r - 10J - 48 + 12[n = 7]. \end{aligned} \tag{6.2}$$

The bracket denotes an indicator. The Gale classification of 4-polytopes with at most seven vertices gives 24 possible rows, on each of which L_1, L_2, L_3, and $g(R) = J - 3n - 3r + 10$ are nonnegative. At $v = 14$ every facet has at most seven vertices, so this classification applies.

Lemma 6.1 (Local-to-global identities). *With global symbols v,h,t,A,G and the facet weights above,*

$$12t = 48v + 33h - 96 + 4A + 6G + \sum_F L_1, \tag{6.3}$$

$$\begin{aligned} 6t = 34v + 14h - 68 - 5q \\ + 2A + 8G + \sum_F L_2, \end{aligned} \tag{6.4}$$

$$\sum_F L_3 + 2A + 22G = 6t - 62v + 124 + 12q. \tag{6.5}$$

Proof. Sum the linear expressions (6.2), use (6.1), and substitute the Euler and flag identities from Sections 2–3. □

Stress localization supplies $2g(R) \le G$ for every facet. Another useful sum is $\sum_F g(R) = 2G + 4v + 4h - I - 8$.

7 Four endpoint exclusions

At fourteen vertices a seven-vertex facet is full: it contains one vertex from every antipodal pair. The full-facet input needed here says that at least eight such facets, with $h \geq 18$, force the target face bound. Therefore a counterexample with these counts has $q \leq 6$: the facet count q is even. Sections 28–32 prove the full-normal rank alternatives and their geometric bounds; Proposition 40.1 combines them to establish this full-facet input.

Proposition 7.1. *The pair (14,26) is excluded by the row sum. The pair (14,22) is excluded using the full-pyramid argument. The pairs (14,24) and (14,20) are excluded using the full-facet input proved in Sections 28–32 and Proposition 40.1.*

Proof. The pair (14,26). Every facet has $L_1 \geq 0$. Equation (6.3) immediately gives

$$12t \geq 48\cdot 14 + 33\cdot 26 - 96 = 1434. \tag{7.1}$$

Thus $t \geq 120$, contradicting $t \leq 118$.

The pair (14,24). If $q \geq 8$, use the geometric lemma. Otherwise $q \leq 6$, and (6.4), nonnegativity, and parity give

$$6t \geq 34\cdot 14 + 14\cdot 24 - 68 - 5\cdot 6 = 714. \tag{7.2}$$

So $t \geq 119$, and its evenness forces $t \geq 120$.

The pair (14,22). Put $M_{22} = L_2 - 5[n = 7]$. It is nonnegative on every catalogue row except (7,15,14,6;29). That exceptional row is a full pyramid. Its geometric consequence, proved in Section 18, makes h divisible by four, which excludes it when $h = 22$. Removing the indicator from (6.4) therefore gives

$$6t \geq 34\cdot 14 + 14\cdot 22 - 68 = 716. \tag{7.3}$$

Again $t \geq 120$. This estimate uses the pyramid parity lemma and avoids a stress-localization case split at this endpoint.

The pair (14,20). Here G is even. If $G \geq 2$, (6.5) and $q \leq 6$ imply $6t \geq 716$. If $G = 0$, stress localization and $g \geq 0$ make every facet's g zero. Its summed identity then gives $I = 128$. But each facet has at most six vertices unless counted in q, so $I \leq 6\cdot 20 + 6 = 126$, a contradiction. □

8 Full facets and the remaining endpoints

Two more fourteen-vertex pairs require full-facet geometry. Equation (3.6) and parity imply $I \geq 108$ at (14,16), and $I \geq 116$ at (14,18). Since each facet has at most seven vertices,

$$I \leq 6h + q. \tag{8.1}$$

Thus $q \geq 12$ at (14,16) and $q \geq 8$ at (14,18). The full-normal arguments in Sections 28–32 exclude both configurations; Section 40.1 gives the complete assembly.

At sixteen vertices a full facet has eight vertices. The endpoint arguments organize these configurations by the rank of their full normals and by the number of full antipodal pairs. They use geometric injections, local incidence estimates, and finite case distinctions. Rank-four configurations are treated by their supporting hyperplanes and exposed-face counts; the other ranks admit their own geometric reductions.

For (16,24), the local L_1 inequality extends to facets with at most eight vertices except the full tetrahedron-prism row (8,16,14,6;32). The full-prism argument is developed in Sections 16–17. When it is absent, (6.3) gives $12t \geq 48\cdot16 + 33\cdot24 - 96 = 1464$, which suffices.

The following table organizes the endpoint arguments. Section 19 supplies the numerical closure at (16,22). The geometric reductions and exact certificates for the other entries are developed in Sections 20–54; Section 55 records their complete coverage.

Pair	**Geometric distinctions**
(16,16), (16,18)	Full facets, rank of full normals, and ridge counts
(16,20)	Number of full pairs, rank-four configurations, and face counts
(16,22)	Facet-row costs, stress localization, and ridge parity
(18,18)	Product facets on either side of polarity and stress bounds
(18,20)	Product facets, edge lower bounds, and facet-row inequalities
(18,22)	Product facets and the possible toric slacks
(20,20)	Product facets on either side of polarity and row exclusions

9 Assembly principle for the endpoint arguments

The endpoint arguments combine with the reductions to give the desired face-count bound. The small vertex case and its polar counterpart are handled by Section 4. The remaining vertex/facet pairs are listed in Proposition 5.1, and the endpoint estimates excluding each of them are assembled in Section 55.

Proposition 9.1. *If the bound holds at each of the fifteen pairs in Proposition 5.1, it holds for every centrally symmetric five-dimensional polytope.*

Proof. Suppose $s(P) < 243$. If $v \le 12$, Section 4 gives a contradiction. Apply the same argument to the polar to exclude $h \le 12$. Parity gives $v, h \ge 14$. Choose P or its polar so that $v \le h$. The chosen polytope still has fewer than 243 faces.

Its face and flag data satisfy the identities and inequalities of Section 3. Proposition 5.1 places its vertex/facet pair in the fifteen-element ledger. The endpoint estimates each give at least 243 faces for the corresponding pair by the hypothesis of this proposition. Every possible pair is therefore contradictory. This proves the bound. □

Section 55 supplies the endpoint hypothesis of this assembly principle by pointing to every complete exclusion in the subsequent geometric arguments and exact tables.

10 The rigidity input and its dimensional extension

For a finite set of points X in a Euclidean space and a set of bars B, an infinitesimal motion assigns a velocity u_x to each point and satisfies

$$\langle u_x - u_y,\, x - y \rangle = 0 \quad \text{for every bar } \{x,y\}. \tag{10.1}$$

In the affine span of a full-dimensional framework, trivial motions have the form $u_x = Ax + b$, with A skew-symmetric. Infinitesimal rigidity means that all motions are trivial. Equivalently, the bars force (10.1) for every pair of vertices.

Classical three-dimensional input. A convex three-dimensional polytope becomes infinitesimally rigid when its polygonal faces are triangulated without new vertices. This is the convex three-dimensional rigidity theorem. Its proof is not reproduced here. The extension to dimensions four and five used below follows from the following slice and gluing argument.

First, the vertices of the 2-faces containing an edge span the affine space of the polytope. Indeed, modulo the edge direction, their directions contain the extreme rays of the tangent cone along the edge; those rays span the quotient. This also explains why a joint inserted on an edge can be braced by joining it to those vertices. After subtracting a trivial motion of the existing framework, the new velocity is orthogonal to spanning bar directions, and so is zero in the affine span. Adding finitely many such joints preserves bracing.

10 The slice and gluing extension, continued

Next, let a be a vertex of a convex polytope of dimension $k+1$. A sufficiently small hyperplane cuts its incident edges to give a k-dimensional vertex figure. Project every other vertex radially from a to this hyperplane:

$$\begin{aligned} z_x &= a + \tau_x(x - a), \quad \tau_x > 0; \\ w_x &= u_a + \tau_x(u_x - u_a). \end{aligned} \tag{10.2}$$

Vertices on a common 2-face through a project to joints on an edge of the vertex figure. Each 2-face of the vertex figure comes from a 3-face through a. The three-dimensional input braces that 3-face, and the projected velocities preserve its slice bar lengths.

$$\begin{aligned} &\langle w_x - w_y, z_x - z_y \rangle \\ &= \tau_x \tau_y \langle u_x - u_y, x - y \rangle. \end{aligned} \tag{10.3}$$

The identity follows by expanding the inner products and using (10.1) for the pairs (a,x) and (a,y). Rigidity in dimension k, including the edge joints just described, therefore braces the entire vertex figure. Since the factors τ are positive, (10.3) braces the vertices of all 2-faces through a, together with a. Call this set the star of a. It spans the original affine space.

Stars of adjacent vertices overlap on the vertices of every 2-face through their edge. That overlap spans the space by the first paragraph. Their trivial motions consequently agree: an affine motion zero on an affinely spanning set is zero everywhere. The edge graph is connected; for example a generic linear functional gives an increasing edge from every vertex except its maximum. Gluing stars along such edges braces all vertices. This proves the induction from dimension three to four, then to five.

A fan triangulation of a polygon braces all pairs of its vertices: its triangles glue along edges in its plane. Thus replacing each polygon by its fan bars provides the bracing assumed in the argument. This works at the actual convex vertex positions; no generic perturbation or simplicial hypothesis is used.

11 Symmetry and the exact stress dimension

Triangulate each polygonal 2-face by a fan, making the choices antipodally. Different polygons share only vertices or original edges, so their added diagonals are distinct. If M is the total number of bars, then

$$M = e + \sum_{Q}(f_0(Q) - 3) = e + x - 3p. \tag{11.1}$$

Every bar lies in a proper face, so no bar joins antipodal vertices and no bar is fixed by the antipodal operation. Choose one representative of each vertex pair and each bar pair. Restrict the rigidity map to even velocity fields, meaning $u_{-x} = u_x$, and keep one equation per bar pair. Its domain and codomain have dimensions $5v/2$ and $M/2$.

A vector in its kernel extends to a motion of the full framework, since the antipodal bar equation is the negative of the chosen one. By Section 10 it is $Ax+b$. Evenness forces $Ax=0$ at every vertex. The vertices span the ambient space, so $A=0$. Conversely, every translation is an even motion. The kernel is therefore exactly five-dimensional.

$$\operatorname{rank} R_{\mathrm{even}} = 5v/2 - 5. \tag{11.2}$$

A stress assigns a real coefficient to every bar so that the sum of the resulting bar forces is zero at each vertex. It is a vector in the kernel of the transpose rigidity map. An antisymmetric stress has opposite coefficients on antipodal bars. It is determined by one coefficient per pair, and its equilibrium equations are precisely the left kernel of R_{even}. Rank–nullity gives

$$\dim S_{\mathrm{anti}} = M/2 - 5v/2 + 5 = G/2. \tag{11.3}$$

Thus G is an even nonnegative integer. The same argument on the polar gives the corresponding facts for G^{*}. This provides the exact numerical meaning of the slacks in Section 3, rather than merely asserting their positivity.

12 Facet stress localization

Let F be a facet with row $(n,e_F,p_F,r_F;J)$. The chosen polygon triangulations restrict to its complete 2-skeleton. Its framework is rigid in its four-dimensional affine span by Section 10.

Summing Euler's relation in its three-dimensional vertex figures gives $f_{02}(F) = J+2e_F-2n$. Euler's relation for F gives $p_F=e_F-n+r_F$. Consequently its bar count is $J+n-3r_F$. The trivial motions in four dimensions have dimension 10, so

$$\begin{aligned}\dim S(F) &= (J+n-3r_F) - (4n-10)\\ &= J-3n-3r_F+10 = g(F).\end{aligned} \tag{12.1}$$

Extend a facet stress ω by zero to the other bars. It remains a global stress: all facet differences lie in its direction space, so its equilibrium equations are unchanged in the ambient space. Let τ copy stress coefficients to antipodal bars. Then

$$\omega \mapsto \omega - \tau\omega \tag{12.2}$$

is an antisymmetric global stress. A proper face cannot contain antipodal vertices, so F and $-F$ have disjoint vertex sets. Restricting (12.2) back to the bars of F recovers ω. The map is injective. Comparing (12.1) with (11.3) proves

$$0 \le 2g(F) \le G. \tag{12.3}$$

If $G=0$, every facet has $g=0$. Summing the definition of g, using (6.1) and the flag identities, gives

$$\sum_F g(F) = 2G + 4v + 4h - I - 8. \tag{12.4}$$

In particular $G=0$ forces $I=4(v+h)-8$. This is the exact incidence equation used at (14,20) and (16,22). The injection concerns a single facet. It does not justify adding stress lower bounds from arbitrary overlapping facets.

13 Polygon localization and the zero-slack case

Let Q be a polygonal 2-face with vertices p_1, ..., p_k, contained in l three-dimensional faces T_j. The two-dimensional face figure is a polygon, so it has l vertices and l edges; equivalently l is also the number of facets containing Q. We prove

$$G \geq 2(k-3)(l-3). \tag{13.1}$$

Choose p_0 in the affine span of Q, and let U be its two-dimensional direction space. The affine dependence space of the polygon vertices is

$$W = \{\alpha : \textstyle\sum_i \alpha_i = 0,\ \sum_i \alpha_i p_i = 0\}, \quad \dim W = k-3. \tag{13.2}$$

For each T_j, choose a vertex q_j outside Q. Its direction $b_j=(q_j-p_0)$ modulo U is nonzero. These directions follow the extreme rays of the full-dimensional tangent cone along Q, and span the three-dimensional quotient. The matrix B with these columns has rank three, and its kernel has dimension $l-3$.

Consider matrices C with each column in W and each row in $\ker B$. Their vector space has dimension $(k-3)(l-3)$. For the j-th column form the virtual bar-row combination

$$s_j = \textstyle\sum_i C_{ij}\ \mathrm{row}(p_i,\ q_j). \tag{13.3}$$

The force at q_j vanishes by the two affine dependence equations. At p_i it is $C_{ij}(p_i-q_j)$. All the virtual bars lie in T_j. Its triangulated framework is rigid in its affine span, so the row space equals the row space of the complete graph on its vertices. This remains true in the ambient space, since bar rows have no normal components. Thus s_j has a representation a_j using actual bars of T_j. Choose the representations linearly.

Summing (13.3), the row-kernel equations make every remaining force lie in U, supported on the polygon vertices. It has zero total force and torque, since it is a sum of bar rows. The polygon fan is rigid in its plane, and its row space is exactly the planar loads orthogonal to translations and rotation. Hence the sum has a representation a_Q using polygon bars. The coefficients

$$\omega(C) = \textstyle\sum_j a_j - a_Q \tag{13.4}$$

form an actual stress supported in the union of the T_j frameworks. Distinct three-faces containing Q meet exactly in Q. A bar from p_i to a vertex of T_j outside Q therefore belongs to no other $T_{j'}$. If $\omega(C)$ vanishes on every bar incident to Q, the normal component of the force of a_j at p_i must vanish. That component is $-C_{ij}b_j$, and b_j is nonzero. Thus $C=0$.

No three-face containing Q contains an antipode of a polygon vertex. Consequently $\tau\omega(C)$ vanishes on bars incident to Q. The map $C \mapsto \omega(C)-\tau\omega(C)$ is injective into the global antisymmetric stress space. Its dimension is $G/2$, which proves (13.1).

The excess A is the sum over polygons of $(k-3)(l-3)$. If $G=0$, (13.1) makes each summand zero, and hence $A=0$. For positive G, this argument bounds each polygon separately; it does not give an additive bound on A.

14 Constructing the small-facet catalogue

Let $a_1, \ldots, a_n$ be the vertices of a four-dimensional polytope with $5 \le n \le 7$. Form the augmented matrix with columns $(a_i,1)$. Its rank is five. A basis of its kernel gives Gale vectors g_i of rank $n-5$, embedded in a plane if necessary.

For a vertex subset F, put S equal to its complement. The face criterion is that the vectors g_i, i in S, admit a dependence with every coefficient strictly positive. To see this, an exposing affine functional has zero slack on F and positive slack on S. Its slack vector lies in the row space of the augmented matrix, which is the orthogonal complement of the Gale matrix columns. This is exactly the displayed dependence condition. The empty complement corresponds to the whole polytope.

$$\dim F = 4 - |S| + \operatorname{rank}\{g_i : i \text{ in } S\}. \tag{14.1}$$

Indeed the affine dependencies supported on F have dimension $(n-5)$ minus the rank on S. Subtract this from $|F|-1$ to obtain (14.1).

Choose a coordinate direction avoiding the finitely many nonzero Gale lines. Rescale each nonzero vector by a positive factor to the form $\pm(1,t)$, and order the distinct slopes. Code the zero vector by 0, and the positive and negative vectors on line j by $2j+1$ and $2j+2$. Sorting all codes only relabels the vertices. Occupied lines are numbered consecutively, so a recursive enumeration may use an existing line or introduce the next line.

The minimal nonnegative dependencies are precisely: a zero singleton; opposite vectors on the same line; or three distinct ordered lines with equal signs on the outside lines and the opposite sign on the middle line. This follows by solving the two equations for $(1,t)$: the middle slope is a strict convex combination of the outside slopes. Rank at most two ensures that no larger minimal support is needed.

A subset admits a dependence positive on every member exactly when it is a union of such minimal supports contained in it. One direction follows by summing their positive dependences. For the other, decompose a nonnegative dependence into minimal ones by subtracting a scaled minimal dependence until a coefficient vanishes, and iterate.

As an independent consistency check, functional sign partitions are enumerated both on an occupied line and between adjacent lines. A subset without a positive dependence must have a functional nonnegative on all its vectors and positive on at least one. Every subset is checked to satisfy exactly one of these alternatives. Every singleton vertex must also pass the face criterion; codes that fail are discarded.

The accompanying standard-library Python program verify_mathematics.py implements these steps with integer masks. It checks 6,364 sorted codes, of which 4,608 have the required rank. Exactly 832 pass all singleton tests. Counting their faces by (14.1) gives exactly the 24 rows in Section 15. These are counts of codes, not counts of distinct combinatorial types.

14 Catalogue profiles and recognition, continued

For the row (7,15,14,6;29), the same face enumeration gives exactly one six-vertex 3-face. For the row (7,18,19,8;35), it gives exactly two vertices lying in six facets. By polarity, the row (8,19,18,7;35) therefore has exactly two six-vertex 3-faces. These profile facts are checked for every valid code producing the indicated row.

The prism row can also be recognized without an affine product assumption. Its dual is (6,14,16,8;32), with a rank-one Gale diagram. Let the numbers of positive, negative and zero Gale vectors be a, b, z. The singleton face tests require $a,b\geq 2$. A facet complement is either one zero vector or one positive and one negative vector, so the facet count is $z+ab$. Together with $a+b+z=6$ and $z+ab=8$, this forces $z=0$ and $\{a,b\}=\{2,4\}$. The eight dual vertices are therefore labelled by a pair in a two-element set times a four-element set. Its six facets are the two tetrahedra and four sides omitting corresponding vertex pairs. The full face criterion gives exactly the tetrahedron-prism face lattice. The program checks these facet witnesses for every code of the dual row.

15 The twenty-four rows and local costs

The columns show $(n,e,p,r;J)$ and the four weights of Section 6. The catalogue is generated by the procedure of Section 14, not obtained by assuming the desired five-dimensional inequality.

n	e	p	r	J	g	L_1	L_2	L_3
5	10	10	5	20	0	0	0	7
6	13	13	6	26	0	5	0	0
6	14	15	7	29	0	15	5	5
6	14	16	8	32	0	21	8	8
6	15	18	9	36	1	30	10	3
7	15	14	6	29	0	0	0	0
7	16	16	7	32	0	10	5	5
7	17	17	7	32	0	14	7	7
7	17	18	8	35	0	20	10	10
7	17	18	8	36	1	19	7	0
7	17	19	9	38	0	26	13	13
7	18	19	8	35	0	24	12	12
7	18	20	9	38	0	30	15	15
7	18	20	9	39	1	29	12	5
7	18	21	10	41	0	36	18	18
7	18	21	10	42	1	35	15	8
7	18	22	11	44	0	42	21	21
7	19	22	10	42	1	39	17	10
7	19	23	11	45	1	45	20	13
7	19	23	11	46	2	44	17	3
7	19	24	12	48	1	51	23	16
7	20	25	12	49	2	54	22	8
7	20	26	13	52	2	60	25	11
7	21	28	14	56	3	69	27	6

All weights are nonnegative. The adjusted cost $L_2-5[n=7]$ is negative only on (7,15,14,6;29), where it is -4. The unadjusted cost L_1 takes values 0 or 5 below 10. With $g=0$, its zero rows are precisely the simplex (5,10,10,5;20) and (7,15,14,6;29).

16 Segment fibres and the parallel full prism

Lemma 16.1 (Three faces over every base face). *Suppose a linear projection of a full-dimensional polytope onto a polytope Q has a one-dimensional kernel, and the fibre above every vertex of Q is a nondegenerate segment. Then $s(P) \geq 3s(Q)$.*

Proof. Fix a nonempty face K of Q. Its inverse-image face P_K is obtained by pulling back an exposing functional; for $K=Q$ it is P itself. If $\dim K=k$, this face has dimension $k+1$, since its vertex fibres are nondegenerate. In fact every fibre is nondegenerate: express a base point as a convex combination of base vertices and combine the two endpoints of their segments.

Over K, the upper endpoint of the fibres is a concave piecewise affine function; the lower endpoint is a convex piecewise affine function. Each has a cell of full dimension k whose relative interior meets that of K. Their graphs are upper and lower faces of P_K, hence faces of P. They have dimension k, so neither is P_K. They cannot be the same face, since fibres above the relative interior of K have positive length.

These three faces for different K are distinct. A chosen upper or lower face projects into K and has a full-dimensional cell there; the smallest face of Q containing its projection is therefore K. It is also never an inverse-image face, because the latter contains entire nondegenerate fibres. Choosing these three faces for every nonempty K proves the injection and the bound. □

Now let a full facet of a sixteen-vertex centrally symmetric five-polytope have the face lattice of a tetrahedron prism. Label its four corresponding edge pairs by E_i. Each pair of their affine lines is coplanar, because the edges are opposite sides of a quadrilateral 2-face. The four lines cannot all lie in one plane, since the facet has dimension four.

Distinct pairwise coplanar lines not contained in one plane are either all parallel or all concurrent. To prove this, take two intersecting lines through p in a plane H. A line coplanar with both that avoids p lies in H. Any further line outside H must pass through p, and cannot then be coplanar with that third line. Thus either all lines pass through p, or all would lie in H, a contradiction.

In the parallel case, project along their common direction. The four edge centres, modulo that direction, are linearly independent: otherwise the vertices of P would span at most four dimensions. The image is the four-dimensional crosspolytope, and each vertex fibre is exactly its corresponding edge or antipodal edge. A crosspolytope has 81 nonempty faces by polarity with the four-cube. Lemma 16.1 gives $3 \cdot 81 = 243$. Equal edge lengths are not required.

17 Thirty-two polar vertices in the concurrent case

In the concurrent case let p be the common point of the four prism-edge lines. It lies outside every closed edge segment: in the convex quadrilateral made by any two opposite edges, the line of one edge leaves the entire other edge strictly on one side. Consequently write the endpoints as $p+s_i u_i$ and $p+t_i u_i$, with $0<s_i<t_i$.

The four directions are independent, since the facet spans four dimensions. Normalize its supporting form to equal 1. It takes value 1 at p and zero on the four directions, so p and those directions are a basis in dimension five. A linear change of coordinates gives

$$P = \operatorname{conv}\{\pm(1,s_i e_i),\ \pm(1,t_i e_i) : 1\le i\le 4\}. \tag{17.1}$$

Its polar is cut out by $|x_0+s_i x_i|\le 1$ and $|x_0+t_i x_i|\le 1$. For every nonempty subset S of $\{1,2,3,4\}$, take

$$x_0=1;\quad x_i=0 \text{ if } i \text{ is in } S, \text{ and } -2/t_i \text{ otherwise.} \tag{17.2}$$

These points are feasible. Inside S both expressions are 1; outside it the t expression is -1 and the s expression is $1-2s/t$, strictly between -1 and 1. Choose an index in S. Its two active equations determine x_0 and its coordinate; one active equation for each remaining index determines all others. Their rank is five, so this is a vertex. The fifteen subsets and their negatives give thirty distinct vertices.

$$\begin{gathered} a=\min_i (t_i+s_i)/(t_i-s_i) > 1; \\ x_0=a,\quad x_i=-(1+a)/t_i. \end{gathered} \tag{17.3}$$

All four t expressions are -1. Each s expression is greater than -1 and at most 1; the latter inequality is equivalent to $a\le (t+s)/(t-s)$. At a minimizing index the expression is 1. Its equation and the four t equations have rank five. Thus (17.3) and its negative are two further vertices, distinct from (17.2).

The polar therefore has at least 32 vertices. A five-dimensional polytope has minimum vertex degree five, so $e\ge 5\cdot 16/2=40$ and, by polarity, $r\ge 5\cdot 32/2=80$. Hence $t\ge 120$ and $s(P)\ge 243$. Together with Section 16, this proves the full tetrahedron-prism statement for every real realization of its face lattice.

18 A full pyramid and divisibility by four

Suppose a full facet is a pyramid with apex a and a three-dimensional base K. Normalize its supporting form to equal 1. The direction space of K has dimension three; its linear span has dimension four, since the supporting form is zero on directions but equals 1 on the base. The apex is outside that linear span: its intersection with the facet hyperplane is the affine span of the base, and an exposed base contains every point of the pyramid in that affine span.

Since the facet is full, all vertices of P are its vertices and their antipodes. Therefore

$$P = \operatorname{conv}(\{\pm a\} \cup R), \quad R = \operatorname{conv}(K \cup -K). \tag{18.1}$$

Here R is a centrally symmetric four-polytope in the complementary four-dimensional subspace. A linear change of coordinates turns (18.1) into a bipyramid over R. Its facets are the pyramids from either apex over a facet of R. Hence

$$f_4(P) = 2f_3(R) \equiv 0 \pmod 4. \tag{18.2}$$

The congruence follows because the facets of R occur in antipodal pairs. This alone excludes a full pyramid at (14,22); no five-dimensional face bound is assumed.

The exceptional seven-vertex row (7,15,14,6;29) is a pyramid over a triangular prism. Its recognition is a face-lattice statement, rather than an affine product assertion. It can be checked on each corresponding Gale code: the six-vertex base facet excludes one apex, and all other vertices lie in that base. A face is the convex hull of its vertices, so this proves the pyramid description.

19 The numerical closure at (16,22)

Assume v=16, h=22 and t≤118. Every facet has at most eight vertices. The tetrahedron-prism row T=(8,16,14,6;32) is excluded by Sections 16–17 after its face-lattice recognition. Put

$$L(F) = L_1(F) + 9[n=8]. \tag{19.1}$$

On nonfull facets the table gives L≥0, with the only values below 10 being 0 and 5. For a full eight-vertex facet other than T, L≥13. If its g is zero, L≥14 and the only value below 20 besides 14 is 18, attained uniquely by Z=(8,19,18,7;35).

Here is the coverage argument for these eight-vertex facts. If the facet has at most seven facets, its dual is in the table. The possibilities with eight vertices are T, (8,18,17,7;35), (8,18,17,7;36), and Z, whose costs are 0,14,13,18. Otherwise let its facet count be r≥8. Euler and J+32≤4e give

$$L = 4e+9r-86-J \geq 9r-54 \geq 18. \tag{19.2}$$

If g=0, then J=14+3r and 4e≥46+3r. For r=8, integer e≥18 gives L=4e−52≥20; for r≥9 the bound is still larger. This proves the stated gaps without a complete eight-vertex catalogue.

Let m count full antipodal facet pairs, and let c be the sum of L over one representative of every facet pair. Dividing (6.3) by two gives the exact identity

$$6t = 699 - 9m + 3G + 2A + c. \tag{19.3}$$

Both A and G are even and nonnegative. The first fact for A follows either from its flag formula or by pairing antipodal polygons. Since c≥13m and 6t≤708, (19.3) yields

$$2A + 4m + 3G \leq 9. \tag{19.4}$$

19 The slack cases and ridge contradiction, continued

If $G>0$, evenness forces $G=2$, $m=0$ and $A=0$. Equation (19.3) gives $c\le 3$, so $c=0$ because a positive pair cost is at least five. It would then give $6t=705$, impossible for integer t.

If $G=0$, Sections 12–13 give $g(F)=0$ for every facet and $A=0$. Thus $c\ge 14m$ and $c\le 9+9m$, so m is 0 or 1. At $m=0$, the only even $t\le 118$ compatible with (19.3) is 118 and $c=9$. A sum of costs 0 or 5 cannot equal 9. At $m=1$, the full pair cost is at least 14; the only possibility is $t=118$, $c=18$. A full cost of 14 would leave four, impossible for the other nonnegative costs. The full pair must therefore be Z, and every other pair has cost zero.

The zero-cost, zero-g rows are the simplex S and the triangular-prism pyramid Y. Equation (12.4) gives $I=144$. The facet count and incidence count now imply

$$\begin{aligned} N_S+N_Y+2&=22,\\ 5N_S+7N_Y+16&=144,\\ N_S=6,\quad N_Y&=14,\quad N_Z=2. \end{aligned} \tag{19.5}$$

A simplex has no six-vertex ridges, Y has one, and Z has two, by the profile counts of Section 14. Thus the number of incidences of six-vertex ridges with facets would be $6\cdot 0+14\cdot 1+2\cdot 2=18$. Every ridge belongs to exactly two facets; these ridges also occur in antipodal pairs. Their incidence count must therefore be divisible by four, contradicting 18. This closes the numerical argument at (16,22) from the stated geometric recognition and rigidity inputs.

20 A product facet: the supporting functionals

We now supply the product-facet geometry and the complete arithmetic closure at (18,22) and (20,20). All quantities in the certificates below are face counts or integer coefficients. The argument does not require a proof assistant or access to its source.

Lemma 20.1. *A four-polytope with row (9,18,15,6;36) has the face lattice of the product of two triangles.* Its polar has six vertices and nine facets. In the rank-one Gale description of Section 14, let a,b,z be the positive, negative and zero multiplicities. One has a,b≥2, a+b+z=6, and ab+z=9. The only possibility is a=b=3 and z=0. Facet complements in the polar consist of one member of each triple. Thus the original nine vertices are x_{ij}, 0≤i,j≤2. Its six facets are R_i and C_j, with x_{ab} in R_i exactly when a≠i and in C_j exactly when b≠j. This establishes the recognition without assuming an affine product realization. □

Let F be such a facet of the five-polytope, exposed at level one by c. Each R_i is a ridge of the five-polytope and belongs to another facet exposed by c_i, also at level one. Put $h_i=c-c_i$. Likewise put $k_j=c-d_j$ for the other facet at C_j. On the vertices of F we have

$$\begin{aligned} &h_i(x_{ab})=0\ (i\neq a),\ h_a(x_{ab})>0;\\ &k_j(x_{ab})=0\ (j\neq b),\ k_b(x_{ab})>0. \end{aligned} \tag{20.1}$$

Lemma 20.2 (Positive representation). *There are positive numbers γ_i,δ_j such that $c=\sum\gamma_i h_i+\sum\delta_j k_j$.* Define $\Psi(\gamma,\delta)$ to be this sum. Evaluation at x_{ab} sees only γ_a and δ_b. A kernel vector with $\gamma_0=0$ consequently has every $\delta_b=0$ and then every $\gamma_a=0$. Thus dim ker $\Psi\leq 1$. The domain has dimension six and the codomain five, so Ψ is onto and its kernel has dimension one.

Normalize a nonzero kernel vector (u,w) by $u_0>0$. Equation (20.1) forces all $u_a>0$ and all $w_b<0$. Take any representation $(\gamma^\circ,\delta^\circ)$ of c. Evaluating it and the kernel vector at x_{ab} gives

$$\begin{aligned} \gamma^\circ_a h_a(x_{ab}) + \delta^\circ_b k_b(x_{ab}) &= 1,\\ u_a h_a(x_{ab}) + w_b k_b(x_{ab}) &= 0. \end{aligned} \tag{20.2}$$

These equations imply $-\gamma^\circ_a/u_a < \delta^\circ_b/(-w_b)$ for every a,b. Choose η strictly between the maximum of the three left endpoints and the minimum of the three right endpoints. Then $\gamma=\gamma^\circ+\eta u$ and $\delta=\delta^\circ+\eta w$ are positive and still represent c. This proves the lemma. □

Set $U_{ab}=\gamma_a h_a(x_{ab})$ and $V_{ab}=\delta_b k_b(x_{ab})$. Both are positive and $U_{ab}+V_{ab}=1$. For potentials u_i,w_j, the functional $L=\sum u_i\gamma_i h_i+\sum w_j\delta_j k_j$ satisfies

$$L(x_{ab})=u_a U_{ab}+w_b V_{ab}. \tag{20.3}$$

20 A product facet: thirty-six side edges, continued

Lemma 20.3. *If v=18 and F has row (9,18,15,6;36), then e≥72.* The vertices of P are exactly the nine x_{ab} and their antipodes, since a facet cannot contain antipodal vertices. Choose i≠i′ and j≠j′. Give u the values 1,−1,0 at i,i′ and the remaining index, and give w the values 1,−1,0 at j,j′ and its remaining index. By (20.3), L has maximum one only at x_{ij} in F and minimum minus one only at $x_{i′j′}$. On −F its maximum is therefore only at $-x_{i′j′}$. The exposed face is the segment joining these two vertices.

There are $(3\cdot2)^2=36$ different such edges: their positive and negative endpoints identify the ordered pairs. None is an edge contained in F or −F. Those two facets contain 18 edges each. The three disjoint sets give 36+18+18=72. □

Corollary 20.4. *A counterexample at (18,22) has no facet of this row.* Every four-dimensional facet has at least five ridges. Each ridge belongs to two facets, so 2r≥5h=110. Lemma 20.3 gives t=e+r≥72+55=127, contrary to t≤118. □

21 A product facet with an extra antipodal pair

Lemma 21.1. *If v=20 and F has row (9,18,15,6;36), then e≥66.* In addition to $\pm x_{ij}$, there is one pair ±z. Write $Z_i=\gamma_i h_i(z)$ and $W_j=\delta_j k_j(z)$. Their sum is c(z), which lies strictly between −1 and 1: z and −z are vertices outside F.

For a candidate $c_0=((i,i'),(j,j'))$ among the same 36 ordered pairs, use potentials 1,−1,α on the three row indices and 1,−1,β on the column indices, where −1<α,β<1. The two intended endpoints remain the unique maxima on F and −F. If k,l are the unused indices, put

$$D=Z_i-Z_{i'}+W_j-W_{j'},\ a=Z_k,\ b=W_l;\qquad L(z)=D+\alpha a+\beta b. \tag{21.1}$$

If the open interval (D−|a|−|b|,D+|a|+|b|) meets (−1,1), choose α,β so that |L(z)|<1, and the candidate edge survives. A candidate is blocked above if D−|a|−|b|≥1, and below if −D−|a|−|b|≥1. The degenerate cases a=b=0,D=±1 are counted as surviving: L exposes a triangle with the two intended endpoints and ±z. Three different vertices of a polytope cannot be collinear, and each side of this triangle is a face of P. These alternatives cover every candidate.

Let B be the set blocked above, after deleting those degenerate cases. Reversing both ordered pairs maps B bijectively to the set blocked below. The two sets are disjoint. Thus at least 36−2|B| candidate edges survive. For $c_0\in B$, one has D>1. Let L_0 use α=β=0 and put ζ=c(z). The functional

$$L_0 + ((D-1)/(1-\zeta))c \tag{21.2}$$

exposes the edge joining z to x_{ij}: on F its unique maximum is 1+(D−1)/(1−ζ), and z has exactly that value; on −F the maximum is smaller. The functional $L_0-((D-1)/(1+\zeta))c$ similarly exposes the edge joining z to $-x_{i'j'}$. Their negatives supply the corresponding edges at −z.

Let f be the number of distinct first cells (i,j) of B and s the number of distinct second cells (i′,j′). These are distinct neighbours of z, so deg(z)≥f+s. Also deg(z)≥5 because its tangent cone spans five-dimensional space. The resulting edge sets are disjoint, giving

$$e \geq 36+(36-2|B|)+2\max(5,f+s). \tag{21.3}$$

21 The extra-pair combinatorics, continued

We prove $|B| \le \max(5, f+s)+3$. Give an ordered pair (i,i') orientation + if $i' = i+1$ modulo three and − otherwise. For two different pairs of the same orientation, the sum of $Z_i - Z_{i'} - |Z_k|$ is at most zero. For example the pairs (0,1),(1,2) give $Z_0 - Z_2 - |Z_2| - |Z_0| \le 0$; the other choices are cyclic permutations or reversals. The same holds for W. Hence two members of B cannot have different row pairs of the same orientation and different column pairs of the same orientation: adding their defining inequalities would give $2 \le 0$.

Within each of the four orientation blocks, the triple (row orientation, column orientation, $j-i$ mod 3) determines a member of B. Indeed equal diagonals and different members force both ordered pairs to differ, which is the forbidden situation just proved. Each block therefore has at most three members; $|B| \le 12$. Also $|B| \le fs$, since a first cell and a second cell uniquely specify the two ordered pairs.

Let m,n be the sizes of the ++ and −− blocks. The two mixed blocks contain at most six members in total, so $|B| \le m+n+6$, where $m,n \le 3$. In each unmixed block, the second cell is the first cell shifted by (1,1) or (−1,−1). These shifts preserve the diagonal. Members on different diagonals cannot share either cell. On a common diagonal, a ++ member and a −− member may share their first cells or their second cells, but cannot share both: the two shifts differ. Consequently the total number of first- or second-cell coincidences is at most min(m,n), and

$$f+s \ge 2(m+n) - \min(m,n). \tag{21.4}$$

If $|B| \ge 11$, then $m+n \ge 5$. For $m+n=5$ the right side of (21.4) is at least eight; for $m+n=6$ it is nine. Together with $|B| \le m+n+6$ this gives $|B| \le f+s+3$. If $|B| \le 10$ and $f+s \ge 7$ the same claim is immediate. If $f+s=6$, then $|B| \le fs \le 9$; if $f+s \le 5$, then $|B| \le fs \le 6 \le 8$. These cases prove the claimed inequality. Substitution in (21.3) gives $e \ge 72-6=66$, completing the proof of Lemma 21.1. □

22 Exact row inequalities and their summation

Here is the arithmetic certificate method, including its whole finite row domain. Write $S=G^*$ and use v,h,t,e,I as global variables. The identities already proved give

$$\begin{gathered} r=t-e,\ p=t+2-v-h,\\ x=G-e+3t+2v-3h-4,\\ y=S+e+2t-3v+2h-4,\\ A=2I-G-S+26-8(v+h),\\ B=-I-G+2S+9e-3t-8v+7h-4,\\ C=-I+2G-S-9e+6t+7v-8h-4. \end{gathered} \tag{22.1}$$

A cost is specified by ten integers $(c_0,c_n,c_e,c_p,c_r,c_j,a_5,a_6,b_5,b_6)$. For a facet row $R=(n,e_0,p_0,r_0;J)$, its value is

$$\begin{gathered} \varphi(R)=c_0+c_n n+c_e e_0+c_p p_0+c_r r_0+c_j J\\ +a_5[n=5]+a_6[n=6]+b_5[r_0=5]+b_6[r_0=6]. \end{gathered} \tag{22.2}$$

Square brackets are zero-one indicators. Let N_5,N_6 count facets with five or six vertices and R_5,R_6 count facets with five or six ridges. Give these counts subscripts F and S on the two sides of polarity. Each is even. The sum of (22.2) over facets is the linear form

$$\begin{gathered} \Phi=c_0 h+c_n I+c_e(I+y-2r)+c_p y+2c_r r\\ +2c_j(I+x-2e)+a_5 N_{F5}+a_6 N_{F6}+b_5 R_{F5}+b_6 R_{F6}. \end{gathered} \tag{22.3}$$

For a second cost ψ, its polar sum Ψ is obtained by v↔h, e↔r, x↔y, G↔S and F↔S. If the two costs are nonnegative on their respective row domains, $\Phi,\Psi\ge 0$.

For completeness, those domains contain every possible facet row. If $n\le 7$ or $r_0\le 7$, use the 24 rows in Section 15 and their duals, subject to $n\le n\max$, $0\le 2g\le \mathrm{cap}$, and the stated exclusions. Otherwise $n,r_0\ge 8$ and all rows are in the following integer box:

$$\begin{gathered} 8\le n\le n\max,\quad 0\le e_0\le n(n-1)/2,\\ 8\le r_0\le \lfloor(4e_0-7n+10)/3\rfloor,\quad p_0=e_0+r_0-n,\\ 3n+3r_0-10\le J\le \min(4e_0-4n,\ 4p_0-4r_0,\ 3n+3r_0-10+\lfloor \mathrm{cap}/2\rfloor). \end{gathered} \tag{22.4}$$

The upper bound on edges counts vertex pairs. In a four-polytope, summing minimum degree three in every three-face gives $J+4n\le 4e_0$; polarity gives $J+4r_0\le 4p_0$. These facts, Euler and $g\ge 0$ prove (22.4). Thus the box is an outer approximation; it need not be realizable to make a cost valid for every actual facet.

22 Checking a cost on the complete row domain, continued

The finite calculation for the large box has a particularly short exact recurrence. After substituting $p_0=e_0+r_0-n$, put $d_n=c_n-c_p$, $d_e=c_e+c_p$, $d_r=c_r+c_p$. For each n,e_0,r_0 in (22.4), set $L=3n+3r_0-10$ and let U be the displayed upper bound. Discard the triple if $L>U$. The minimum over J is attained at L when $c_j\geq 0$ and at U when $c_j<0$. Hence the minimum cost on the whole large box is exactly

$$\min\{c_0+d_n n+d_e e_0+d_r r_0+c_j \text{ (L if } c_j\geq 0\text{, U otherwise)}\}. \tag{22.5}$$

There are no indicator terms here, since $n,r_0\geq 8$. For the small rows substitute the printed row entries in (22.2). Section 24 prints both minima for every cost. The symbol "—" denotes an empty large box. The integers are obtained by the finite ranges (22.4) and the endpoint substitution (22.5), so neither rounding nor a numerical optimization assumption enters this step.

For each case below, label the following nonnegative linear forms $Q_1,\dots,Q_{23}$, in the printed order:

Labels	Nonnegative forms
Q_1–Q_{11}	e, p, r, I, x, y, G, S, A, B, C
Q_{12}–Q_{19}	N_{F5}, N_{F6}, R_{F5}, R_{F6}, N_{S5}, N_{S6}, R_{S5}, R_{S6}
Q_{20}, Q_{21}	$vh-2I$, $118-t$
Q_{22}, Q_{23}	Φ, Ψ from the two case costs

List the additional conditions of a case in order. A condition $z\geq b$ supplies $H=z-b$; $z\leq b$ supplies $H=b-z$; an equality supplies both forms, with $z-b$ first. Number these $H_1,H_2,\dots$ in that order. The final line of each case prints nonnegative integer multipliers and a positive integer κ. After the substitutions (22.1), the indicated sum is identically $-\kappa$. Since every summand is nonnegative, that identity is a contradiction. A reader can check it by collecting coefficients of t,e,I,G,S and the eight row counts.

23 Complete closure at (18,22) and (20,20)

Proposition 23.1. *Neither (18,22) nor (20,20) is the vertex/facet pair of a counterexample.* At (18,22), Corollary 20.4 excludes the product row on the facet side. The vertex limits for facets of P and its polar are nine and eleven. Use the 69 cases labelled A1–A69 in Section 24, excluding (9,18,15,6;36) only on the facet side.

Here is their coverage. First separate G≥22, 12≤G≤20, and G≤10. In the last range separate S≥32, 20≤S≤30, and S≤18. This leaves just the sixty even pairs G∈{0,2,4,6,8,10}, S∈{0,2,...,18}. Nine rectangles cover all but the following nine individual pairs:

$$(0,10),\ (2,4),\ (2,6),\ (2,8),\ (2,10),\ (2,12),\ (4,10),\ (4,12),\ (4,14). \tag{23.1}$$

The conditions printed for these pairs split only on the even quantities $t,e,I,N_{F5},N_{F6},\ R_{S5},R_{S6}$. Every split is z≤b or z≥b+2, which exhausts the even integers. For example, (0,10) first splits t≤116 or t=118; in the latter part it splits e≤54 or e≥56; if e≤54, it splits R_{S5}≤2 or ≥4; in the last part it splits R_{S6}≤10 or ≥12. All five resulting cases are printed. The other pairs are resolved by the full condition lists in the same way. Whenever G=0 or S=0, Section 13 supplies A=0. There is therefore no omitted slack value or omitted boundary equality.

At (20,20), a product facet gives e≥66≥62 by Lemma 21.1, and the case B1 applies. A product facet of the polar gives r≥66≥62 and B2 applies. If neither side has that row, use B3–B7. Their five conditions exhaust G,S≥0 even: G≥12; G≤10,S≥20; G≤10,6≤S≤18; G≤10,2≤S≤4; G≤10,S=0. In the last case A=0. The respective vertex limits are ten on both sides.

The cap 1000 in a case denotes the uniform generous bound. Indeed (3.6) and 2I≤vh give G+S≤vh−8(v+h)+26, namely 102 or 106 for these two endpoints. By (12.3), every local 2g is at most its global slack. Smaller caps in the tables are the upper bounds of the corresponding slack cells. Thus all local cost hypotheses are established. Every case has nonnegative costs and a strictly negative constant sum in Section 24, which proves the proposition. □

The companion arithmetic check also partitions each even axis at every printed boundary, rather than testing random configurations. It checks 32,196 complete axis cells for the first endpoint and 3,025 for the non-product branch of the second, using all 69 and all five applicable cases, respectively. This is an optional independent check of the printed coverage, not a hypothesis of the argument.

24 The printed integer certificates

A cases use (v,h)=(18,22), nmax=(9,11), excluding (9,18,15,6;36) on F only. B1 and B2 use (20,20), nmax=(10,10), without exclusions. B3–B7 exclude that row on both sides. Every case assumes t≤118, already Q_{21}; an additional printed t≤118 is harmless. F and S after “cap” specify the two local stress caps. The cost tuples have exactly the coefficient order in (22.2). Minima are reported as (small rows, large box). All multipliers refer to Q and H as defined in Section 22.

A1. cap F=0, S=10. Conditions: G≥0, G≤0, S≥10, S≤10, A=0, t≤116, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−270, 22, 14, 0, 72, −17, 0, 0, 0, 0); minima (0, 0).

Identity: Q23 + 15·H2 + 33·H3 + H6 + 42·H7 = −24.

A2. cap F=0, S=10. Conditions: G≥0, G≤0, S≥10, S≤10, A=0, t≥118, e≤54, R_{S5}≤2, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−150, 20, 0, 0, 31, −6, 0, 0, 15, 0); minima (0, 0).

Identity: Q23 + 4·H2 + 8·H3 + 4·H6 + 26·H8 + 15·H9 = −2.

A3. cap F=0, S=10. Conditions: G≥0, G≤0, S≥10, S≤10, A=0, t≥118, e≤54, R_{S5}≥4, R_{S6}≤10, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−65, 7, 0, 0, 18, −3, 0, 0, 0, 2); minima (0, 0).

Identity: 12·Q23 + 6·H2 + 66·H3 + 6·H6 + 216·H8 + 24·H10 = −24.

A4. cap F=0, S=10. Conditions: G≥0, G≤0, S≥10, S≤10, A=0, t≥118, e≤54, R_{S5}≥4, R_{S6}≥12, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(111, −17, 6, 0, 24, −5, 0, 0, −106, −30); minima (0, 0).

Identity: 9·Q7 + 21·Q9 + 36·Q21 + 2·Q23 + 41·H3 + 212·H9 + 60·H10 = −36.

A5. cap F=0, S=10. Conditions: G≥0, G≤0, S≥10, S≤10, A=0, t≥118, e≥56, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−180, −8, 32, 0, 48, −17, 0, 0, 0, 0); minima (0, 0).

Identity: Q9 + 2·Q11 + 12·Q21 + 3·H2 + 3·H3 + 18·H8 = −24.

A6. cap F=2, S=4. Conditions: G≥2, G≤2, S≥4, S≤4, e≤54, t≤118.

φ=(−110, 11, 7, 0, 33, −9, 0, 0, 0, 0); minima (0, 15).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 12·Q21 + Q22 + 18·H1 + 7·H4 + 9·H5 = −12.

A7. cap F=2, S=4. Conditions: G≥2, G≤2, S≥4, S≤4, e≥56, t≤118.

φ=(−420, 68, −3, 0, 126, −26, 0, 0, 0, 0); minima (0, 0).

ψ=(0, −52, 52, 0, 0, −13, 0, 0, 0, 0); minima (0, 0).

Identity: 13·Q11 + 174·Q21 + Q22 + 26·H1 + 16·H3 + 222·H5 = −148.

A8. cap F=2, S=6. Conditions: G≥2, G≤2, S≥6, S≤6, t≤116, t≤118.

φ=(−120, 12, 6, 0, 36, −9, 0, 0, 0, 0); minima (0, 18).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: Q22 + 18·H1 + 6·H4 + 18·H5 = −24.

A9. cap F=2, S=6. Conditions: G≥2, G≤2, S≥6, S≤6, t≥118, e≤54, t≤118.

φ=(−450, 69, 47, 0, 135, −52, 0, 0, 0, 0); minima (0, 0).

ψ=(0, −4, 4, 0, 0, −1, 0, 0, 0, 0); minima (0, 0).

Identity: 12·Q10 + Q22 + 116·H1 + 71·H4 + 78·H5 + 291·H6 = −246.

A10. cap F=2, S=6. Conditions: G≥2, G≤2, S≥6, S≤6, t≥118, e≥56, t≤118.

φ=(−30, 5, 0, 0, 9, −2, 0, 0, 0, 0); minima (0, 3).

ψ=(0, −4, 4, 0, 0, −1, 0, 0, 0, 0); minima (0, 0).

Identity: Q11 + 12·Q21 + Q22 + 2·H1 + H3 + 15·H6 = −12.

A11. cap F=2, S=8. Conditions: G≥2, G≤2, S≥8, S≤8, t≤116, t≤118.

φ=(−120, 12, 6, 0, 36, −9, 0, 0, 0, 0); minima (0, 18).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: Q22 + 18·H1 + 6·H4 + 18·H5 = −12.

A12. cap F=2, S=8. Conditions: G≥2, G≤2, S≥8, S≤8, t≥118, e≤54, t≤118.

φ=(−30, 3, 5, 0, 9, −4, 0, 0, 0, 0); minima (0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: Q22 + 8·H1 + 5·H4 + 6·H5 + 21·H6 = −8.

A13. cap F=2, S=8. Conditions: G≥2, G≤2, S≥8, S≤8, t≥118, e≥56, t≤118.

φ=(−50, 9, 1, 0, 15, −4, 0, 0, 0, 0); minima (0, 3).

ψ=(0, −4, 4, 0, 0, −1, 0, 0, 0, 0); minima (0, 0).

Identity: 2·Q11 + 18·Q21 + Q22 + 4·H1 + H3 + 21·H6 = −18.

A14. cap F=4, S=10. Conditions: G≥4, G≤4, S≥10, S≤10, e≤54, t≤118.

φ=(−70, 7, 5, 0, 21, −6, 0, 0, 0, 0); minima (0, 3).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 6·Q21 + Q22 + 12·H1 + 5·H4 + 9·H5 = −6.

A15. cap F=4, S=10. Conditions: G≥4, G≤4, S≥10, S≤10, e≥56, t≤118.

φ=(−50, 5, 1, 0, 15, −3, 0, 0, 0, 0); minima (0, 6).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 2·Q9 + 4·Q11 + 36·Q21 + Q22 + 5·H3 + 45·H5 = −36.

A16. cap F=4, S=12. Conditions: G≥4, G≤4, S≥12, S≤12, t≤116, t≤118.

φ=(−5100, 888, 57, 0, 1530, −378, 0, 0, 0, 0); minima (0, 18).

ψ=(0, −378, −9, 0, 0, 99, 0, 0, 0, 0); minima (0, 0).

Identity: Q22 + Q23 + 765·H1 + 255·H4 + 765·H5 = −1020.

A17. cap F=4, S=12. Conditions: G≥4, G≤4, S≥12, S≤12, t≥118, e≤54, R_{S5}≤0, t≤118.

φ=(0, −60, 60, 0, 0, −15, 0, 0, 0, 0); minima (0, 0).

ψ=(−1296, 150, −2, 0, 261, −44, 0, 0, 141, 0); minima (0, 0).

Identity: 60·Q10 + Q23 + 62·H1 + 32·H4 + 186·H5 + 804·H6 + 141·H7 = −64.

A18. cap F=4, S=12. Conditions: G≥4, G≤4, S≥12, S≤12, t≥118, e≤54, R_{S5}≥2, I≤156, t≤118.

φ=(−110, 17, 4, 0, 33, −9, 0, 0, 0, 0); minima (0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 12·Q21 + Q22 + 18·H1 + 4·H4 + 3·H8 = −4.

A19. cap F=4, S=12. Conditions: G≥4, G≤4, S≥12, S≤12, t≥118, e≤54, R_{S5}≥2, I≥158, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−105, −27, −2, 0, 64, 0, 0, 0, −60, 0); minima (0, 0).

Identity: Q23 + 2·H1 + 6·H5 + 134·H6 + 60·H7 + 29·H8 = −4.

A20. cap F=4, S=12. Conditions: G≥4, G≤4, S≥12, S≤12, t≥118, e≥56, t≤118.

φ=(−70, 7, 1, 0, 21, −4, 0, 0, 0, 0); minima (0, 9).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 8·Q9 + 16·Q11 + 150·Q21 + 3·Q22 + 21·H3 + 189·H6 = −198.

A21. cap F=4, S=14. Conditions: G≥4, G≤4, S≥14, S≤14, t≤116, t≤118.

φ=(−2400, 396, −6, 0, 720, −156, 0, 0, 0, 0); minima (0, 0).

ψ=(−43740, 2292, 2238, 0, 11664, −2421, 0, 0, 0, 0); minima (0, 0).

Identity: 317900·Q9 + 558008·Q10 + 353210·Q11 + 8667·Q22 + 1284·Q23 + 5824224·H3 + 13434180·H5 = −18893400.

A22. cap F=4, S=14. Conditions: G≥4, G≤4, S≥14, S≤14, t≥118, e≤54, t≤118.

φ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); minima (0, 0).

ψ=(−285, 25, 2, 0, 76, −12, 0, 0, 0, 0); minima (0, 0).

Identity: 6·Q21 + 3·Q22 + Q23 + 8·H2 + 24·H3 + 74·H6 = −6.

A23. cap F=4, S=14. Conditions: G≥4, G≤4, S≥14, S≤14, t≥118, e≥56, t≤118.

φ=(−4400, 726, −11, 0, 1320, −286, 0, 0, 0, 0); minima (0, 0).

ψ=(−6735, 163, 582, 0, 1796, −444, 0, 0, 0, 0); minima (0, 0).

Identity: 143·Q11 + 1782·Q21 + Q22 + 286·H1 + 154·H3 + 2244·H6 = −3608.

A24. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≤52, t≤118.

φ=(−850, 131, 88, 0, 255, −98, 0, 0, 0, 0); minima (0, 0).

ψ=(0, −92, 92, 0, 0, −23, 0, 0, 0, 0); minima (0, 0).

Identity: 20·Q10 + 49·Q11 + 3·Q22 + 510·H1 + 255·H4 + 765·H5 = −1020.

A25. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≤116, t≤118.

φ=(−1260, 204, −9, 0, 378, −78, 0, 0, 0, 0); minima (0, 0).

ψ=(0, −156, 156, 0, 0, −39, 0, 0, 0, 0); minima (0, 0).

Identity: 39·Q11 + Q22 + 78·H1 + 48·H3 + 666·H5 + 522·H6 = −444.

A26. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≥118, e≤54, R_{S5}≤2, I≤154, R_{S6}≤14, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−119, 10, 0, 0, 21, −3, 0, 0, 24, 11); minima (0, 0).

Identity: Q23 + 6·H3 + 24·H7 + 24·H8 + 4·H9 + 11·H10 = −4.

A27. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≥118, e≤54, R_{S5}≤2, I≤154, R_{S6}≥16, R_{S5}≤0, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−870, 100, 16, 0, 175, −38, 0, 0, 95, 0); minima (0, 0).

Identity: 48·Q21 + Q23 + 16·H2 + 76·H3 + 74·H7 + 40·H9 + 95·H11 = −48.

A28. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≥118, e≤54, R_{S5}≤2, I≤154, R_{S6}≥16, R_{S5}≥2, t≤118.

φ=(−420, 68, −3, 0, 126, −26, 0, 0, 0, 0); minima (0, 0).

ψ=(719, 179, 0, 0, 288, −96, 0, 0, −1134, −602); minima (0, 0).

Identity: 13·Q9 + 2·Q23 + 13·H1 + 397·H3 + 1204·H10 + 2268·H11 = −358.

A29. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≥118, e≤54, R_{S5}≤2, I≥156, N_{F5}≤6, N_{F6}≤0, t≤118.

φ=(−22, −5, 5, 0, −3, 0, 12, 6, 0, 0); minima (0, 3).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: Q22 + 5·H4 + 6·H6 + 21·H7 + 12·H10 + 6·H11 = −6.

A30. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≥118, e≤54, R_{S5}≤2, I≥156, N_{F5}≤6, N_{F6}≥2, t≤118.

φ=(−60, 6, 8, 0, 18, −7, 0, −6, 0, 0); minima (0, 3).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: Q22 + 14·H1 + 8·H4 + 6·H6 + 30·H7 + 6·H11 = −4.

A31. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≥118, e≤54, R_{S5}≤2, I≥156, N_{F5}≥8, t≤118.

φ=(0, −1, 1, 0, 4, −1, −5, 0, 0, 0); minima (0, 1).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 2·Q21 + Q22 + 2·H1 + H4 + H7 + 2·H9 + 5·H10 = −2.

A32. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≥118, e≤54, R_{S5}≥4, t≤118.

φ=(−580, 116, 29, 0, 174, −58, 0, 0, 0, 0); minima (0, 0).

ψ=(−105, −29, 0, −2, 66, 0, 0, 0, −60, 0); minima (0, 0).

Identity: Q22 + Q23 + 118·H1 + 29·H4 + 6·H6 + 221·H7 + 60·H8 = −4.

A33. cap F=2, S=10. Conditions: G≥2, G≤2, S≥10, S≤10, e≥54, t≥118, e≥56, t≤118.

φ=(−70, 13, 2, 0, 21, −6, 0, 0, 0, 0); minima (0, 3).

ψ=(0, −4, 4, 0, 0, −1, 0, 0, 0, 0); minima (0, 0).

Identity: 2·Q9 + 7·Q11 + 48·Q21 + Q22 + 7·H3 + 63·H7 = −60.

A34. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≤116, t≤118.

φ=(30, −9, 0, 0, −9, 3, 0, 0, 0, 0); minima (0, 0).

ψ=(−675, 51, 36, 0, 180, −42, 0, 0, 0, 0); minima (0, 0).

Identity: Q22 + Q23 + 42·H2 + 84·H3 + 108·H5 = −138.

A35. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≤0, t≤118.

φ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); minima (0, 0).

ψ=(−120, 10, 4, 0, 23, −5, 0, 0, 15, 0); minima (0, 0).

Identity: 12·Q21 + 4·Q22 + Q23 + 12·H2 + 10·H3 + 4·H6 + 15·H9 = −12.

A36. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≤52, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−684, 54, 26, 0, 195, −40, 0, 0, −21, 0); minima (0, 0).

Identity: 80·Q9 + 160·Q10 + 3·Q23 + 240·H1 + 78·H2 + 246·H5 + 63·H9 + 1656·H10 = −2154.

A37. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≤154, N_{F5}≤6, t≤118.

φ=(−28, 5, 1, 0, 6, −2, 3, 1, 0, 0); minima (0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: Q22 + 4·H1 + H4 + 3·H6 + H8 + 2·H11 + 3·H12 = −2.

A38. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≤154, N_{F5}≥8, R_{S5}≤2, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−48, 6, 0, 0, 11, −2, 0, 0, 3, 0); minima (0, 0).

Identity: Q23 + 4·H3 + 10·H6 + 2·H11 + 3·H13 = −2.

A39. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≤154, N_{F5}≥8, R_{S5}≥4, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−270, 24, 4, 0, 81, −14, 0, 0, −15, 0); minima (0, 0).

Identity: 12·Q21 + Q23 + 4·H2 + 28·H3 + 66·H6 + 15·H13 = −12.

A40. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≥156, N_{F5}≤6, R_{S6}≤14, R_{S5}≤2, t≤118.

φ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); minima (0, 0).

ψ=(−154, 13, 0, 2, 29, −5, 0, 0, 24, 13); minima (0, 0).

Identity: 6·Q21 + 3·Q22 + Q23 + 8·H2 + 10·H3 + 26·H6 + 13·H13 + 24·H14 = −6.

A41. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≥156, N_{F5}≤6, R_{S6}≤14, R_{S5}≥4, t≤118.

φ=(−16, 3, 0, 0, 4, −1, 1, 0, 0, 0); minima (0, 1).

ψ=(−63, 3, 2, 0, 24, −4, 0, 0, −12, 0); minima (0, 0).

Identity: 2·Q10 + Q23 + 8·H3 + 4·H4 + 36·H6 + 5·H11 + 12·H14 = −10.

A42. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≥156, N_{F5}≤6, R_{S6}≥16, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−7, 6, 0, 2, 7, −3, 0, 0, −18, −11); minima (0, 0).

Identity: 6·Q21 + Q23 + 2·H2 + 6·H3 + 18·H9 + 6·H10 + 11·H13 = −6.

A43. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≥156, N_{F5}≥8, R_{S6}≤14, R_{S5}≤2, t≤118.

φ=(2, −1, 0, 0, 1, 0, −2, 0, 0, 0); minima (0, 1).

ψ=(−36, 3, 0, 0, 7, −1, 0, 0, 6, 3); minima (0, 0).

Identity: 8·Q21 + 4·Q22 + Q23 + 2·H3 + 3·H11 + 8·H12 + 3·H13 + 6·H14 = −2.

A44. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≥156, N_{F5}≥8, R_{S6}≤14, R_{S5}≥4, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−270, 24, 4, 0, 81, −14, 0, 0, −15, 0); minima (0, 0).

Identity: 12·Q21 + Q23 + 4·H2 + 28·H3 + 66·H6 + 15·H14 = −12.

A45. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≤0, R_{S5}≥2, e≥54, I≥156, N_{F5}≥8, R_{S6}≥16, t≤118.

φ=(2, −1, 0, 0, 1, 0, −2, 0, 0, 0); minima (0, 1).

ψ=(−12, 3, 0, 0, 4, −1, 0, 0, −3, −3); minima (0, 0).

Identity: 2·Q21 + Q22 + Q23 + 2·H3 + 3·H9 + 2·H12 + 3·H13 = −2.

A46. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≤154, N_{F5}≤6, R_{S5}≤2, t≤118.

φ=(−16, 3, 0, 0, 4, −1, 1, 0, 0, 0); minima (0, 1).

ψ=(−48, 6, 0, 0, 11, −2, 0, 0, 3, 0); minima (0, 0).

Identity: Q23 + 4·H3 + 10·H6 + 2·H9 + 3·H11 = −2.

A47. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≤154, N_{F5}≤6, R_{S5}≥4, t≤118.

φ=(−16, 3, 0, 0, 4, −1, 1, 0, 0, 0); minima (0, 1).

ψ=(−63, 3, 2, 0, 24, −4, 0, 0, −12, 0); minima (0, 0).

Identity: 12·Q9 + 60·Q21 + 3·Q22 + 9·Q23 + 84·H3 + 156·H6 + 3·H10 + 108·H11 = −108.

A48. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≤154, N_{F5}≥8, t≤118.

φ=(−34, 5, 2, 0, 15, −4, −6, −8, 0, 0); minima (0, 3).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: Q9 + 12·Q21 + 2·Q22 + 17·H1 + 3·H4 + 16·H8 + 12·H10 = −12.

A49. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≥156, R_{S5}≤0, t≤118.

φ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); minima (0, 0).

ψ=(−120, 10, 4, 0, 23, −5, 0, 0, 15, 0); minima (0, 0).

Identity: 12·Q21 + 4·Q22 + Q23 + 12·H2 + 10·H3 + 4·H6 + 15·H10 = −12.

A50. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≥156, R_{S5}≥2, R_{S6}≤14, N_{F5}≤4, R_{S5}≤2, t≤118.

φ=(4, −3, 0, 0, −2, 1, 1, 0, 0, 0); minima (0, 2).

ψ=(−140, 10, 0, 0, 24, −3, 0, 0, 30, 14); minima (0, 0).

Identity: 8·Q21 + 4·Q22 + Q23 + 8·H2 + 6·H3 + 22·H6 + 14·H11 + 4·H12 + 30·H13 = −8.

A51. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≥156, R_{S5}≥2, R_{S6}≤14, N_{F5}≤4, R_{S5}≥4, t≤118.

φ=(−48, 9, 0, 0, 12, −3, 3, 0, 0, 0); minima (0, 3).

ψ=(−135, 3, 8, 0, 54, −10, 0, 0, −30, 0); minima (0, 0).

Identity: 8·Q10 + Q23 + 20·H3 + 16·H4 + 96·H6 + 17·H9 + 30·H13 = −10.

A52. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≥156, R_{S5}≥2, R_{S6}≤14, N_{F5}≥6, R_{S5}≤2, t≤118.

φ=(2, −1, 0, 0, 1, 0, −2, −2, 0, 0); minima (0, 1).

ψ=(−140, 10, 0, 0, 24, −3, 0, 0, 30, 14); minima (0, 0).

Identity: 30·Q21 + 15·Q22 + Q23 + 6·H3 + 30·H8 + 11·H9 + 14·H11 + 30·H12 + 30·H13 = −8.

A53. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≥156, R_{S5}≥2, R_{S6}≤14, N_{F5}≥6, R_{S5}≥4, t≤118.

φ=(−420, 68, −3, 0, 126, −26, 0, −29, 0, 0); minima (0, 0).

ψ=(−5166, −98, 504, 0, 2226, −455, 0, 0, −1414, 0); minima (0, 0).

Identity: 1512·Q21 + Q23 + 504·H2 + 910·H3 + 210·H6 + 504·H9 + 1414·H13 = −1820.

A54. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≤8, N_{F6}≥2, I≥156, R_{S5}≥2, R_{S6}≥16, t≤118.

φ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); minima (0, 0).

ψ=(−50, 11, 2, 0, 18, −5, 0, 0, −15, −13); minima (0, 0).

Identity: 6·Q21 + 3·Q22 + Q23 + 8·H2 + 10·H3 + 15·H10 + 13·H11 = −6.

A55. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≤54, N_{F5}≥10, t≤118.

φ=(2, −1, 0, 0, 1, 0, −2, 0, 0, 0); minima (0, 1).

ψ=(−45, 5, 0, 0, 12, −2, 0, 0, 0, 0); minima (0, 0).

Identity: 11·Q9 + 4·Q10 + 36·Q21 + 18·Q22 + 15·H1 + 3·H3 + 12·H5 + 36·H7 = −12.

A56. cap F=2, S=12. Conditions: G≥2, G≤2, S≥12, S≤12, t≥118, e≥56, t≤118.

φ=(−420, 68, −3, 0, 126, −26, 0, 0, 0, 0); minima (0, 0).

ψ=(−1020, 16, 120, 0, 272, −81, 0, 0, 0, 0); minima (0, 0).

Identity: 13·Q11 + 174·Q21 + Q22 + 26·H1 + 16·H3 + 222·H6 = −276.

A57. cap F=1000, S=1000. Conditions: G≥22, S≥0, t≤118.

φ=(−1357665, −152627, 81764, 0, 381540, −30227, 0, 0, 0, 0); minima (0, 0).

ψ=(−196845, −8351, −2386, 0, 52492, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 7927265092·Q9 + 54317336532·Q21 + 96953·Q22 + 290859·Q23 + 14482451328·H1 = −54317336532.

A58. cap F=20, S=1000. Conditions: G≥12, G≤20, S≥0, t≤118.

φ=(−24460, 69, 858, 0, 7295, −1047, 0, 0, 0, 0); minima (0, 0).

ψ=(−10065, −427, −122, 0, 2684, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 3564132·Q9 + 32991068·Q21 + 4154·Q22 + 4154·Q23 + 12769396·H1 = −32991068.

A59. cap F=10, S=30. Conditions: G≥0, G≤10, S≥20, S≤30, t≤118.

φ=(−700, 70, 0, 0, 210, −35, 0, 0, 0, 0); minima (0, 0).

ψ=(−3780, −124, 234, 0, 1008, −149, 0, 0, 0, 0); minima (0, 0).

Identity: 94·Q9 + 1122·Q21 + 2·Q22 + Q23 + 392·H3 = −944.

A60. cap F=10, S=1000. Conditions: G≥0, G≤10, S≥32, t≤118.

φ=(−65875, 10637, 0, 0, 16870, −3583, 0, 0, 0, 0); minima (0, 0).

ψ=(−134475, −20009, 12674, 0, 35860, −3576, 0, 0, 0, 0); minima (0, 0).

Identity: 5508·Q9 + 50264·Q21 + Q22 + Q23 + 12660·H3 = −50264.

A61. cap F=10, S=12. Conditions: G≥6, G≤10, S≥0, S≤12, t≤118.

φ=(−160, 8, 10, 0, 48, −11, 0, 0, 0, 0); minima (0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 2·Q9 + 30·Q21 + Q22 + 24·H1 + 8·H4 = −8.

A62. cap F=8, S=18. Conditions: G≥0, G≤8, S≥16, S≤18, t≤118.

φ=(−90, 11, 0, 0, 27, −5, 0, 0, 0, 0); minima (0, 0).

ψ=(−3645, 53, 190, 0, 972, −169, 0, 0, 0, 0); minima (0, 0).

Identity: 40·Q9 + 930·Q21 + 15·Q22 + Q23 + 378·H3 = −6.

A63. cap F=0, S=8. Conditions: G≥0, G≤0, S≥0, S≤8, A=0, t≤118.

φ=(−410, 123, 0, 0, 123, −41, 0, 0, 0, 0); minima (0, 0).

ψ=(−540, 68, 22, 0, 144, −37, 0, 0, 0, 0); minima (0, 0).

Identity: 123·Q7 + 2·Q22 + 41·H4 + 41·H6 = −82.

A64. cap F=4, S=8. Conditions: G≥4, G≤4, S≥0, S≤8, t≤118.

φ=(−1700, 296, 19, 0, 510, −126, 0, 0, 0, 0); minima (0, 6).

ψ=(0, −42, −1, 0, 0, 11, 0, 0, 0, 0); minima (0, 0).

Identity: 255·Q21 + Q22 + 3·Q23 + 255·H1 + 85·H4 = −170.

A65. cap F=10, S=16. Conditions: G≥6, G≤10, S≥14, S≤16, t≤118.

φ=(−2560, 232, 30, 0, 768, −137, 0, 0, 0, 0); minima (0, 0).

ψ=(−330, 26, 0, 0, 88, −12, 0, 0, 0, 0); minima (0, 0).

Identity: 714·Q21 + Q22 + 6·Q23 + 274·H1 + 114·H3 = −36.

A66. cap F=0, S=14. Conditions: G≥0, G≤0, S≥12, S≤14, A=0, t≤118.

φ=(490, −147, 0, 0, −147, 49, 0, 0, 0, 0); minima (0, 0).

ψ=(−360, 16, 22, 0, 96, −21, 0, 0, 0, 0); minima (0, 0).

Identity: 49·Q9 + 2·Q22 + 147·H2 + 49·H3 = −98.

A67. cap F=2, S=2. Conditions: G≥2, G≤2, S≥0, S≤2, t≤118.

φ=(−1700, 296, 19, 0, 510, −126, 0, 0, 0, 0); minima (0, 132).

ψ=(0, −42, −1, 0, 0, 11, 0, 0, 0, 0); minima (0, 0).

Identity: 255·Q21 + Q22 + 3·Q23 + 255·H1 + 85·H4 = −170.

A68. cap F=2, S=14. Conditions: G≥0, G≤2, S≥14, S≤14, t≤118.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(−360, 16, 22, 0, 96, −21, 0, 0, 0, 0); minima (0, 0).

Identity: 2·Q9 + 66·Q21 + Q23 + 20·H2 + 44·H3 = −16.

A69. cap F=10, S=18. Conditions: G≥0, G≤10, S≥18, S≤18, t≤118.

φ=(−20, 2, 0, 0, 6, −1, 0, 0, 0, 0); minima (0, 0).

ψ=(−810, 14, 40, 0, 216, −37, 0, 0, 0, 0); minima (0, 0).

Identity: 10·Q9 + 210·Q21 + 15·Q22 + Q23 + 84·H3 = −156.

B1. cap F=1000, S=1000. Conditions: G≥0, S≥0, e≥62, t≤118.

φ=(−375, −11, −7, 0, 100, 0, 0, 0, 0, 0); minima (0, 0).

ψ=(1500, −622, 172, 0, −450, 107, 0, 0, 0, 0); minima (0, 0).

Identity: 25·Q8 + 12·Q9 + 6·Q11 + 236·Q21 + Q22 + 275·H3 = −206.

B2. cap F=1000, S=1000. Conditions: G≥0, S≥0, r≥62, t≤118.

φ=(1500, −622, 172, 0, −450, 107, 0, 0, 0, 0); minima (0, 0).

ψ=(−375, −11, −7, 0, 100, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 172·Q9 + 1716·Q21 + Q22 + 6·Q23 + 2100·H3 = −1536.

B3. cap F=1000, S=1000. Conditions: G≥12, S≥0, t≤118.

φ=(−609000, −99416, 48260, 0, 182700, −14501, 0, 0, 0, 0); minima (0, 0).

ψ=(−34875, −2154, −573, 0, 10275, 0, 0, 0, 0, 0); minima (0, 0).

Identity: 2239215740·Q9 + 12438643920·Q21 + 46399·Q22 + 278394·Q23 + 3744399300·H1 = −12438643920.

B4. cap F=10, S=1000. Conditions: G≥0, G≤10, S≥20, t≤118.

φ=(−122375, 22270, 0, 0, 28905, −6675, 0, 0, 0, 0); minima (0, 0).

ψ=(−177750, −34723, 19792, 0, 53325, −5659, 0, 0, 0, 0); minima (0, 0).

Identity: 39118·Q9 + 290784·Q21 + 3·Q22 + 4·Q23 + 84390·H3 = −290784.

B5. cap F=10, S=18. Conditions: G≥0, G≤10, S≥6, S≤18, t≤118.

φ=(−90, 7, 0, 0, 27, −4, 0, 0, 0, 0); minima (0, 0).

ψ=(−810, −25, 52, 0, 243, −40, 0, 0, 0, 0); minima (0, 0).

Identity: 56·Q9 + 492·Q21 + 6·Q22 + 2·Q23 + 216·H3 = −408.

B6. cap F=10, S=4. Conditions: G≥0, G≤10, S≥2, S≤4, t≤118.

φ=(−180, 14, 0, 0, 54, −8, 0, 0, 0, 0); minima (0, 0).

ψ=(−1080, 108, 34, 0, 324, −71, 0, 0, 0, 0); minima (0, 74).

Identity: 2·Q9 + 222·Q21 + 2·Q22 + Q23 + 144·H3 = −120.

B7. cap F=10, S=0. Conditions: G≥0, G≤10, S≥0, S≤0, A=0, t≤118.

φ=(−810, 27, 54, 0, 243, −54, 0, 0, 0, 0); minima (0, 0).

ψ=(−2430, 729, 0, 0, 729, −243, 0, 0, 0, 0); minima (0, 0).

Identity: 324·Q8 + 162·Q21 + Q22 + Q23 + 108·H6 = −972.

25 Two-facet stress localization

Lemma 25.1. *If F,H are different facets and H≠−F, then 2g(F)+2g(H)≤G.* The polygon fans chosen in Section 11 are used throughout, so every common face has the same bar set in the two containing facets. For a three-dimensional face T, its triangulated graph has 3n−6 bars: summing polygon vertex counts gives 2e, and Euler gives e+(2e−3p)=3n−6. Its rigidity rank is also 3n−6 by Section 10. It has no nonzero stress. A polygon fan has 2n−3 bars and rigidity rank 2n−3, so it too has no stress. The same fact holds for edges and vertices.

Extend stresses on F,H by zero and antisymmetrize, as in (12.2). Suppose a linear combination of these two antisymmetric stresses vanishes. Then the stress on F is supported on the union of the bar sets of F∩H and F∩(−H). These intersections are faces of dimension at most three, and their vertex sets are disjoint because H has no antipodal vertices. The equilibrium equations therefore separate into a stress on each intersection. Both stresses vanish by the preceding bar counts and rigidity. The stress on F is zero, and restriction to H similarly makes the other stress zero. Thus the direct sum of the two facet stress spaces injects into the global antisymmetric stress space. Its dimension is G/2, proving the lemma. □

Corollary 25.2. *If some facet has g≥j, at most two facets have 2g>G−2j.* Any facet other than that facet and its antipode satisfies 2g≤G−2j by Lemma 25.1. This proves the count bound. If instead every facet has 2g≤τ, the number with 2g>τ is zero. These are the two ways to justify threshold counts below. □

In particular a row cost may add a term $a_g[2g>\tau]$. Its sum adds a_gK, where K is the number of exceptional rows. When the preceding corollary applies, 0≤K≤2 is an additional pair of valid linear inequalities. One must retain K as a variable in the contradiction; the sign of a_g cannot be ignored. The argument allows two nonantipodal facets, and asserts no unrestricted additivity for three or more facets.

26 The ridge capacity of full facets

A facet is full when it contains exactly one vertex from every antipodal pair. Suppose v=16, and let a be the number of full facets. Define R to be the sum of their numbers of ridges. Then

$$2R \le 2a(a-2)+a(h-a). \tag{26.1}$$

To prove this, associate to a ridge of a fixed facet the other facet containing it. This association is injective: if two facets shared two different ridges, their intersection would be a face containing both and would have to be the entire first facet, which is impossible. A full facet has at most a−2 full partners, excluding itself and its antipode. Summing these directed incidences gives at most a(a−2).

A nonfull facet contains at most seven vertices. For a full antipodal pair F,−F, it cannot meet both members in ridges. Each ridge has dimension three and hence at least four vertices, while F and −F have disjoint vertex sets. Two such ridges would require at least eight vertices. Thus each nonfull facet has at most a/2 full ridge partners. Their directed contribution to R is at most (h−a)a/2. Adding the two bounds proves (26.1).

Write a=2k. The useful form, and its numerical instances, are

$$R \le k(h+2k-4). \tag{26.2}$$

h	k	Upper bound on R
16	1, 2, 3	14, 32, 54
20	7, 8, 9, 10	210, 256, 306, 360

A full-row cost $b_0[\,n=8\,]+b_1r_0[\,n=8\,]$ therefore sums to b_0a+b_1R, with a=2k and the bound (26.2). This supplies the geometric inequalities for the corresponding sixteen-vertex certificates.

27 A four-dimensional bound needed for fibre splitting

Lemma 27.1. *Every centrally symmetric four-polytope has at least 81 nonempty faces.* Here is a proof using the rigidity and cube-section arguments already developed, so no separate dimension-four result is needed. Write its face counts as v,e,p,h and its vertex–facet incidences as I. Euler gives $v-e+p-h=0$, and hence $s=1+2(e+h)$.

If $v\le10$, the polar embeds as a central hyperplane section of a five-cube. The three-state orbit argument of Section 4, with five coordinates, gives at least $3^5-2\cdot3^4=81$ faces. Polarity also handles $h\le10$. Otherwise $v,h\ge12$ and, after polarity if necessary, $v\le h$. Minimum vertex degree four gives $e\ge2v$. Its polar gives $p\ge2h$; with Euler this implies $e\ge v+h$. If $s<81$, parity gives $e+h\le38$, so $v+2h\le38$. The only remaining even pair with $v\le h$ and $v,h\ge12$ is $v=h=12$.

It remains to exclude that pair. The three-dimensional vertex figures give $f_{02}=I+2e-2v$. The symmetric rigidity calculation in dimension four gives $M\ge4v-8$, where $M=e+f_{02}-3p$. Substituting Euler gives

$$I\ge3(v+h)-8. \tag{27.1}$$

Minimum degree three in every facet gives $f_{13}\ge3I/2$. Applying the vertex-figure identity to the polar gives $f_{13}=I+2p-2h$, so $p\ge h+I/4$. At $v=h=12$ Euler gives $p=e$, and (27.1) gives $I\ge64$. Therefore $e+h\ge2h+I/4\ge40$, contrary to $e+h\le38$. This proves the lemma. □

For centrally symmetric polytopes in complementary subspaces, the free sum $\operatorname{conv}(Q\cup R)$ is polar to $Q^\circ\times R^\circ$. Faces of a product are products of faces, including either whole factor, so $s(\operatorname{conv}(Q\cup R))=s(Q)s(R)$. In particular splitting off a segment multiplies the face count by three.

28 Full normals and the rank-three reduction

Normalize every facet functional to support value one. For a full facet, its values on all vertices are ± 1. If the full functionals span a space of dimension d, there are at most 2^d of them: choose d vertex evaluations forming a basis of the dual of that space. Those evaluations are injective on the space, and a full functional has only two possible values at each chosen vertex. Vertex evaluations separate functionals because the vertices span the ambient space.

Thus at least eight full facets force rank at least three, and more than eight force rank at least four. In the rank-three case there are exactly eight. Choose three independent full functionals as the first three coordinates. Every vertex projects to $\{\pm 1\}^3$. There are four antipodal classes with representatives $(1,1,1),(1,1,-1),(1,-1,1),(1,-1,-1)$. If all four occur, a further full functional within their span has coordinates c satisfying $c_1 \pm c_2 \pm c_3 \in \{-1,1\}$ for all four choices. Taking differences, or solving the sixteen sign choices, forces $c = \pm e_1, \pm e_2, \pm e_3$. That would give only six full functionals. Consequently one class is absent. At least three independent classes occur, since the projection has rank three. An invertible change of the three coordinates sends these classes to $\pm e_1, \pm e_2, \pm e_3$.

Completing the coordinates gives the normal form

$$P = \operatorname{conv} \cup_{i=1}^{3} (\{e_i\} \times K_i \cup \{-e_i\} \times (-K_i)), \quad K_i \subset \mathbb{R}^2. \tag{28.1}$$

Each K_i is a point, segment or convex polygon whose vertices are exactly the corresponding vertices of P. Their differences span the fibre plane; otherwise P would have dimension at most four. If a fibre is a point, shear its coordinates to zero. Its antipodal pair splits off as a segment, and the other two fibre pairs form a centrally symmetric four-polytope. Lemma 27.1 then gives $s(P) \geq 3 \cdot 81 = 243$.

29 Fourteen vertices in the rank-three case

For fourteen vertices the three fibre sizes sum to seven. If none is a point they are 2,2,3. Thus K_1,K_2 are segments and K_3 is a triangle T. This paragraph proves the full rank-three branch, including all relative positions of the segments and triangle.

There are exactly eight full facets, each with seven vertices. Let n_5,n_6 be the numbers of the other facets with five and six vertices. Then $n_5+n_6=h-8$ and $I=56+5n_5+6n_6=5h+16+n_6$. Equation (3.6), nonnegativity and even I give $I\geq 4(14+h)-12$, hence

$$n_6\geq 28-h. \tag{29.1}$$

Assume first that the two segment directions are independent. The fibres and their antipodes contain $2(1+1+3)=10$ edges. The inverse image of an octahedral edge joining the two segment classes is a tetrahedron, giving four cross edges for each of four sign choices. Each segment/triangle inverse image is a three-polytope with five vertices, hence at least eight edges. After subtracting its four fibre edges, at least four cross edges remain. The eight segment/triangle sign choices therefore give 32 more. These edges have different projections or lie in different fibres, so $e\geq 58$.

Every full facet is a Cayley polytope of the three selected fibres over a triangle. It has three vertical ridges. In addition, each distinct outward edge-normal direction of any selected fibre gives a ridge by choosing the supporting faces of all three fibres. The two segments give four directions, and the triangle three. At most two coincide, because triangle edge normals have no antipodal pair. Thus each full facet has at least eight ridges. The nonfull facets have at least five or six ridges according to their vertex counts. Using (29.1),

$$\begin{aligned} 2r&\geq 8\cdot 8+5(h-8)+n_6\geq 4h+52,\\ t&\geq 58+2h+26=2h+84\geq 120 \quad (h\geq 18). \end{aligned} \tag{29.2}$$

29 The parallel segments, continued

Suppose the two segments are parallel and no triangle edge is parallel to them. Their four octahedral edge fibres are quadrilaterals, with two cross edges each. A segment/triangle edge fibre is now a simplicial three-polytope with five vertices: a four-vertex coplanar facet would contain the segment and a parallel triangle edge, which is excluded. It has nine edges, five of them cross edges. Thus $e \geq 10+8+8\cdot 5=58$.

A full facet has two vertical ridges, those between the triangle and the segments. Its nonvertical ridges come from the two segment normals and three triangle normals, all different, giving at least seven ridges. We also need $h \geq 20$. At a common segment normal, both segment widths vanish and the triangle width is positive. Choosing each segment sign freely and both supporting levels of the triangle gives eight nonfull facets, or four antipodal pairs. At a triangle edge normal, give both signs to the fibre classes of largest width and one chosen sign to the other classes. This exposes a facet: the projected face has dimension three and the triangle support edge adds one fibre dimension. If only a minimum edge is exposed, choose the negative triangle sign. At least two distinct triangle normal lines give further pairs; their fibre components are not proportional, so their facet functionals cannot define the same pair. Together with the four full pairs this gives $h \geq 2(4+4+2)=20$. Now (29.1) gives

$$\begin{aligned} 2r &\geq 8\cdot 7+5(h-8)+n_6 \geq 4h+44, \\ t &\geq 58+2h+22=2h+80 \geq 120. \end{aligned} \tag{29.3}$$

Finally suppose the segment direction is parallel to an edge of T. Shears and an invertible change of the fibre plane put the fibres in the form $K_1=[-A,A]\times\{0\}$, $K_2=[-B,B]\times\{0\}$, $T=\operatorname{conv}\{(-1,0),(1,0),(0,1)\}$, with $A,B>0$. Replace the coordinates (q_3,y) by $(u,w)=(q_3-y,y)$. The two triangle apices become the pair $\pm e_w$; all other vertices have $w=0$. They span a centrally symmetric four-polytope. Hence P is its free sum with a segment, and Lemma 27.1 gives $s(P) \geq 243$. This covers the last parallel case.

At $h=16$ the rank-three case is impossible already by (29.1): $n_6 \geq 12$ while $n_5+n_6=8$. Thus all fourteen-vertex rank-three branches are now proved. Rank at least four requires the separate lifting enumeration.

30 Four full normals: reduction covering every real lift

A full facet contains exactly one vertex from each antipodal pair. Its normalized normal u satisfies u(x)=±1 at every vertex. Choose four linearly independent full normals as the first four coordinates, and complete them to linear coordinates in dimension five. For m=7 or 8 antipodal pairs, write the vertices as

$$\pm y_i=(\pm q_i,\pm h_i),\ q_i\in\{-1,1\}^4,\ 0\le i<m. \tag{30.1}$$

Change a representative to its negative when necessary, so its first base coordinate is one. There are eight base classes $(1,(-1)^{b_0},(-1)^{b_1},(-1)^{b_2})$, indexed by $b_0+2b_1+4b_2$. A class cannot occur three times: among three points on its vertical line, one lies between the other two and cannot be a vertex. Sorting the m class indices therefore gives exactly the nondecreasing sequences with no entry equal to the entry two places later. Direct enumeration gives 1,016 sequences for m=7 and 1,107 for m=8. This argument includes repeated bases and all coincidences among the remaining real heights.

For each sequence c the reduction uses one of two integer identities. Either a nonzero vector n satisfies $n\cdot q_i=0$ for every i, which contradicts full dimension, or an invertible rational map sends its signed, permuted base list onto a listed family. The latter identity has the form

$$BA=eI_4,\ de\neq0;\quad A(\varepsilon_i q_{\tau(i)})=dQ_i,\ \varepsilon_i\in\{-1,1\},\ \tau \text{ a permutation}. \tag{30.2}$$

Equation (30.2) is an invertible linear change on the base coordinates. Next choose four independent base points Q_b. The inverse of their matrix allows a shear $h\mapsto h-l(q)$ making their heights zero. The remaining m−4 heights are arbitrary real parameters a. Thus every realization becomes a listed family

$$y_i(a)=(Q_i,\zeta_i\cdot a),\ a\in\mathbb{R}^{m-4}. \tag{30.3}$$

The basis inverse is checked as an integer matrix identity $MN=DI_4$, $D\neq0$. The four basis height forms are zero and the free height forms are the coordinate unit vectors. The zero parameter is excluded because all vertices would then lie in a four-dimensional subspace. Neither generic positions nor rational coordinates are assumed.

ID	Base class list	Sequences reduced	Forms / records
14.1	7,3,5,6,7,3,5	232	9 / 308
14.2	7,3,5,1,6,7,3	288	16 / 304
14.3	7,3,5,1,6,7,6	192	16 / 260
14.4	7,3,5,6,0,7,3	32	22 / 232
14.5	7,3,5,6,0,3,5	48	24 / 224
14.6	7,3,5,1,6,2,7	96	20 / 290
14.7	7,3,5,1,6,0,7	24	23 / 242

ID	Base class list	Sequences reduced	Forms / records
14.8	7,3,5,1,6,0,3	48	30 / 230
14.9	7,3,5,1,6,2,4	8	33 / 280

The 9 families cover 968 spanning sequences; 48 sequences are excluded by a displayed nonzero annihilator in the integer reduction data. The ordering of the bases in this table is retained when attaching free heights; equal class multisets alone are not used to identify families.

ID	Base class list	Sequences reduced	Forms / records
16.1	7,3,5,6,7,3,5,6	32	16 / 365
16.2	7,3,5,2,7,3,5,2	24	16 / 351
16.3	7,1,2,4,7,1,2,4	2	16 / 369
16.4	7,3,5,1,6,7,3,5	48	41 / 493
16.5	7,3,5,1,6,7,3,1	48	41 / 492
16.6	7,3,5,1,6,7,3,6	48	37 / 575
16.7	7,3,5,1,6,7,5,1	48	41 / 489
16.8	7,3,5,1,6,7,5,6	48	36 / 540
16.9	7,3,5,1,6,7,1,6	48	37 / 608
16.10	7,3,5,1,6,3,5,1	48	41 / 487
16.11	7,3,5,1,6,3,5,6	48	37 / 543
16.12	7,3,5,1,6,3,1,6	48	37 / 569
16.13	7,3,5,1,6,5,1,6	48	37 / 582
16.14	7,3,5,6,0,7,3,5	8	61 / 550
16.15	7,3,5,6,0,7,3,6	8	60 / 526
16.16	7,3,5,6,0,7,3,0	8	61 / 628
16.17	7,3,5,6,0,7,5,6	8	61 / 575
16.18	7,3,5,6,0,7,5,0	8	61 / 672
16.19	7,3,5,6,0,7,6,0	8	61 / 592
16.20	7,3,5,6,0,3,5,6	8	65 / 644

ID	Base class list	Sequences reduced	Forms / records
16.21	7,3,5,6,0,3,5,0	8	65 / 621
16.22	7,3,5,6,0,3,6,0	8	65 / 653
16.23	7,3,5,6,0,5,6,0	8	65 / 663
16.24	7,3,5,1,6,2,7,3	16	46 / 565
16.25	7,3,5,1,6,2,7,5	16	48 / 490
16.26	7,3,5,1,6,2,7,1	16	51 / 541
16.27	7,3,5,1,6,2,7,6	16	49 / 508
16.28	7,3,5,1,6,2,7,2	16	52 / 597
16.29	7,3,5,1,6,2,3,5	16	51 / 510
16.30	7,3,5,1,6,2,3,1	16	52 / 549
16.31	7,3,5,1,6,2,3,6	16	51 / 528
16.32	7,3,5,1,6,2,3,2	16	51 / 553
16.33	7,3,5,1,6,2,5,1	16	48 / 602
16.34	7,3,5,1,6,2,5,6	16	46 / 530
16.35	7,3,5,1,6,2,5,2	16	52 / 605
16.36	7,3,5,1,6,2,1,6	16	51 / 522
16.37	7,3,5,1,6,2,1,2	16	51 / 614
16.38	7,3,5,1,6,2,6,2	16	47 / 571
16.39	7,3,5,1,6,0,7,3	12	57 / 475
16.40	7,3,5,1,6,0,7,5	12	62 / 500
16.41	7,3,5,1,6,0,7,1	12	54 / 631
16.42	7,3,5,1,6,0,7,6	12	60 / 469
16.43	7,3,5,1,6,0,7,0	12	68 / 652
16.44	7,3,5,1,6,0,3,5	12	77 / 607
16.45	7,3,5,1,6,0,3,1	12	55 / 476
16.46	7,3,5,1,6,0,3,6	12	84 / 698
16.47	7,3,5,1,6,0,3,0	12	83 / 686

ID	Base class list	Sequences reduced	Forms / records
16.48	7,3,5,1,6,0,5,1	12	63 / 502
16.49	7,3,5,1,6,0,5,6	12	84 / 663
16.50	7,3,5,1,6,0,5,0	12	83 / 603
16.51	7,3,5,1,6,0,1,6	12	68 / 650
16.52	7,3,5,1,6,0,1,0	12	68 / 614
16.53	7,3,5,1,6,0,6,0	12	81 / 678
16.54	7,3,5,1,6,2,4,7	8	82 / 575
16.55	7,3,5,1,6,2,4,3	8	75 / 537
16.56	7,3,5,1,6,2,4,5	8	76 / 502
16.57	7,3,5,1,6,2,4,1	8	85 / 623
16.58	7,3,5,1,6,2,4,6	8	75 / 554
16.59	7,3,5,1,6,2,4,2	8	91 / 689
16.60	7,3,5,1,6,2,4,4	8	87 / 630
16.61	7,3,5,1,6,2,4,0	1	115 / 748

The 61 families cover 1095 spanning sequences; 12 sequences are excluded by a displayed nonzero annihilator in the integer reduction data. The ordering of the bases in this table is retained when attaching free heights; equal class multisets alone are not used to identify families.

31 Why the finite sign certificates prove a continuous statement

Here are the complete mathematical rules of the calculation. A plane form is $l_p(a)=L_p\cdot a$ with integer coefficients. Every branch separates $l_p<0$, $l_p=0$ and $l_p>0$. All three children are checked. This is a partition of all real parameter space, including every wall and lower-dimensional intersection.

A supporting-function record is an affine function

$$\varphi_a(q,h)=\sum_{j=1}^{4}(C_j\cdot a)q_j+c_4h+C_5\cdot a. \tag{31.1}$$

All its values $\varphi_a(y_w)$ are first verified as exact integer vector identities: each is zero or a recorded integer multiple of one plane form. Thus the signs along a branch determine whether φ has the same weak sign at every vertex. If it does, its zero set is an exposed face and its zero vertices give its vertex mask. In the fourteen-vertex calculation this alone suffices; the supporting faces need not be facets.

In the sixteen-vertex calculation an upper facet is certified by five zero vertices T. Four of their base vectors are independent; an integer dependency $\sum\lambda_iQ_{T_i}=0$, $\lambda_k\neq0$, has nonzero height $\sum\lambda_i\zeta_{T_i}\cdot a$ on the branch. Hence the five vectors y_{T_i} span dimension five. Since they lie in a supporting hyperplane with nonzero constant, their affine span has dimension four. The sign of $C_5\cdot a$ orients (31.1); requiring the resulting last normal coordinate positive makes this an upper facet. Its antipode is a distinct lower facet.

Vertical facets use an integer inequality $n\cdot Q\le d$, $d>0$. The tight base mask is checked by direct dot products. If it is full, it contains one of each vertex pair and spans the facet hyperplane: their linear span is that of all the vertices. Otherwise the data identify two distinct tight points with equal base, and four independent tight base vectors. The height difference is nonzero because the vertices are distinct; together these again span dimension five, hence a facet.

For face-count terminals, let $g_1,\dots,g_s$ be certified supporting masks. For each proposed nonzero mask M, check

$$M=\cap\{g_i:\ M\subseteq g_i\}. \tag{31.2}$$

The intersection of nonempty exposed faces is exposed: normalize their support inequalities and add them; equality in the sum holds exactly when equality holds in each summand. Its vertices are exactly the intersection of the vertex masks. Distinct masks therefore give distinct faces. For fourteen vertices the list contains at least 243 masks, with the whole-polytope mask permitted. For sixteen vertices at least 242 distinct proper nonempty masks are supplied; adding the polytope gives 243.

There are four other terminal rules. A nonnegative linear combination of oriented strict inequalities, with at least one positive coefficient, cannot be the zero form. Such a record excludes its branch. A nonzero determinant of vanishing plane forms forces $a=0$ and excludes full dimension. If two equal bases have a height difference equal to a combination of vanishing forms, their vertices coincide and the branch is invalid. Finally, a full upper facet yields a fifth independent full normal, since its last coordinate is positive; the five-normal calculation below applies.

$$\sum c_i\sigma_i L_{p_i}=0,\ c_i\geq 0 \text{ for } \sigma_i\neq 0,\ \text{some } c_i>0 \text{ with } \sigma_i\neq 0. \tag{31.3}$$

For vanishing facts take $\sigma=1$ and allow signed coefficients. An implied sign uses the same combination equal to $D\sigma L_p$, $D>0$, rather than zero. An implied equality uses only vanishing facts and $D\neq 0$. Shared subtrees must have exactly the same hypotheses and cannot refer back to themselves. These rules justify induction from every terminal to the root of every family.

Facet-count terminals at sixteen vertices certify V vertical facets and N distinct upper facets, with $V+2N\geq 21$. Their antipodes are lower facets, and neither is vertical. This contradicts $h\leq 20$. Thus every admissible real lift either has at least 21 facets, has at least 243 nonempty faces, or has a fifth independent full normal. At fourteen vertices every admissible real lift directly has at least 243 nonempty faces.

m	Three-way branches	Face/facet terminals	Other terminals
7	1666	860 / 0	full: 0; coincident: 22; zero: 53; refuted: 2406
8	33863	63803 / 448	full: 2073; coincident: 1455; zero: 8; refuted: 0

These are counts of checks, not sampled real polytopes. The supplement contains only integer vectors, rational-map identities and finite sign trees. The independent Python calculation checks every identity above using ordinary rational arithmetic. The proof of continuous coverage is the normalization and three-way induction just given, rather than the output of an optimizer or checks of generic sample coordinates.

32 Five full normals: a finite sign-vector calculation

With five independent full normals the vertices become cube sign vectors in $\{-1,1\}^5$. For sixteen vertices there are eight distinct antipodal classes, chosen from the sixteen classes having first coordinate one. Thus there are C(16,8)=12,870 subsets. Coordinate permutations and coordinate sign changes act invertibly, preserving all faces. They reduce the list to the following 25 representatives.

Representative	**Eight classes**	**Certified nonempty faces**
1	0,1,2,3,4,5,6,7	not spanning
2	0,1,2,3,4,5,6,8	279
3	0,1,2,3,4,5,6,9	279
4	0,1,2,3,4,5,8,9	243
5	0,1,2,3,4,5,8,10	259
6	0,1,2,3,4,5,8,11	315
7	0,1,2,3,4,5,8,14	331
8	0,1,2,3,4,5,8,15	367
9	0,1,2,3,4,5,10,11	243
10	0,1,2,3,4,5,10,12	259
11	0,1,2,3,4,5,10,13	315
12	0,1,2,3,4,7,8,11	307
13	0,1,2,3,4,7,8,13	323
14	0,1,2,3,4,7,8,15	395
15	0,1,2,3,4,7,9,10	339
16	0,1,2,3,4,7,9,13	315
17	0,1,2,3,4,7,9,14	403
18	0,1,2,4,7,8,11,13	295
19	0,1,2,4,7,9,10,11	331
20	0,1,2,4,7,9,10,12	351
21	0,1,2,4,7,9,10,13	379
22	0,1,2,4,9,10,11,12	259

Representative	Eight classes	Certified nonempty faces
23	0,1,2,4,9,10,11,13	259
24	0,1,2,5,6,7,11,12	391
25	0,1,6,7,10,11,12,13	243

Coverage is checked on every eight-bit subset mask. Its listed signed permutation carries its classes onto its representative. The first representative has a nonzero annihilator and is excluded by full dimension. For each other representative, the integer witnesses (d,z,M) satisfy d>0 and

$$|z\cdot q_i|\le d;\quad M=\{i:z\cdot q_i=d\}\cup\{i+8:z\cdot q_i=-d\}. \tag{32.1}$$

Each nonzero M is the vertex set of the face exposed by z/d. The masks are strictly increasing and there are at least 242; none contains both members of a vertex pair, so all are proper. Adding the polytope proves s≥243. All inequalities are integer dot products. This supplies the last terminal rule in Section 31.

33 Sixteen vertices and full-normal rank three

When there are at least four full pairs, their normal span has dimension at least three by $q \le 2^d$ in Section 28. If its dimension is at least four, Sections 30–32 apply. For rank exactly three the same argument as Section 28 gives three base classes, transformed to $\pm e_1, \pm e_2, \pm e_3$. The vertices have the form $\pm(e_i, k)$, $k \in K_i \subset \mathbb{R}^2$. Each K_i is the vertex set of a convex polygon, segment or point. Their sizes sum to eight.

If one class is a point, shear it to zero and split off its segment as in Section 28. The remaining centrally symmetric four-polytope has at least 81 faces by Section 27; the free sum has at least $3 \cdot 81 = 243$. Otherwise the sizes, up to order, are (4,2,2) or (3,3,2). We prove both without a lifting enumeration.

Label each class O,+,− or B, according as neither sign, only the positive sign, only the negative sign, or both signs occur in an exposed proper face. Different labels give different vertex sets. For a nonempty label containing no B, put M equal to the Minkowski sum of the signed active K's. Every face of M, including M, gives an exposed face of the large polytope: choose the base coefficient for each active class so its selected support equals a common large H, and the opposite and inactive supports are strictly smaller. The vertex pattern determines the individual summand faces, and hence their Minkowski face, so this construction is injective. A polygon with m edge normals contributes $2m+1$ patterns; a segment contributes three.

Edge normal sets add by union under Minkowski addition. A triangle has three normals and contains no antipodal normal pair. A quadrilateral has four; a segment has its two opposite perpendicular normals. Write $w_i(b)$ for half the width of K_i in direction b.

Equatorial counting rule. Suppose a segment K_j has zero width at α and another class i is strictly widest there. For each choice of O,+,− on the remaining class, the label with B at i and a fixed sign at j has at least six patterns. Indeed take $b = \alpha$ and $b = -\alpha$, tie the two signs of i by choosing its base coefficient to be minus the midpoint of its two support values, and use $H = w_i(b)$. The strict width gaps allow the other labels to be selected freely. At j its entire segment is exposed. Perturb b slightly on either side to expose each endpoint separately. The two choices of ±α remain distinct on class i because its width is positive. Thus there are $2 \cdot 3$ patterns. If both other classes are unused, the two directions still give two distinct B patterns. This proves the rule, including all width degeneracies not ruled out by a strict gap.

A further full normal exists whenever, in a nonzero direction b, every class is either zero-width or has one common positive width W and all its vertices lie on the two extreme support levels. Center each nonzero-width class by its base coefficient and scale by 1/W; at a zero-width class choose the base coefficient to give value one. The result is ±1 at every vertex and has a nonzero fibre component. This contradicts full-normal rank three. We will use precisely this explicit criterion to rule out two equality cases.

Case (4,2,2). Let Q be the quadrilateral and let $\pm\alpha, \pm\gamma$ be the two segment normal pairs. If the segments are parallel, Q cannot have both normals ±α; otherwise the preceding criterion supplies a fourth full normal. Every signed sum involving Q and a segment has at least five normals. The octahedral labels give at least $8 \cdot 11 + 4 \cdot 11 + 4 \cdot 11 + 2 \cdot 9 + 6 \cdot 3 + 2 \cdot 3 = 218$ faces. Q is strictly widest at α; the equatorial

rule supplies $6\cdot6+2\cdot6+2=50$ more, hence at least 269 including the polytope.

If the segments are nonparallel and Q contains neither complete normal pair, sums with one segment have at least five normals and those with both at least six. There are at least $8\cdot13+4\cdot11+4\cdot11+2\cdot9+4\cdot9+4\cdot3=258$ octahedral faces, hence 259 including P. If Q contains $\pm\alpha$, compare its width there with that of the other segment. They are unequal: equality gives a fourth full normal, since Q has two opposite extreme edges containing all its vertices and the other segment has only two vertices. The wider class is strictly widest, while the first segment has zero width. The crude octahedral count is 210, and the six labels with the widest class B and the first segment signed give 36 equatorial faces. Thus $s\geq247$. The case where Q contains $\pm\gamma$ is identical with the segment indices interchanged.

Case (3,3,2). Let N_1,N_2 be the triangle normal sets and $\pm\alpha$ the segment normal pair. Put $x=|N_1\cup N_2|$ and $y=|N_1\cup(-N_2)|$. The intersections $N_1\cap N_2$ and $N_1\cap(-N_2)$ are disjoint, so $x+y\geq9$. Two signed triangles contribute $4(x+y)+4$ patterns, and the same labels with the segment signed contribute $8(x+y)+8$. Triangle-segment labels contribute $8\cdot9=72$, single triangles $4\cdot7=28$, and a single segment $2\cdot3=6$. Including P gives the initial bound

$$s\geq12(x+y)+119\geq227. \tag{33.1}$$

If the triangle widths at α differ, the wider triangle is strictly widest, giving six equatorial labels with six patterns each. Therefore $s\geq263$. If they are equal, at least one triangle contains neither α nor $-\alpha$ as an edge normal. Otherwise each triangle lies on its two extreme levels in that direction, giving a fourth full normal. A triangle has no antipodal normal pair, so each of the four signed triangle-normal unions lacks at least one of $\alpha,-\alpha$. Adding the segment gives one extra normal, hence two extra patterns at each of the eight three-signed labels. The gain of 16 in (33.1) proves $s\geq243$. These alternatives exhaust the rank-three case.

34 Further endpoint costs and the completed (16,16) case

We extend the affine cost of Section 22 by two terms: $a\cdot[2g>\tau]$ and $[n=8](b+c\,r)$. Their sums are aK and bN_8+cR_8, where K counts facets above the stress threshold, N_8 counts eight-vertex facets and R_8 is their total ridge count. On any leaf where $a\neq0$ we use $0\le K\le2$. If M is the largest local g, this is justified whenever either $2M\le\tau$, giving $K=0$, or $2M\ge G-\tau$: all facet pairs other than a maximizing pair then have $2g\le G-2M\le\tau$ by Section 25. The maximizing pair contains two facets. The same rule applies to the polar.

For a fixed n,e,r the extended local cost is affine in J on each of the two integer intervals split by $2(J-3n-3r+10)\le\tau$. Check its endpoints, together with the elementary minimum recurrence of Section 22. The full-facet term is constant in J. Thus the local nonnegativity test is exact on the whole row domain; the discontinuous threshold is not minimized by assuming a single affine interval.

For full facets at $v=16$, Section 26 gives $N_8=2k$ and $R_8\le k(h+2k-4)$. The numbers for $h=16,k=1,2,3$ are 14,32,54; those for $h=20,k=7,8,9,10$ are 210,256,306,360. These are bounds for all realizations and do not require generic incidences.

Proposition 34.1. *A centrally symmetric five-polytope with $v=h=16$ has at least 243 nonempty faces.* If no facet is full, the one E case below applies. With one to three full pairs, a tetrahedron-prism facet on either side of polarity is excluded by Sections 16–17; otherwise the 25 G cases cover all slacks and local maxima. With at least four full pairs, full-normal rank is at least three by Section 28. Rank three is covered by Section 33; rank at least four by Sections 30–32, since $h\le20$. These alternatives exhaust every number of full pairs and every possible normal rank.

The same tables close (16,18) without full facets, and (16,20) with at least seven full pairs, after excluding the full tetrahedron-prism using Sections 16–17. For (18,20), the product row on P gives $e\ge72$ and $r\ge50$, hence $t\ge122$. A product row only on the polar gives $r\ge66$ by Section 21 and is included in C. Its remaining C cases require the tetrahedron-prism row to be absent on P or the indicated edge bound. D similarly handles the non-product, non-prism branches at (18,18), conditional on excluding row (9,19,17,7;38). Those two geometric exclusions are kept separate from the numerical computation.

Group	Endpoint and geometric conditions	Cases
C	(18,20); product/prism rows as individually specified	25
D	(18,18); product/prism/U rows as individually specified	216
E	(16,16), no full facet	1
F	(16,18), no full facet	14
G	(16,16), k=1,2,3 full pairs, no prism on either side	25
H	(16,20), k=7,8,9,10 full pairs, no prism on P	4

Here is a finite coverage check written in terms of the actual integer quantities. First G,S≥0 are even and G+S≤vh−8(v+h)+26, namely 82,62,26,42,26,58 for C,D,E,F,G,H respectively. Enumerate each such pair and each integer M_F∈[0,G/2], M_S∈[0,S/2]. Retain the cases whose caps cover 2M and whose threshold satisfies the two alternatives proved above. If G=0 or S=0, include A=0 from Section 13. Split each remaining even nonnegative count axis at every printed boundary b and b+2; equality boundaries therefore stand in a separate cell. Each case condition is constant on a cell. The unbounded last cell is represented by its least value. This is an exhaustive interval partition, not a sampling argument.

For G and H fix N_8=2k. Every condition on R_8 is an upper bound, so it suffices to substitute the worst value k(h+2k−4); any smaller nonnegative value satisfies the same upper bounds. This partitions 2,380 cells/maxima for E, 12,650 for F, 7,140 for G and 163,680 for H, and covers every one. The conditional non-product/non-prism branches of C and D cover 149,361 and 52,883 respectively. The tables below specify the full conditions, so the coverage calculation can be reconstructed using only integer intervals.

Each leaf retains $Q_1,\ldots,Q_{23}$ from Section 22; Q_{22},Q_{23} now include the extra threshold and full-facet terms. The following U list first contains the threshold count bounds 0≤K≤2, then nonnegativity of N_8,R_8 when used, then the printed case conditions in order. An equality supplies its lower and upper form, lower first. As before z≥b means z−b, z≤b means b−z. The final identity for each leaf is a nonnegative integer combination equal to a negative constant. All local costs were checked on their complete finite row domains.

C1. nmax F/S=9/10; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36); S: none.

φ=(−2160, −29404, 29248, 0, 648, −7327, 0, 0, 0, 0).

ψ=(−303375, −8899, −5663, 0, 80900, 0, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: S ≥ 0; 3: r ≥ 66; 4: t ≤ 118.

Identity: 20225·Q7 + 9708·Q9 + 4854·Q10 + 190924·Q21 + Q23 + 222475·U3 = −194160.

C2. nmax F/S=9/10; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36); S: (9, 18, 15, 6, 36).

φ=(−180, −8, −5, 0, 54, 0, 0, 0, 0, 0).

ψ=(230, −69, 0, 0, −69, 23, 0, 0, 0, 0).

U: 1: G ≥ 12; 2: S ≥ 0; 3: e ≥ 60; 4: t ≤ 118.

Identity: 345·Q8 + 598·Q9 + 7452·Q21 + 69·Q22 + 13·Q23 + 598·U1 + 8487·U3 = −2438.

C3. nmax F/S=9/10; cap F/S=10/1000. Exclude F: (9, 18, 15, 6, 36); S: (9, 18, 15, 6, 36).

φ=(−2280, 228, 0, 0, 684, −114, 0, 0, 0, 0).

ψ=(−25500, −7354, 5212, 0, 7650, −1405, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 10; 3: S ≥ 20; 4: e ≥ 60; 5: t ≤ 118.

Identity: 76·Q9 + 152·Q11 + 1596·Q21 + Q22 + 228·U3 + 2052·U4 = −5472.

C4. nmax F/S=9/10; cap F/S=10/18. Exclude F: (9, 18, 15, 6, 36); S: (9, 18, 15, 6, 36).

φ=(−20, 2, 0, 0, 6, −1, 0, 0, 0, 0).

ψ=(0, −4, 4, 0, 0, −1, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 10; 3: S ≥ 0; 4: S ≤ 18; 5: e ≥ 60; 6: t ≤ 118.

Identity: 6·Q8 + 2·Q9 + 4·Q11 + 42·Q21 + 3·Q22 + 54·U5 = −24.

C5. nmax F/S=9/10; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−1491615, −264435, 0, 205958, 389986, −59786, 0, 0, 0, 0).

ψ=(−110750, −7531, −1772, 0, 33225, 0, 0, 0, 0, 0).

U: 1: G ≥ 12; 2: S ≥ 0; 3: t ≤ 118.

Identity: 14299869898·Q9 + 56740679544·Q21 + 69431·Q22 + 208293·Q23 + 22970968626·U1 = −56740679544.

C6. nmax F/S=9/10; cap F/S=10/1000. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(445, −846, 0, 0, 9, 187, 0, 0, 0, 0).

ψ=(−204, −44, 0, 0, 98, −1, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 10; 3: S ≥ 20; 4: t ≤ 118.

Identity: 374·Q9 + 1140·Q21 + Q22 + 6·Q23 + 386·U3 = −560.

C7. nmax F/S=9/10; cap F/S=10/16. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−835, −57, 68, 0, 312, −56, 0, 0, 0, 0).

ψ=(−210, 7, 0, 0, 63, −7, 0, 0, 0, 0).

U: 1: G ≥ 4; 2: G ≤ 10; 3: S ≥ 0; 4: S ≤ 16; 5: t ≤ 118.

Identity: 108·Q9 + 576·Q21 + 2·Q22 + 2·Q23 + 332·U1 = −16.

C8. nmax F/S=9/10; cap F/S=10/18. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−35, −12, 0, 0, 15, 1, 0, 0, 0, 0).

ψ=(−90, −1, 4, 0, 27, −4, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 10; 3: S ≥ 16; 4: S ≤ 18; 5: t ≤ 118.

Identity: 10·Q9 + 60·Q21 + Q22 + 2·Q23 + 26·U3 = −48.

C9. nmax F/S=9/10; cap F/S=0/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−205, 54, 0, 0, 63, −19, 0, 0, 0, 0).

ψ=(−20, 2, 0, 0, 6, −1, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 0; 3: S ≥ 0; 4: S ≤ 8; 5: A ≥ 0; 6: A ≤ 0; 7: t ≤ 118.

Identity: 30·Q7 + 12·Q21 + Q22 + 2·Q23 + 4·U4 + 8·U6 = −12.

C10. nmax F/S=9/10; cap F/S=10/16. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−5, −21, 0, 0, 6, 4, 0, 0, 0, 0).

ψ=(−30, 1, 0, 0, 9, −1, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 10; 3: S ≥ 14; 4: S ≤ 16; 5: t ≤ 118.

Identity: 8·Q9 + 36·Q21 + Q22 + 3·Q23 + 14·U3 = −8.

C11. nmax F/S=9/10; cap F/S=10/12. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−20, −4, 0, 0, 8, 0, 0, 0, 0, 0).

ψ=(−560, 16, 22, 0, 168, −29, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 10; 3: S ≥ 12; 4: S ≤ 12; 5: t ≤ 118.

Identity: 22·Q9 + 162·Q21 + 6·Q22 + Q23 + 80·U3 = −12.

C12. nmax F/S=9/10; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−25, 0, 2, 0, 9, −2, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 4; 3: S ≥ 0; 4: S ≤ 2; 5: t ≤ 118.

Identity: Q9 + 6·Q21 + Q22 + 5·U1 + U4 = −2.

C13. nmax F/S=9/10; cap F/S=0/10. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

ψ=(−570, 31, 20, 0, 171, −32, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 0; 3: S ≥ 10; 4: S ≤ 10; 5: A ≥ 0; 6: A ≤ 0; 7: e ≤ 56; 8: t ≤ 118.

Identity: 13·Q9 + 120·Q21 + 2·Q23 + 27·U2 + 141·U3 + 180·U7 = −120.

C14. nmax F/S=9/10; cap F/S=0/10. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−25, 0, 0, 0, 9, −1, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 0; 3: S ≥ 10; 4: S ≤ 10; 5: A ≥ 0; 6: A ≤ 0; 7: e ≥ 58; 8: t ≤ 118.

Identity: 2·Q9 + 2·Q11 + 24·Q21 + Q22 + 4·U3 + 30·U7 = −24.

C15. nmax F/S=9/10; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−110, 3, 10, 0, 39, −10, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: e ≤ 56; 6: t ≤ 118.

Identity: 7·Q9 + 36·Q21 + 2·Q22 + 47·U1 + 13·U4 + 24·U5 = −36.

C16. nmax F/S=9/10; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−5, −1, 0, 0, 2, 0, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: e ≥ 58; 6: t ≤ 118.

Identity: Q9 + 8·Q21 + 2·Q22 + U1 + U3 + 8·U5 = −4.

C17. nmax F/S=9/10; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−75, 6, 0, 6, 21, −6, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 116; 6: t ≤ 118.

Identity: 3·Q9 + Q22 + 15·U1 + 3·U4 + 18·U5 = −30.

C18. nmax F/S=9/10; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−185, 73, −47, 65, 0, −18, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≤ 56; 7: t ≤ 118.

Identity: 5·Q9 + 22·Q21 + Q22 + 41·U1 + 13·U4 + 32·U6 = −22.

C19. nmax F/S=9/10; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−5, 1, −2, 2, 0, 0, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≥ 58; 7: t ≤ 118.

Identity: Q9 + 8·Q21 + 2·Q22 + U1 + U3 + 8·U6 = −6.

C20. nmax F/S=9/10; cap F/S=2/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−45, −9, 0, 0, 18, 0, 0, 0, 0, 0).

ψ=(−60, 6, 0, 0, 18, −3, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 116; 6: t ≤ 118.

Identity: 9·Q9 + 2·Q22 + 4·Q23 + 9·U1 + 33·U3 + 72·U5 = −120.

C21. nmax F/S=9/10; cap F/S=2/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−5, 1, −2, 2, 0, 0, 0, 0, 0, 0).

ψ=(−20, 2, 0, 0, 6, −1, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≥ 118; 6: e ≤ 56; 7: t ≤ 118.

Identity: Q23 + 2·U3 + 6·U6 = −4.

C22. nmax F/S=9/10; cap F/S=2/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−5, 1, −2, 2, 0, 0, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≥ 118; 6: e ≥ 58; 7: t ≤ 118.

Identity: Q9 + 8·Q21 + 2·Q22 + U1 + U3 + 8·U6 = −8.

C23. nmax F/S=9/10; cap F/S=2/10. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−25, −5, 0, 0, 10, 0, 0, 0, 0, 0).

ψ=(−180, 14, 0, 0, 54, −8, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 10; 4: S ≤ 10; 5: t ≤ 116; 6: t ≤ 118.

Identity: 17·Q9 + 6·Q22 + 2·Q23 + 17·U1 + 49·U3 + 120·U5 = −234.

C24. nmax F/S=9/10; cap F/S=2/10. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−5, 1, −2, 2, 0, 0, 0, 0, 0, 0).

ψ=(−90, 7, 0, 0, 27, −4, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 10; 4: S ≤ 10; 5: t ≥ 118; 6: e ≤ 56; 7: t ≤ 118.

Identity: Q9 + 2·Q23 + U1 + 17·U3 + 60·U6 = −42.

C25. nmax F/S=9/10; cap F/S=2/10. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(−5, 1, −2, 2, 0, 0, 0, 0, 0, 0).

ψ=(−90, 7, 0, 0, 27, −4, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 10; 4: S ≤ 10; 5: t ≥ 118; 6: e ≥ 58; 7: t ≤ 118.

Identity: Q9 + 8·Q21 + 2·Q22 + U1 + U3 + 8·U6 = −10.

D1. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: none; S: none.

φ=(−12015, −445, 0, −356, 3560, 0, 0, 0, 0, 0).

ψ=(−41985, −1555, 0, 3528, 7668, −1193, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: S ≥ 0; 3: e ≥ 72; 4: t ≤ 118.

Identity: 2403·Q8 + 890·Q9 + 445·Q11 + 21894·Q21 + 3·Q22 + 26433·U3 = −187968.

D2. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36); S: none.

φ=(−5040, −364, 0, 0, 1512, −35, 0, 0, 0, 0).

ψ=(−29835, 1399, −965, 0, 6498, 0, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: S ≥ 0; 3: r ≥ 72; 4: t ≤ 118.

Identity: 93605·Q7 + 418810·Q8 + 418810·Q11 + 11284710·Q21 + 965·Q23 + 11565525·U3 = −8082840.

D3. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36); S: (9, 18, 15, 6, 36).

φ=(−180, −13, 0, −5, 59, 0, 0, 0, 0, 0).

ψ=(1080, −324, 0, 124, −448, 77, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: S ≥ 0; 3: e ≥ 64; 4: t ≤ 118.

Identity: 124·Q9 + 1020·Q21 + 6·Q22 + Q23 + 1296·U3 = −528.

D4. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32); S: (9, 18, 15, 6, 36).

φ=(515, −170, 0, 62, −201, 36, 0, 0, 0, 0).

ψ=(−180, −13, 0, −5, 59, 0, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: S ≥ 0; 3: r ≥ 64; 4: t ≤ 118.

Identity: 62·Q9 + 402·Q21 + Q22 + 2·Q23 + 494·U3 = −362.

D5. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1185, 0, 224, −174, 585, −112, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 150·Q21 + Q23 + 50·U2 + 224·U3 = −150.

D6. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−523, 78, 22, 0, 183, −50, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 66·Q21 + Q22 + 100·U1 + 22·U4 = −66.

D7. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−85, −156, 50, 0, 81, −2, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−110, 33, 0, 0, 33, −11, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 150·Q21 + Q22 + 10·Q23 + 4·U1 + 170·U3 = −150.

D8. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−150, 31, 8, 0, 51, −17, 0, 0, 0, 0); threshold (a,τ)=(17, 0).

ψ=(−215, −9, 40, 0, 84, −28, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 120·Q21 + 5·Q22 + Q23 + 85·U2 + 130·U3 + 16·U5 = −120.

D9. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(850, −255, 0, 0, −255, 85, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−850, 255, 0, 0, 255, −85, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: Q22 + Q23 + 170·U2 + 170·U3 = −340.

D10. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−991, 299, 0, 0, 300, −100, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−5, −13, 4, 0, 6, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 132·Q21 + Q22 + 11·Q23 + 156·U1 = −132.

D11. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−4675, 1105, 170, 0, 1530, −510, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(425, −425, 170, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 510·Q21 + Q22 + Q23 + 850·U1 + 170·U4 = −2380.

D12. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(310, −101, 8, 0, −93, 29, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−390, 109, 8, 0, 117, −41, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 12·Q21 + Q22 + Q23 + 66·U2 + 74·U3 = −12.

D13. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−410, 81, 24, 0, 141, −47, 0, 0, 0, 0); threshold (a,τ)=(47, 0).

ψ=(−165, 5, 24, 0, 60, −20, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 72·Q21 + Q22 + Q23 + 47·U2 + 70·U3 + 16·U5 = −72.

D14. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−475, 84, 27, 0, 165, −52, 0, 0, 0, 0); threshold (a,τ)=(95, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 7·Q10 + 60·Q21 + Q23 + 95·U2 + 20·U4 + 90·U5 = −60.

D15. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−330, −251, 0, 0, 169, 37, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−3490, 739, 0, −16, 1115, −281, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 512·Q21 + Q22 + Q23 + 58·U2 + 562·U3 = −512.

D16. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−190, 57, 0, 0, 57, −19, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−45, −52, 18, 0, 33, −2, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 54·Q21 + 2·Q22 + Q23 + 58·U1 + 4·U3 = −54.

D17. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−2990, 897, 0, 0, 897, −299, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−185, −427, 130, 0, 204, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 2·Q11 + 402·Q21 + Q22 + Q23 + 464·U1 + 2·U3 = −402.

D18. nmax F/S=9/9; cap F/S=0/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1240, 217, 50, 0, 423, −123, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(35, −35, 14, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 150·Q21 + Q22 + Q23 + 232·U1 + 50·U4 = −150.

D19. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(250, −91, 16, 0, −75, 21, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−410, 107, 16, 0, 123, −45, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 24·Q21 + Q22 + Q23 + 58·U2 + 74·U3 = −24.

D20. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−70, 7, 8, 0, 27, −9, 0, 0, 0, 0); threshold (a,τ)=(9, 0).

ψ=(−115, 19, 8, 0, 36, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 24·Q21 + Q22 + Q23 + 9·U2 + 10·U3 + 16·U5 = −24.

D21. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1630, 391, 56, 0, 531, −177, 0, 0, 0, 0); threshold (a,τ)=(177, 0).

ψ=(−295, −101, 80, 0, 144, −36, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 240·Q21 + Q22 + Q23 + 177·U2 + 274·U3 + 16·U5 = −240.

D22. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−15, −10, 4, 0, 9, −1, 0, 0, 0, 0); threshold (a,τ)=(1, 0).

ψ=(−20, 6, 0, 0, 6, −2, 0, 0, 0, 0); threshold (a,τ)=(6, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 2; 6: G ≤ 2; 7: S ≥ 6; 8: S ≤ 6; 9: t ≤ 118.

Identity: 12·Q21 + Q22 + 4·Q23 + U2 + 24·U4 + 2·U5 + 12·U7 = −12.

D23. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(850, −255, 0, 0, −255, 85, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−850, 255, 0, 0, 255, −85, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: Q22 + Q23 + 170·U2 + 170·U3 = −1020.

D24. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−991, 299, 0, 0, 300, −100, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−5, −13, 4, 0, 6, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 132·Q21 + Q22 + 11·Q23 + 156·U1 = −132.

D25. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−12495, 3710, 0, −2, 3757, −1241, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−380, −614, 0, 269, 152, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 1678·Q21 + Q22 + 2·Q23 + 1944·U1 + 2·U3 = −1678.

D26. nmax F/S=9/9; cap F/S=0/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1270, 241, 50, 0, 429, −129, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(55, −55, 22, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 150·Q21 + Q22 + Q23 + 236·U1 + 50·U4 = −150.

D27. nmax F/S=9/9; cap F/S=0/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−29335, 8764, −2, 0, 8807, −2925, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−10905, −883, 1111, 0, 4210, −842, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 6730·Q21 + Q22 + 2·Q23 + 3628·U1 + 3370·U3 = −6730.

D28. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(190, −81, 24, 0, −57, 13, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−430, 105, 24, 0, 129, −49, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 36·Q21 + Q22 + Q23 + 50·U2 + 74·U3 = −36.

D29. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−1345, 172, 62, 0, 429, −114, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 186·Q21 + 2·Q22 + Q23 + 62·U2 + 224·U3 = −186.

D30. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−250, 61, 8, 0, 81, −27, 0, 0, 0, 0); threshold (a,τ)=(27, 0).

ψ=(−125, −7, 16, 0, 48, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 48·Q21 + Q22 + Q23 + 27·U2 + 38·U3 + 16·U5 = −48.

D31. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1310, 323, 40, 0, 423, −141, 0, 0, 0, 0); threshold (a,τ)=(141, 0).

ψ=(−55, −113, 56, 0, 60, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 168·Q21 + Q22 + Q23 + 141·U2 + 226·U3 + 16·U6 = −168.

D32. nmax F/S=9/9; cap F/S=2/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1150, 323, −1, 0, 337, −107, 0, 0, 0, 0); threshold (a,τ)=(77, 0).

ψ=(−3045, −218, 288, 0, 1255, −251, 0, 0, 0, 0); threshold (a,τ)=(251, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 2; 6: G ≤ 2; 7: S ≥ 8; 8: S ≤ 8; 9: t ≤ 118.

Identity: 992·Q21 + 4·Q22 + Q23 + 308·U2 + 251·U4 + 568·U5 + 506·U7 = −992.

D33. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1050, 315, 0, 0, 315, −105, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(12580, −3778, 4, 0, −3774, 1257, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 210·Q10 + 630·Q21 + 1262·Q22 + 105·Q23 + 264810·U1 + 264390·U4 = −522480.

D34. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−6650, 951, 288, 0, 2319, −629, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 19·Q9 + 1728·Q21 + 2·Q22 + 2535·U1 + 557·U4 = −6820.

D35. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1210, 243, 0, 50, 367, −117, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 150·Q21 + 3·Q22 + Q23 + 50·U2 + 228·U3 = −150.

D36. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−165, 5, 24, 0, 60, −20, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−410, 81, 24, 0, 141, −47, 0, 0, 0, 0); threshold (a,τ)=(47, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 72·Q21 + Q22 + Q23 + 47·U2 + 16·U3 + 70·U5 = −72.

D37. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−1210, 193, 50, 0, 417, −117, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 150·Q21 + 3·Q22 + Q23 + 50·U2 + 228·U3 = −150.

D38. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−385, 112, 2, 0, 117, −39, 0, 0, 0, 0); threshold (a,τ)=(78, 0).

ψ=(−20, −79, 25, 0, 33, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 150·Q21 + 3·Q22 + 2·Q23 + 234·U2 + 184·U3 + 6·U6 = −150.

D39. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−415, 0, 76, −58, 199, −38, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 54·Q21 + Q23 + 18·U2 + 76·U3 = −54.

D40. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−510, 153, 0, 0, 153, −51, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−85, −221, 68, 0, 102, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 204·Q21 + 3·Q22 + Q23 + 238·U1 = −714.

D41. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−2770, 703, 128, 0, 831, −309, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(280, −355, 172, 0, 15, −15, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 324·Q21 + Q22 + Q23 + 446·U1 + 98·U4 = −324.

D42. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(45, −5, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−335, 69, 0, 12, 102, −32, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 5·Q10 + 5·Q11 + 87·Q21 + 2·Q23 + 29·U2 + 123·U3 = −102.

D43. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −3, 0, 2, −2, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 116; 6: t ≤ 118.

Identity: 3·Q9 + Q22 + Q23 + U1 + U3 + 6·U5 = −6.

D44. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 3·U1 + U4 + 12·U6 = −6.

D45. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −3, 0, 2, −2, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −6.

D46. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−380, 92, −1, 0, 106, −30, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−10995, 1522, 478, 0, 3845, −1031, 0, 0, 0, 0); threshold (a,τ)=(1310, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 314·Q9 + 70·Q11 + 13326·Q21 + 9·Q23 + 11790·U2 + 4128·U4 + 18942·U5 = −14574.

D47. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−85, −11, −2, 0, 32, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−2650, 439, 62, 0, 891, −231, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 378·Q21 + 3·Q22 + Q23 + 62·U2 + 468·U3 = −378.

D48. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−10995, 1522, 478, 0, 3845, −1031, 0, 0, 0, 0); threshold (a,τ)=(1310, 0).

ψ=(−380, 92, −1, 0, 106, −30, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 314·Q9 + 70·Q10 + 13326·Q21 + 9·Q22 + 11790·U2 + 18942·U3 + 4128·U6 = −14574.

D49. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−25, 5, 0, 0, 8, −2, 0, 0, 0, 0); threshold (a,τ)=(2, 0).

ψ=(−395, 53, 16, 0, 138, −36, 0, 0, 0, 0); threshold (a,τ)=(42, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 4; 6: G ≤ 4; 7: S ≥ 4; 8: S ≤ 4; 9: t ≤ 118.

Identity: 60·Q21 + 3·Q22 + Q23 + 6·U2 + 42·U4 + 4·U6 + 72·U7 = −60.

D50. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−110, 33, 0, 0, 33, −11, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(140, −71, 8, 0, −33, 15, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 24·Q21 + 3·Q22 + Q23 + 58·U1 + 30·U4 = −24.

D51. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−510, 153, 0, 0, 153, −51, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−85, −221, 68, 0, 102, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 204·Q21 + 3·Q22 + Q23 + 238·U1 = −714.

D52. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−8075, 1190, 340, 0, 2805, −765, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 1020·Q21 + Q22 + 1530·U1 + 340·U4 = −2890.

D53. nmax F/S=9/9; cap F/S=0/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−20840, 2883, 918, 0, 7293, −1961, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−495, 143, −22, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 2754·Q21 + Q22 + Q23 + 3944·U1 + 918·U4 = −2754.

D54. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−50, −131, −8, 0, 41, 29, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1860, 383, 0, 0, 593, −151, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 256·Q21 + Q22 + Q23 + 58·U2 + 310·U3 = −256.

D55. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(60, −36, 0, 24, −24, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1305, 270, 54, 0, 387, −126, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 116; 6: t ≤ 118.

Identity: 2·Q22 + Q23 + 54·U2 + 204·U3 + 162·U5 = −246.

D56. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 3·U1 + U4 + 12·U6 = −4.

D57. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−565, 129, 0, 44, 136, −60, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 132·Q21 + 3·Q22 + Q23 + 44·U2 + 114·U3 + 114·U6 = −132.

D58. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 116; 6: t ≤ 118.

Identity: 9·Q9 + Q22 + Q23 + 3·U1 + 3·U3 + 18·U5 = −24.

D59. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 3·U1 + U4 + 12·U6 = −4.

D60. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −12.

D61. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−760, 184, −2, 0, 212, −60, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−33805, 3802, 1604, 0, 12051, −3075, 0, 0, 0, 0); threshold (a,τ)=(6399, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 1256·Q9 + 280·Q11 + 16116·Q21 + 3·Q23 + 19197·U2 + 4116·U4 + 19986·U5 = −24036.

D62. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−745, −52, 74, 0, 285, −58, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−470, 141, 0, 0, 141, −47, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 222·Q21 + Q22 + 2·Q23 + 116·U1 + 114·U3 = −222.

D63. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−145, 12, 18, 0, 57, −19, 0, 0, 0, 0); threshold (a,τ)=(38, 0).

ψ=(−135, 23, 9, 0, 42, −14, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 54·Q21 + Q22 + 2·Q23 + 38·U2 + 20·U3 + 38·U5 = −54.

D64. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−240, 37, 10, 0, 83, −23, 0, 0, 0, 0); threshold (a,τ)=(32, 0).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 90·Q21 + 3·Q22 + Q23 + 96·U2 + 136·U3 + 30·U6 = −90.

D65. nmax F/S=9/9; cap F/S=4/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−45, 11, 0, 0, 14, −4, 0, 0, 0, 0); threshold (a,τ)=(6, 0).

ψ=(−65, 0, 4, 0, 25, −5, 0, 0, 0, 0); threshold (a,τ)=(5, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 4; 6: G ≤ 4; 7: S ≥ 6; 8: S ≤ 6; 9: t ≤ 118.

Identity: 20·Q21 + 2·Q22 + Q23 + 12·U2 + 5·U4 + 12·U5 + 10·U7 = −20.

D66. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1590, 127, 0, 0, 547, −89, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−2230, 361, 0, −16, 737, −155, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 512·Q21 + Q22 + Q23 + 194·U1 + 310·U3 = −512.

D67. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−275, 91, 0, 0, 96, −32, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−5, −13, 4, 0, 6, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 36·Q21 + Q22 + 3·Q23 + 52·U1 = −36.

D68. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−21190, 3921, 0, 918, 6477, −1999, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−315, 91, −14, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 2754·Q21 + Q22 + Q23 + 4012·U1 + 918·U4 = −2754.

D69. nmax F/S=9/9; cap F/S=0/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−15570, 3691, 560, 0, 5091, −1697, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(4285, −4285, 1714, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 29138·Q10 + 16012188·Q21 + 9427·Q22 + 3131·Q23 + 26657842·U1 + 5337396·U4 = −105961194.

D70. nmax F/S=9/9; cap F/S=0/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−15570, 3691, 560, 0, 5091, −1697, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(4285, −4285, 1714, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 29138·Q10 + 16012188·Q21 + 9427·Q22 + 3131·Q23 + 26657842·U1 + 5337396·U4 = −105961194.

D71. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−45, 13, −2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1210, 189, 0, 62, 373, −111, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 35·Q9 + 4·Q11 + 396·Q21 + 2·Q23 + 97·U2 + 483·U3 = −2526.

D72. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(10, −6, 0, 4, −4, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1405, 288, 0, 62, 373, −126, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 186·Q21 + 6·Q22 + Q23 + 62·U2 + 228·U3 = −186.

D73. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−145, 21, 0, 8, 40, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 116; 6: t ≤ 118.

Identity: 3·Q9 + 2·Q23 + 13·U2 + 51·U3 + 48·U5 = −90.

D74. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−265, 45, 4, 0, 80, −20, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 306·Q9 + 36·Q21 + 213·Q22 + 3·Q23 + 294·U1 + 1362·U6 = −240.

D75. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−50, −16, 13, 0, 24, −6, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 15·Q9 + 78·Q21 + 2·Q23 + 11·U2 + 39·U3 + 54·U6 = −78.

D76. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1305, 270, 54, 0, 387, −126, 0, 0, 0, 0); threshold (a,τ)=(243, 2).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 116; 8: t ≤ 118.

Identity: 8·Q22 + Q23 + 243·U2 + 54·U4 + 204·U5 + 162·U7 = −168.

D77. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 3·U1 + U4 + 12·U6 = −2.

D78. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−785, 113, 64, 0, 252, −84, 0, 0, 0, 0); threshold (a,τ)=(168, 2).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 8; 6: S ≤ 8; 7: t ≥ 118; 8: e ≥ 60; 9: t ≤ 118.

Identity: 192·Q21 + 3·Q22 + Q23 + 168·U2 + 64·U4 + 162·U5 + 174·U8 = −192.

D79. nmax F/S=9/9; cap F/S=2/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−9500, 2300, −25, 0, 2650, −750, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−33195, 2434, 1774, 0, 12173, −2879, 0, 0, 0, 0); threshold (a,τ)=(6206, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 7850·Q9 + 1750·Q11 + 58398·Q21 + 9·Q23 + 55854·U2 + 11616·U4 + 61422·U5 = −126198.

D80. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1295, −3625, 1122, 0, 1640, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−25030, 7509, 0, 0, 7509, −2503, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 215258·Q11 + 75180108·Q21 + 22527·Q22 + 22613·Q23 + 430516·U2 + 88140642·U3 = −615692946.

D81. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(10, −10, 4, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−1405, 226, 62, 0, 435, −126, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 4; 2: G ≤ 4; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 186·Q21 + 6·Q22 + Q23 + 62·U2 + 228·U3 = −186.

D82. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−215, 54, 6, 0, 69, −23, 0, 0, 0, 0); threshold (a,τ)=(46, 0).

ψ=(−340, 10, 29, 0, 120, −30, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 174·Q21 + 3·Q22 + 2·Q23 + 138·U2 + 80·U3 + 102·U5 = −174.

D83. nmax F/S=9/9; cap F/S=4/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−125, 34, 2, 0, 39, −13, 0, 0, 0, 0); threshold (a,τ)=(26, 0).

ψ=(−70, −14, 11, 0, 30, −6, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 66·Q21 + 3·Q22 + 2·Q23 + 78·U2 + 56·U3 + 18·U5 = −66.

D84. nmax F/S=9/9; cap F/S=4/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−625, 125, 0, 0, 200, −50, 0, 0, 0, 0); threshold (a,τ)=(50, 0).

ψ=(−505, 31, 16, 0, 182, −36, 0, 0, 0, 0); threshold (a,τ)=(54, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 4; 6: G ≤ 4; 7: S ≥ 8; 8: S ≤ 8; 9: t ≤ 118.

Identity: 148·Q21 + Q22 + Q23 + 50·U2 + 54·U4 + 84·U5 + 72·U7 = −148.

D85. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−6370, 1907, 4, 0, 1911, −638, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(6350, −1905, 0, 0, −1905, 635, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 1270·Q11 + 3810·Q21 + 635·Q22 + 633·Q23 + 807720·U1 + 805180·U4 = −3187700.

D86. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−180, 25, 8, 0, 63, −17, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: Q9 + 48·Q21 + 2·Q22 + 69·U1 + 15·U4 = −318.

D87. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−45, −73, 0, 32, 18, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−745, 221, 0, 0, 224, −74, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 100·Q21 + Q22 + Q23 + 116·U3 = −100.

D88. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−115, 19, 8, 0, 36, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−70, 7, 8, 0, 27, −9, 0, 0, 0, 0); threshold (a,τ)=(9, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 24·Q21 + Q22 + Q23 + 9·U2 + 16·U3 + 10·U5 = −24.

D89. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−5, −1, 2, 0, 2, −1, 0, 0, 0, 0); threshold (a,τ)=(1, 2).

ψ=(−75, 13, 4, 0, 26, −8, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 10·Q21 + Q22 + Q23 + U2 + 2·U4 + 14·U5 = −10.

D90. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−295, −101, 80, 0, 144, −36, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1630, 391, 56, 0, 531, −177, 0, 0, 0, 0); threshold (a,τ)=(177, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 240·Q21 + Q22 + Q23 + 177·U2 + 16·U3 + 274·U5 = −240.

D91. nmax F/S=9/9; cap F/S=6/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−385, −896, 273, 0, 427, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−12675, 3766, −2, 0, 3809, −1259, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 1702·Q21 + 2·Q22 + Q23 + 2·U1 + 1972·U3 = −1702.

D92. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−180, −19, 20, 0, 75, −15, 0, 0, 0, 0); threshold (a,τ)=(11, 0).

ψ=(−290, 87, 0, 0, 87, −29, 0, 0, 0, 0); threshold (a,τ)=(29, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 6; 6: G ≤ 6; 7: S ≥ 2; 8: S ≤ 2; 9: t ≤ 118.

Identity: 60·Q21 + Q22 + Q23 + 11·U2 + 29·U4 + 30·U5 + 38·U7 = −60.

D93. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−12740, 3818, 4, 0, 3822, −1275, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(530, −159, 0, 0, −159, 53, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 106·Q11 + 318·Q21 + 53·Q22 + 1270·Q23 + 134938·U1 + 134726·U4 = −266696.

D94. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−7650, 2295, 0, 0, 2295, −765, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−425, −1105, 340, 0, 510, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 1020·Q21 + Q22 + Q23 + 1190·U1 = −5950.

D95. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−8470, 2053, 488, 0, 2541, −969, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(170, −335, 200, 0, 33, −33, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 1068·Q21 + Q22 + 3·Q23 + 1338·U1 + 290·U4 = −1068.

D96. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−385, 12, 0, 38, 109, −30, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−80, 24, 0, 0, 24, −8, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 114·Q21 + Q22 + 6·Q23 + 60·U1 + 58·U3 = −114.

D97. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1305, 324, 0, 54, 333, −126, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(60, −60, 24, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 116; 6: t ≤ 118.

Identity: Q22 + 2·Q23 + 204·U1 + 54·U4 + 162·U5 = −246.

D98. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−650, 122, 37, 0, 198, −66, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −3, 0, 2, −2, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 54·Q21 + Q22 + 9·Q23 + 114·U1 + 37·U4 + 57·U6 = −54.

D99. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −3, 0, 2, −2, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −4.

D100. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−135, 23, 9, 0, 42, −14, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−145, 12, 18, 0, 57, −19, 0, 0, 0, 0); threshold (a,τ)=(38, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 54·Q21 + 2·Q22 + Q23 + 38·U2 + 38·U3 + 20·U5 = −54.

D101. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−660, 154, 22, 0, 154, −55, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−5565, 724, 276, 0, 1969, −533, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 341·Q9 + 88·Q10 + 7188·Q21 + 9·Q23 + 2055·U2 + 9759·U3 = −14910.

D102. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 116; 6: t ≤ 118.

Identity: 9·Q9 + Q22 + Q23 + 3·U1 + 3·U3 + 18·U5 = −24.

D103. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 3·U1 + U4 + 12·U6 = −12.

D104. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −4.

D105. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−240, 37, 10, 0, 83, −23, 0, 0, 0, 0); threshold (a,τ)=(32, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 90·Q21 + Q22 + 3·Q23 + 96·U2 + 30·U4 + 136·U5 = −90.

D106. nmax F/S=9/9; cap F/S=6/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(140, −63, 12, 0, −33, 11, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−330, 91, 8, 0, 99, −35, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 24·Q21 + Q22 + Q23 + 30·U2 + 58·U3 = −24.

D107. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−33805, 3802, 1604, 0, 12051, −3075, 0, 0, 0, 0); threshold (a,τ)=(6399, 0).

ψ=(−760, 184, −2, 0, 212, −60, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 1256·Q9 + 280·Q10 + 16116·Q21 + 3·Q22 + 19197·U2 + 19986·U3 + 4116·U6 = −24036.

D108. nmax F/S=9/9; cap F/S=6/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−110, 11, 4, 0, 39, −9, 0, 0, 0, 0); threshold (a,τ)=(15, 0).

ψ=(−25, 5, 0, 0, 8, −2, 0, 0, 0, 0); threshold (a,τ)=(2, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 6; 6: G ≤ 6; 7: S ≥ 4; 8: S ≤ 4; 9: t ≤ 118.

Identity: 936·Q9 + 576·Q10 + 14472·Q21 + 675·Q22 + 729·Q23 + 10125·U2 + 1458·U4 + 13662·U5 = −1512.

D109. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−45, 0, 8, −6, 21, −4, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 6·Q21 + Q23 + 2·U2 + 8·U3 = −6.

D110. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−7650, 2295, 0, 0, 2295, −765, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−425, −1105, 340, 0, 510, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 1020·Q21 + Q22 + Q23 + 1190·U1 = −5950.

D111. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−10, 3, 0, 0, 3, −1, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−25, 0, 2, 0, 9, −2, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 6·Q21 + 2·Q22 + Q23 + 2·U1 + 4·U3 = −6.

D112. nmax F/S=9/9; cap F/S=0/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−25190, 7557, 0, 0, 7557, −2519, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−1335, −3645, 1126, 0, 1660, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 146102·Q10 + 76144332·Q21 + 22729·Q22 + 22671·Q23 + 89127258·U1 + 292204·U4 = −444175270.

D113. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(261, −29, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−955, 213, 0, 28, 290, −92, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 29·Q10 + 29·Q11 + 255·Q21 + 2·Q23 + 85·U2 + 339·U3 = −226.

D114. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−300, 75, 0, 15, 75, −30, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−30, −18, 15, 0, 18, −6, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 116; 6: t ≤ 118.

Identity: Q22 + Q23 + 45·U1 + 3·U4 + 45·U5 = −48.

D115. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−55, 7, 5, 0, 18, −6, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: Q22 + 12·U1 + 5·U4 + 15·U6 = −8.

D116. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −3, 0, 2, −2, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −10.

D117. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−120, 24, 6, 0, 36, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 116; 6: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 2·Q23 + 39·U1 + 9·U4 + 36·U5 = −42.

D118. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(65, −65, 26, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 39·Q9 + 2·Q22 + 39·U1 + 13·U4 + 156·U6 = −130.

D119. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −10.

D120. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−425, 73, 28, 0, 132, −44, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−220, 17, 28, 0, 87, −29, 0, 0, 0, 0); threshold (a,τ)=(87, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 84·Q21 + Q22 + Q23 + 87·U2 + 60·U3 + 30·U5 = −84.

D121. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−45, −12, 1, 0, 19, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−435, 94, −2, 0, 137, −35, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 70·Q21 + 2·Q22 + Q23 + 2·U1 + 68·U3 = −70.

D122. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−120, 24, 6, 0, 36, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 116; 6: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 2·Q23 + 3·U1 + 12·U2 + 39·U3 + 36·U5 = −42.

D123. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(65, −65, 26, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 39·Q9 + 2·Q22 + 39·U1 + 13·U4 + 156·U6 = −130.

D124. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −10.

D125. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 116; 6: t ≤ 118.

Identity: 9·Q9 + Q22 + Q23 + 3·U1 + 3·U3 + 18·U5 = −30.

D126. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 3·U1 + U4 + 12·U6 = −10.

D127. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −10.

D128. nmax F/S=9/9; cap F/S=4/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−55, 11, 0, 0, 16, −4, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1045, 134, 44, 0, 367, −95, 0, 0, 0, 0); threshold (a,τ)=(203, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 164·Q21 + 4·Q22 + Q23 + 203·U2 + 12·U4 + 190·U5 = −164.

D129. nmax F/S=9/9; cap F/S=6/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−85, −11, −2, 0, 32, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−440, 81, 2, 0, 143, −35, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 70·Q21 + Q22 + Q23 + 2·U2 + 72·U3 = −70.

D130. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−220, 17, 28, 0, 87, −29, 0, 0, 0, 0); threshold (a,τ)=(87, 0).

ψ=(−425, 73, 28, 0, 132, −44, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 84·Q21 + Q22 + Q23 + 87·U2 + 30·U3 + 60·U5 = −84.

D131. nmax F/S=9/9; cap F/S=6/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1045, 134, 44, 0, 367, −95, 0, 0, 0, 0); threshold (a,τ)=(203, 0).

ψ=(−55, 11, 0, 0, 16, −4, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 164·Q21 + Q22 + 4·Q23 + 203·U2 + 190·U3 + 12·U6 = −164.

D132. nmax F/S=9/9; cap F/S=6/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−15, 2, 0, 0, 5, −1, 0, 0, 0, 0); threshold (a,τ)=(1, 0).

ψ=(−155, 22, 4, 0, 53, −13, 0, 0, 0, 0); threshold (a,τ)=(25, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 6; 6: G ≤ 6; 7: S ≥ 6; 8: S ≤ 6; 9: t ≤ 118.

Identity: 28·Q21 + 4·Q22 + Q23 + 4·U2 + 25·U4 + 4·U5 + 26·U7 = −28.

D133. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−470, 141, 0, 0, 141, −47, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(280, −113, 8, 0, −75, 29, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 24·Q21 + Q22 + Q23 + 86·U1 + 58·U4 = −24.

D134. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−12435, 2614, 0, −58, 3977, −997, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−1300, −310, 0, 25, 520, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 2006·Q21 + Q22 + 2·Q23 + 1944·U1 + 58·U3 = −2006.

D135. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−16715, 2846, 0, 838, 5113, −1573, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−1875, 500, 0, 0, 275, −100, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 1375·Q9 + 50·Q11 + 22476·Q21 + 9·Q22 + 29664·U1 + 75·U2 + 6117·U4 = −88626.

D136. nmax F/S=9/9; cap F/S=0/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−46790, 11111, 1672, 0, 15291, −5097, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(4315, −4315, 1726, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 139806·Q10 + 48455724·Q21 + 9493·Q22 + 9439·Q23 + 80619734·U1 + 16151908·U4 = −480502866.

D137. nmax F/S=9/9; cap F/S=0/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−46790, 11111, 1672, 0, 15291, −5097, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(4315, −4315, 1726, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 139806·Q10 + 48455724·Q21 + 9493·Q22 + 9439·Q23 + 80619734·U1 + 16151908·U4 = −480502866.

D138. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−7225, −935, −170, 0, 2720, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−66725, 12580, 0, 0, 21505, −5185, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 10880·Q21 + 2·Q22 + Q23 + 10710·U3 = −27370.

D139. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(55, −33, 0, 22, −22, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−1465, 280, 62, 0, 441, −138, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 186·Q21 + 2·Q22 + Q23 + 62·U2 + 232·U3 = −186.

D140. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(25, −25, 10, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −3, 0, 2, −2, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: e ≤ 58; 6: t ≤ 118.

Identity: 15·Q9 + 2·Q22 + 15·U1 + 5·U4 + 60·U5 = −40.

D141. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−15, −13, 0, 16, 0, −4, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: e ≥ 60; 6: t ≤ 118.

Identity: 21·Q9 + 96·Q21 + 2·Q23 + 11·U2 + 37·U3 + 80·U5 = −96.

D142. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(25, −25, 10, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: e ≤ 58; 6: t ≤ 118.

Identity: 15·Q9 + 2·Q22 + 15·U1 + 5·U4 + 60·U5 = −40.

D143. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−25, −7, 8, 0, 12, −4, 0, 0, 0, 0); threshold (a,τ)=(8, 2).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 8; 6: S ≤ 8; 7: e ≥ 60; 8: t ≤ 118.

Identity: 7·Q9 + 48·Q21 + 2·Q23 + 16·U2 + 9·U4 + 23·U5 + 48·U7 = −48.

D144. nmax F/S=9/9; cap F/S=2/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1115, 234, 42, 0, 329, −106, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−340, 4, 36, 0, 136, −36, 0, 0, 0, 0); threshold (a,τ)=(80, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 154700·Q9 + 47320·Q10 + 900354·Q21 + 663·Q22 + 9516·Q23 + 761280·U2 + 717366·U5 = −1075074.

D145. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−760, −1847, 565, 0, 869, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−6410, 1923, 0, 0, 1923, −641, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 55126·Q10 + 19887666·Q21 + 11538·Q22 + 23162·Q23 + 55126·U1 + 23064462·U3 = −159742328.

D146. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(110, −140, 54, 0, 10, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−3005, 630, 114, 0, 887, −286, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 382·Q21 + 2·Q22 + Q23 + 114·U2 + 464·U3 = −382.

D147. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(25, −25, 10, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: e ≤ 58; 6: t ≤ 118.

Identity: 15·Q9 + 2·Q22 + 15·U1 + 5·U4 + 60·U5 = −40.

D148. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−15, −29, 16, 0, 16, −4, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: e ≥ 60; 6: t ≤ 118.

Identity: 21·Q9 + 96·Q21 + 2·Q23 + 11·U2 + 37·U3 + 80·U5 = −96.

D149. nmax F/S=9/9; cap F/S=4/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(25, −25, 10, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: e ≤ 58; 6: t ≤ 118.

Identity: 15·Q9 + 2·Q22 + 15·U1 + 5·U4 + 60·U5 = −40.

D150. nmax F/S=9/9; cap F/S=4/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−25, −7, 8, 0, 12, −4, 0, 0, 0, 0); threshold (a,τ)=(8, 2).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 8; 6: S ≤ 8; 7: e ≥ 60; 8: t ≤ 118.

Identity: 7·Q9 + 48·Q21 + 2·Q23 + 16·U2 + 9·U4 + 23·U5 + 48·U7 = −48.

D151. nmax F/S=9/9; cap F/S=4/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−385, 77, 0, 0, 112, −28, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−2185, 217, 92, 0, 780, −186, 0, 0, 0, 0); threshold (a,τ)=(430, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 444·Q21 + 3·Q22 + Q23 + 430·U2 + 76·U3 + 372·U5 = −444.

D152. nmax F/S=9/9; cap F/S=6/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(345, −533, 326, 0, 0, −47, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−4410, 1043, 280, 0, 1323, −511, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 558·Q21 + Q22 + Q23 + 186·U2 + 696·U3 = −558.

D153. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(110, −140, 54, 0, 10, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−3005, 630, 114, 0, 887, −286, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 6; 2: G ≤ 6; 3: S ≥ 8; 4: S ≤ 8; 5: t ≤ 118.

Identity: 382·Q21 + 2·Q22 + Q23 + 114·U2 + 464·U3 = −382.

D154. nmax F/S=9/9; cap F/S=6/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1040, 263, 28, 0, 333, −111, 0, 0, 0, 0); threshold (a,τ)=(333, 0).

ψ=(−1235, 47, 100, 0, 432, −108, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 300·Q21 + Q22 + Q23 + 333·U2 + 122·U3 + 188·U5 = −300.

D155. nmax F/S=9/9; cap F/S=6/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1145, 164, 44, 0, 397, −105, 0, 0, 0, 0); threshold (a,τ)=(233, 0).

ψ=(−65, 9, 0, 0, 20, −4, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 6; 4: G ≤ 6; 5: S ≥ 8; 6: S ≤ 8; 7: t ≤ 118.

Identity: 164·Q21 + Q22 + 2·Q23 + 233·U2 + 210·U3 + 28·U6 = −164.

D156. nmax F/S=9/9; cap F/S=6/8. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−105, 14, 0, 0, 35, −7, 0, 0, 0, 0); threshold (a,τ)=(7, 0).

ψ=(−205, 28, 2, 0, 69, −15, 0, 0, 0, 0); threshold (a,τ)=(34, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 6; 6: G ≤ 6; 7: S ≥ 8; 8: S ≤ 8; 9: t ≤ 118.

Identity: 96·Q21 + 3·Q22 + 2·Q23 + 21·U2 + 68·U4 + 38·U5 + 60·U7 = −96.

D157. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−12820, 3834, 12, 0, 3846, −1285, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(3190, −957, 0, 0, −957, 319, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 1914·Q11 + 5742·Q21 + 319·Q22 + 1270·Q23 + 816002·U1 + 812174·U4 = −4830936.

D158. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−21470, 6441, 0, 0, 6441, −2147, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(20180, −6273, −12, 0, −6015, 2069, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 300580·Q11 + 1571604·Q21 + 6581·Q22 + 6441·Q23 + 27734946·U1 + 26352278·U4 = −156971464.

D159. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−25, 2, 0, 2, 7, −2, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−10, 3, 0, 0, 3, −1, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 6·Q21 + Q22 + 2·Q23 + 4·U1 + 2·U3 = −6.

D160. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1345, 172, 62, 0, 429, −114, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 186·Q21 + Q22 + 2·Q23 + 224·U1 + 62·U4 = −186.

D161. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(55, −55, 22, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−1270, 291, 0, 50, 379, −129, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 150·Q21 + Q22 + Q23 + 50·U2 + 236·U3 = −150.

D162. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−125, −7, 16, 0, 48, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−250, 61, 8, 0, 81, −27, 0, 0, 0, 0); threshold (a,τ)=(27, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 48·Q21 + Q22 + Q23 + 27·U2 + 16·U3 + 38·U5 = −48.

D163. nmax F/S=9/9; cap F/S=6/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(55, −55, 22, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1270, 241, 50, 0, 429, −129, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 150·Q21 + Q22 + Q23 + 50·U2 + 236·U3 = −150.

D164. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−55, −113, 56, 0, 60, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1310, 323, 40, 0, 423, −141, 0, 0, 0, 0); threshold (a,τ)=(141, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 168·Q21 + Q22 + Q23 + 141·U2 + 16·U4 + 226·U5 = −168.

D165. nmax F/S=9/9; cap F/S=8/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(55, −55, 22, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−1270, 241, 50, 0, 429, −129, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 150·Q21 + Q22 + Q23 + 50·U2 + 236·U3 = −150.

D166. nmax F/S=9/9; cap F/S=8/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−75, −7, 8, 0, 30, −6, 0, 0, 0, 0); threshold (a,τ)=(2, 0).

ψ=(−110, 33, 0, 0, 33, −11, 0, 0, 0, 0); threshold (a,τ)=(11, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 8; 6: G ≤ 8; 7: S ≥ 2; 8: S ≤ 2; 9: t ≤ 118.

Identity: 154·Q9 + 792·Q21 + 33·Q22 + 5·Q23 + 66·U2 + 55·U4 + 550·U5 = −176.

D167. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−6410, 1919, 4, 0, 1923, −642, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(2130, −639, 0, 0, −639, 213, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 426·Q11 + 1278·Q21 + 213·Q22 + 637·Q23 + 272640·U1 + 271788·U4 = −1077780.

D168. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−6710, 1215, 0, 304, 2051, −633, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 51·Q9 + 1824·Q21 + 2·Q22 + 2583·U1 + 557·U4 = −15798.

D169. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−8630, 1997, 592, 0, 2589, −1011, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(60, −315, 228, 0, 51, −51, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 1164·Q21 + Q22 + 3·Q23 + 1338·U1 + 286·U4 = −1164.

D170. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−280, −203, 49, 0, 161, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−4155, 1054, 0, −10, 1289, −373, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 614·Q21 + 2·Q22 + Q23 + 10·U1 + 648·U3 = −614.

D171. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1405, 288, 0, 62, 373, −126, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(10, −6, 0, 4, −4, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 186·Q21 + Q22 + 6·Q23 + 228·U1 + 62·U4 = −186.

D172. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−3455, 738, 114, 0, 1013, −322, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(80, −164, 60, 0, 28, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 454·Q21 + Q22 + 2·Q23 + 524·U1 + 114·U4 = −454.

D173. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−1265, 236, 0, 54, 381, −118, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 162·Q21 + Q23 + 54·U2 + 236·U3 = −162.

D174. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−145, 13, 8, 0, 48, −12, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 116; 6: t ≤ 118.

Identity: 3·Q9 + 2·Q22 + 51·U1 + 13·U4 + 48·U5 = −90.

D175. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−65, 6, 0, 7, 17, −6, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≤ 58; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + Q22 + 15·U1 + 4·U4 + 9·U6 = −12.

D176. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−55, 11, 0, 0, 16, −4, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 14·Q21 + Q22 + Q23 + 6·U1 + 14·U6 = −4.

D177. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−340, 10, 29, 0, 120, −30, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−215, 54, 6, 0, 69, −23, 0, 0, 0, 0); threshold (a,τ)=(46, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 174·Q21 + 2·Q22 + 3·Q23 + 138·U2 + 102·U3 + 80·U5 = −174.

D178. nmax F/S=9/9; cap F/S=6/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−280, −203, 49, 0, 161, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−4155, 1064, −10, 0, 1279, −373, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 614·Q21 + 2·Q22 + Q23 + 10·U1 + 648·U3 = −614.

D179. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1305, 270, 54, 0, 387, −126, 0, 0, 0, 0); threshold (a,τ)=(243, 2).

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 116; 8: t ≤ 118.

Identity: Q22 + 8·Q23 + 243·U2 + 204·U3 + 54·U6 + 162·U7 = −168.

D180. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−760, 136, 47, 0, 234, −78, 0, 0, 0, 0); threshold (a,τ)=(156, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 4; 6: S ≤ 4; 7: t ≥ 118; 8: e ≤ 58; 9: t ≤ 118.

Identity: 54·Q21 + Q22 + 9·Q23 + 156·U2 + 138·U3 + 47·U6 + 87·U8 = −54.

D181. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≥ 118; 6: e ≥ 60; 7: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U6 = −2.

D182. nmax F/S=9/9; cap F/S=6/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−70, −14, 11, 0, 30, −6, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−125, 34, 2, 0, 39, −13, 0, 0, 0, 0); threshold (a,τ)=(26, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 66·Q21 + 2·Q22 + 3·Q23 + 78·U2 + 18·U3 + 56·U5 = −66.

D183. nmax F/S=9/9; cap F/S=8/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−26405, 3892, 1114, 0, 9173, −2503, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 7·Q9 + 14·Q10 + 30036·Q21 + 9·Q23 + 10005·U2 + 45033·U3 = −64980.

D184. nmax F/S=9/9; cap F/S=8/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−33195, 2434, 1774, 0, 12173, −2879, 0, 0, 0, 0); threshold (a,τ)=(6206, 0).

ψ=(−9500, 2300, −25, 0, 2650, −750, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 7850·Q9 + 1750·Q10 + 58398·Q21 + 9·Q22 + 55854·U2 + 61422·U3 + 11616·U6 = −126198.

D185. nmax F/S=9/9; cap F/S=8/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−785, 17, 32, 0, 292, −54, 0, 0, 0, 0); threshold (a,τ)=(54, 0).

ψ=(−1205, 259, 0, 0, 382, −100, 0, 0, 0, 0); threshold (a,τ)=(118, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 8; 6: G ≤ 8; 7: S ≥ 4; 8: S ≤ 4; 9: t ≤ 118.

Identity: 260·Q21 + Q22 + Q23 + 54·U2 + 118·U4 + 108·U5 + 168·U7 = −260.

D186. nmax F/S=9/9; cap F/S=0/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−12820, 3842, 4, 0, 3846, −1283, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(800, −240, 0, 0, −240, 80, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 80·Q11 + 240·Q21 + 40·Q22 + 639·Q23 + 102480·U1 + 102320·U4 = −202880.

D187. nmax F/S=9/9; cap F/S=0/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−3910, 1173, 0, 296, 877, −465, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(−535, −311, 350, 0, 282, −141, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 606·Q21 + Q22 + Q23 + 580·U1 + 14·U4 = −606.

D188. nmax F/S=9/9; cap F/S=0/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−8570, 2003, 568, 0, 2571, −999, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(250, −955, 664, 0, 141, −141, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 1140·Q21 + Q22 + Q23 + 1334·U1 + 286·U4 = −1140.

D189. nmax F/S=9/9; cap F/S=0/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−13500, 1987, 561, 0, 4687, −1274, 0, 0, 0, 0); threshold (a,τ)=(0, 8).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 47·Q9 + 94·Q10 + 30858·Q21 + 18·Q22 + 46005·U1 + 10239·U4 = −266892.

D190. nmax F/S=9/9; cap F/S=2/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(113, −46, 0, 14, 7, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(−230, 69, 0, 0, 69, −23, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 42·Q21 + Q22 + 2·Q23 + 78·U3 = −42.

D191. nmax F/S=9/9; cap F/S=2/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1465, 342, 0, 62, 379, −138, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(55, −55, 22, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 186·Q21 + Q22 + 2·Q23 + 232·U1 + 62·U4 = −186.

D192. nmax F/S=9/9; cap F/S=2/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1465, 280, 62, 0, 441, −138, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(55, −55, 22, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 186·Q21 + Q22 + 2·Q23 + 232·U1 + 62·U4 = −186.

D193. nmax F/S=9/9; cap F/S=2/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1465, 280, 62, 0, 441, −138, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

ψ=(55, −55, 22, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 186·Q21 + Q22 + 2·Q23 + 232·U1 + 62·U4 = −186.

D194. nmax F/S=9/9; cap F/S=4/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−1315, 236, 0, 58, 383, −118, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 174·Q21 + Q23 + 58·U2 + 236·U3 = −174.

D195. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−20, −2, 3, 0, 8, −2, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: e ≤ 58; 6: t ≤ 118.

Identity: 3·Q9 + 8·Q21 + 2·Q22 + 11·U1 + 3·U4 + 10·U5 = −8.

D196. nmax F/S=9/9; cap F/S=4/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: e ≥ 60; 6: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U5 = −8.

D197. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−20, −2, 3, 0, 8, −2, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: e ≤ 58; 6: t ≤ 118.

Identity: 3·Q9 + 8·Q21 + 2·Q22 + 11·U1 + 3·U4 + 10·U5 = −8.

D198. nmax F/S=9/9; cap F/S=4/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: e ≥ 60; 6: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U5 = −8.

D199. nmax F/S=9/9; cap F/S=4/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1235, 47, 100, 0, 432, −108, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

ψ=(−1040, 263, 28, 0, 333, −111, 0, 0, 0, 0); threshold (a,τ)=(333, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 300·Q21 + Q22 + Q23 + 333·U2 + 188·U3 + 122·U5 = −300.

D200. nmax F/S=9/9; cap F/S=6/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−1300, −335, 25, 0, 545, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−12435, 2672, −58, 0, 3919, −997, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 2006·Q21 + 2·Q22 + Q23 + 58·U1 + 1944·U3 = −2006.

D201. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(35, −35, 14, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: e ≤ 58; 6: t ≤ 118.

Identity: 21·Q9 + 2·Q22 + 21·U1 + 7·U4 + 84·U5 = −112.

D202. nmax F/S=9/9; cap F/S=6/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: e ≥ 60; 6: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U5 = −8.

D203. nmax F/S=9/9; cap F/S=6/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(35, −35, 14, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: e ≤ 58; 6: t ≤ 118.

Identity: 21·Q9 + 2·Q22 + 21·U1 + 7·U4 + 84·U5 = −112.

D204. nmax F/S=9/9; cap F/S=6/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: e ≥ 60; 6: t ≤ 118.

Identity: 3·Q9 + 12·Q21 + 2·Q23 + U2 + 3·U3 + 12·U5 = −8.

D205. nmax F/S=9/9; cap F/S=6/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−65, 9, 0, 0, 20, −4, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−1145, 164, 44, 0, 397, −105, 0, 0, 0, 0); threshold (a,τ)=(233, 0).

U: 1: kS ≥ 0; 2: kS ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 164·Q21 + 2·Q22 + Q23 + 233·U2 + 28·U4 + 210·U5 = −164.

D206. nmax F/S=9/9; cap F/S=8/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(280, −91, 4, 0, −81, 27, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−380, 107, 4, 0, 117, −39, 0, 0, 0, 0); threshold (a,τ)=(0, 6).

U: 1: G ≥ 8; 2: G ≤ 8; 3: S ≥ 6; 4: S ≤ 6; 5: t ≤ 118.

Identity: 12·Q21 + Q22 + Q23 + 58·U2 + 74·U3 = −12.

D207. nmax F/S=9/9; cap F/S=8/2. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−340, 4, 36, 0, 136, −36, 0, 0, 0, 0); threshold (a,τ)=(80, 0).

ψ=(−1115, 234, 42, 0, 329, −106, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 8; 4: G ≤ 8; 5: S ≥ 6; 6: S ≤ 6; 7: t ≤ 118.

Identity: 154700·Q9 + 47320·Q11 + 900354·Q21 + 9516·Q22 + 663·Q23 + 761280·U2 + 717366·U3 = −1075074.

D208. nmax F/S=9/9; cap F/S=8/4. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−415, 28, 23, 0, 153, −36, 0, 0, 0, 0); threshold (a,τ)=(76, 0).

ψ=(−175, 42, 0, 0, 49, −14, 0, 0, 0, 0); threshold (a,τ)=(7, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 8; 6: G ≤ 8; 7: S ≥ 6; 8: S ≤ 6; 9: t ≤ 118.

Identity: 19950·Q9 + 3990·Q10 + 271530·Q21 + 3150·Q22 + 2160·Q23 + 239400·U2 + 15120·U4 + 250740·U5 = −379260.

D209. nmax F/S=9/9; cap F/S=8/6. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−345, 17, 8, 0, 124, −22, 0, 0, 0, 0); threshold (a,τ)=(22, 0).

ψ=(−535, 103, 0, 0, 172, −42, 0, 0, 0, 0); threshold (a,τ)=(80, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 8; 6: G ≤ 8; 7: S ≥ 6; 8: S ≤ 6; 9: t ≤ 118.

Identity: 116·Q21 + Q22 + Q23 + 22·U2 + 80·U4 + 44·U5 + 76·U7 = −116.

D210. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −3, 0, 2, −2, 0, 0, 0, 0, 0).

ψ=(−25, −31, 16, 0, 20, −4, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: S ≥ 10; 3: t ≤ 118.

Identity: 117·Q9 + 288·Q21 + 32·Q22 + 6·Q23 + 21·U1 + 101·U2 = −6.

D211. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−25, −15, 0, 16, 4, −4, 0, 0, 0, 0).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 10; 2: S ≥ 2; 3: t ≤ 118.

Identity: 117·Q9 + 288·Q21 + 6·Q22 + 32·Q23 + 101·U1 + 21·U2 = −6.

D212. nmax F/S=9/9; cap F/S=0/12. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−290, 87, 0, 0, 87, −29, 0, 0, 0, 0).

ψ=(−55, −13, 8, 0, 24, −4, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 0; 3: S ≥ 0; 4: S ≤ 12; 5: A ≥ 0; 6: A ≤ 0; 7: t ≤ 118.

Identity: 84·Q7 + 13·Q9 + 48·Q21 + 2·Q22 + 2·Q23 + 29·U6 = −12.

D213. nmax F/S=9/9; cap F/S=12/0. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−55, −5, 0, 8, 16, −4, 0, 0, 0, 0).

ψ=(−290, 87, 0, 0, 87, −29, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 12; 3: S ≥ 0; 4: S ≤ 0; 5: A ≥ 0; 6: A ≤ 0; 7: t ≤ 118.

Identity: 84·Q8 + 13·Q9 + 48·Q21 + 2·Q22 + 2·Q23 + 29·U6 = −12.

D214. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(150, −69, 0, 32, −53, 7, 0, 0, 0, 0).

ψ=(−25, −31, 16, 0, 20, −4, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: S ≥ 12; 3: t ≤ 118.

Identity: 806·Q9 + 1872·Q21 + 13·Q22 + 39·Q23 + 702·U2 = −1170.

D215. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(5, −3, 0, 2, −2, 0, 0, 0, 0, 0).

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 8; 2: S ≥ 8; 3: t ≤ 118.

Identity: 3·Q9 + 6·Q21 + Q22 + Q23 + U1 + U2 = −2.

D216. nmax F/S=9/9; cap F/S=1000/1000. Exclude F: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38); S: (9, 18, 15, 6, 36), (8, 16, 14, 6, 32), (9, 19, 17, 7, 38).

φ=(−25, −15, 0, 16, 4, −4, 0, 0, 0, 0).

ψ=(150, −69, 0, 32, −53, 7, 0, 0, 0, 0).

U: 1: G ≥ 12; 2: S ≥ 0; 3: t ≤ 118.

Identity: 62·Q9 + 144·Q21 + 3·Q22 + Q23 + 54·U1 = −90.

E1. nmax F/S=7/8; cap F/S=1000/1000. Exclude F: none; S: none.

φ=(158, 4, 6, 0, 99, −30, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: S ≥ 0; 3: t ≤ 118.

Identity: 85·Q7 + 19·Q8 + 25·Q9 + 18·Q21 + Q22 = −18.

F1. nmax F/S=7/9; cap F/S=1000/1000. Exclude F: none; S: none.

φ=(−141, 54, 0, 10, 116, −37, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 4; 2: S ≥ 0; 3: t ≤ 118.

Identity: 10·Q9 + 30·Q21 + Q22 + 84·U1 = −30.

F2. nmax F/S=7/9; cap F/S=0/1000. Exclude F: none; S: none.

φ=(−172, 106, 5, 0, 168, −57, 0, 0, 0, 0).

ψ=(0, 0, 0, 28, −28, −7, 0, 0, 0, 0).

U: 1: G ≥ 0; 2: G ≤ 0; 3: S ≥ 0; 4: A ≥ 0; 5: A ≤ 0; 6: t ≤ 118.

Identity: 336·Q7 + 8·Q9 + 7·Q11 + 24·Q21 + 3·Q22 = −24.

F3. nmax F/S=7/9; cap F/S=2/0. Exclude F: none; S: none.

φ=(−205, 47, 4, 0, 66, −20, 0, 0, 0, 0).

ψ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 0; 4: S ≤ 0; 5: A ≥ 0; 6: A ≤ 0; 7: t ≤ 118.

Identity: 24·Q21 + 2·Q22 + 69·U1 + 19·U4 + 11·U6 = −28.

F4. nmax F/S=7/9; cap F/S=2/1000. Exclude F: none; S: none.

φ=(90, −4, 2, 0, 33, −10, 0, 0, 0, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 6; 4: t ≤ 118.

Identity: 11·Q9 + 6·Q21 + Q22 + 31·U1 + 9·U3 = −6.

F5. nmax F/S=7/9; cap F/S=0/0. Exclude F: none; S: none.

φ=(167, −40, 0, 6, 0, 1, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−190, 57, 0, 0, 57, −19, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 18·Q21 + Q22 + 2·Q23 + 2·U2 + 70·U3 = −18.

F6. nmax F/S=7/9; cap F/S=0/2. Exclude F: none; S: none.

φ=(−213, 68, 6, 0, 120, −37, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 18·Q21 + Q22 + 74·U1 + 6·U4 = −18.

F7. nmax F/S=7/9; cap F/S=2/0. Exclude F: none; S: none.

φ=(485, −23, 0, 54, 81, −54, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−1550, 465, 0, 48, 417, −167, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 2; 4: S ≤ 2; 5: t ≤ 118.

Identity: 90·Q21 + Q22 + Q23 + 60·U1 + 280·U3 = −90.

F8. nmax F/S=7/9; cap F/S=2/2. Exclude F: none; S: none.

φ=(−149, 50, 4, 0, 87, −27, 0, 0, 0, 0); threshold (a,τ)=(7, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 118.

Identity: 12·Q21 + Q22 + 7·U2 + 54·U3 + 4·U6 = −12.

F9. nmax F/S=7/9; cap F/S=0/0. Exclude F: none; S: none.

φ=(320, −117, 0, 8, −95, 33, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(−170, 51, 0, 0, 51, −17, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 24·Q21 + Q22 + 3·Q23 + 66·U2 + 94·U3 = −24.

F10. nmax F/S=7/9; cap F/S=0/2. Exclude F: none; S: none.

φ=(−3164, 1390, 0, 0, 2115, −664, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, −67, 0, −5, 5, 18, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 216·Q21 + Q22 + 2·Q23 + 1338·U1 + 72·U4 = −216.

F11. nmax F/S=7/9; cap F/S=0/4. Exclude F: none; S: none.

φ=(−73, 24, 2, 0, 42, −13, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 6·Q21 + Q22 + 26·U1 + 2·U4 = −6.

F12. nmax F/S=7/9; cap F/S=2/0. Exclude F: none; S: none.

φ=(320, −117, 0, 8, −95, 33, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

ψ=(−170, 102, −51, 51, 0, −17, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: G ≥ 2; 2: G ≤ 2; 3: S ≥ 4; 4: S ≤ 4; 5: t ≤ 118.

Identity: 24·Q21 + Q22 + 3·Q23 + 66·U2 + 94·U3 = −24.

F13. nmax F/S=7/9; cap F/S=2/2. Exclude F: none; S: none.

φ=(−155, 54, 4, 0, 93, −29, 0, 0, 0, 0); threshold (a,τ)=(9, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 12·Q21 + Q22 + 9·U2 + 58·U3 + 4·U6 = −12.

F14. nmax F/S=7/9; cap F/S=2/4. Exclude F: none; S: none.

φ=(−155, 54, 4, 0, 93, −29, 0, 0, 0, 0); threshold (a,τ)=(9, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: S ≤ 4; 7: t ≤ 118.

Identity: 12·Q21 + Q22 + 9·U2 + 58·U3 + 4·U6 = −12.

G1. nmax F/S=8/8; cap F/S=1000/1000. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(51955, −7661, 187, 33, 0, −55, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(6077, 187).

ψ=(70, −28, 0, 21, −28, 0, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 0; 4: S ≥ 0; 5: nF8 ≥ 2; 6: nF8 ≤ 2; 7: rF8 ≤ 14; 8: t ≤ 118.

Identity: 11244·Q7 + 11178·Q8 + 11530·Q9 + 308·Q11 + 1056·Q21 + 3·Q22 + 18231·U6 + 561·U7 = −1896.

G2. nmax F/S=8/8; cap F/S=0/1000. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−10797, 4756, 0, 0, 6135, −2045, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(1379, 0).

ψ=(224, −195, 0, 82, 110, −23, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 0; 4: G ≤ 0; 5: S ≥ 0; 6: A ≥ 0; 7: A ≤ 0; 8: nF8 ≥ 4; 9: nF8 ≤ 4; 10: rF8 ≤ 32; 11: t ≤ 118.

Identity: 7840530·Q7 + 1006140·Q21 + 2045·Q22 + 4090·Q23 + 188140·U7 + 2820055·U9 = −1006140.

G3. nmax F/S=8/8; cap F/S=1000/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(133, −19, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(19, 0).

ψ=(−585, 165, 0, 4, 176, −58, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: S ≥ 0; 5: S ≤ 0; 6: A ≥ 0; 7: A ≤ 0; 8: nF8 ≥ 4; 9: nF8 ≤ 4; 10: rF8 ≤ 32; 11: t ≤ 118.

Identity: 120·Q8 + 4·Q9 + 12·Q21 + 3·Q22 + Q23 + 57·U9 = −12.

G4. nmax F/S=8/8; cap F/S=1000/1000. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(1275, −190, 4, 0, 4, −1, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(181, 0).

ψ=(30, −10, 0, 5, −5, 0, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: S ≥ 2; 5: nF8 ≥ 4; 6: nF8 ≤ 4; 7: rF8 ≤ 32; 8: t ≤ 118.

Identity: 851·Q9 + 10·Q11 + 78·Q21 + 9·Q22 + 849·U3 + 825·U4 + 1629·U6 = −318.

G5. nmax F/S=8/8; cap F/S=0/1000. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−7467, 3757, 0, 0, 5136, −1712, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(1379, 0).

ψ=(224, −195, 0, 82, 110, −23, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 0; 4: G ≤ 0; 5: S ≥ 0; 6: A ≥ 0; 7: A ≤ 0; 8: nF8 ≥ 6; 9: nF8 ≤ 6; 10: rF8 ≤ 54; 11: t ≤ 118.

Identity: 352672·Q7 + 26322·Q21 + 107·Q22 + 107·Q23 + 4922·U7 + 147553·U9 = −26322.

G6. nmax F/S=8/8; cap F/S=1000/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(9877, −1574, 0, 82, 110, −23, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(1379, 0).

ψ=(−8560, 2568, 0, 0, 2568, −856, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: S ≥ 0; 5: S ≤ 0; 6: A ≥ 0; 7: A ≤ 0; 8: nF8 ≥ 6; 9: nF8 ≤ 6; 10: rF8 ≤ 54; 11: t ≤ 118.

Identity: 3296·Q8 + 246·Q21 + Q22 + 2·Q23 + 46·U7 + 1379·U9 = −246.

G7. nmax F/S=8/8; cap F/S=1000/1000. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(1352, −197, 0, 4, 0, −1, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(192, 0).

ψ=(8, −9, 4, 0, 4, −1, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 4; 4: S ≥ 4; 5: nF8 ≥ 6; 6: nF8 ≤ 6; 7: rF8 ≤ 54; 8: t ≤ 118.

Identity: 1801·Q9 + 20·Q11 + 156·Q21 + 18·Q22 + 1797·U3 + 1749·U4 + 3456·U6 = −1014.

G8. nmax F/S=8/8; cap F/S=2/1000. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(3399, −507, 10, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(477, 0).

ψ=(8, −5, 0, 4, 0, −1, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 6; 6: nF8 ≥ 6; 7: nF8 ≤ 6; 8: rF8 ≤ 54; 9: t ≤ 118.

Identity: 1501·Q9 + 20·Q11 + 120·Q21 + 6·Q22 + 1461·U3 + 1461·U5 + 2862·U7 = −882.

G9. nmax F/S=8/8; cap F/S=1000/2. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(1121, −164, 0, 4, 0, −1, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(159, 0).

ψ=(30, −10, 0, 5, −5, 0, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 6; 4: S ≥ 2; 5: S ≤ 2; 6: nF8 ≥ 6; 7: nF8 ≤ 6; 8: rF8 ≤ 54; 9: t ≤ 118.

Identity: 752·Q9 + 10·Q11 + 78·Q21 + 9·Q22 + 750·U3 + 726·U4 + 1431·U7 = −258.

G10. nmax F/S=8/8; cap F/S=0/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−195, 86, 0, 0, 111, −37, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(25, 0).

ψ=(−15, 0, 2, 0, 12, −3, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 2; 6: S ≤ 2; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 6·Q20 + Q22 + 74·U3 + 25·U8 = −102.

G11. nmax F/S=8/8; cap F/S=0/2. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−2103, 764, 66, 0, 1068, −367, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(205, 0).

ψ=(40, −28, 12, 0, −4, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 2; 6: S ≤ 2; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 150·Q21 + Q22 + 6·Q23 + 662·U3 + 66·U6 + 205·U8 = −150.

G12. nmax F/S=8/8; cap F/S=2/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(175, −25, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(25, 0).

ψ=(−415, 117, 0, 6, 141, −46, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 2; 6: S ≤ 2; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 18·Q21 + Q22 + Q23 + 6·U4 + 92·U5 + 25·U8 = −18.

G13. nmax F/S=8/8; cap F/S=2/2. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(605, −107, 12, 0, −6, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(83, 0).

ψ=(−1174, 349, 0, 16, 365, −127, 0, 0, 0, 0); threshold (a,τ)=(127, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 2; 6: G ≤ 2; 7: S ≥ 2; 8: S ≤ 2; 9: nF8 ≥ 6; 10: nF8 ≤ 6; 11: rF8 ≤ 54; 12: t ≤ 118.

Identity: 36·Q21 + Q22 + Q23 + 127·U4 + 16·U6 + 242·U7 + 83·U10 = −36.

G14. nmax F/S=8/8; cap F/S=0/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(670, −201, 0, 8, −209, 65, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(0, 0).

ψ=(−750, 225, 0, 8, 217, −77, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: S ≤ 4; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 12·Q21 + Q22 + Q23 + 138·U4 + 146·U5 = −12.

G15. nmax F/S=8/8; cap F/S=0/2. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−90, 37, 0, 0, 54, −17, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(11, 0).

ψ=(1, −1, 0, 0, 1, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: S ≤ 4; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 6·Q21 + Q22 + 3·Q23 + 34·U3 + 11·U8 = −6.

G16. nmax F/S=8/8; cap F/S=0/4. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−82, 32, 0, 2, 49, −16, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(9, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: S ≤ 4; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 6·Q21 + Q22 + 32·U3 + 2·U6 + 9·U8 = −6.

G17. nmax F/S=8/8; cap F/S=2/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(310, −93, 0, 0, −93, 31, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(0, 0).

ψ=(−405, 111, 0, 4, 122, −40, 0, 0, 0, 0); threshold (a,τ)=(0, 4).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: S ≤ 4; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 12·Q21 + Q22 + Q23 + 66·U4 + 80·U5 = −12.

G18. nmax F/S=8/8; cap F/S=2/2. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−183, 82, 4, 0, 123, −41, 0, 0, 0, 0); threshold (a,τ)=(41, 0); full (b,c)=(29, 0).

ψ=(−5, 0, 0, 2, 1, −1, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: G ≥ 2; 6: G ≤ 2; 7: S ≥ 4; 8: S ≤ 4; 9: nF8 ≥ 6; 10: nF8 ≤ 6; 11: rF8 ≤ 54; 12: t ≤ 118.

Identity: 12·Q21 + Q22 + 2·Q23 + 41·U2 + 78·U5 + 29·U10 = −12.

G19. nmax F/S=8/8; cap F/S=2/4. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−46, 27, 0, 0, 39, −13, 0, 0, 0, 0); threshold (a,τ)=(13, 0); full (b,c)=(12, 0).

ψ=(−65, 5, 4, 0, 24, −6, 0, 0, 0, 0); threshold (a,τ)=(1, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: kS ≥ 0; 6: kS ≤ 2; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 54; 14: t ≤ 118.

Identity: Q20 + 2·Q22 + 26·U2 + 52·U7 + 24·U12 = −84.

G20. nmax F/S=8/8; cap F/S=0/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−339, 226, 0, 0, 339, −113, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(113, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 2; 6: S ≤ 2; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: Q22 + 226·U3 + 113·U8 = −1130.

G21. nmax F/S=8/8; cap F/S=0/2. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(−90, 27, 0, 0, 27, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(0, 0).

ψ=(160, −57, 4, 0, −33, 13, 0, 0, 0, 0); threshold (a,τ)=(0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 2; 6: S ≤ 2; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 12·Q21 + 3·Q22 + Q23 + 50·U3 + 26·U6 = −12.

G22. nmax F/S=8/8; cap F/S=2/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(21, −3, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(3, 0).

ψ=(−145, 39, 2, 0, 51, −16, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 2; 6: S ≤ 2; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 6·Q21 + 3·Q22 + Q23 + 2·U4 + 32·U5 + 9·U8 = −6.

G23. nmax F/S=8/8; cap F/S=2/2. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(27, 5, 2, 0, 33, −10, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(26, 0).

ψ=(−130, 39, 0, 0, 39, −13, 0, 0, 0, 0); threshold (a,τ)=(13, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: kS ≥ 0; 4: kS ≤ 2; 5: G ≥ 4; 6: G ≤ 4; 7: S ≥ 2; 8: S ≤ 2; 9: nF8 ≥ 6; 10: nF8 ≤ 6; 11: rF8 ≤ 54; 12: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 13·U4 + 20·U5 + 24·U7 + 26·U10 = −6.

G24. nmax F/S=8/8; cap F/S=4/0. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(289, −56, 10, 0, −3, −1, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(43, 0).

ψ=(−338, 93, 7, 0, 114, −38, 0, 0, 0, 0); threshold (a,τ)=(0, 2).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 4; 4: G ≤ 4; 5: S ≥ 2; 6: S ≤ 2; 7: nF8 ≥ 6; 8: nF8 ≤ 6; 9: rF8 ≤ 54; 10: t ≤ 118.

Identity: 30·Q21 + Q22 + 2·Q23 + 12·U4 + 142·U5 + 43·U8 = −30.

G25. nmax F/S=8/8; cap F/S=4/2. Exclude F: (8, 16, 14, 6, 32); S: (8, 16, 14, 6, 32).

φ=(76, −11, 0, 2, 10, −3, 0, 0, 0, 0); threshold (a,τ)=(3, 0); full (b,c)=(13, 0).

ψ=(−170, 51, 0, 0, 51, −17, 0, 0, 0, 0); threshold (a,τ)=(17, 0).

U: 1: kF ≥ 0; 2: kF ≤ 2; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: kS ≥ 0; 6: kS ≤ 2; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 54; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 3·U2 + 17·U6 + 6·U7 + 32·U9 + 13·U12 = −6.

H1. nmax F/S=8/10; cap F/S=1000/1000. Exclude F: (8, 16, 14, 6, 32); S: none.

φ=(0, −6, 0, 2, −2, 1, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−7, 0).

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 0; 4: S ≥ 0; 5: nF8 ≥ 14; 6: nF8 ≤ 14; 7: rF8 ≤ 210; 8: t ≤ 118.

Identity: 2·Q9 + 6·Q21 + Q22 + 7·U5 = −2.

H2. nmax F/S=8/10; cap F/S=1000/1000. Exclude F: (8, 16, 14, 6, 32); S: none.

φ=(−2200, −542, 0, 244, −14, 127, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−1959, 0).

ψ=(255, −100, 0, −5, 120, 0, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 0; 4: S ≥ 0; 5: nF8 ≥ 16; 6: nF8 ≤ 16; 7: rF8 ≤ 256; 8: t ≤ 118.

Identity: 149084·Q9 + 728312·Q21 + 611·Q22 + 1222·Q23 + 1196949·U5 = −728312.

H3. nmax F/S=8/10; cap F/S=1000/1000. Exclude F: (8, 16, 14, 6, 32); S: none.

φ=(−1595, 68, 0, 0, 223, 7, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−755, 0).

ψ=(−474, −68, −14, 0, 223, 0, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 0; 4: S ≥ 0; 5: nF8 ≥ 18; 6: nF8 ≤ 18; 7: rF8 ≤ 306; 8: t ≤ 118.

Identity: 446·Q21 + Q22 + Q23 + 755·U5 = −446.

H4. nmax F/S=8/10; cap F/S=1000/1000. Exclude F: (8, 16, 14, 6, 32); S: none.

φ=(−2210, 382, −370, 370, 0, 15, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−1001, 0).

ψ=(−745, −21, 0, −15, 200, 0, 0, 0, 0, 0).

U: 1: nF8 ≥ 0; 2: rF8 ≥ 0; 3: G ≥ 0; 4: S ≥ 0; 5: nF8 ≥ 20; 6: nF8 ≤ 20; 7: rF8 ≤ 360; 8: t ≤ 118.

Identity: 740·Q21 + Q22 + 2·Q23 + 1001·U5 = −740.

35 Quadrilateral links and (16,20) without full facets

For a four-dimensional facet H put $q_4(H)=\#\{\text{quadrilateral faces of H}\}$ and $\eta(H)=\Sigma_{Q\subset H,\dim Q=2}(\#\text{vert } Q-3)$. Counting vertex–polygon incidences inside H gives $\eta=J+n-e-3r$. In particular $q_4\le\eta$. Call H good if its row is Y=(7,15,14,6;29) or E=(7,16,16,7;32). The small catalogue gives $\eta(Y)=q_4(Y)=3$ and $\eta(E)=2$. It also gives the following incidence property: every quadrilateral in a good H belongs to exactly one five-vertex three-face of H.

Here is an explicit way to check the last assertion, including its incidence content. For every admissible seven-point signed Gale code of Section 7 form the face masks by the positive-circuit criterion. Retain the codes with row Y or E. For each two-face mask Q with four bits, count the three-face masks T with five bits and $Q\subset T$; the count is one. This is an exhaustive combinatorial check of all realizable codes, rather than a test on chosen coordinates. The three-face profiles are, for Y, two tetrahedra, three square pyramids and one triangular prism; for E they are either four tetrahedra, two square pyramids and one three-face with (n,e,p)=(6,10,6), or five tetrahedra and two three-faces with (n,e,p)=(6,10,6). In the latter profile there is no quadrilateral. The mask check specifies which of these faces meet, so it proves the stated incidence count as well as their numbers.

Fix a quadrilateral Q of the five-polytope. Write l for the number of facets through Q and a for the number of good ones. The link of Q is a polygon, so $l\ge3$. If all l facets are good, the number of pairs (H,T), where $Q\subset T\subset H$ and T has five vertices, is l. Every T is in exactly two facets; thus l is even and $l\ge4$. In this case $a-2(l-a)=l\le4(l-3)$. Otherwise $a\le l-1$, giving $3a-2l\le l-3\le4(l-3)$. Consequently, in all cases

$$a-2(l-a)\le 4(l-3). \tag{35.1}$$

Sum (35.1) over quadrilaterals. The left side becomes $\Sigma_{H\ \text{good}}q_4(H)-2\Sigma_{H\ \text{not good}}q_4(H)$. The right side is at most 4A, because $A=\Sigma_Q(\#\text{vert } Q-3)(l(Q)-3)$, and the terms for other polygons are nonnegative. Add $2\Sigma\eta(H)$. A Y facet now contributes $3+2\cdot3=9$; an E facet contributes $q_4+4\ge4$; every other facet contributes $-2q_4+2\eta\ge0$. Hence, with $W=9\#Y+4\#E$,

$$\begin{gathered}0\le W\le 4A+2(f_{024}-3y),\\ f_{024}=2f_{03}+2f_{14}-2I.\end{gathered} \tag{35.2}$$

The second equality follows by summing $f_{02}(H)=J(H)+2e(H)-2n(H)$ over H: $\Sigma J=2f_{03}$, $\Sigma e(H)=f_{14}$ and $\Sigma n(H)=I$. Thus (35.2) is entirely in the global variables of Section 22.

Proposition 35.1. *If v=16,h=20 and no facet is full, then s(P)≥243.* All facet rows have n≤7, and all polar facet rows have n≤10. Extend a facet cost φ of Section 22 by μ(9[Y]+4[E]). Its sum is the ordinary sum of φ plus μW. The polar cost is unchanged. The following seven exact certificates prove a contradiction under t≤118.

For each leaf use $Q_1,\dots,Q_{23}$ as before, with μW included in Q_{22}. Set $U_1=W$ and $U_2=4A+4f_{03}+4f_{14}-4I-6y-W$. These are nonnegative by (35.2). The subsequent U entries are the case conditions below, written as their nonnegative differences; an equality is written lower then upper. Every identity is an exact integer identity in

the displayed global variables.

I1. Cap F/S=1000/1000; φ=(−27, −6, 6, 0, 24, −6, 0, 0, 0, 0); μ=1; ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: G ≥ 4; 4: S ≥ 0; 5: t ≤ 118.

Identity: 12·Q21 + Q22 + U2 + 12·U3 = −12.

I2. Cap F/S=0/1000; φ=(−342, −8, 38, 40, 299, −93, 0, 0, 0, 0); μ=19; ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: G ≥ 0; 4: G ≤ 0; 5: S ≥ 0; 6: A ≥ 0; 7: A ≤ 0; 8: t ≤ 118.

Identity: 4650·Q7 + 3720·Q21 + 31·Q22 + 589·U2 + 1116·U7 = −3720.

I3. Cap F/S=2/0; φ=(−90, 24, 0, 69, 0, −36, 0, 0, 0, 0); μ=0; ψ=(−12360, 3328, 380, 0, 3708, −1331, 0, 0, 0, 0).

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 0; 6: S ≤ 0; 7: A ≥ 0; 8: A ≤ 0; 9: t ≤ 118.

Identity: 1842834·Q8 + 662592·Q21 + 5916·Q22 + 986·Q23 + 14214·U3 + 336636·U4 + 373694·U8 = −662592.

I4. Cap F/S=2/1000; φ=(234, −102, 24, 0, 24, −3, 0, 0, 0, 0); μ=7; ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 4; 6: t ≤ 118.

Identity: 30·Q21 + Q22 + 7·U2 + 6·U3 + 18·U5 = −24.

I5. Cap F/S=2/2; φ=(−615, 153, 0, 12, 186, −60, 0, 0, 0, 0); μ=0; ψ=(330, −99, 0, 0, −99, 33, 0, 0, 0, 0).

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 2; 6: S ≤ 2; 7: t ≤ 116; 8: t ≤ 118.

Identity: Q22 + Q23 + 120·U3 + 78·U6 + 36·U7 = −24.

I6. Cap F/S=2/2; φ=(−30, 5, 3, 0, 11, −4, 0, 0, 0, 0); μ=0; ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 2; 6: S ≤ 2; 7: t ≥ 118; 8: e ≤ 56; 9: t ≤ 118.

Identity: Q22 + 8·U3 + 3·U6 + 2·U7 + 11·U8 = −10.

I7. Cap F/S=2/2; φ=(−5, 0, −2, 0, 1, 1, 0, 0, 0, 0); μ=0; ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0).

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: G ≥ 2; 4: G ≤ 2; 5: S ≥ 2; 6: S ≤ 2; 7: t ≥ 118; 8: e ≥ 58; 9: t ≤ 118.

Identity: 8·Q21 + Q22 + 2·U4 + 2·U5 + 14·U8 = −8.

To see the coverage without any numerical assumption, G and S are nonnegative even integers. If G≥4 use I1. If G=0 use I2; Section 13 gives A=0. Otherwise G=2. The alternatives S=0 and S≥4 use I3 and I4, with A=0 in the former. In the remaining case G=S=2, even t is either at most 116 or is 118. Use I5 in the first

case. In the second, even e is either at most 56 or at least 58; use I6 or I7 respectively. The caps are valid since 2g(H)≤G and, on the polar, 2g(H*)≤S. A cap of 1000 safely covers G+S≤58. All local costs are nonnegative on the complete row domain by the catalogue and the finite affine check of Section 22. This proves Proposition 35.1.

36 A quadrilateral budget at sixteen vertices

Let Q be a quadrilateral contained in a facet H. Exactly two three-faces R,S of H contain Q, and R∩S=Q. Each has at least five vertices, and |vert R|+|vert S|≤|vert H|+4≤12 when v(P)=16. If both have at least six vertices, equality forces H to be full and vert H=vert R∪vert S. We first show that this full-cover situation forces at least four facets through Q.

Suppose instead that there are exactly three, H,H_R, H_S, where H_R and H_S are the other facets through R and S. They are distinct: otherwise that facet contains vert R∪vert S=vert H and equals H. The polygon link of Q is a triangle. Its third three-face T belongs to H_R and H_S and is not in H. A vertex w of T outside H has −w in H, since H is full. The cover puts −w in R or S, hence in H_R or H_S. That proper facet already contains w, giving an antipodal pair, which is impossible. Thus every vertex of T lies in H, contradicting T⊄H. This proves the assertion.

Define $b_{55}(H)$ to be the number of quadrilaterals of H whose two containing three-faces both have five vertices, and $w(H)=q_4(H)-3b_{55}(H)$. For a Q in exactly three facets, each facet has at most one containing three-face with other than five vertices, by the preceding cover argument. The sum of these numbers is even, since every three-face through Q occurs twice. Of three numbers in {0,1}, at least one is zero. Thus at least one of the three facets contributes −2 to w at Q; the other two contribute at most 1, and their sum is at most zero. For $l\geq 4$ the sum is at most $l\leq 4(l-3)$. Summing gives

$$\begin{aligned} \Sigma_H w(H) &\leq 4A, \\ \Sigma_H (w(H)+2\eta(H)) &\leq 4A+2(f_{024}-3y). \end{aligned} \tag{36.1}$$

The same nonnegative quantity w+2η admits useful lower bounds depending only on the row. In general $b_{55}\leq q_4\leq \eta$, so w+2η≥0. In a five-vertex three-polytope, a quadrilateral can occur only in the square pyramid (5,8,5), whose base is its unique quadrilateral. Hence, if a is the number of square-pyramidal three-faces and c the number of triangular-prismatic ones,

$$2b_{55}\leq a,\ 2q_4\geq a+3c. \tag{36.2}$$

The first inequality counts two pyramids at each (5,5) quadrilateral. The second counts their bases and the three quadrilaterals of each triangular prism, with every quadrilateral counted at most twice.

We also need to exclude six purely numerical rows with eight vertices. Every proper three-face of such a facet has at most seven vertices. Euler and the degree bounds 2e≥3n and 2e≥3p leave exactly the following possible three-face number triples. The ordering is fixed for the count vectors below.

1: (4, 6, 4); 2: (5, 8, 5); 3: (5, 9, 6); 4: (6, 9, 5); 5: (6, 10, 6); 6: (6, 11, 7); 7: (6, 12, 8); 8: (7, 11, 6); 9: (7, 12, 7); 10: (7, 13, 8); 11: (7, 14, 9); 12: (7, 15, 10).

Write c_i for these twelve counts, and N for the number of three-faces having a nontriangular polygon. The exact equations are $\Sigma c_i=r$, $\Sigma n_i c_i=J$ and $\Sigma p_i c_i=2p$. Moreover $N=c_2+c_4+c_5+c_6+c_8+c_9+c_{10}+c_{11}$. Three elementary incidence restrictions suffice:

$$\begin{aligned} c_4>0 &\Rightarrow N\geq 4;\\ c_8>0 &\Rightarrow N\geq 3;\\ c_4\geq 2 \text{ and } c_2>0 &\Rightarrow N\geq 5. \end{aligned} \tag{36.3}$$

For their proofs, every nontriangular polygon of a three-face T has a unique partner three-face of H. Different polygons of T have different partners, since two distinct three-faces meet in a single proper face. Thus N is at least one plus the number of nontriangular polygons of T. A triangular prism has three, proving the first restriction. A (7,11,6) three-polytope has polygon excess $2\cdot 11-3\cdot 6=4$. Each of its polygons has at most five sides, since its distinct edges require distinct other facets among the remaining five; it therefore has at least two nontriangular polygons. This proves the second restriction. Finally, if two prisms and a square pyramid belonged to at most four nonsimplicial three-faces, each prism would have all three other faces as partners. Each prism would share a quadrilateral with the pyramid. Those quadrilaterals are distinct, since each polygon lies in exactly two three-faces. This contradicts the pyramid’s unique quadrilateral and proves the third restriction.

Solving the three count equations with $c_i\geq 0$ for each row below gives exactly the listed vectors. Every one violates (36.3), so none of the six rows is geometrically possible. The lists are short enough to check by substitution; a systematic enumeration chooses c_1 from 0 to r, subtracts its contributions, then c_2, and so on, rejecting a negative remainder. At the twelfth step all three remainders must be zero.

Excluded row (8, 18, 18, 8, 38). (4, 2, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0); (5, 0, 0, 2, 1, 0, 0, 0, 0, 0, 0, 0); (5, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0); (6, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0).

Excluded row (8, 18, 18, 8, 39). (4, 1, 0, 3, 0, 0, 0, 0, 0, 0, 0, 0); (5, 0, 0, 2, 0, 0, 0, 1, 0, 0, 0, 0).

Excluded row (8, 19, 21, 10, 44). (8, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0).

Excluded row (8, 20, 23, 11, 47). (9, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0); (10, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0).

Excluded row (8, 20, 23, 11, 48). (9, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0).

Excluded row (8, 19, 20, 9, 42). (5, 2, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0); (6, 0, 0, 2, 1, 0, 0, 0, 0, 0, 0, 0); (6, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0); (7, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0).

One additional obstruction comes from vertex parity. At each vertex x of H, the degrees of x in its incident three-faces sum to twice the number of polygons of H through x: this is the face–edge count in the three-dimensional vertex figure. A tetrahedron contributes degree 3. A square pyramid contributes 4 minus the indicator that x belongs to its quadrilateral base. These base indicators sum to an even number because each quadrilateral belongs to two three-faces. If every three-face is a tetrahedron or square pyramid, the number of tetrahedra through each vertex is therefore even. Exactly two tetrahedra would have identical vertex sets, a contradiction. The same argument outside one exceptional three-face shows that a single tetrahedron and square pyramids cannot occur with just that exception: all tetrahedron vertices would lie in the distinct exceptional three-face.

Here is the row-weight table used below. Unlisted rows have weight zero. Each entry $\lambda(R)$ satisfies $\lambda(R)\leq w(H)+2\eta(H)$ for a facet with row R.

Row R	**λ(R)**
(7, 15, 14, 6, 29)	9
(7, 16, 16, 7, 32)	4
(8, 18, 17, 7, 36)	15
(8, 18, 17, 7, 35)	10
(8, 19, 18, 7, 35)	9
(8, 19, 19, 8, 38)	1
(8, 18, 18, 8, 40)	18
(8, 19, 19, 8, 39)	4
(8, 19, 19, 8, 40)	8

The Y and E entries follow from Section 35, because a good facet has $b_{55}=0$. The next three rows have at most seven polar vertices, so the Gale catalogue of Section 7 computes their whole face lattice. For each original face mask M, its polar mask has one bit for every facet mask containing M; its dimension is $3-\dim M$. Count the four-bit two-face masks and those with two five-bit three-face supersets. Direct substitution into $q_4-3b_{55}+2\eta$ gives the bounds 15,10,9 for every admissible code with the stated row. This is the same exhaustive sign-code proof, now on the polar; no eight-point classification is used.

For the last four rows use the twelve count equations and (36.3), together with (36.2), $0\le b_{55}\le q_4\le\eta$. At fixed c the least possible weight is the minimum of $q-3b+2\eta$ over integers $0\le q\le\eta$, $0\le b\le\min(q,\lfloor c_2/2\rfloor)$, $2q\ge c_2+3c_4$. For (8,18,18,8;40) the only count vector is (4,0,0,4,0,0,0,0,0,0,0,0), giving $q\ge6$, $b=0,\eta=6$ and weight at least 18. For the other three rows the same finite minimum is 2,4,8 respectively, after excluding the two-tetrahedron profile by the parity argument. The weaker bound 1 for the first is sufficient here. The finite recurrence just described supplies all count vectors and all q,b values, so these minima do not presume realizability of any unclassified row.

Consequently, with $W=\Sigma_H\lambda(\text{row H})$, (36.1) implies $0\le W\le 4A+2(f_{024}-3y)$. We also use $J\ge4r$ in every local row, since each three-face has at least four vertices. These inequalities and the six exclusions are now proved mathematical inputs to the remaining sixteen-vertex cost calculation.

37 Completion of the (16,20) endpoint

Proposition 37.1. *Every centrally symmetric five-polytope with v=16 and h=20 has at least 243 nonempty faces.* Section 35 proves the no-full case. With at least four full pairs the full-normal rank is at least three: rank three is covered by Section 33, and rank at least four by Sections 30–32. It remains to treat one, two or three full pairs. A full tetrahedron-prism gives the bound by Sections 16–17, so exclude that row on the facet side. The six impossible rows of Section 36 are excluded on both sides.

Extend the costs of Section 34 by $\mu\lambda(R)$, where λ is the row weight of Section 36. The facet sum acquires μW. Every polar cost below has $\mu=0$, so no quadrilateral budget for the twenty-vertex polar is assumed. On both sides use $J\ge 4r$ in the local row domain. The full-ridge bound is $N_8=2k$ and $R_8\le 18,40,66$ for $k=1,2,3$. As in Section 34, threshold counts are at most two when their printed cap covers a maximizing local g and the threshold satisfies $2M\le\tau$ or $2M\ge G-\tau$.

Below $Q_1,\dots,Q_{23}$ retain their meaning, with threshold, full and μW terms included in the two costs. The U list first contains $W\ge 0$ and the budget of Section 36, then the threshold and full-count bounds, then the case conditions. Equalities are lower then upper. All integers multiplying Q or U are nonnegative. Each printed identity is a negative constant, proving its branch impossible under $t\le 118$.

J1. nmax F/S=8/10; cap F/S=1000/1000.

φ=(−75, −98, 8, 44, 61, −13, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−45, 0); μ=4.

ψ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: S ≥ 0; 9: nF8 ≥ 2; 10: nF8 ≤ 2; 11: rF8 ≤ 18; 12: t ≤ 118.

Identity: 36·Q9 + 132·Q21 + Q22 + 4·Q23 + 4·U2 + 62·U7 + 45·U9 = −46.

J2. nmax F/S=8/10; cap F/S=1000/1000.

φ=(998, −472, 58, 62, −2, 11, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(67, 0); μ=29.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: S ≥ 12; 9: nF8 ≥ 2; 10: nF8 ≤ 2; 11: rF8 ≤ 18; 12: t ≤ 118.

Identity: 22·Q9 + 186·Q21 + Q22 + 29·U2 + 76·U8 + 67·U10 = −186.

J3. nmax F/S=8/10; cap F/S=1000/1000.

φ=(274, −104, 14, 10, 2, 1, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(45, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: S ≥ 10; 9: nF8 ≥ 2; 10: nF8 ≤ 2; 11: rF8 ≤ 18; 12: t ≤ 118.

Identity: 2·Q9 + 30·Q21 + Q22 + 7·U2 + 20·U8 + 45·U10 = −30.

J4. nmax F/S=8/10; cap F/S=0/1000.

φ=(−162, 2, 32, 0, 168, −49, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(47, 0); μ=9.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 0; 10: A ≥ 0; 11: A ≤ 0; 12: nF8 ≥ 2; 13: nF8 ≤ 2; 14: rF8 ≤ 18; 15: t ≤ 118.

Identity: 3724·Q7 + 2058·Q21 + 49·Q22 + 441·U2 + 1078·U11 + 2303·U13 = −2058.

J5. nmax F/S=8/10; cap F/S=4/1000.

φ=(−45, −18, 12, 0, 39, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−3, 0); μ=2.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 0; 10: nF8 ≥ 2; 11: nF8 ≤ 2; 12: rF8 ≤ 18; 13: t ≤ 118.

Identity: 24·Q21 + Q22 + 2·U2 + 18·U7 + 3·U10 = −18.

J6. nmax F/S=8/10; cap F/S=6/1000.

φ=(−45, −18, 12, 0, 39, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(18, 0); μ=2.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 0; 10: nF8 ≥ 2; 11: nF8 ≤ 2; 12: rF8 ≤ 18; 13: t ≤ 118.

Identity: 24·Q21 + Q22 + 2·U2 + 18·U7 + 18·U11 = −12.

J7. nmax F/S=8/10; cap F/S=8/1000.

φ=(−25830, 6564, 1902, 0, 20808, −6081, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(26118, 0); μ=317.

ψ=(−375, −18, 0, −7, 107, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 0; 10: nF8 ≥ 2; 11: nF8 ≤ 2; 12: rF8 ≤ 18; 13: t ≤ 118.

Identity: 5004·Q21 + Q22 + 6·Q23 + 317·U2 + 12204·U7 + 26118·U11 = −5004.

J8. nmax F/S=8/10; cap F/S=10/1000.

φ=(−122256, 34584, 4872, 0, 83181, −24114, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(124470, 0); μ=812.

ψ=(−3375, −162, 0, −63, 963, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: nF8 ≥ 2; 11: nF8 ≤ 2; 12: rF8 ≤ 18; 13: t ≤ 118.

Identity: 20544·Q21 + Q22 + 6·Q23 + 812·U2 + 48606·U7 + 124470·U11 = −20544.

J9. nmax F/S=8/10; cap F/S=2/0.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

ψ=(−224970, 67299, 0, −8, 67499, −22447, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: 67491·Q8 + 128·Q9 + 72·Q11 + 384·Q21 + 2·Q23 + 22497·U8 + 22497·U12 = −173832.

J10. nmax F/S=8/10; cap F/S=2/2.

φ=(−64, 4, 0, 2, 11, −2, 9, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−7, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 2; 10: S ≤ 2; 11: nF5 ≤ 0; 12: r ≤ 60; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: Q9 + 2·Q11 + 12·Q21 + 3·Q22 + 12·U7 + 3·U8 + 3·U10 + 27·U11 + 42·U12 + 21·U13 = −12.

J11. nmax F/S=8/10; cap F/S=2/2.

φ=(−126, 28, 2, 22, 32, −21, 6, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(9, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 2; 10: S ≤ 2; 11: nF5 ≤ 0; 12: r ≥ 62; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: 66·Q21 + Q22 + U2 + 42·U7 + 18·U10 + 6·U11 + 84·U12 + 9·U14 = −66.

J12. nmax F/S=8/10; cap F/S=2/2.

φ=(−14, −4, 2, 2, 20, −5, −6, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(1, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 2; 10: S ≤ 2; 11: nF5 ≥ 2; 12: r ≤ 60; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 10·U7 + 2·U9 + 6·U11 + 8·U12 + U14 = −6.

J13. nmax F/S=8/10; cap F/S=2/2.

φ=(−60, 12, 2, 13, 22, −13, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(6, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 2; 10: S ≤ 2; 11: nF5 ≥ 2; 12: r ≥ 62; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: 39·Q21 + Q22 + U2 + 26·U7 + 9·U10 + 47·U12 + 6·U14 = −34.

J14. nmax F/S=8/10; cap F/S=2/4.

φ=(110, −116, 18, 14, 50, −5, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−9, 0); μ=9.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 4; 10: S ≤ 4; 11: r ≤ 60; 12: nF8 ≥ 2; 13: nF8 ≤ 2; 14: rF8 ≤ 18; 15: t ≤ 118.

Identity: 42·Q21 + Q22 + 9·U2 + 10·U7 + 22·U9 + 56·U11 + 9·U12 = −42.

J15. nmax F/S=8/10; cap F/S=2/4.

φ=(−30, 4, 2, 10, 14, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(5, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 4; 10: S ≤ 4; 11: r ≥ 62; 12: nF8 ≥ 2; 13: nF8 ≤ 2; 14: rF8 ≤ 18; 15: t ≤ 118.

Identity: 30·Q21 + Q22 + U2 + 18·U7 + 6·U10 + 36·U11 + 5·U13 = −14.

J16. nmax F/S=8/10; cap F/S=2/6.

φ=(170, −236, 42, 50, 134, −29, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−21, 0); μ=21.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≤ 12; 12: x ≤ 266; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: 31·Q9 + 62·Q11 + 21·Q22 + 441·U2 + 2700·U7 + 93·U8 + 807·U9 + 1482·U12 + 441·U13 = −1488.

J17. nmax F/S=8/10; cap F/S=2/6.

φ=(0, −8, 2, 4, 8, −3, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−1, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≤ 12; 12: x ≥ 268; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: 24·Q21 + Q22 + U2 + 6·U12 + U13 = −2.

J18. nmax F/S=8/10; cap F/S=2/6.

φ=(170, −236, 42, 50, 134, −29, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−21, 0); μ=21.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≥ 14; 12: x ≤ 266; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: 31·Q9 + 62·Q11 + 21·Q22 + 441·U2 + 2700·U7 + 93·U8 + 807·U9 + 1482·U12 + 441·U13 = −1488.

J19. nmax F/S=8/10; cap F/S=2/6.

φ=(0, −8, 2, 4, 8, −3, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−1, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≥ 14; 12: x ≥ 268; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 18; 16: t ≤ 118.

Identity: 24·Q21 + Q22 + U2 + 6·U12 + U13 = −2.

J20. nmax F/S=8/10; cap F/S=2/8.

φ=(234, −92, 14, 10, 14, −3, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(41, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 18; 14: t ≤ 118.

Identity: 30·Q21 + Q22 + 7·U2 + 6·U7 + 18·U9 + 41·U12 = −14.

J21. nmax F/S=8/10; cap F/S=1000/1000.

φ=(−1590, −2365, 880, 0, 1815, −223, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−810, 0); μ=58.

ψ=(190, −57, 0, 0, −57, 19, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: S ≥ 0; 9: nF8 ≥ 4; 10: nF8 ≤ 4; 11: rF8 ≤ 40; 12: t ≤ 118.

Identity: 646·Q9 + 2292·Q21 + Q22 + 3·Q23 + 58·U2 + 1092·U7 + 810·U9 = −2292.

J22. nmax F/S=8/10; cap F/S=1000/1000.

φ=(150, −102, 24, 0, 12, 3, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: S ≥ 12; 9: nF8 ≥ 4; 10: nF8 ≤ 4; 11: rF8 ≤ 40; 12: t ≤ 118.

Identity: 6·Q9 + 42·Q21 + Q22 + 5·U2 + 12·U8 + 5·U9 = −20.

J23. nmax F/S=8/10; cap F/S=4/1000.

φ=(−45, −18, 12, 0, 39, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−3, 0); μ=2.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 0; 10: nF8 ≥ 4; 11: nF8 ≤ 4; 12: rF8 ≤ 40; 13: t ≤ 118.

Identity: 24·Q21 + Q22 + 2·U2 + 18·U7 + 3·U10 = −24.

J24. nmax F/S=8/10; cap F/S=8/1000.

φ=(−405, −228, 108, 0, 285, −48, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−87, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 0; 10: nF8 ≥ 4; 11: nF8 ≤ 4; 12: rF8 ≤ 40; 13: t ≤ 118.

Identity: 66·Q9 + 282·Q21 + Q22 + 7·U2 + 162·U7 + 87·U10 = −48.

J25. nmax F/S=8/10; cap F/S=10/1000.

φ=(−75, −44, 20, 0, 51, −8, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−17, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: nF8 ≥ 4; 11: nF8 ≤ 4; 12: rF8 ≤ 40; 13: t ≤ 118.

Identity: 14·Q9 + 54·Q21 + Q22 + U2 + 30·U7 + 17·U10 = −44.

J26. nmax F/S=8/10; cap F/S=0/0.

φ=(−100, 0, −33, 57, 0, −7, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−30, 0); μ=0.

ψ=(−4510, 1353, 0, 0, 1353, −451, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 1353·Q8 + 2·Q23 + 451·U8 + 451·U12 = −4510.

J27. nmax F/S=8/10; cap F/S=0/2.

φ=(−10785, 6469, −3234, 3238, 0, −1080, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 2; 10: S ≤ 2; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 192780·Q6 + 630438600·Q7 + 456220·Q9 + 711980·Q11 + 3103680·Q21 + 388440·Q22 + 1262670·U8 + 209465070·U10 + 208886730·U12 + 388440·U13 = −1603112400.

J28. nmax F/S=8/10; cap F/S=0/4.

φ=(−805, 198, 12, 0, 255, −79, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−15, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 4; 10: S ≤ 4; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 132·Q7 + 132·Q9 + 36·Q21 + Q22 + 38·U10 + 158·U12 + 15·U13 = −36.

J29. nmax F/S=8/10; cap F/S=0/6.

φ=(−395, 104, 4, 0, 123, −39, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 6; 10: S ≤ 6; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 63·Q7 + 63·Q9 + 12·Q21 + Q22 + 19·U10 + 78·U12 + 5·U13 = −12.

J30. nmax F/S=8/10; cap F/S=0/8.

φ=(−775, 218, 4, 0, 237, −77, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 8; 10: S ≤ 8; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 120·Q7 + 120·Q9 + 12·Q21 + Q22 + 38·U10 + 154·U12 + 5·U13 = −12.

J31. nmax F/S=8/10; cap F/S=0/10.

φ=(−580, −38, 82, 0, 386, −99, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−19, 0); μ=19.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 10; 10: S ≤ 10; 11: A ≥ 0; 12: A ≤ 0; 13: x ≤ 266; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 40; 17: t ≤ 118.

Identity: 4647·Q7 + 20·Q9 + 40·Q11 + 21·Q22 + 399·U2 + 45·U10 + 777·U12 + 1326·U13 + 399·U14 = −600.

J32. nmax F/S=8/10; cap F/S=0/10.

φ=(−110, −6, 24, 0, 72, −23, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−3, 0); μ=3.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 10; 10: S ≤ 10; 11: A ≥ 0; 12: A ≤ 0; 13: x ≥ 268; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 40; 17: t ≤ 118.

Identity: 966·Q7 + 3450·Q21 + 23·Q22 + 69·U2 + 1104·U8 + 230·U10 + 92·U12 + 1104·U13 + 69·U14 = −828.

J33. nmax F/S=8/10; cap F/S=2/0.

φ=(0, −5, 5, 0, 3, −2, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(1, 0); μ=0.

ψ=(−56890, 17067, 0, 0, 17067, −5689, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 34134·Q8 + 4·Q23 + 11378·U8 + 11378·U12 = −91024.

J34. nmax F/S=8/10; cap F/S=2/2.

φ=(−45, −17, 15, 0, 40, −11, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−3, 0); μ=2.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 2; 10: S ≤ 2; 11: e ≤ 56; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 40; 15: t ≤ 118.

Identity: 14·Q21 + Q22 + 2·U2 + 22·U7 + 3·U10 + 19·U11 + 3·U12 = −14.

J35. nmax F/S=8/10; cap F/S=2/2.

φ=(−80, −3, 1, 0, 37, −5, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−8, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 2; 10: S ≤ 2; 11: e ≥ 58; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 40; 15: t ≤ 118.

Identity: 44·Q21 + Q22 + U2 + 10·U7 + 5·U9 + 47·U11 + 8·U12 = −40.

J36. nmax F/S=8/10; cap F/S=2/4.

φ=(160, −260, 82, 0, 164, −25, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−19, 0); μ=19.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 4; 10: S ≤ 4; 11: x ≤ 266; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 40; 15: t ≤ 118.

Identity: 40·Q21 + Q22 + 19·U2 + 96·U7 + 32·U9 + 46·U11 + 19·U12 = −40.

J37. nmax F/S=8/10; cap F/S=2/4.

φ=(0, −12, 6, 0, 12, −3, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−1, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 4; 10: S ≤ 4; 11: x ≥ 268; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 40; 15: t ≤ 118.

Identity: 24·Q21 + Q22 + U2 + 6·U11 + U12 = −4.

J38. nmax F/S=8/10; cap F/S=2/6.

φ=(30, −66, 24, 0, 48, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 42·Q21 + Q22 + 5·U2 + 18·U7 + 6·U9 + 5·U11 = −8.

J39. nmax F/S=8/10; cap F/S=2/8.

φ=(30, −66, 24, 0, 48, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 42·Q21 + Q22 + 5·U2 + 18·U7 + 6·U9 + 5·U11 = −20.

J40. nmax F/S=8/10; cap F/S=2/10.

φ=(30, −66, 24, 0, 48, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 10; 10: S ≤ 10; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40;

14: t ≤ 118.

Identity: 42·Q21 + Q22 + 5·U2 + 18·U7 + 6·U9 + 5·U11 = −32.

J41. nmax F/S=8/10; cap F/S=6/0.

φ=(30, −14, 5, 0, −6, 1, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

ψ=(−19290, 5787, 0, 0, 5787, −1929, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 11574·Q8 + 4·Q23 + 3858·U8 + 3858·U12 = −15432.

J42. nmax F/S=8/10; cap F/S=6/2.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(12, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 18·U7 + 4·U10 + 12·U12 = −8.

J43. nmax F/S=8/10; cap F/S=0/0.

φ=(−165, 0, 44, 0, 77, −33, 0, 0, 0, 0); threshold (a,τ)=(0, 6); full (b,c)=(−11, 0); μ=0.

ψ=(−310, 65, 24, 0, 109, −39, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 100·Q21 + Q22 + 2·Q23 + 18·U7 + 112·U9 + 11·U11 = −100.

J44. nmax F/S=8/10; cap F/S=0/2.

φ=(−6845, 1008, 286, 0, 2377, −647, 0, 0, 0, 0); threshold (a,τ)=(0, 6); full (b,c)=(−361, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 7·Q9 + 14·Q10 + 7806·Q21 + 9·Q22 + 11667·U7 + 2595·U10 + 3249·U11 = −43548.

J45. nmax F/S=8/10; cap F/S=0/4.

φ=(−7390, 1602, 615, 0, 2586, −985, 0, 0, 0, 0); threshold (a,τ)=(0, 6); full (b,c)=(0, 0); μ=0.

ψ=(0, 0, 0, 1132, −1276, −247, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 20·Q9 + 287·Q11 + 984·Q21 + Q22 + 1416·U7 + 308·U10 = −3456.

J46. nmax F/S=8/10; cap F/S=2/0.

φ=(−1765, 411, 32, 0, 558, −170, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(115, 0); μ=0.

ψ=(1030, −309, 0, 0, −309, 103, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 96·Q21 + Q22 + Q23 + 340·U7 + 238·U10 + 115·U12 = −96.

J47. nmax F/S=8/10; cap F/S=2/2.

φ=(−14765, 3386, 287, 0, 4753, −1440, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(1080, 0); μ=0.

ψ=(7930, −2379, 0, 0, −2379, 793, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 3965·Q10 + 6168747·Q21 + 7137·Q22 + 7132·Q23 + 20558525·U7 + 13367601·U10 + 7707960·U12 = −16323112.

J48. nmax F/S=8/10; cap F/S=2/4.

φ=(−14765, 3386, 287, 0, 4753, −1440, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(1080, 0); μ=0.

ψ=(7930, −2379, 0, 0, −2379, 793, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 3965·Q10 + 6168747·Q21 + 7137·Q22 + 7132·Q23 + 20558525·U7 + 13367601·U10 + 7707960·U12 = −16323112.

J49. nmax F/S=8/10; cap F/S=4/0.

φ=(−2435, −3271, 928, 0, 1902, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(−1415, 0); μ=0.

ψ=(−26570, 7449, 0, −20, 8159, −2553, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 3744·Q21 + Q22 + Q23 + 20·U7 + 4178·U9 + 1415·U11 = −3744.

J50. nmax F/S=8/10; cap F/S=4/2.

φ=(−760, 126, 39, 0, 264, −79, 0, 0, 0, 0); threshold (a,τ)=(52, 2); full (b,c)=(46, 0); μ=0.

ψ=(0, 0, 0, 28, −28, −7, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 7·Q11 + 96·Q21 + Q22 + 52·U4 + 144·U9 + 32·U12 + 46·U14 = −96.

J51. nmax F/S=8/10; cap F/S=4/4.

φ=(−760, 126, 39, 0, 264, −79, 0, 0, 0, 0); threshold (a,τ)=(52, 2); full (b,c)=(46, 0); μ=0.

ψ=(0, 0, 0, 28, −28, −7, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 7·Q11 + 96·Q21 + Q22 + 52·U4 + 144·U9 + 32·U12 + 46·U14 = −96.

J52. nmax F/S=8/10; cap F/S=6/0.

φ=(155, −116, 44, 0, −15, 3, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(−19, 0); μ=0.

ψ=(−410, 107, 16, 0, 123, −45, 0, 0, 0, 0); threshold (a,τ)=(0, 4); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 84·Q21 + Q22 + 2·Q23 + 38·U8 + 136·U9 + 19·U11 = −84.

J53. nmax F/S=8/10; cap F/S=6/2.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold (a,τ)=(6, 0); full (b,c)=(6, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 2); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 6·U4 + 18·U9 + 4·U12 + 6·U14 = −12.

J54. nmax F/S=8/10; cap F/S=6/4.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold (a,τ)=(6, 0); full (b,c)=(6, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 0); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 6·U4 + 18·U9 + 4·U12 + 6·U14 = −12.

J55. nmax F/S=8/10; cap F/S=6/6.

φ=(−215, 2, 20, 0, 85, −21, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 6; 10: S ≤ 6; 11: y ≤ 284; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 40; 15: t ≤ 118.

Identity: 10·Q9 + 12·Q21 + Q22 + 52·U7 + 6·U9 + 16·U11 = −12.

J56. nmax F/S=8/10; cap F/S=6/6.

φ=(−55, 2, 4, 0, 21, −5, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 6; 10: S ≤ 6; 11: y ≥ 286; 12: I ≤ 136; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 40; 16: t ≤ 118.

Identity: Q9 + U7 + U9 + 2·U12 = −2.

J57. nmax F/S=8/10; cap F/S=6/6.

φ=(−378, 6, 24, 0, 96, −21, 48, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−33, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 6; 10: S ≤ 6; 11: y ≥ 286; 12: I ≥ 138; 13: nF5 ≤ 2; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 40; 17: t ≤ 118.

Identity: 66·Q21 + Q22 + U2 + 42·U7 + 18·U10 + 48·U13 + 33·U14 = −48.

J58. nmax F/S=8/10; cap F/S=6/6.

φ=(57, −14, 4, 0, 21, −5, −32, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(16, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 6; 10: S ≤ 6; 11: y ≥ 286; 12: I ≥ 138; 13: nF5 ≥ 4; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 40; 17: t ≤ 118.

Identity: 20·Q9 + 60·Q21 + 5·Q22 + 70·U7 + 60·U12 + 160·U13 + 80·U15 = −120.

J59. nmax F/S=8/10; cap F/S=6/8.

φ=(−450, −82, 56, 0, 144, −25, 80, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−85, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 8; 10: S ≤ 8; 11: nF5 ≤ 2; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 40; 15: t ≤ 118.

Identity: 8·Q9 + 138·Q21 + Q22 + 5·U2 + 58·U7 + 18·U10 + 80·U11 + 85·U12 = −28.

J60. nmax F/S=8/10; cap F/S=6/8.

φ=(15, −8, 4, 0, 21, −5, −20, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(10, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 8; 10: S ≤ 8; 11: nF5 ≥ 4; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 40; 15: t ≤ 118.

Identity: 7·Q9 + 12·Q21 + Q22 + 17·U7 + 3·U9 + 20·U11 + 10·U13 = −12.

J61. nmax F/S=8/10; cap F/S=6/10.

φ=(410, −294, 72, 0, 48, 5, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−15, 0); μ=15.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 10; 10: S ≤ 10; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 40; 14: t ≤ 118.

Identity: 16·Q9 + 126·Q21 + Q22 + 15·U2 + 6·U7 + 34·U9 + 15·U11 = −24.

J62. nmax F/S=8/10; cap F/S=1000/1000.

φ=(−1798661089940650, −983156041597462, 406214273493176, 0, 989719957352340, −114815061188275, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−442644998845296, 0); μ=0.

ψ=(−26625, −1278, 0, −497, 7597, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: S ≥ 0; 9: nF8 ≥ 6; 10: nF8 ≤ 6; 11: rF8 ≤ 66; 12: t ≤ 118.

Identity: 114396032629780759240·Q9 + 361513901464907703720·Q21 + 281615·Q22 + 1290549547575030·Q23 + 179704722667997677400·U7 + 124655471349818033040·U9 = −180756950732443713720.

J63. nmax F/S=8/10; cap F/S=1000/1000.

φ=(150, −102, 24, 0, 12, 3, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: S ≥ 12; 9: nF8 ≥ 6; 10: nF8 ≤ 6; 11: rF8 ≤ 66; 12: t ≤ 118.

Identity: 6·Q9 + 42·Q21 + Q22 + 5·U2 + 12·U8 + 5·U9 = −30.

J64. nmax F/S=8/10; cap F/S=1000/1000.

φ=(410, −294, 72, 0, 48, 5, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−15, 0); μ=15.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: S ≥ 10; 9: nF8 ≥ 6; 10: nF8 ≤ 6; 11: rF8 ≤ 66; 12: t ≤ 118.

Identity: 6·Q7 + 16·Q9 + 126·Q21 + Q22 + 15·U2 + 34·U8 + 15·U9 = −18.

J65. nmax F/S=8/10; cap F/S=0/1000.

φ=(−90, −30, 24, 0, 84, −21, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 0; 10: A ≥ 0; 11: A ≤ 0; 12: nF8 ≥ 6; 13: nF8 ≤ 6; 14: rF8 ≤ 66; 15: t ≤ 118.

Identity: 252·Q7 + 294·Q21 + 7·Q22 + 35·U2 + 42·U11 + 35·U12 = −42.

J66. nmax F/S=8/10; cap F/S=2/1000.

φ=(−45, −18, 12, 0, 39, −9, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−5, 0); μ=2.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 0; 10: nF8 ≥ 6; 11: nF8 ≤ 6; 12: rF8 ≤ 66; 13: t ≤ 118.

Identity: 24·Q21 + Q22 + 2·U2 + 18·U7 + 5·U10 = −6.

J67. nmax F/S=8/10; cap F/S=4/1000.

φ=(−186345, −64596, 43774, 0, 146977, −33165, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−21583, 0); μ=6870.

ψ=(−4125, −198, 0, −77, 1177, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 0; 10: nF8 ≥ 6; 11: nF8 ≤ 6; 12: rF8 ≤ 66; 13: t ≤ 118.

Identity: 60634514·Q9 + 2243571982·Q21 + 23741·Q22 + 47482·Q23 + 163100670·U2 + 1639031158·U7 + 512402003·U10 = −2243571982.

J68. nmax F/S=8/10; cap F/S=6/1000.

φ=(−45, −24, 12, 0, 33, −6, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−9, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 0; 10: nF8 ≥ 6; 11: nF8 ≤ 6; 12: rF8 ≤ 66; 13: t ≤ 118.

Identity: 6·Q9 + 30·Q21 + Q22 + U2 + 18·U7 + 9·U10 = −6.

J69. nmax F/S=8/10; cap F/S=8/1000.

φ=(−405, −228, 108, 0, 285, −48, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−87, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 0; 10: nF8 ≥ 6; 11: nF8 ≤ 6; 12: rF8 ≤ 66; 13: t ≤ 118.

Identity: 66·Q9 + 282·Q21 + Q22 + 7·U2 + 162·U7 + 87·U10 = −222.

J70. nmax F/S=8/10; cap F/S=10/1000.

φ=(−145785, −68252, 32420, 0, 89721, −14288, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(−27563, 0); μ=1475.

ψ=(−1500, −44, −28, 0, 400, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: nF8 ≥ 6; 11: nF8 ≤ 6; 12: rF8 ≤ 66; 13: t ≤ 118.

Identity: 368139830·Q9 + 1455846990·Q21 + 15619·Q22 + 93714·Q23 + 23038025·U2 + 817092366·U7 + 430506497·U10 = −1455846990.

J71. nmax F/S=8/10; cap F/S=0/1000.

φ=(−162, 2, 32, 0, 168, −49, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(47, 0); μ=9.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold (a,τ)=(0, 1000000); full (b,c)=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 0; 10: A ≥ 0; 11: A ≤ 0; 12: nF8 ≥ 2; 13: nF8 ≤ 2; 14: rF8 ≤ 18; 15: t ≤ 118.

Identity: 3724·Q7 + 2058·Q21 + 49·Q22 + 441·U2 + 1078·U11 + 2303·U13 = −2058.

For completeness, coverage is checked by the following explicit integer partition. For each k=1,2,3 set $N_8=2k$ and $R_8=k(20+2k-4)$; every R_8 condition is an upper bound, so a smaller actual value satisfies the same leaves. Enumerate nonnegative even G,S with G+S≤58 and maxima $0\le M_F\le G/2, 0\le M_S\le S/2$. Discard a leaf whose cap is below 2M, or whose nonzero threshold coefficient does not satisfy the preceding alternatives. When G=0 or S=0 use A=0; otherwise discard leaves requiring A=0. On every remaining even count axis, split at all printed boundaries

b and b+2, retaining the unbounded last interval. Each condition is constant on each interval, so testing its least value is exact. This partition has 124,058 cells/maxima, all covered by the printed leaves. Together with their exact local nonnegativity and negative global identities this completes the few-full case and proves Proposition 37.1.

38 The nine-vertex U facet forces at least twenty facets

Proposition 38.1. *If v(P)=18 and P contains a facet of row U=(9,19,17,7;38), then h(P)≥20.* Its polar as a four-polytope has row (7,17,19,9;38). The small Gale catalogue gives exactly ten sorted codes with that row. The following permutations carry all their facet masks to 15,23,27,29,46,86,102,122,124.

Signed Gale code	Permutation i ↦ π(i)
(1, 1, 2, 3, 6, 7, 10)	(1, 2, 0, 3, 5, 6, 4)
(1, 2, 2, 4, 5, 8, 9)	(0, 1, 2, 3, 5, 6, 4)
(1, 3, 3, 4, 5, 8, 9)	(3, 1, 2, 0, 4, 6, 5)
(1, 4, 5, 6, 6, 8, 9)	(5, 3, 0, 1, 2, 4, 6)
(1, 4, 5, 7, 7, 8, 9)	(5, 6, 4, 1, 2, 0, 3)
(1, 4, 5, 8, 9, 10, 10)	(3, 5, 6, 4, 0, 1, 2)
(2, 3, 4, 4, 6, 7, 10)	(3, 0, 1, 2, 4, 6, 5)
(2, 3, 5, 5, 6, 7, 10)	(5, 3, 1, 2, 0, 4, 6)
(2, 3, 6, 7, 8, 8, 10)	(5, 6, 4, 0, 1, 2, 3)
(2, 3, 6, 7, 9, 9, 10)	(3, 5, 6, 4, 1, 2, 0)

The face criterion of Section 7 checks each mask directly. Completeness follows by exhausting all seven-point sorted codes in that criterion. Dualizing gives nine vertices $x_0,\dots,x_8$ of U; the bits of the ith mask specify which of its seven three-faces $G_0,\dots,G_6$ contain x_i. Since U is full, $P=\operatorname{conv}\{\pm x_0,\dots,\pm x_8\}$. In particular G_3 and G_4 are triangular prisms with product edges

$$A=[x_2,x_7],\ B=[x_3,x_8],\ C=[x_0,x_4],\ D=[x_1,x_5]. \tag{38.1}$$

The pairs AB,AC,BC,AD,BD lie respectively in quadrilateral planes with vertex sets {2,3,7,8}, {0,2,4,7}, {0,3,4,8}, {1,2,5,7}, {1,3,5,8}. The C,D endpoints and $z=x_6$ lie in the pentagon $\{0,1,4,5,6\}=G_1\cap G_2$. Thus the four supporting lines in (38.1) are all parallel or concurrent. Here is the linear-algebra proof, to include every real realization. If A,B have independent directions, their common plane makes them meet at p. If C did not pass through p, the planes containing A,C and B,C would both equal the plane spanned by C and p; G_3 would then be planar, contradicting its dimension three. Hence C passes through p, and the same argument in G_4 applies to D. If A,B are parallel, a direction of C independent of theirs would make those same two planes equal and again collapse G_3 to dimension two. Thus C and similarly D are parallel to them.

Parallel case. Put $d=x_7-x_2$. Write $x_8=x_3+\mu_B d$, $x_4=x_0+\mu_C d$ and $x_5=x_1+\mu_D d$, with all μ nonzero. The pentagon plane gives $z=x_0+\alpha d+\beta(x_1-x_0)$. A functional zero on G_3 and strictly positive at the remaining U vertices gives $\beta>0$; the analogous functional for G_4 gives $\beta<1$. The five vectors x_2,x_3,x_0,x_1,d form a basis: their span contains all nine x_i by these formulas, hence is $\mathbb{R}^5$.

For every $(a,b,c,e)\in\{-1,1\}^4$ define the functional y with values a,b,c,e on x_2,x_3,x_0,x_1 and value zero on d. All $|y(x_i)|\le 1$, since $y(z)=(1-\beta)c+\beta e$. At least five independent constraints are tight: the constraints on x_2,x_7 determine d, and those on x_3,x_0,x_1 determine the other basis directions. Thus these are sixteen distinct vertices of P*. Those with $y(z)=1$ necessarily have $c=e=1$, and annihilate both d and x_0-x_1.

The facet of P* exposed by the vertex z is four-dimensional and its vertices span $\mathbb{R}^5$ linearly, since its affine hyperplane has value one at z. The sixteen vertices just constructed that belong to this facet annihilate the two-dimensional space $\operatorname{span}\{d,x_0-x_1\}$. At most one further vertex in that facet cannot remove this common kernel: there remains a nonzero vector annihilated by all its vertices. Therefore there are at least two further vertices tight at z. Their negatives are distinct further vertices, none among the original sixteen. This gives at least twenty vertices of P*, as required.

Concurrent case. Write the endpoints on the four lines as $p+u_i$ and $p+\tau_i u_i$ (i=A,B,C,D). A supporting form of G_2 is zero on line B and positive on both A endpoints, so $\tau_A>0$. Use G_1 for B, G_4 for C and G_3 for D to get $\tau_i>0$ for all i. Distinct endpoints give $\tau_i\neq 1$. The pentagon plane gives $z=p+w_C u_C+w_D u_D$; the positive supporting forms of G_4 and G_3 give $w_C,w_D>0$. Again p,u_A,u_B, u_C,u_D is a basis because it spans all vertices.

$$m_i=\max(1,\tau_i),\ M_C=\max(1,\tau_C,w_C),\qquad M_D=\max(1,\tau_D,w_D). \tag{38.2}$$

Construct twelve functionals y by $y(p)=1$; independently choose $y(u_A)\in\{0,-2/m_A\}$ and $y(u_B)\in\{0,-2/m_B\}$; choose the pair of C,D values from $(0,0)$, $(-2/M_C,0)$, $(0,-2/M_D)$. Every endpoint and z then has value in $[-1,1]$, since $0<t\le M$ implies $-1\le 1-2t/M\le 1$. Thus $y\in P^*$. We verify that each is a vertex, including ties among the maxima.

Let a linear perturbation vanish on all tight constraints. At least one of C,D is unlowered; its two endpoints have value one and their equations force the perturbation to vanish on p and on that line's u, because $\tau\neq 1$. If the other is lowered, a maximum among $1,\tau,w$ is attained. The corresponding endpoint or z has value -1 and forces its u component to vanish as well. For A and B an unlowered pair gives both equations; a lowered line has an endpoint attaining m and value -1. With the p component already zero, each forces its u component zero. The perturbation therefore vanishes on the basis. This proves extremality without assuming unique maxima or generic coordinates.

The twelve functionals are distinct by their basis values, all have value one at p, and their negatives have value -1 there. Hence P* has at least twenty-four vertices in the concurrent case. Both cases prove $h(P)\ge 20$ and Proposition 38.1. In particular U is absent on both sides at (18,18), and on the polar side at (16,18).

39 A tetrahedral-prism facet at eighteen vertices

Proposition 39.1. *A centrally symmetric five-polytope with eighteen vertices and a facet combinatorially equivalent to $\Delta_1\times\Delta_3$ has at least sixty-four edges.* Label the eight vertices of the facet F as x_{sk}, $s\in\{0,1\}$, $k\in\{0,1,2,3\}$. The remaining antipodal pair is $\pm z$. Normalize its supporting form c by $c(F)=1$. Thus $|c(z)|<1$. For a form c_R supporting a ridge opposite row s, put $h_s=c-c_R$; similarly define g_k for the ridge opposite column k. These forms are positive at their own row or column and zero at all others. The product recognition was proved in Section 16.

A corner representation. The map from six coefficients to forms, $(\alpha,\beta)\mapsto\Sigma\alpha_s h_s+\Sigma\beta_k g_k$, has a one-dimensional kernel. Indeed a kernel vector satisfies, at every vertex, $\alpha_s h_s(x_{sk})+\beta_k g_k(x_{sk})=0$. If any α is zero all β and then all α are zero. Otherwise the α entries have the same sign and all β have the opposite sign. Fixing one α determines every β and the other α; the kernel therefore has dimension at most one. Six forms in five dimensions have a nonzero relation, so the kernel has dimension exactly one and the map is onto. Choose its vector (κ,λ) with $\kappa_s>0$ and $\lambda_k<0$.

Take any representation $c=\Sigma\alpha h+\Sigma\beta g$. Since $c(x_{sk})=1$, the kernel relation implies $-\alpha_s/\kappa_s<\beta_k/(-\lambda_k)$ for all s,k. Translate the coefficients along the kernel until the largest lower endpoint is attained. The resulting coefficients γ,δ satisfy $\gamma\ge0$, $\gamma_0=0$ after relabeling the rows, and $\delta>0$. For all sufficiently small $\tau>0$ both $\gamma(\tau)=\gamma+\tau\kappa$ and $\delta(\tau)=\delta+\tau\lambda$ are strictly positive.

$$\begin{aligned} c&=\Sigma\gamma_s(\tau)h_s+\Sigma\delta_k(\tau)g_k,\\ L_\tau(u,v)&=\Sigma u_s\gamma_s(\tau)h_s+\Sigma v_k\delta_k(\tau)g_k. \end{aligned} \tag{39.1}$$

At x_{sk}, L_τ is a strictly positive convex combination of u_s and v_k when $\tau>0$. Put $\xi=\gamma_1 h_1(z)$, $W_k=\delta_k g_k(z)$. Then $|\xi+\Sigma W_k|<1$. For ordered pairs $a=(k,l)$, $k\ne l$, set

$$R(a)=\Sigma_{j\ne k,l}|W_j|,\quad \psi(a)=W_k-W_l-R(a). \tag{39.2}$$

Candidate edges and blocked candidates. There are twenty-four opposite-row candidates $[x_{sk},-x_{1-s,l}]$, $k\ne l$. Choose $u_s=1, u_{1-s}=-1$, $v_k=1, v_l=-1$ and all other v entries in $(-1,1)$. At $\tau>0$ only these two signed vertices have value one. At $\tau=0$ the z-value ranges over the open interval $W_k-W_l-\xi\pm R$ for $s=0$ and $W_k-W_l+\xi\pm R$ for $s=1$; when $R=0$ the interval denotes its single value. If a value is strictly between -1 and 1, continuity gives an edge at sufficiently small $\tau>0$. Thus an actual nonedge has the whole interval above one or below minus one. Let U_0,U_1 be the actual nonedges of the first kind. Their antipodes are exactly the nonedges of the second kind, and hence there are at least $24-2(|U_0|+|U_1|)$ opposite-row edges. Moreover

$$a\in U_0\Rightarrow \psi(a)-\xi\ge1,\quad a\in U_1\Rightarrow \psi(a)+\xi\ge1. \tag{39.3}$$

If $\gamma_1>0$ there are also twelve same-row candidates $[x_{0k},-x_{0l}]$. At $\tau=0$ choose $u_0=0, u_1\in(-1,1)$ and the same column signs. Only the two stated signed vertices are tight: the positive γ_1 makes the other row strict. The z-values range over $W_k-W_l\pm(|\xi|+R)$. Writing U_S for their upwards nonedges gives at least $12-2|U_S|$ further edges and

$$a\in U_S \Rightarrow \psi(a)-|\xi|\geq 1. \tag{39.4}$$

Each upwards blocked candidate gives two edges incident with z, one to each of its endpoints. To see this explicitly, choose an admissible form L with only those two signed F vertices tight and with $L(z)\geq 1$. If $L(z)>1$, L+a c, with $a=(L(z)-1)/(1-c(z))$, exposes z and the positive F endpoint; L−b c, with $b=(L(z)-1)/(1+c(z))$, exposes z and the negative F endpoint. All other vertices satisfy a strict inequality after normalization. If $L(z)=1$ the exposed face is the triangle of the two endpoints and z, whose sides are edges. Three distinct vertices cannot be collinear in a polytope. An interval with upper endpoint strictly greater than one permits such an L even when the candidate itself is an edge.

For opposite-row candidates use sufficiently small $\tau>0$ in this argument. If the interval merely touches one at $\tau=0$ and the candidate is a nonedge, no small positive τ can move the whole interval into $(-1,1)$; choosing a fixed interior set of parameters therefore still gives $L(z)\geq 1$. For same-row candidates $\tau=0$ already has exactly the required tight endpoints.

Let d be the degree of z; always $d\geq 5$. Write H(V),T(V) for the sets of heads and tails of ordered pairs in V. The adjacency just proved gives, in the case $\gamma_1>0$,

$$d\geq |H(U_0\cup U_S)|+|H(U_1)|+|T(U_0)|+|T(U_1\cup U_S)|. \tag{39.5}$$

The four terms count positive row-zero, positive row-one, negative row-one and negative row-zero endpoints respectively. For the loose same-row set $L_S=\{a:W_k-W_l+|\xi|+R(a)>1\}$, we also have

$$\begin{aligned} d&\geq |H(U_0)|+|T(U_0)|+|T(U_0\cap L_S)|,\\ d&\geq |H(U_1)|+|T(U_1)|+|H(U_1\cap L_S)|. \end{aligned} \tag{39.6}$$

These inequalities count distinct signed row endpoints. They will compensate for missing candidate edges. The following elementary identity supplies the only compatibility rule:

$$\psi(k,l)+\psi(k',l')\leq 0 \text{ if } k\neq k' \text{ and } l\neq l'. \tag{39.7}$$

Indeed any positive occurrence of W_k or $W_{k'}$ cancels a negative tail occurrence if it is the other tail; otherwise it is bounded by an absolute-value term in the other remainder. The same applies to negative tail occurrences. Every uncancelled term is therefore nonpositive. A set of ordered pairs that pairwise share a head or a tail lies in one row or one column: two distinct pairs sharing a head force every third compatible pair to share that head; the tail case is identical. Such a set has at most three elements.

The parallel corner $\gamma_1=0$. Here $\xi=0$. Put $U=|U_0|+|U_1|$. The set $\Psi=\{a:\psi(a)\geq 1\}$ lies in a row or column by (39.7), so $U\leq 6$. We prove $d\geq U+4$. If $U\leq 1$ this follows from $d\geq 5$. Otherwise take (k,l) in $U_0\cup U_1$, and denote the other indices by i,j. The inequalities $\psi(k,l)\geq 1$ and $|\Sigma W|<1$ give $W_k>0, W_l<0$. Put $L=\{a:W_{head}-W_{tail}+R(a)>1\}$. Every pair in L gives both row versions of the two z-adjacencies; hence $d\geq 2(|H(L)|+|T(L)|)$.

If $W_j\neq 0$, the pairs (k,i) and (i,l) are in L, since their upper values are at least $\psi(k,l)+2|W_j|$. If $W_j<0$, (k,j) is also in L; if $W_j>0$, (j,l) is in L. Thus $|H(L)|+|T(L)|\geq 5$ and $d\geq 10\geq U+4$. The same argument interchanges i,j if only W_i is nonzero. If both

are zero and $W_k - W_l > 1$, all four pairs (k,i),(k,j),(i,l),(j,l) are in L, giving $d \geq 12$.

The only remaining case has $W_i = W_j = 0$ and $W_k - W_l = 1$. The signs above show that Ψ consists of (k,l) alone. Since $U \geq 2$ this pair would have to be in both U_0 and U_1. This tie is impossible. With the other column potentials zero, the derivatives of the two opposite-row z-values have sum $2(W'_k - W'_l)$, where $W'_a = (\lambda_a/\delta_a)W_a$. It is strictly negative because $W_k > 0$, $W_l < 0$ and $\lambda < 0$. At least one candidate consequently moves from z-value one to a value in $(-1,1)$ for small $\tau > 0$ and is an edge. Thus $d \geq U+4$ in every parallel case.

The other corner $\gamma_1 > 0$. Put $U = |U_0| + |U_1| + |U_S|$. We prove $U \leq d+2$. This is immediate for $U \leq 7$. For $U \geq 8$ define the weak and strong sets $\Psi_1 = \{a : \psi(a) \geq 1 - |\xi|\}$ and $\Psi_3 = \{a : \psi(a) \geq 1 + |\xi|\}$. Every strong pair is compatible with every weak pair by (39.7), since their ψ-sum is at least two. All three U sets are weak; U_S and, according to the sign of ξ, either U_0 or U_1 are strong.

If Ψ_3 contains exactly one pair, there are at most five weak pairs compatible with it. Thus one U set has size at most five and each of the other two has size at most one, contradicting $U \geq 8$. If Ψ_3 contains two distinct pairs, they share a head or a tail and force every weak pair into the same row or column. Each U set then has size at most three. In a common row, $U \geq 8$ ensures U_1 and $U_0 \cup U_S$ are nonempty. Equation (39.5) gives $d \geq 2 + |U_0| + |U_1 \cup U_S| \geq U - 1$, because $|U_1 \cap U_S| \leq 3$. In a common column the analogous calculation gives $d \geq U - 1$. Both are stronger than required.

It remains that Ψ_3 is empty. Then U_S is empty; when $\xi < 0$ only U_0 can be nonempty, when $\xi > 0$ only U_1, and when $\xi = 0$ neither. Consider U_0; the U_1 argument exchanges heads and tails. Let $a = |H(U_0)|$ and $b = |T(U_0)|$. There are at most $ab - \max(0, a+b-4)$ distinct off-diagonal ordered pairs between these sets. For $0 \leq a,b \leq 4$ this is at most $a+b+2$ unless $a = b = 4$. Hence (39.5) proves $|U_0| \leq d+2$ unless $a = b = 4$ and $|U_0| \geq 11$. For that last case at most one of the twelve ordered pairs is absent.

For any pair in U_0 outside L_S, (39.3) and $\xi < 0$ give $W_k - W_l + |\xi| - R \geq 1$ while the loose upper value is at most one. Thus $R = 0$. Suppose fewer than three tails occur in $U_0 \cap L_S$. Choose two distinct omitted tails l, l'. At least two pairs in U_0 have each of these tails, since at most one ordered pair is absent. Their heads are distinct. Their two zero remainders force every W except possibly W_l to vanish. The omitted tail l' similarly forces every W except possibly $W_{l'}$ to vanish. Hence $W = 0$. But a pair in U_0 would then imply $|\xi| \geq 1$, contrary to $|\xi + \Sigma W| < 1$. Therefore $|T(U_0 \cap L_S)| \geq 3$. Equation (39.6) gives $d \geq 4+4+3 = 11$, and $|U_0| \leq 12 \leq d+2$. This completes the second corner as well.

The edge count. The two prism facets F and $-F$ provide thirty-two edges. The candidate edges cross between these facets; none joins an antipodal pair. The edges incident with z or $-z$ provide 2d distinct additional edges. All three collections are disjoint. In the parallel corner we obtain

$$e \geq 32 + 24 - 2U + 2d \geq 56 + 8 = 64. \tag{39.8}$$

In the other corner the same-row candidates add twelve more before the blocked pairs are removed, giving

$$e \geq 32 + 24 + 12 - 2U + 2d \geq 68 - 4 = 64. \tag{39.9}$$

The proof treats every real realization and the interval-endpoint ties explicitly; it does not assume generic coordinates.

40 Assembly of the fourteen- and eighteen-vertex endpoints

Proposition 40.1. *The six endpoints with v=14 are excluded.* If there are at least four full facet pairs, their normals have rank at least three by Section 28. Rank three is excluded by Section 29, including its point-fibre alternative. Rank at least four is excluded by Sections 30–32, which apply to arbitrary real coordinates. Thus a counterexample has at most six full facets. At (14,16) the incidence lower bound $I \geq 4(v+h)-12=108$, and $I \leq 6h+q=96+q$, forces $q \geq 12$. At (14,18) these bounds force $q \geq 8$. Both contradict $q \leq 6$. The four remaining $h=20,22,24,26$ cases now use the row sums of Section 7 with $q \leq 6$. Their inequalities give respectively $6t \geq 716$, $6t \geq 716$, $6t \geq 714$ and $12t \geq 1434$. Since t is even, each forces $t \geq 120$. In the zero-stress (14,20) alternative, the separately proved incidence contradiction in Section 7 applies instead. This completes all six fourteen-vertex branches.

Proposition 40.2. *The endpoint (18,18) is excluded.* Let Q denote the row (9,18,15,6;36), T the row (8,16,14,6;32), and U the row (9,19,17,7;38). These names refer to numerical rows whose required geometric recognition has already been proved. Use the following ordered alternatives; each new alternative retains the exclusions in the earlier ones.

Alternative	Geometric consequence	Certificate in §34
Q occurs on P	$e \geq 72$ by §20	D1
Q occurs only on P*	$r \geq 72$ by §20 on P*	D2
No Q; T occurs on P	$e \geq 64$ by §39	D3
No Q; T only on P*	$r \geq 64$ by §39 on P*	D4
No Q or T on either side	U is also absent on both sides by §38	Remaining D cases

Every D cost is nonnegative on its printed local row domain and its global sum gives the displayed strictly negative combination. The first four cases require only $G,S \geq 0$ and the displayed edge or ridge bound. For the last case the complete even-integer coverage calculation of Section 34 includes all local maximum stresses; every threshold count is justified by the two-facet stress bound. Hence none of the five alternatives can satisfy $t \leq 118$.

Proposition 40.3. *The endpoint (18,20) is excluded.* If Q occurs on P, $e \geq 72$ and $2r \geq 5h=100$, so $t \geq 122$. If Q occurs only on P*, its twenty vertices and the extra-pair argument of Section 21 give $r \geq 66$; case C1 of Section 34 applies. Otherwise Q is absent on both sides. If T occurs on P, Proposition 39.1 gives $e \geq 64$, which implies the $e \geq 60$ hypothesis of C2–C4. Their stress alternatives $G \geq 12$; $G \leq 10, S \geq 20$; and $G \leq 10, S \leq 18$ exhaust every nonnegative even G,S. If T does not occur on P, the remaining twenty-one C cases cover all even slacks and local maxima as proved in Section 34. T is permitted on the polar in this last alternative; no additional twenty-vertex prism bound is needed. These alternatives exclude every (18,20) counterexample.

For completeness, (18,22) was proved in Section 23 and (20,20) in Section 24. The sixteen-vertex h=24 case uses 12t≥1464 after excluding the full prism, as in Section 8; h=22 was proved in Section 19; h=20 in Sections 35–37, including its rank-three and rank-four alternatives; h=16 in Section 34. Thus the only endpoint left for Sections 41–54 is (16,18).

41 Stronger quadrilateral weights from ridge counts

Use q for the number of quadrilaterals in a four-dimensional facet, b for those between two five-vertex ridges, and η for its polygon excess. Section 36 proved the global budget for $w=q-3b+2\eta$. We now prove stronger local lower bounds without classifying all eight-vertex four-polytopes. The twelve ridge types and their count variables $c_1,\dots,c_{12}$ are exactly those of Section 36.

Reserved quadrilaterals. A quadrilateral inside a ridge with other than five vertices does not contribute to b. A triangular prism therefore reserves three such quadrilaterals, so $b+3\le q$. Two distinct prisms share at most one two-face: their intersection is a single proper face of each, and any two-dimensional face in that intersection must equal it. Thus two prisms reserve at least five distinct quadrilaterals and $b+5\le q$.

An eight-vertex facet with a seven-vertex ridge is a pyramid: its remaining vertex is outside the ridge hyperplane and the facet is their convex hull. Every two-face outside the base is a triangle over a base edge. Hence every quadrilateral lies in the seven-vertex base. In particular $b=0$. Each square-pyramidal ridge has a different quadrilateral base shared with that seven-vertex ridge, giving $q\ge c_2$.

Excess and long polygons. A quadrilateral contributes one to η, and a polygon with at least five vertices contributes at least two. Consequently $\eta-q$ cannot equal one. If $q=\eta$ all polygons are triangles or quadrilaterals. For a three-dimensional ridge of row (n,e,r), its polygon excess is $2e-3r$, by counting edge–polygon incidences. Thus in this last case it has exactly $2e-3r$ quadrilaterals. For every non-five-vertex ridge these are reserved, and $b+(2e-3r)\le q$.

The arithmetic is now completely specified. For each count vector in the recurrence of Section 36, retain the three local polygon restrictions P1–P3 and the two tetrahedron–pyramid parity exclusions proved there. Enumerate integers q,b with $0\le b\le q\le\eta$, $2b\le c_2$ and $2q\ge c_2+3c_4$. Delete $\eta-q=1$; apply $b=0,q\ge c_2$ if a seven-vertex ridge occurs, $b+3\le q$ if $c_4\ge1$, $b+5\le q$ if $c_4\ge2$, and the reserved excess inequalities when $q=\eta$. These are all necessary conditions proved above, so minimization over this enlarged integer set is a valid lower bound for every real facet.

Facet row	η	Raw count vectors	Minimum w	Used bound
(8, 19, 19, 8, 38)	3	19	4	4
(8, 20, 20, 8, 39)	3	40	3	3
(8, 19, 19, 8, 39)	4	16	8	8
(8, 19, 19, 8, 40)	5	12	10	10
(8, 19, 20, 9, 41)	3	6	9	8
(8, 19, 19, 8, 41)	6	8	12	12

For transparency, the complete raw count vectors are printed below in the twelve-type order. The count equations determine each finite list; no realization

assumption is made about any vector. The supplied independent integer recurrence produces precisely these lists and checks the displayed minima.

(8, 19, 19, 8, 38). (2, 6, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0); (3, 3, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0); (3, 4, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0); (4, 0, 2, 2, 0, 0, 0, 0, 0, 0, 0, 0); (4, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0); (4, 2, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0); (4, 2, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0); (4, 2, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0); (4, 3, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0); (5, 0, 0, 0, 3, 0, 0, 0, 0, 0, 0, 0); (5, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0); (5, 0, 0, 2, 0, 0, 1, 0, 0, 0, 0, 0); (5, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 0); (5, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0); (5, 1, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0); (5, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0); (5, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0); (6, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0); (6, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0).

(8, 20, 20, 8, 39). (1, 6, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0); (2, 3, 2, 1, 0, 0, 0, 0, 0, 0, 0, 0); (2, 4, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0); (2, 5, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0); (3, 0, 3, 2, 0, 0, 0, 0, 0, 0, 0, 0); (3, 1, 2, 1, 1, 0, 0, 0, 0, 0, 0, 0); (3, 2, 1, 0, 2, 0, 0, 0, 0, 0, 0, 0); (3, 2, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0); (3, 2, 2, 0, 0, 0, 0, 1, 0, 0, 0, 0); (3, 3, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0); (3, 3, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0); (3, 3, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0); (3, 4, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0); (4, 0, 1, 0, 3, 0, 0, 0, 0, 0, 0, 0); (4, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0); (4, 0, 1, 2, 0, 0, 1, 0, 0, 0, 0, 0); (4, 0, 2, 0, 1, 0, 0, 1, 0, 0, 0, 0); (4, 0, 2, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0); (4, 1, 0, 0, 2, 1, 0, 0, 0, 0, 0, 0); (4, 1, 0, 1, 0, 2, 0, 0, 0, 0, 0, 0); (4, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0); (4, 1, 1, 0, 0, 1, 0, 1, 0, 0, 0, 0); (4, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0, 0); (4, 1, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0); (4, 2, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0); (4, 2, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0); (4, 2, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0); (4, 2, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0); (5, 0, 0, 0, 2, 0, 1, 0, 0, 0, 0, 0); (5, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0); (5, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0); (5, 0, 0, 0, 2, 0, 0, 0, 0, 1, 0, 0); (5, 0, 0, 1, 0, 0, 1, 0, 1, 0, 0, 0); (5, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 0); (5, 0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0); (5, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 1); (5, 0, 1, 0, 0, 0, 0, 0, 2, 0, 0, 0); (5, 0, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0); (5, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0); (5, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0).

(8, 19, 19, 8, 39). (2, 5, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0); (3, 2, 1, 2, 0, 0, 0, 0, 0, 0, 0, 0); (3, 3, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0); (3, 4, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0); (4, 0, 1, 2, 1, 0, 0, 0, 0, 0, 0, 0); (4, 1, 0, 1, 2, 0, 0, 0, 0, 0, 0, 0); (4, 1, 0, 2, 0, 1, 0, 0, 0, 0, 0, 0); (4, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0); (4, 2, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0); (4, 2, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0); (5, 0, 0, 0, 2, 0, 0, 1, 0, 0, 0, 0); (5, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0); (5, 0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0); (5, 0, 0, 2, 0, 0, 0, 0, 0, 1, 0, 0); (5, 0, 1, 0, 0, 0, 0, 2, 0, 0, 0, 0); (5, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0).

(8, 19, 19, 8, 40). (2, 4, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0); (3, 1, 1, 3, 0, 0, 0, 0, 0, 0, 0, 0); (3, 2, 0, 2, 1, 0, 0, 0, 0, 0, 0, 0); (3, 3, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0); (4, 0, 0, 2, 2, 0, 0, 0, 0, 0, 0, 0); (4, 0, 0, 3, 0, 1, 0, 0, 0, 0, 0, 0); (4, 0, 1, 2, 0, 0, 0, 1, 0, 0, 0, 0); (4, 1, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0); (4, 1, 0, 2, 0, 0, 0, 0, 1, 0, 0, 0); (4, 2, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0); (5, 0, 0, 0, 1, 0, 0, 2, 0, 0, 0, 0); (5, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0).

(8, 19, 20, 9, 41). (5, 3, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0); (6, 0, 1, 2, 0, 0, 0, 0, 0, 0, 0, 0); (6, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0); (6, 2, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0); (7, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0); (7, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0).

(8, 19, 19, 8, 41). (2, 3, 0, 3, 0, 0, 0, 0, 0, 0, 0, 0); (3, 0, 1, 4, 0, 0, 0, 0, 0, 0, 0, 0); (3, 1, 0, 3, 1, 0, 0, 0, 0, 0, 0, 0); (3, 2, 0, 2, 0, 0, 0, 1, 0, 0, 0, 0); (4, 0, 0, 2, 1, 0, 0, 1, 0, 0, 0, 0); (4, 0, 0, 3, 0, 0, 0, 0, 1, 0, 0, 0); (4, 1, 0, 1, 0, 0, 0, 2, 0, 0, 0, 0); (5, 0, 0, 0, 0, 0, 0, 3, 0, 0, 0, 0).

A useful short explanation of the weight-three bounds is also available. For $\eta=3$, $w<3$ forces $q\geq 2$ because $b\leq q$. Since $\eta-q\neq 1$, necessarily $q=3$. Then $w=9-3b<3$

forces b=3 and at least six square pyramids. For row (8,19,19,8;38), the count equations leave two tetrahedra and six square pyramids; for (8,20,20,8;39), one tetrahedron, one triangular bipyramid and six square pyramids. Both violate the vertex parity argument of Section 36. The reserved-quadrilateral inequality further raises the first row’s bound to four, as the full integer table records.

Let ρ be the Section 36 weight with the six entries above replaced by 3,3,8,10,8,12 respectively in the displayed row order. Let ρ′ replace only its first entry by four. All earlier entries retain their proved bounds. Both satisfy, at sixteen vertices,

$$\begin{aligned} \Sigma_F \rho(F) &\le 4A + 2(f_{024} - 3y), \\ \Sigma_F \rho'(F) &\le 4A + 2(f_{024} - 3y). \end{aligned} \tag{41.1}$$

The second inequality is stronger. The first is retained where a certificate uses only the smaller entry; this cannot invalidate its conclusion.

42 Exact reduction of the few-full (16,18) branch

Assume v=16,h=18,t≤118 and k=1,2,3 full facet pairs. A full tetrahedral prism has already been excluded. Both sides exclude the six impossible rows of Section 36. Every facet row also satisfies J≥4r, since every three-dimensional ridge has at least four vertices. The exact costs below use ρ or ρ′ from Section 41. No stronger weight is silently assumed.

For full facets $N_8=2k$ and $R_8\le k(18+2k-4)$, giving respectively 16,36,60. As in Section 34, every stress-threshold coefficient is summed using K≤2 only when 2M≤τ or 2M≥G−τ, where M is the largest local stress. The latter alternative is precisely the two-facet bound. Local nonnegativity is checked on the Section 22 row domain intersected with J≥4r.

Enumerate even G,S≥0 with G+S≤42 and both integer local maxima in [0,G/2] and [0,S/2]. Partition every remaining even count axis at b,b+2 as in Section 34. The following 276 cases cover every cell except the explicitly listed residues. The coverage check visits 46941 cells/maxima outside those residues, including all cases G≥16 or S≥16. Only 275 of the printed leaves are needed by this enlarged interval coverage; extra valid leaves do not weaken the result.

k	Remaining (G,S) pairs
1	(0, 10)
2	(0, 10), (2, 2), (2, 4), (2, 6), (2, 8), (2, 10), (4, 2), (4, 4), (4, 6)
3	(0, 10), (2, 2), (2, 4), (2, 6), (2, 8), (2, 10), (2, 12), (4, 2), (4, 4), (4, 6), (4, 8), (4, 10), (4, 12), (4, 14), (6, 2), (6, 4), (6, 6), (6, 8), (6, 10), (8, 6), (8, 8), (8, 10)

This is a necessary-cell reduction. Membership in the list does not assert that a polytope exists. At k=1 only (G,S)=(0,10) remains. At k=2 nine cells remain; at k=3 twenty-two. The r≥64 certificates below additionally show r≤62 for a counterexample. Hence Section 20 and Proposition 39.1 exclude both product rows Q and T on the polar. Section 38 excludes U on the polar as well. These exclusions are consequences, not hypotheses in the initial reduction.

Use $Q_1,\dots,Q_{23}$ from Section 22, with their two summed costs now including threshold, full and quadrilateral terms. The extra list U first gives QW≥0 and the quadrilateral budget $4A+4f_{03}+4f_{14}-4I-6y-QW\ge0$; then the nonzero threshold-count bounds, nonnegativity of the full counts, and each printed bound in order. An equality supplies its lower then upper inequality. Every displayed integer combination has been recomputed using fractions and equals the printed negative constant exactly.

L1. ρ; cap F/S=2/12.

φ=(198, −78, 24, 0, 60, −15, 0, 0, 0, 0); threshold=(0, 1000000); full=(39, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 30·Q21 + Q22 + 7·U2 + 30·U7 + 18·U9 + 39·U12 = −12.

L2. ρ; cap F/S=0/0.

φ=(−70, −37, 16, 0, 39, −5, 0, 0, 0, 0); threshold=(0, 4); full=(−20, 0); μ=0.

ψ=(−310, 93, 0, 0, 93, −31, 0, 0, 0, 0); threshold=(0, 8); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 48·Q21 + Q22 + Q23 + 10·U7 + 46·U9 + 20·U11 = −48.

L3. ρ; cap F/S=0/2.

φ=(−189, 52, 6, 0, 96, −29, 0, 0, 0, 0); threshold=(0, 4); full=(8, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 18·Q21 + Q22 + 58·U7 + 6·U10 + 8·U12 = −18.

L4. ρ; cap F/S=0/4.

φ=(−189, 52, 6, 0, 96, −29, 0, 0, 0, 0); threshold=(0, 4); full=(8, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 18·Q21 + Q22 + 58·U7 + 6·U10 + 8·U12 = −18.

L5. ρ; cap F/S=0/6.

φ=(−189, 52, 6, 0, 96, −29, 0, 0, 0, 0); threshold=(0, 4); full=(8, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 18·Q21 + Q22 + 58·U7 + 6·U10 + 8·U12 = −18.

L6. ρ; cap F/S=0/8.

φ=(−189, 52, 6, 0, 96, −29, 0, 0, 0, 0); threshold=(0, 4); full=(8, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 18·Q21 + Q22 + 58·U7 + 6·U10 + 8·U12 = −18.

L7. ρ; cap F/S=2/0.

φ=(−80, −63, 24, 0, 51, −5, 0, 0, 0, 0); threshold=(0, 2); full=(−25, 0); μ=0.

ψ=(−490, 147, 0, 0, 147, −49, 0, 0, 0, 0); threshold=(0, 8); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 72·Q21 + Q22 + Q23 + 10·U7 + 74·U9 + 25·U11 = −72.

L8. ρ; cap F/S=2/2.

φ=(10, −6, 4, 0, 18, −5, 0, 0, 0, 0); threshold=(0, 2); full=(6, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 10·U7 + 2·U9 + 6·U12 = −4.

L9. ρ; cap F/S=2/4.

φ=(10, −6, 4, 0, 18, −5, 0, 0, 0, 0); threshold=(0, 2); full=(6, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 10·U7 + 2·U9 + 6·U12 = −4.

L10. ρ; cap F/S=2/6.

φ=(10, −6, 4, 0, 18, −5, 0, 0, 0, 0); threshold=(0, 2); full=(6, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 10·U7 + 2·U9 + 6·U12 = −4.

L11. ρ; cap F/S=2/8.

φ=(−59, 35, 6, 0, 99, −30, 0, 0, 0, 0); threshold=(0, 2); full=(58, 0); μ=0.

ψ=(−190, 57, 0, 0, 57, −19, 0, 0, 0, 0); threshold=(76, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: kS ≥ 0; 6: kS ≤ 2; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 4; 10: G ≤ 4; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 18·Q21 + Q22 + Q23 + 76·U6 + 60·U9 + 32·U11 + 58·U14 = −18.

L12. ρ; cap F/S=4/0.

φ=(121723, −17389, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 0); full=(17389, 0); μ=0.

ψ=(−176060, 52349, 0, −35, 52853, −17480, 0, 0, 0, 0); threshold=(0, 8); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 4990643·Q11 + 24466323·Q21 + 51880·Q22 + 52167·Q23 + 8155441·U8 + 1828748963·U9 + 902141320·U12 = −12336382604.

L13. ρ; cap F/S=4/2.

φ=(−112, 50, 2, 0, 81, −26, 0, 0, 0, 0); threshold=(49, 0); full=(19, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 4; 10: G ≤ 4; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 49·U4 + 52·U9 + 2·U12 + 19·U14 = −6.

L14. ρ; cap F/S=4/4.

φ=(−112, 50, 2, 0, 81, −26, 0, 0, 0, 0); threshold=(49, 0); full=(19, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 4; 10: G ≤ 4; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 49·U4 + 52·U9 + 2·U12 + 19·U14 = −6.

L15. ρ; cap F/S=4/6.

φ=(−112, 50, 2, 0, 81, −26, 0, 0, 0, 0); threshold=(49, 0); full=(19, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 4; 10: G ≤ 4; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 49·U4 + 52·U9 + 2·U12 + 19·U14 = −6.

L16. ρ; cap F/S=4/8.

φ=(32, −11, 6, 0, 36, −9, 0, 0, 0, 0); threshold=(9, 0); full=(11, 0); μ=0.

ψ=(−230, 69, 0, 0, 69, −23, 0, 0, 0, 0); threshold=(92, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: kS ≥ 0; 8: kS ≤ 2; 9: nS8 ≥ 0; 10: rS8 ≥ 0; 11: G ≥ 4; 12: G ≤ 4; 13: S ≥ 8; 14: S ≤ 8; 15: nF8 ≥ 4; 16: nF8 ≤ 4; 17: rF8 ≤ 36; 18: t ≤ 118.

Identity: 18·Q21 + Q22 + Q23 + 9·U4 + 92·U8 + 18·U11 + 40·U13 + 11·U16 = −18.

L17. ρ; cap F/S=0/0.

φ=(290, −87, 0, 0, −87, 29, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

ψ=(−290, 87, 0, 0, 87, −29, 0, 0, 0, 0); threshold=(0, 12); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: Q22 + Q23 + 58·U8 + 58·U9 = −348.

L18. ρ; cap F/S=0/2.

φ=(−571, 166, 36, 0, 261, −93, 0, 0, 0, 0); threshold=(0, 6); full=(23, 0); μ=0.

ψ=(0, −8, 8, 0, 0, −2, 0, 0, 0, 0); threshold=(0, 10); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 16·Q11 + 60·Q21 + Q22 + 154·U7 + 20·U10 + 23·U12 = −60.

L19. ρ; cap F/S=0/4.

φ=(−571, 166, 36, 0, 261, −93, 0, 0, 0, 0); threshold=(0, 6); full=(23, 0); μ=0.

ψ=(0, −8, 8, 0, 0, −2, 0, 0, 0, 0); threshold=(0, 8); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 16·Q11 + 60·Q21 + Q22 + 154·U7 + 20·U10 + 23·U12 = −60.

L20. ρ; cap F/S=0/6.

φ=(−571, 166, 36, 0, 261, −93, 0, 0, 0, 0); threshold=(0, 6); full=(23, 0); μ=0.

ψ=(0, −8, 8, 0, 0, −2, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 16·Q11 + 60·Q21 + Q22 + 154·U7 + 20·U10 + 23·U12 = −60.

L21. ρ; cap F/S=0/8.

φ=(−107, 22, 4, 0, 45, −13, 0, 0, 0, 0); threshold=(0, 6); full=(−1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 26·U7 + 4·U10 + U11 = −12.

L22. ρ; cap F/S=0/10.

φ=(−571, 166, 36, 0, 261, −93, 0, 0, 0, 0); threshold=(0, 6); full=(23, 0); μ=0.

ψ=(0, −8, 8, 0, 0, −2, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 16·Q11 + 60·Q21 + Q22 + 154·U7 + 20·U10 + 23·U12 = −60.

L23. ρ; cap F/S=0/12.

φ=(−571, 166, 36, 0, 261, −93, 0, 0, 0, 0); threshold=(0, 6); full=(23, 0); μ=0.

ψ=(0, −8, 8, 0, 0, −2, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 16·Q11 + 60·Q21 + Q22 + 154·U7 + 20·U10 + 23·U12 = −60.

L24. ρ; cap F/S=2/0.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

ψ=(−87, 174, 0, 0, 261, −87, 0, 0, 0, 0); threshold=(0, 12); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: Q23 + 174·U9 = −696.

L25. ρ; cap F/S=2/2.

φ=(57, −15, 2, 0, 3, 0, 0, 0, 0, 0); threshold=(0, 4); full=(6, 0); μ=0.

ψ=(−117, 39, 0, 0, 39, −13, 0, 0, 0, 0); threshold=(0, 10); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 24·U9 + 6·U12 = −6.

L26. ρ; cap F/S=2/4.

φ=(−240, 42, 42, 0, 309, −90, 0, 0, 0, 0); threshold=(0, 4); full=(105, 0); μ=8.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 8); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 78·Q21 + Q22 + 8·U2 + 180·U7 + 6·U9 + 105·U12 = −78.

L27. ρ; cap F/S=2/6.

φ=(−240, 42, 42, 0, 309, −90, 0, 0, 0, 0); threshold=(0, 4); full=(105, 0); μ=8.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 78·Q21 + Q22 + 8·U2 + 180·U7 + 6·U9 + 105·U12 = −78.

L28. ρ; cap F/S=2/8.

φ=(−240, 42, 42, 0, 309, −90, 0, 0, 0, 0); threshold=(0, 4); full=(105, 0); μ=8.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 78·Q21 + Q22 + 8·U2 + 180·U7 + 6·U9 + 105·U12 = −78.

L29. ρ; cap F/S=2/10.

φ=(−240, 42, 42, 0, 309, −90, 0, 0, 0, 0); threshold=(0, 4); full=(105, 0); μ=8.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 78·Q21 + Q22 + 8·U2 + 180·U7 + 6·U9 + 105·U12 = −78.

L30. ρ; cap F/S=2/12.

φ=(−36, 14, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(0, 4); full=(17, 0); μ=0.

ψ=(−20, 6, 0, 0, 6, −2, 0, 0, 0, 0); threshold=(12, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: kS ≥ 0; 6: kS ≤ 2; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 2·Q23 + 24·U6 + 20·U9 + 6·U11 + 17·U14 = −6.

L31. ρ; cap F/S=4/0.

φ=(−875, 175, 0, 0, 280, −70, 0, 0, 0, 0); threshold=(0, 2); full=(35, 0); μ=0.

ψ=(−465, 504, 0, −8, 997, −287, 0, 0, 0, 0); threshold=(0, 12); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 256·Q21 + 2·Q22 + Q23 + 288·U7 + 574·U9 + 70·U12 = −256.

L32. ρ; cap F/S=4/2.

φ=(57, −15, 2, 0, 3, 0, 0, 0, 0, 0); threshold=(0, 2); full=(6, 0); μ=0.

ψ=(−117, 39, 0, 0, 39, −13, 0, 0, 0, 0); threshold=(0, 10); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 24·U9 + 6·U12 = −6.

L33. ρ; cap F/S=4/4.

φ=(13, −8, 4, 0, 15, −4, 0, 0, 0, 0); threshold=(8, 2); full=(4, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 8); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 8·U4 + 8·U9 + 2·U11 + 4·U14 = −2.

L34. ρ; cap F/S=4/6.

φ=(13, −8, 4, 0, 15, −4, 0, 0, 0, 0); threshold=(8, 2); full=(4, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 8·U4 + 8·U9 + 2·U11 + 4·U14 = −2.

L35. ρ; cap F/S=4/8.

φ=(13, −8, 4, 0, 15, −4, 0, 0, 0, 0); threshold=(8, 2); full=(4, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 8·U4 + 8·U9 + 2·U11 + 4·U14 = −2.

L36. ρ; cap F/S=4/10.

φ=(13, −8, 4, 0, 15, −4, 0, 0, 0, 0); threshold=(8, 2); full=(4, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 8·U4 + 8·U9 + 2·U11 + 4·U14 = −2.

L37. ρ; cap F/S=4/12.

φ=(−11, −7, 8, 0, 48, −12, 0, 0, 0, 0); threshold=(0, 2); full=(19, 0); μ=0.

ψ=(−230, 69, 0, 0, 69, −23, 0, 0, 0, 0); threshold=(138, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: kS ≥ 0; 6: kS ≤ 2; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14:

nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 24·Q21 + Q22 + Q23 + 138·U6 + 24·U9 + 38·U11 + 19·U14 = −24.

L38. ρ; cap F/S=6/0.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

ψ=(−538, 130, 0, 24, 207, −65, 0, 0, 0, 0); threshold=(0, 12); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 72·Q21 + Q23 + 24·U8 + 130·U9 = −72.

L39. ρ; cap F/S=6/2.

φ=(57, −15, 2, 0, 3, 0, 0, 0, 0, 0); threshold=(0, 0); full=(6, 0); μ=0.

ψ=(−117, 39, 0, 0, 39, −13, 0, 0, 0, 0); threshold=(0, 10); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 24·U9 + 6·U12 = −6.

L40. ρ; cap F/S=6/4.

φ=(−97, 40, 2, 0, 66, −21, 0, 0, 0, 0); threshold=(58, 0); full=(14, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 8); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 58·U4 + 42·U9 + 2·U12 + 14·U14 = −6.

L41. ρ; cap F/S=6/6.

φ=(−97, 40, 2, 0, 66, −21, 0, 0, 0, 0); threshold=(58, 0); full=(14, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 58·U4 + 42·U9 + 2·U12 + 14·U14 = −6.

L42. ρ; cap F/S=6/8.

φ=(−97, 40, 2, 0, 66, −21, 0, 0, 0, 0); threshold=(58, 0); full=(14, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 58·U4 + 42·U9 + 2·U12 + 14·U14 = −6.

L43. ρ; cap F/S=6/10.

φ=(−97, 40, 2, 0, 66, −21, 0, 0, 0, 0); threshold=(58, 0); full=(14, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 12; 12: S ≤ 12; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 58·U4 + 42·U9 + 2·U12 + 14·U14 = −6.

L44. ρ; cap F/S=6/12.

φ=(−73, −13, 16, 0, 84, −20, 0, 0, 0, 0); threshold=(20, 0); full=(1, 0); μ=0.

ψ=(−370, 111, 0, 0, 111, −37, 0, 0, 0, 0); threshold=(222, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: kS ≥ 0; 8: kS ≤ 2; 9: nS8 ≥ 0; 10: rS8 ≥ 0; 11: G ≥ 6; 12: G ≤ 6; 13: S ≥ 12; 14: S ≤ 12; 15: nF8 ≥ 6; 16: nF8 ≤ 6; 17: rF8 ≤ 60; 18: t ≤ 118.

Identity: 48·Q21 + Q22 + Q23 + 20·U4 + 222·U8 + 40·U11 + 58·U13 + U16 = −48.

L45. ρ; cap F/S=0/0.

φ=(−6470, 1941, 0, 0, 1941, −647, 0, 0, 0, 0); threshold=(0, 8); full=(0, 0); μ=0.

ψ=(6450, −1939, 4, 0, −1935, 644, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 11646·Q3 + 3882·Q10 + 15528·Q21 + 1300·Q22 + 1294·Q23 + 1680906·U7 + 1674436·U10 = −5861820.

L46. ρ; cap F/S=0/2.

φ=(−13775, 2958, 1044, 0, 4785, −1769, 0, 0, 0, 0); threshold=(0, 8); full=(−261, 0); μ=0.

ψ=(0, −116, 116, 0, 0, −29, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 464·Q11 + 1740·Q21 + Q22 + 2610·U7 + 580·U10 + 261·U11 = −4176.

L47. ρ; cap F/S=0/4.

φ=(−2675, 360, 120, 0, 939, −251, 0, 0, 0, 0); threshold=(0, 8); full=(−153, 0); μ=0.

ψ=(−45, 13, −2, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 11160·Q21 + 31·Q22 + 62·Q23 + 15686·U7 + 3720·U10 + 4743·U11 = −11160.

L48. ρ; cap F/S=2/0.

φ=(−615, 141, 12, 0, 198, −60, 0, 0, 0, 0); threshold=(0, 6); full=(47, 0); μ=0.

ψ=(110, −33, 0, 0, −33, 11, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 36·Q21 + Q22 + 3·Q23 + 120·U7 + 78·U10 + 47·U12 = −36.

L49. ρ; cap F/S=2/2.

φ=(−1856, 522, 58, 0, 957, −290, 0, 0, 0, 0); threshold=(0, 6); full=(377, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 174·Q21 + Q22 + 580·U7 + 58·U10 + 377·U12 = −870.

L50. ρ; cap F/S=2/4.

φ=(−1255, 289, 24, 0, 402, −122, 0, 0, 0, 0); threshold=(0, 6); full=(89, 0); μ=0.

ψ=(230, −69, 0, 0, −69, 23, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 72·Q21 + Q22 + 3·Q23 + 244·U7 + 162·U10 + 89·U12 = −72.

L51. ρ; cap F/S=4/0.

φ=(−23, −1, 4, 0, 24, −6, 0, 0, 0, 0); threshold=(0, 4); full=(7, 0); μ=0.

ψ=(−30, 9, 0, 0, 9, −3, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 3·Q23 + 12·U7 + 14·U9 + 7·U12 = −12.

L52. ρ; cap F/S=4/2.

φ=(−201, 12, 18, 0, 102, −27, 0, 0, 0, 0); threshold=(0, 4); full=(9, 0); μ=2.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 42·Q21 + Q22 + 2·U2 + 54·U7 + 6·U10 + 9·U12 = −12.

L53. ρ; cap F/S=4/4.

φ=(−201, 12, 18, 0, 102, −27, 0, 0, 0, 0); threshold=(0, 4); full=(9, 0); μ=2.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 42·Q21 + Q22 + 2·U2 + 54·U7 + 6·U10 + 9·U12 = −12.

L54. ρ; cap F/S=6/0.

φ=(103, 3, 4, 0, 66, −20, 0, 0, 0, 0); threshold=(0, 2); full=(89, 0); μ=0.

ψ=(−110, 33, 0, 0, 33, −11, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 3·Q23 + 40·U7 + 62·U9 + 89·U12 = −12.

L55. ρ; cap F/S=6/2.

φ=(−246, 66, 12, 0, 150, −45, 0, 0, 0, 0); threshold=(66, 2); full=(57, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 30·Q21 + Q22 + U2 + 66·U4 + 90·U9 + 6·U12 + 57·U14 = −18.

L56. ρ; cap F/S=6/4.

φ=(−246, 66, 12, 0, 150, −45, 0, 0, 0, 0); threshold=(66, 2); full=(57, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 30·Q21 + Q22 + U2 + 66·U4 + 90·U9 + 6·U12 + 57·U14 = −18.

L57. ρ; cap F/S=8/0.

φ=(39, −31, 12, 0, 48, −10, 0, 0, 0, 0); threshold=(0, 0); full=(17, 0); μ=0.

ψ=(−130, 39, 0, 0, 39, −13, 0, 0, 0, 0); threshold=(0, 4); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60;

14: t ≤ 118.

Identity: 36·Q21 + Q22 + 3·Q23 + 20·U7 + 66·U9 + 17·U12 = −36.

L58. ρ; cap F/S=8/2.

φ=(−107, 22, 4, 0, 45, −13, 0, 0, 0, 0); threshold=(42, 0); full=(−1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 42·U4 + 26·U9 + 4·U12 + U13 = −12.

L59. ρ; cap F/S=8/4.

φ=(−107, 22, 4, 0, 45, −13, 0, 0, 0, 0); threshold=(42, 0); full=(−1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 42·U4 + 26·U9 + 4·U12 + U13 = −12.

L60. ρ; cap F/S=1000/1000.

φ=(3654, −1484, 194, 160, 17, 21, 0, 0, 0, 0); threshold=(0, 1000000); full=(474, 0); μ=97.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: S ≥ 16; 9: nF8 ≥ 2; 10: nF8 ≤ 2; 11: rF8 ≤ 16; 12: t ≤ 118.

Identity: 42·Q9 + 480·Q21 + Q22 + 97·U2 + 270·U8 + 474·U10 = −480.

L61. ρ; cap F/S=1000/1000.

φ=(1907, −728, 134, 0, 12, 27, 0, 0, 0, 0); threshold=(0, 1000000); full=(141, 0); μ=36.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: S ≥ 16; 9: nF8 ≥ 4; 10: nF8 ≤ 4; 11: rF8 ≤ 36; 12: t ≤ 118.

Identity: 54·Q9 + 186·Q21 + Q22 + 36·U2 + 136·U8 + 141·U10 = −186.

L62. ρ; cap F/S=1000/1000.

φ=(1623, −558, 96, 0, −3, 21, 0, 0, 0, 0); threshold=(0, 1000000); full=(159, 0); μ=28.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: S ≥ 16; 9: nF8 ≥ 6; 10: nF8 ≤ 6; 11: rF8 ≤ 60; 12: t ≤ 118.

Identity: 42·Q9 + 120·Q21 + Q22 + 28·U2 + 114·U8 + 159·U10 = −24.

L63. ρ; cap F/S=1000/1000.

φ=(891, −236, 8, 44, 61, −13, 0, 0, 0, 0); threshold=(0, 1000000); full=(93, 0); μ=4.

ψ=(250, −75, 0, 0, −75, 25, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: S ≥ 0; 9: nF8 ≥ 2; 10: nF8 ≤ 2; 11: rF8 ≤ 16; 12: t ≤ 118.

Identity: 128·Q9 + 132·Q21 + Q22 + 2·Q23 + 4·U2 + 154·U7 + 93·U10 = −132.

L64. ρ; cap F/S=1000/1000.

φ=(1542, −373, 52, 0, 105, −13, 0, 0, 0, 0); threshold=(0, 1000000); full=(186, 0); μ=4.

ψ=(270, −81, 0, 0, −81, 27, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: S ≥ 0; 9: nF8 ≥ 4; 10: nF8 ≤ 4; 11: rF8 ≤ 36; 12: t ≤ 118.

Identity: 190·Q9 + 132·Q21 + Q22 + 3·Q23 + 4·U2 + 216·U7 + 186·U10 = −132.

L65. ρ; cap F/S=1000/1000.

φ=(10536, −2824, 446, 0, 945, −115, 0, 0, 0, 0); threshold=(0, 1000000); full=(1208, 0); μ=23.

ψ=(1430, −429, 0, 0, −429, 143, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 18; 8: S ≥ 0; 9: nF8 ≥ 6; 10: nF8 ≤ 6; 11: rF8 ≤ 60; 12: t ≤ 118.

Identity: 1452·Q9 + 1200·Q21 + Q22 + 4·Q23 + 23·U2 + 1682·U7 + 1208·U10 = −1200.

L66. ρ; cap F/S=16/0.

φ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

ψ=(17180, −5158, 4, 0, −5154, 1717, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 18·Q3 + 1723·Q9 + 6·Q10 + 24·Q21 + 2·Q23 + 1721·U7 + 5157·U10 = −8934.

L67. ρ; cap F/S=0/0.

φ=(−270, 81, 0, 0, 81, −27, 0, 0, 0, 0); threshold=(0, 16); full=(0, 0); μ=0.

ψ=(12820, −3874, 28, 0, −3846, 1275, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 27·Q20 + 2·Q22 + 108·U7 = −864.

L68. ρ; cap F/S=0/2.

φ=(−11790, 3537, 0, 0, 3537, −1179, 0, 0, 0, 0); threshold=(0, 16); full=(0, 0); μ=0.

ψ=(10194, −3159, −14, 0, −2767, 997, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 1179·Q20 + 2·Q22 + 4716·U7 = −37728.

L69. ρ; cap F/S=2/0.

φ=(−375, −47, 44, 0, 162, −32, 0, 0, 0, 0); threshold=(0, 14); full=(−23, 0); μ=0.

ψ=(−670, 201, 0, 0, 201, −67, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 2010·Q9 + 8844·Q21 + 67·Q22 + 7·Q23 + 6298·U7 + 1541·U11 = −8844.

L70. ρ; cap F/S=2/2.

φ=(−24960, 4242, 2414, 0, 9070, −3287, 0, 0, 0, 0); threshold=(0, 14); full=(2414, 0); μ=0.

ψ=(−9330, 458, 2030, 0, 2488, −1285, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 4043·Q9 + 8824·Q11 + 38706·Q21 + 9·Q22 + 45561·U7 + 8859·U10 + 21726·U12 = −178656.

L71. ρ; cap F/S=4/0.

φ=(−1745, 452, 8, 0, 405, −131, 0, 0, 0, 0); threshold=(0, 12); full=(−15, 0); μ=0.

ψ=(330, −99, 0, 0, −99, 33, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 24·Q21 + Q22 + 6·Q23 + 262·U7 + 404·U10 + 15·U11 = −24.

L72. ρ; cap F/S=4/2.

φ=(−1745, 452, 8, 0, 405, −131, 0, 0, 0, 0); threshold=(0, 12); full=(−15, 0); μ=0.

ψ=(330, −99, 0, 0, −99, 33, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 24·Q21 + Q22 + 6·Q23 + 262·U7 + 404·U10 + 15·U11 = −24.

L73. ρ; cap F/S=6/0.

φ=(−2625, 649, 26, 0, 684, −215, 0, 0, 0, 0); threshold=(0, 10); full=(282, 0); μ=0.

ψ=(2450, −735, 0, 0, −735, 245, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 78·Q21 + Q22 + Q23 + 430·U7 + 516·U10 + 282·U12 = −78.

L74. ρ; cap F/S=6/2.

φ=(−2625, 649, 26, 0, 684, −215, 0, 0, 0, 0); threshold=(0, 10); full=(282, 0); μ=0.

ψ=(2450, −735, 0, 0, −735, 245, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 78·Q21 + Q22 + Q23 + 430·U7 + 516·U10 + 282·U12 = −78.

L75. ρ; cap F/S=8/0.

φ=(−4725, 1151, 64, 0, 1386, −430, 0, 0, 0, 0); threshold=(0, 8); full=(1273, 0); μ=0.

ψ=(3550, −1065, 0, 0, −1065, 355, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 192·Q21 + Q22 + Q23 + 860·U7 + 774·U10 + 1273·U12 = −192.

L76. ρ; cap F/S=8/2.

φ=(−4725, 1151, 64, 0, 1386, −430, 0, 0, 0, 0); threshold=(0, 8); full=(1273, 0); μ=0.

ψ=(3550, −1065, 0, 0, −1065, 355, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 192·Q21 + Q22 + Q23 + 860·U7 + 774·U10 + 1273·U12 = −192.

L77. ρ; cap F/S=10/0.

φ=(−255, −25, 28, 0, 108, −22, 0, 0, 0, 0); threshold=(18, 6); full=(−13, 0); μ=0.

ψ=(−410, 123, 0, 0, 123, −41, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 16; 10: G ≤ 16; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 738·Q9 + 3444·Q21 + 41·Q22 + 5·Q23 + 738·U4 + 2542·U9 + 533·U13 = −3444.

L78. ρ; cap F/S=10/2.

φ=(−190, 28, 8, 0, 66, −18, 0, 0, 0, 0); threshold=(11, 6); full=(49, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 16; 10: G ≤ 16; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 24·Q21 + Q22 + 11·U4 + 36·U9 + 8·U12 + 49·U14 = −24.

L79. ρ; cap F/S=12/0.

φ=(−125, −6, 12, 0, 51, −11, 0, 0, 0, 0); threshold=(25, 4); full=(−4, 0); μ=0.

ψ=(−40, 12, 0, 0, 12, −4, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 16; 10: G ≤ 16; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 20·Q9 + 108·Q21 + 3·Q22 + 2·Q23 + 75·U4 + 86·U9 + 12·U13 = −108.

L80. ρ; cap F/S=12/2.

φ=(−345, 2, 28, 0, 135, −31, 0, 0, 0, 0); threshold=(85, 4); full=(−4, 0); μ=0.

ψ=(−80, 24, 0, 0, 24, −8, 0, 0, 0, 0); threshold=(8, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: kS ≥ 0; 8: kS ≤ 2; 9: nS8 ≥ 0; 10: rS8 ≥ 0; 11: G ≥ 16; 12: G ≤ 16; 13: S ≥ 2; 14: S ≤ 2; 15: nF8 ≥ 6; 16: nF8 ≤ 6; 17: rF8 ≤ 60; 18: t ≤ 118.

Identity: 12·Q9 + 84·Q21 + Q22 + Q23 + 85·U4 + 8·U8 + 74·U11 + 4·U15 = −132.

L81. ρ; cap F/S=14/0.

φ=(−125, −6, 12, 0, 51, −11, 0, 0, 0, 0); threshold=(25, 2); full=(−4, 0); μ=0.

ψ=(−40, 12, 0, 0, 12, −4, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 16; 10: G ≤ 16; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 20·Q9 + 108·Q21 + 3·Q22 + 2·Q23 + 75·U4 + 86·U9 + 12·U13 = −108.

L82. ρ; cap F/S=14/2.

φ=(−100, 1, 8, 0, 39, −9, 0, 0, 0, 0); threshold=(26, 2); full=(−1, 0); μ=0.

ψ=(−30, 9, 0, 0, 9, −3, 0, 0, 0, 0); threshold=(3, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: kS ≥ 0; 8: kS ≤ 2; 9: nS8 ≥ 0; 10: rS8 ≥ 0; 11: G ≥ 16; 12: G ≤ 16; 13: S ≥ 2; 14: S ≤ 2; 15: nF8 ≥ 6; 16: nF8 ≤ 6; 17: rF8 ≤ 60; 18: t ≤ 118.

Identity: 90·Q9 + 648·Q21 + 27·Q22 + 21·Q23 + 702·U4 + 63·U8 + 576·U11 + 27·U15 = −1008.

L83. ρ; cap F/S=16/0.

φ=(−35, −4, 4, 0, 15, −3, 0, 0, 0, 0); threshold=(10, 0); full=(−5, 0); μ=0.

ψ=(−10, 3, 0, 0, 3, −1, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 16; 10: G ≤ 16; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 24·Q9 + 108·Q21 + 9·Q22 + 6·Q23 + 90·U4 + 78·U9 + 45·U13 = −108.

L84. ρ; cap F/S=16/2.

φ=(−305, −10, 28, 0, 123, −27, 0, 0, 0, 0); threshold=(118, 0); full=(−35, 0); μ=0.

ψ=(−30, 9, 0, 0, 9, −3, 0, 0, 0, 0); threshold=(3, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: kS ≥ 0; 8: kS ≤ 2; 9: nS8 ≥ 0; 10: rS8 ≥ 0; 11: G ≥ 16; 12: G ≤ 16; 13: S ≥ 2; 14: S ≤ 2; 15: nF8 ≥ 6; 16: nF8 ≤ 6; 17: rF8 ≤ 60; 18: t ≤ 118.

Identity: 396·Q9 + 2268·Q21 + 27·Q22 + 60·Q23 + 3186·U4 + 180·U8 + 1854·U11 + 945·U15 = −3852.

L85. ρ; cap F/S=16/4.

φ=(61, −14, 2, 0, 6, −1, 0, 0, 0, 0); threshold=(0, 1000000); full=(9, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 7·Q9 + 6·Q21 + Q22 + 9·U7 + 5·U9 + 9·U12 = −6.

L86. ρ; cap F/S=16/6.

φ=(26, −9, 2, 0, 6, −1, 0, 0, 0, 0); threshold=(0, 1000000); full=(4, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 9·Q9 + 12·Q21 + 2·Q22 + 13·U7 + 5·U9 + 8·U12 = −12.

L87. ρ; cap F/S=16/8.

φ=(43, −22, 6, 0, 18, −3, 0, 0, 0, 0); threshold=(0, 1000000); full=(7, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 11·Q9 + 18·Q21 + Q22 + 17·U7 + 5·U9 + 7·U12 = −18.

L88. ρ; cap F/S=16/10.

φ=(17, −13, 4, 0, 12, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(3, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 10; 10: S ≤ 10; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 13·Q9 + 24·Q21 + 2·Q22 + 21·U7 + 5·U9 + 6·U12 = −24.

L89. ρ; cap F/S=16/12.

φ=(5, −6, 2, 0, 6, −1, 0, 0, 0, 0); threshold=(0, 1000000); full=(1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 12; 10: S ≤ 12; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 3·Q9 + 6·Q21 + Q22 + 5·U7 + U9 + U12 = −6.

L90. ρ; cap F/S=16/14.

φ=(8, −17, 6, 0, 18, −3, 0, 0, 0, 0); threshold=(0, 1000000); full=(2, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 16; 8: G ≤ 16; 9: S ≥ 14; 10: S ≤ 14; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 17·Q9 + 36·Q21 + 2·Q22 + 29·U7 + 5·U9 + 4·U12 = −36.

L91. ρ; cap F/S=14/14.

φ=(1293, −258, 24, 0, 15, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(195, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 14; 9: S ≥ 6; 10: S ≤ 14; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 75·Q9 + 30·Q21 + Q22 + 7·U2 + 75·U7 + 93·U9 + 195·U12 = −30.

L92. ρ; cap F/S=14/14.

φ=(−51, 10, 6, 0, 48, −14, 0, 0, 0, 0); threshold=(0, 1000000); full=(97, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 14; 9: S ≥ 0; 10: S ≤ 14; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + U2 + 28·U7 + 97·U12 = −12.

L93. ρ; cap F/S=14/14.

φ=(2631, −702, 96, 0, −3, 21, 0, 0, 0, 0); threshold=(0, 1000000); full=(303, 0); μ=28.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 14; 9: S ≥ 10; 10: S ≤ 14; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 114·Q9 + 120·Q21 + Q22 + 28·U2 + 72·U7 + 186·U9 + 303·U12 = −120.

L94. ρ; cap F/S=14/0.

φ=(−145, 33, 0, 4, 44, −14, 0, 0, 0, 0); threshold=(0, 1000000); full=(79, 0); μ=0.

ψ=(−170, 51, 0, 0, 51, −17, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 14; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 70·Q8 + 12·Q21 + Q22 + 3·Q23 + 28·U12 + 79·U14 = −12.

L95. ρ; cap F/S=0/8.

φ=(−486, 114, 24, 0, 240, −75, 0, 0, 0, 0); threshold=(0, 1000000); full=(27, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 0; 10: S ≤ 8; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 114·Q7 + 114·Q9 + 30·Q21 + Q22 + 7·U2 + 18·U10 + 150·U12 + 27·U14 = −30.

L96. ρ; cap F/S=14/14.

φ=(703, −258, 48, 0, 9, 7, 0, 0, 0, 0); threshold=(0, 1000000); full=(83, 0); μ=14.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 14; 9: S ≥ 12; 10: S ≤ 14; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 14·Q9 + 60·Q21 + Q22 + 14·U2 + 50·U9 + 83·U12 = −60.

L97. ρ; cap F/S=10/14.

φ=(−378, 102, 36, 0, 354, −105, 0, 0, 0, 0); threshold=(0, 1000000); full=(561, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 14; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 6·Q9 + 78·Q21 + Q22 + 5·U2 + 216·U7 + 561·U12 = −78.

L98. ρ; cap F/S=6/14.

φ=(−1273, 383, 40, 84, 1044, −332, 0, 0, 0, 0); threshold=(0, 1000000); full=(989, 0); μ=20.

ψ=(−45, 5, 0, 0, 12, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 0; 10: S ≤ 14; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 264·Q21 + Q22 + Q23 + 20·U2 + 664·U7 + 989·U12 = −264.

L99. ρ; cap F/S=14/14.

φ=(4227, −930, 96, 0, −3, 21, 0, 0, 0, 0); threshold=(0, 1000000); full=(531, 0); μ=28.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 14; 9: S ≥ 8; 10: S ≤ 14; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 228·Q9 + 120·Q21 + Q22 + 28·U2 + 186·U7 + 300·U9 + 531·U12 = −120.

L100. ρ; cap F/S=2/14.

φ=(198, −68, 14, 10, 50, −15, 0, 0, 0, 0); threshold=(0, 1000000); full=(39, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 8; 10: S ≤ 14; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 30·Q21 + Q22 + 7·U2 + 30·U7 + 18·U9 + 39·U12 = −18.

L101. ρ; cap F/S=4/2.

φ=(−280, 68, 4, 0, 84, −26, 0, 0, 0, 0); threshold=(0, 6); full=(49, 0); μ=0.

ψ=(50, −15, 0, 0, −15, 5, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 4·Q23 + 52·U7 + 44·U10 + 49·U12 = −12.

L102. ρ; cap F/S=6/2.

φ=(−1415, 339, 22, 0, 432, −133, 0, 0, 0, 0); threshold=(0, 4); full=(282, 0); μ=0.

ψ=(950, −285, 0, 0, −285, 95, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 66·Q21 + Q22 + Q23 + 266·U7 + 212·U10 + 282·U12 = −66.

L103. ρ; cap F/S=8/2.

φ=(−64, 18, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(3, 2); full=(35, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 3·U4 + 20·U9 + 2·U12 + 35·U14 = −6.

L104. ρ; cap F/S=10/2.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(35, 0); full=(−5, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 35·U4 + 18·U9 + 4·U12 + 5·U13 = −12.

L105. ρ; cap F/S=0/4.

φ=(−5510, 1218, 435, 0, 1914, −725, 0, 0, 0, 0); threshold=(0, 10); full=(0, 0); μ=0.

ψ=(0, −812, 812, 0, 0, −203, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 203·Q11 + 696·Q21 + Q22 + 1044·U7 + 232·U10 = −3132.

L106. ρ; cap F/S=2/4.

φ=(−5945, 1363, 116, 0, 1914, −580, 0, 0, 0, 0); threshold=(0, 8); full=(435, 0); μ=0.

ψ=(3190, −957, 0, 0, −957, 319, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36;

14: t ≤ 118.

Identity: 319·Q20 + 696·Q21 + 2·Q22 + 2320·U7 + 232·U10 + 870·U12 = −2204.

L107. ρ; cap F/S=4/4.

φ=(−5568, 1566, 174, 0, 2871, −870, 0, 0, 0, 0); threshold=(0, 6); full=(1740, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 522·Q21 + Q22 + 1740·U7 + 174·U10 + 1740·U12 = −5916.

L108. ρ; cap F/S=6/4.

φ=(−99, 23, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(0, 4); full=(23, 0); μ=0.

ψ=(50, −15, 0, 0, −15, 5, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 20·U7 + 12·U10 + 23·U12 = −6.

L109. ρ; cap F/S=8/4.

φ=(−64, 18, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(5, 2); full=(33, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 5·U4 + 20·U9 + 2·U12 + 33·U14 = −6.

L110. ρ; cap F/S=10/4.

φ=(−49, 8, 2, 0, 18, −5, 0, 0, 0, 0); threshold=(20, 0); full=(−2, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 20·U4 + 10·U9 + 2·U12 + 2·U13 = −6.

L111. ρ; cap F/S=0/2.

φ=(−41325, 8874, 3132, 0, 14355, −5307, 0, 0, 0, 0); threshold=(0, 12); full=(−783, 0); μ=0.

ψ=(0, −1392, 1392, 0, 0, −348, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 1392·Q11 + 5220·Q21 + Q22 + 7830·U7 + 1740·U10 + 783·U11 = −45762.

L112. ρ; cap F/S=2/2.

φ=(−5945, 1363, 116, 0, 1914, −580, 0, 0, 0, 0); threshold=(0, 10); full=(435, 0); μ=0.

ψ=(3190, −957, 0, 0, −957, 319, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 319·Q20 + 696·Q21 + 2·Q22 + 2320·U7 + 232·U10 + 870·U12 = −7308.

L113. ρ; cap F/S=4/2.

φ=(−280, 68, 4, 0, 84, −26, 0, 0, 0, 0); threshold=(0, 8); full=(75, 0); μ=0.

ψ=(50, −15, 0, 0, −15, 5, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 4·Q23 + 52·U7 + 44·U10 + 75·U12 = −12.

L114. ρ; cap F/S=6/2.

φ=(−280, 68, 4, 0, 84, −26, 0, 0, 0, 0); threshold=(0, 6); full=(75, 0); μ=0.

ψ=(50, −15, 0, 0, −15, 5, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 4·Q23 + 52·U7 + 44·U10 + 75·U12 = −12.

L115. ρ; cap F/S=8/2.

φ=(−1415, 339, 22, 0, 432, −133, 0, 0, 0, 0); threshold=(0, 4); full=(415, 0); μ=0.

ψ=(950, −285, 0, 0, −285, 95, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 66·Q21 + Q22 + Q23 + 266·U7 + 212·U10 + 415·U12 = −66.

L116. ρ; cap F/S=10/2.

φ=(−64, 18, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(3, 2); full=(45, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 3·U4 + 20·U9 + 2·U12 + 45·U14 = −6.

L117. ρ; cap F/S=12/2.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(35, 0); full=(4, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 35·U4 + 18·U9 + 4·U12 + 4·U14 = −12.

L118. ρ; cap F/S=0/2.

φ=(−16790, 3579, 1296, 0, 5847, −2165, 0, 0, 0, 0); threshold=(0, 8); full=(−324, 0); μ=0.

ψ=(0, −8, 8, 0, 0, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 31·Q9 + 1152·Q11 + 4320·Q21 + 2·Q22 + 6480·U7 + 93·U8 + 1409·U10 + 648·U11 = −12256.

L119. ρ; cap F/S=2/2.

φ=(−965, 239, 12, 0, 282, −88, 0, 0, 0, 0); threshold=(0, 6); full=(31, 0); μ=0.

ψ=(250, −75, 0, 0, −75, 25, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 36·Q21 + Q22 + 3·Q23 + 176·U7 + 162·U10 + 31·U12 = −36.

L120. ρ; cap F/S=4/2.

φ=(−1125, 231, 36, 0, 402, −120, 0, 0, 0, 0); threshold=(0, 4); full=(141, 0); μ=4.

ψ=(250, −75, 0, 0, −75, 25, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 84·Q21 + Q22 + 3·Q23 + 4·U2 + 240·U7 + 162·U10 + 141·U12 = −48.

L121. ρ; cap F/S=6/2.

φ=(−435, 93, 12, 0, 150, −45, 0, 0, 0, 0); threshold=(66, 2); full=(30, 0); μ=1.

ψ=(90, −27, 0, 0, −27, 9, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 30·Q21 + Q22 + 3·Q23 + U2 + 66·U4 + 90·U9 + 60·U12 + 30·U14 = −30.

L122. ρ; cap F/S=8/2.

φ=(−115, 20, 4, 0, 39, −11, 0, 0, 0, 0); threshold=(34, 0); full=(−5, 0); μ=0.

ψ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 2·Q23 + 34·U4 + 22·U9 + 8·U12 + 5·U13 = −12.

L123. ρ; cap F/S=0/0.

φ=(−2255, 672, 4, 0, 679, −227, 0, 0, 0, 0); threshold=(0, 10); full=(−1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 9810·Q6 + 24490·Q9 + 37250·Q11 + 161760·Q21 + 20340·Q22 + 6986910·U7 + 66165·U8 + 2293035·U10 + 2263605·U12 + 20340·U13 = −87726480.

L124. ρ; cap F/S=2/0.

φ=(−1235, 235, 86, 0, 432, −148, 0, 0, 0, 0); threshold=(0, 8); full=(111, 0); μ=0.

ψ=(0, −188, 188, 0, 0, −47, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 25338·Q8 + 3116·Q9 + 30832·Q11 + 161376·Q21 + 984·Q22 + 283392·U7 + 50676·U8 + 76014·U10 + 109224·U14 = −161376.

L125. ρ; cap F/S=4/0.

φ=(−1185, 296, 12, 0, 321, −101, 0, 0, 0, 0); threshold=(0, 6); full=(73, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 149·Q9 + 36·Q21 + Q22 + 149·U7 + 65·U10 + 202·U12 + 73·U14 = −36.

L126. ρ; cap F/S=6/0.

φ=(−435, 101, 8, 0, 138, −42, 0, 0, 0, 0); threshold=(0, 4); full=(93, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 143·Q9 + 48·Q21 + 2·Q22 + 143·U7 + 41·U10 + 168·U12 + 186·U14 = −48.

L127. ρ; cap F/S=8/0.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(17, 2); full=(4, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 0; 12: S ≤ 0; 13: A ≥ 0; 14: A ≤ 0; 15: nF8 ≥ 6; 16: nF8 ≤ 6; 17: rF8 ≤ 60; 18: t ≤ 118.

Identity: 12·Q21 + Q22 + 17·U4 + 18·U9 + 4·U12 + 4·U16 = −12.

L128. ρ; cap F/S=10/0.

φ=(−385, 43, 20, 0, 138, −36, 0, 0, 0, 0); threshold=(130, 0); full=(−25, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 0; 12: S ≤ 0; 13: A ≥ 0; 14: A ≤ 0; 15: nF8 ≥ 6; 16: nF8 ≤ 6; 17: rF8 ≤ 60; 18: t ≤ 118.

Identity: 9·Q9 + 120·Q21 + 2·Q22 + 260·U4 + 153·U9 + 31·U12 + 50·U15 = −120.

L129. ρ; cap F/S=0/2.

φ=(−41325, 8874, 3132, 0, 14355, −5307, 0, 0, 0, 0); threshold=(0, 10); full=(−783, 0); μ=0.

ψ=(0, −696, 696, 0, 0, −174, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 1392·Q11 + 5220·Q21 + Q22 + 7830·U7 + 1740·U10 + 783·U11 = −31668.

L130. ρ; cap F/S=2/2.

φ=(−447, 131, 2, 0, 135, −44, 0, 0, 0, 0); threshold=(0, 8); full=(2, 0); μ=0.

ψ=(150, −45, 0, 0, −45, 15, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + 3·Q23 + 88·U7 + 92·U10 + 2·U12 = −6.

L131. ρ; cap F/S=4/2.

φ=(−1415, 344, 20, 0, 423, −131, 0, 0, 0, 0); threshold=(0, 6); full=(159, 0); μ=0.

ψ=(170, −51, 0, 0, −51, 17, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 60·Q21 + Q22 + 6·Q23 + 262·U7 + 224·U10 + 159·U12 = −60.

L132. ρ; cap F/S=6/2.

φ=(−205, 47, 4, 0, 66, −20, 0, 0, 0, 0); threshold=(11, 4); full=(34, 0); μ=0.

ψ=(110, −33, 0, 0, −33, 11, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + Q23 + 11·U4 + 40·U9 + 26·U12 + 34·U14 = −12.

L133. ρ; cap F/S=8/2.

φ=(−115, 20, 4, 0, 39, −11, 0, 0, 0, 0); threshold=(23, 2); full=(6, 0); μ=0.

ψ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 2·Q23 + 23·U4 + 22·U9 + 8·U12 + 6·U14 = −12.

L134. ρ; cap F/S=10/2.

φ=(−98505, 13726, 4326, 0, 34447, −9281, 0, 0, 0, 0); threshold=(35475, 0); full=(−5465, 0); μ=0.

ψ=(−2550, 675, 0, −5, 685, −210, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 2410·Q9 + 230·Q10 + 118182·Q21 + 9·Q22 + 319275·U4 + 169698·U9 + 36984·U12 + 49185·U13 = −165582.

L135. ρ; cap F/S=0/4.

φ=(−2755, 406, 116, 0, 957, −261, 0, 0, 0, 0); threshold=(0, 10); full=(−145, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 348·Q21 + Q22 + 522·U7 + 116·U10 + 145·U11 = −2436.

L136. ρ; cap F/S=2/4.

φ=(−5568, 1566, 174, 0, 2871, −870, 0, 0, 0, 0); threshold=(0, 8); full=(1131, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 522·Q21 + Q22 + 1740·U7 + 174·U10 + 1131·U12 = −6090.

L137. ρ; cap F/S=4/4.

φ=(−205, 47, 4, 0, 66, −20, 0, 0, 0, 0); threshold=(0, 6); full=(29, 0); μ=0.

ψ=(110, −33, 0, 0, −33, 11, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + Q23 + 40·U7 + 26·U10 + 29·U12 = −12.

L138. ρ; cap F/S=6/4.

φ=(−55, 12, 2, 0, 24, −7, 0, 0, 0, 0); threshold=(5, 4); full=(11, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 5·U4 + 14·U9 + 2·U12 + 11·U14 = −6.

L139. ρ; cap F/S=8/4.

φ=(−107, 22, 4, 0, 45, −13, 0, 0, 0, 0); threshold=(29, 2); full=(12, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 29·U4 + 26·U9 + 4·U12 + 12·U14 = −12.

L140. ρ; cap F/S=10/4.

φ=(−6405, 936, 271, 0, 2227, −606, 0, 0, 0, 0); threshold=(2350, 0); full=(−340, 0); μ=0.

ψ=(−120, 28, 0, 0, 32, −9, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 1636·Q21 + 2·Q22 + Q23 + 4700·U4 + 2424·U9 + 524·U12 + 680·U13 = −1636.

L141. ρ; cap F/S=0/6.

φ=(−8265, 1218, 348, 0, 2871, −783, 0, 0, 0, 0); threshold=(0, 10); full=(−435, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 1044·Q21 + Q22 + 1566·U7 + 348·U10 + 435·U11 = −6612.

L142. ρ; cap F/S=2/6.

φ=(−1856, 522, 58, 0, 957, −290, 0, 0, 0, 0); threshold=(0, 8); full=(377, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 174·Q21 + Q22 + 580·U7 + 58·U10 + 377·U12 = −1914.

L143. ρ; cap F/S=4/6.

φ=(−1856, 522, 58, 0, 957, −290, 0, 0, 0, 0); threshold=(0, 6); full=(580, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 174·Q21 + Q22 + 580·U7 + 58·U10 + 580·U12 = −696.

L144. ρ; cap F/S=6/6.

φ=(−128, 36, 4, 0, 66, −20, 0, 0, 0, 0); threshold=(15, 4); full=(41, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 15·U4 + 40·U9 + 4·U12 + 41·U14 = −12.

L145. ρ; cap F/S=8/6.

φ=(−113, 26, 4, 0, 51, −15, 0, 0, 0, 0); threshold=(35, 2); full=(16, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 35·U4 + 30·U9 + 4·U12 + 16·U14 = −12.

L146. ρ; cap F/S=10/6.

φ=(−49, 8, 2, 0, 18, −5, 0, 0, 0, 0); threshold=(20, 0); full=(−2, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 20·U4 + 10·U9 + 2·U12 + 2·U13 = −6.

L147. ρ; cap F/S=0/8.

φ=(−1205, 258, 92, 0, 419, −155, 0, 0, 0, 0); threshold=(0, 10); full=(−23, 0); μ=0.

ψ=(0, −20, 20, 0, 0, −5, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 4·Q9 + 368·Q11 + 1380·Q21 + 9·Q22 + 2070·U7 + 12·U8 + 456·U10 + 207·U11 = −5484.

L148. ρ; cap F/S=2/8.

φ=(−120, 74, 2, 0, 117, −38, 0, 0, 0, 0); threshold=(0, 8); full=(41, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + 76·U7 + 2·U10 + 41·U12 = −6.

L149. ρ; cap F/S=4/8.

φ=(−107, 22, 4, 0, 45, −13, 0, 0, 0, 0); threshold=(0, 6); full=(19, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 26·U7 + 4·U10 + 19·U12 = −12.

L150. ρ; cap F/S=6/8.

φ=(−1398, 390, 48, 0, 744, −225, 0, 0, 0, 0); threshold=(183, 4); full=(444, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 6; 14:

nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 138·Q21 + Q22 + U2 + 183·U4 + 450·U9 + 42·U12 + 444·U14 = −138.

L151. ρ; cap F/S=8/8.

φ=(−119, 30, 4, 0, 57, −17, 0, 0, 0, 0); threshold=(41, 2); full=(20, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 41·U4 + 34·U9 + 4·U12 + 20·U14 = −12.

L152. ρ; cap F/S=10/8.

φ=(−153, 28, 6, 0, 60, −17, 0, 0, 0, 0); threshold=(70, 0); full=(−4, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 18·Q21 + Q22 + 70·U4 + 34·U9 + 6·U12 + 4·U13 = −18.

L153. ρ; cap F/S=0/2.

φ=(−41325, 8874, 3132, 0, 14355, −5307, 0, 0, 0, 0); threshold=(0, 12); full=(−783, 0); μ=0.

ψ=(0, −348, 348, 0, 0, −87, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 1392·Q11 + 5220·Q21 + Q22 + 7830·U7 + 1740·U10 + 783·U11 = −47328.

L154. ρ; cap F/S=2/2.

φ=(−5945, 1363, 116, 0, 1914, −580, 0, 0, 0, 0); threshold=(0, 10); full=(435, 0); μ=0.

ψ=(3190, −957, 0, 0, −957, 319, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 319·Q20 + 696·Q21 + 2·Q22 + 2320·U7 + 232·U10 + 870·U12 = −5568.

L155. ρ; cap F/S=4/2.

φ=(−1525, 380, 16, 0, 417, −131, 0, 0, 0, 0); threshold=(0, 8); full=(101, 0); μ=0.

ψ=(670, −201, 0, 0, −201, 67, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 48·Q21 + Q22 + 2·Q23 + 262·U7 + 284·U10 + 101·U12 = −48.

L156. ρ; cap F/S=6/2.

φ=(−625, 149, 10, 0, 192, −59, 0, 0, 0, 0); threshold=(0, 6); full=(126, 0); μ=0.

ψ=(410, −123, 0, 0, −123, 41, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 30·Q21 + Q22 + Q23 + 118·U7 + 92·U10 + 126·U12 = −30.

L157. ρ; cap F/S=8/2.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(10, 4); full=(11, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 10·U4 + 18·U9 + 4·U12 + 11·U14 = −12.

L158. ρ; cap F/S=10/2.

φ=(−190, 28, 8, 0, 66, −18, 0, 0, 0, 0); threshold=(47, 2); full=(13, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 24·Q21 + Q22 + 47·U4 + 36·U9 + 8·U12 + 13·U14 = −24.

L159. ρ; cap F/S=12/2.

φ=(−99285, 12418, 4680, 0, 35131, −9263, 0, 0, 0, 0); threshold=(43556, 0); full=(−6011, 0); μ=0.

ψ=(−3570, 952, −7, 0, 952, −294, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6748·Q9 + 644·Q10 + 130224·Q21 + 9·Q22 + 392004·U4 + 174126·U9 + 36660·U12 + 54099·U13 = −277224.

L160. ρ; cap F/S=0/4.

φ=(−2755, 406, 116, 0, 957, −261, 0, 0, 0, 0); threshold=(0, 12); full=(−145, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 348·Q21 + Q22 + 522·U7 + 116·U10 + 145·U11 = −3480.

L161. ρ; cap F/S=2/4.

φ=(−17835, 4089, 348, 0, 5742, −1740, 0, 0, 0, 0); threshold=(0, 10); full=(1305, 0); μ=0.

ψ=(9570, −2871, 0, 0, −2871, 957, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 957·Q20 + 2088·Q21 + 2·Q22 + 6960·U7 + 696·U10 + 2610·U12 = −15312.

L162. ρ; cap F/S=4/4.

φ=(−420, 102, 6, 0, 126, −39, 0, 0, 0, 0); threshold=(0, 8); full=(53, 0); μ=0.

ψ=(50, −15, 0, 0, −15, 5, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 18·Q21 + Q22 + 6·Q23 + 78·U7 + 66·U10 + 53·U12 = −18.

L163. ρ; cap F/S=6/4.

φ=(−125, 23, 4, 0, 42, −12, 0, 0, 0, 0); threshold=(0, 6); full=(21, 0); μ=0.

ψ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 3·Q23 + 24·U7 + 10·U10 + 21·U12 = −12.

L164. ρ; cap F/S=8/4.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(12, 4); full=(9, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 12·U4 + 18·U9 + 4·U12 + 9·U14 = −12.

L165. ρ; cap F/S=10/4.

φ=(−190, 28, 8, 0, 66, −18, 0, 0, 0, 0); threshold=(51, 2); full=(9, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 24·Q21 + Q22 + 51·U4 + 36·U9 + 8·U12 + 9·U14 = −24.

L166. ρ; cap F/S=12/4.

φ=(−3240, 438, 143, 0, 1136, −303, 0, 0, 0, 0); threshold=(1448, 0); full=(−185, 0); μ=0.

ψ=(−600, 140, 0, 0, 160, −45, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 908·Q21 + 2·Q22 + Q23 + 2896·U4 + 1212·U9 + 196·U12 + 370·U13 = −908.

L167. ρ; cap F/S=0/6.

φ=(−8265, 1218, 348, 0, 2871, −783, 0, 0, 0, 0); threshold=(0, 12); full=(−435, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 1044·Q21 + Q22 + 1566·U7 + 348·U10 + 435·U11 = −9744.

L168. ρ; cap F/S=2/6.

φ=(−615, 141, 12, 0, 198, −60, 0, 0, 0, 0); threshold=(0, 10); full=(101, 0); μ=0.

ψ=(110, −33, 0, 0, −33, 11, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 36·Q21 + Q22 + 3·Q23 + 120·U7 + 78·U10 + 101·U12 = −36.

L169. ρ; cap F/S=4/6.

φ=(−210, 72, 6, 0, 126, −39, 0, 0, 0, 0); threshold=(0, 8); full=(101, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 18·Q21 + Q22 + 78·U7 + 6·U10 + 101·U12 = −18.

L170. ρ; cap F/S=6/6.

φ=(−66, 20, 2, 0, 36, −11, 0, 0, 0, 0); threshold=(0, 6); full=(31, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 12; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 6·Q21 + Q22 + 22·U7 + 2·U10 + 31·U12 = −6.

L171. ρ; cap F/S=8/6.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(14, 4); full=(7, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 14·U4 + 18·U9 + 4·U12 + 7·U14 = −12.

L172. ρ; cap F/S=10/6.

φ=(−49, 8, 2, 0, 18, −5, 0, 0, 0, 0); threshold=(15, 2); full=(3, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 15·U4 + 10·U9 + 2·U12 + 3·U14 = −6.

L173. ρ; cap F/S=12/6.

φ=(−895, 127, 38, 0, 312, −84, 0, 0, 0, 0); threshold=(406, 0); full=(−49, 0); μ=0.

ψ=(−15, 3, 0, 0, 4, −1, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 12; 10: G ≤ 12; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 120·Q21 + Q22 + 3·Q23 + 406·U4 + 168·U9 + 32·U12 + 49·U13 = −120.

L174. ρ; cap F/S=0/2.

φ=(−290, 87, 0, 0, 87, −29, 0, 0, 0, 0); threshold=(0, 14); full=(0, 0); μ=0.

ψ=(290, −87, 0, 0, −87, 29, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 29·Q20 + 2·Q22 + 116·U7 = −696.

L175. ρ; cap F/S=2/2.

φ=(−17835, 4089, 348, 0, 5742, −1740, 0, 0, 0, 0); threshold=(0, 12); full=(1305, 0); μ=0.

ψ=(9570, −2871, 0, 0, −2871, 957, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 957·Q20 + 2088·Q21 + 2·Q22 + 6960·U7 + 696·U10 + 2610·U12 = −30624.

L176. ρ; cap F/S=4/2.

φ=(−17835, 4089, 348, 0, 5742, −1740, 0, 0, 0, 0); threshold=(0, 10); full=(2523, 0); μ=0.

ψ=(9570, −2871, 0, 0, −2871, 957, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 957·Q20 + 2088·Q21 + 2·Q22 + 6960·U7 + 696·U10 + 5046·U12 = −16008.

L177. ρ; cap F/S=6/2.

φ=(−490, 120, 6, 0, 138, −43, 0, 0, 0, 0); threshold=(0, 8); full=(75, 0); μ=0.

ψ=(200, −60, 0, 0, −60, 20, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 18·Q21 + Q22 + 2·Q23 + 86·U7 + 86·U10 + 75·U12 = −18.

L178. ρ; cap F/S=8/2.

φ=(−205, 47, 4, 0, 66, −20, 0, 0, 0, 0); threshold=(1, 6); full=(64, 0); μ=0.

ψ=(110, −33, 0, 0, −33, 11, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 14; 10: G ≤ 14; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + Q23 + U4 + 40·U9 + 26·U12 + 64·U14 = −12.

L179. ρ; cap F/S=10/2.

φ=(−190, 28, 8, 0, 66, −18, 0, 0, 0, 0); threshold=(29, 4); full=(31, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 14; 10: G ≤ 14; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 24·Q21 + Q22 + 29·U4 + 36·U9 + 8·U12 + 31·U14 = −24.

L180. ρ; cap F/S=12/2.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(28, 2); full=(11, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 14; 10: G ≤ 14; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 28·U4 + 18·U9 + 4·U12 + 11·U14 = −12.

L181. ρ; cap F/S=14/2.

φ=(−103255, 10618, 5402, 0, 37209, −9495, 0, 0, 0, 0); threshold=(46937, 0); full=(−7063, 0); μ=0.

ψ=(−4905, 1314, −9, 0, 1305, −405, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 14; 10: G ≤ 14; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 3195846090·Q11 + 32792159880·Q20 + 3706884081000·Q21 + 213552045·Q22 + 1420262100·Q23 + 10023492336165·U4 + 4061744001270·U9 + 1508318093835·U13 = −1773397196550.

L182. ρ; cap F/S=0/4.

φ=(−13775, 2958, 1044, 0, 4785, −1769, 0, 0, 0, 0); threshold=(0, 14); full=(−261, 0); μ=0.

ψ=(0, −232, 232, 0, 0, −58, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 580·Q9 + 464·Q11 + 580·Q20 + 1740·Q21 + Q22 + 3190·U7 + 261·U11 = −5916.

L183. ρ; cap F/S=2/4.

φ=(−17835, 4089, 348, 0, 5742, −1740, 0, 0, 0, 0); threshold=(0, 12); full=(1305, 0); μ=0.

ψ=(9570, −2871, 0, 0, −2871, 957, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 957·Q20 + 2088·Q21 + 2·Q22 + 6960·U7 + 696·U10 + 2610·U12 = −29232.

L184. ρ; cap F/S=4/4.

φ=(−1865, 464, 20, 0, 513, −161, 0, 0, 0, 0); threshold=(0, 10); full=(129, 0); μ=0.

ψ=(270, −81, 0, 0, −81, 27, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 60·Q21 + Q22 + 6·Q23 + 322·U7 + 344·U10 + 129·U12 = −60.

L185. ρ; cap F/S=6/4.

φ=(−815, 197, 12, 0, 246, −76, 0, 0, 0, 0); threshold=(0, 8); full=(159, 0); μ=0.

ψ=(190, −57, 0, 0, −57, 19, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 36·Q21 + Q22 + 3·Q23 + 152·U7 + 126·U10 + 159·U12 = −36.

L186. ρ; cap F/S=8/4.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(3, 6); full=(18, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 14; 10: G ≤ 14; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 3·U4 + 18·U9 + 4·U12 + 18·U14 = −12.

L187. ρ; cap F/S=10/4.

φ=(−190, 28, 8, 0, 66, −18, 0, 0, 0, 0); threshold=(33, 4); full=(27, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 14; 10: G ≤ 14; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 24·Q21 + Q22 + 33·U4 + 36·U9 + 8·U12 + 27·U14 = −24.

L188. ρ; cap F/S=12/4.

φ=(−95, 14, 4, 0, 33, −9, 0, 0, 0, 0); threshold=(30, 2); full=(9, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 14; 10: G ≤ 14; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 30·U4 + 18·U9 + 4·U12 + 9·U14 = −12.

L189. ρ; cap F/S=14/4.

φ=(−6430, 766, 301, 0, 2282, −591, 0, 0, 0, 0); threshold=(3026, 0); full=(−405, 0); μ=0.

ψ=(−2760, 644, 0, 0, 736, −207, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 14; 10: G ≤ 14; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 2036·Q21 + 2·Q22 + Q23 + 6052·U4 + 2364·U9 + 188·U12 + 810·U13 = −2036.

L190. ρ; cap F/S=14/14.

φ=(2540, −1085, 322, 0, −400, 84, 0, 0, 0, 0); threshold=(0, 1000000); full=(10, 0); μ=21.

ψ=(−45, 5, 0, 0, 12, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 14; 9: S ≥ 0; 10: S ≤ 14; 11: r ≥ 64; 12: nF8 ≥ 2; 13: nF8 ≤ 2; 14: rF8 ≤ 16; 15: t ≤ 118.

Identity: 168·Q9 + 924·Q21 + Q22 + 7·Q23 + 21·U2 + 1220·U11 + 10·U13 = −924.

L191. ρ; cap F/S=14/14.

φ=(3068, −1337, 394, 0, −470, 102, 0, 0, 0, 0); threshold=(0, 1000000); full=(−14, 0); μ=27.

ψ=(−315, 35, 0, 0, 84, −14, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 14; 9: S ≥ 0; 10: S ≤ 14; 11: r ≥ 64; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 204·Q9 + 1104·Q21 + Q22 + Q23 + 27·U2 + 1432·U11 + 14·U12 = −1104.

L192. ρ; cap F/S=14/14.

φ=(2876, −1253, 370, 0, −444, 96, 0, 0, 0, 0); threshold=(0, 1000000); full=(−14, 0); μ=25.

ψ=(−315, 35, 0, 0, 84, −14, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 14; 9: S ≥ 0; 10: S ≤ 14; 11: r ≥ 64; 12: nF8 ≥ 6; 13: nF8 ≤ 6; 14: rF8 ≤ 60; 15: t ≤ 118.

Identity: 192·Q9 + 1044·Q21 + Q22 + Q23 + 25·U2 + 1356·U11 + 14·U12 = −1044.

L193. ρ; cap F/S=0/10.

φ=(−178, −42, 72, 0, 336, −97, 0, 0, 0, 0); threshold=(0, 1000000); full=(81, 0); μ=21.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 10; 10: S ≤ 10; 11: A ≥ 0; 12: A ≤ 0; 13: I ≤ 126; 14: nF8 ≥ 2; 15: nF8 ≤ 2; 16: rF8 ≤ 16; 17: t ≤ 118.

Identity: Q7 + Q9 + U9 + 2·U13 = −4.

L194. ρ; cap F/S=14/14.

φ=(2221, −400, 26, 0, 24, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(293, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 14; 9: S ≥ 8; 10: S ≤ 14; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 157·Q9 + 48·Q21 + Q22 + 5·U2 + 157·U7 + 161·U9 + 293·U12 = −48.

L195. ρ; cap F/S=0/8.

φ=(−1026, 276, 24, 0, 402, −129, 0, 0, 0, 0); threshold=(0, 1000000); full=(27, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 0; 10: S ≤ 8; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 195·Q7 + 195·Q9 + 30·Q21 + Q22 + 7·U2 + 45·U10 + 258·U12 + 27·U14 = −30.

L196. ρ; cap F/S=14/14.

φ=(3261, −792, 96, 0, −3, 21, 0, 0, 0, 0); threshold=(0, 1000000); full=(393, 0); μ=28.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 14; 9: S ≥ 12; 10: S ≤ 14; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 159·Q9 + 120·Q21 + Q22 + 28·U2 + 117·U7 + 231·U9 + 393·U12 = −120.

L197. ρ; cap F/S=14/14.

φ=(187, −32, 2, 0, 6, −1, 0, 0, 0, 0); threshold=(0, 1000000); full=(27, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 14; 9: S ≥ 4; 10: S ≤ 14; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 16·Q9 + 6·Q21 + Q22 + 18·U7 + 14·U9 + 27·U12 = −6.

L198. ρ; cap F/S=8/0.

φ=(−6435, 1475, 126, 0, 2072, −628, 0, 0, 0, 0); threshold=(0, 1000000); full=(2041, 0); μ=0.

ψ=(−21630, 6489, 0, 0, 6489, −2163, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 8; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 12978·Q8 + 4·Q23 + 4326·U8 + 4326·U12 = −8652.

L199. ρ; cap F/S=14/14.

φ=(429, −144, 24, 0, −3, 6, 0, 0, 0, 0); threshold=(0, 1000000); full=(42, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 14; 9: S ≥ 14; 10: S ≤ 14; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 12·Q9 + 30·Q21 + Q22 + 7·U2 + 30·U9 + 42·U12 = −30.

L200. ρ; cap F/S=14/14.

φ=(177, −29, 2, 0, 9, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(33, 0); μ=0.

ψ=(30, −9, 0, 0, −9, 3, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 0; 10: S ≤ 14; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 20·Q9 + 6·Q21 + Q22 + 3·Q23 + 24·U7 + 33·U12 = −6.

L201. ρ; cap F/S=14/14.

φ=(4857, −1020, 96, 0, −3, 21, 0, 0, 0, 0); threshold=(0, 1000000); full=(621, 0); μ=28.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 14; 9: S ≥ 10; 10: S ≤ 14; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 273·Q9 + 120·Q21 + Q22 + 28·U2 + 231·U7 + 345·U9 + 621·U12 = −120.

L202. ρ; cap F/S=14/14.

φ=(243, −40, 2, 0, 6, −1, 0, 0, 0, 0); threshold=(0, 1000000); full=(35, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 14; 9: S ≥ 6; 10: S ≤ 14; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 20·Q9 + 6·Q21 + Q22 + 22·U7 + 18·U9 + 35·U12 = −6.

L203. ρ; cap F/S=0/14.

φ=(774, −264, 24, 0, −138, 51, 0, 0, 0, 0); threshold=(0, 1000000); full=(27, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 12; 10: S ≤ 14; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 27·Q7 + 27·Q9 + 30·Q21 + Q22 + 7·U2 + 102·U8 + 45·U9 + 27·U14 = −30.

L204. ρ; cap F/S=2/14.

φ=(198, −78, 24, 0, 60, −15, 0, 0, 0, 0); threshold=(0, 1000000); full=(39, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 12; 10: S ≤ 14; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 30·Q21 + Q22 + 7·U2 + 30·U7 + 18·U9 + 39·U12 = −12.

L205. ρ; cap F/S=14/0.

φ=(−25, −7, 4, 0, 12, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(−3, 0); μ=0.

ψ=(550, −165, 0, 0, −165, 55, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 14; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 55·Q9 + 2·Q23 + 55·U7 + 165·U10 = −110.

L206. ρ; cap F/S=14/14.

φ=(8056, −1400, 86, 0, 189, −25, 0, 0, 0, 0); threshold=(0, 1000000); full=(1098, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 14; 9: S ≥ 10; 10: S ≤ 14; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 652·Q9 + 228·Q21 + Q22 + 5·U2 + 702·U7 + 596·U9 + 1098·U12 = −228.

L207. ρ; cap F/S=0/8.

φ=(−298, 82, 4, 0, 102, −33, 0, 0, 0, 0); threshold=(0, 1000000); full=(1, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 0; 10: S ≤ 8; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 50·Q7 + 50·Q9 + 6·Q21 + Q22 + U2 + 14·U10 + 66·U12 + U14 = −6.

L208. ρ; cap F/S=8/0.

φ=(−115, −9, 12, 0, 48, −10, 0, 0, 0, 0); threshold=(0, 1000000); full=(−5, 0); μ=0.

ψ=(−1930, 579, 0, 0, 579, −193, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 8; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 579·Q8 + 2·Q23 + 193·U8 + 193·U12 = −386.

L209. ρ; cap F/S=14/14.

φ=(921, −186, 16, 0, 3, 3, 0, 0, 0, 0); threshold=(0, 1000000); full=(113, 0); μ=4.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 14; 9: S ≥ 12; 10: S ≤ 14; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 58·Q9 + 24·Q21 + Q22 + 4·U2 + 52·U7 + 66·U9 + 113·U12 = −24.

L210. ρ; cap F/S=14/14.

φ=(6572, −1188, 86, 0, 189, −25, 0, 0, 0, 0); threshold=(0, 1000000); full=(886, 0); μ=5.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 14; 9: S ≥ 8; 10: S ≤ 14; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 546·Q9 + 228·Q21 + Q22 + 5·U2 + 596·U7 + 490·U9 + 886·U12 = −228.

L211. ρ; cap F/S=0/14.

φ=(262, −86, 4, 0, −66, 23, 0, 0, 0, 0); threshold=(0, 1000000); full=(1, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 0; 8: G ≤ 0; 9: S ≥ 12; 10: S ≤ 14; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 276·Q9 + 138·Q21 + 23·Q22 + 23·U2 + 782·U8 + 322·U9 + 23·U14 = −138.

L212. ρ; cap F/S=14/0.

φ=(−25, −7, 4, 0, 12, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(−3, 0); μ=0.

ψ=(490, −147, 0, 0, −147, 49, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 12; 8: G ≤ 14; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 49·Q9 + 2·Q23 + 49·U7 + 147·U10 = −98.

L213. ρ; cap F/S=14/14.

φ=(725, −158, 16, 0, 3, 3, 0, 0, 0, 0); threshold=(0, 1000000); full=(85, 0); μ=4.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 14; 9: S ≥ 14; 10: S ≤ 14; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 44·Q9 + 24·Q21 + Q22 + 4·U2 + 38·U7 + 52·U9 + 85·U12 = −24.

L214. ρ; cap F/S=14/14.

φ=(117, −22, 2, 0, 6, −1, 0, 0, 0, 0); threshold=(0, 1000000); full=(17, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 14; 8: G ≤ 14; 9: S ≥ 6; 10: S ≤ 14; 11: nF8 ≥ 6; 12: nF8 ≤ 6; 13: rF8 ≤ 60; 14: t ≤ 118.

Identity: 11·Q9 + 6·Q21 + Q22 + 13·U7 + 9·U9 + 17·U12 = −6.

L215. ρ; cap F/S=4/6.

φ=(−60, 9, 9, 0, 48, −15, 0, 0, 0, 0); threshold=(0, 1000000); full=(−24, 6); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≤ 10; 12: e ≤ 56; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 30·U7 + 3·U10 + 15·U12 + 24·U13 + 6·U15 = −6.

L216. ρ; cap F/S=4/6.

φ=(−6, 6, 0, 0, 36, −9, 0, 0, 0, 0); threshold=(0, 1000000); full=(30, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≤ 10; 12: e ≥ 58; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 18·Q21 + Q22 + U2 + 18·U7 + 6·U9 + 24·U12 + 30·U14 = −12.

L217. ρ; cap F/S=4/6.

φ=(−60, 9, 9, 0, 48, −15, 0, 0, 0, 0); threshold=(0, 1000000); full=(−24, 6); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≥ 12; 12: rS5 ≤ 0; 13: e ≤ 56; 14:

nF8 ≥ 2; 15: nF8 ≤ 2; 16: rF8 ≤ 16; 17: t ≤ 118.

Identity: 6·Q21 + Q22 + U2 + 30·U7 + 3·U10 + 15·U13 + 24·U14 + 6·U16 = −6.

L218. ρ; cap F/S=4/6.

φ=(−6, 6, 0, 0, 36, −9, 0, 0, 0, 0); threshold=(0, 1000000); full=(30, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≥ 12; 12: rS5 ≤ 0; 13: e ≥ 58; 14: nF8 ≥ 2; 15: nF8 ≤ 2; 16: rF8 ≤ 16; 17: t ≤ 118.

Identity: 18·Q21 + Q22 + U2 + 18·U7 + 6·U9 + 24·U13 + 30·U15 = −12.

L219. ρ; cap F/S=4/6.

φ=(228, −93, 27, 0, 36, −9, 0, 0, 0, 0); threshold=(0, 1000000); full=(−96, 18); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≥ 12; 12: rS5 ≥ 2; 13: e ≤ 56; 14: nF8 ≥ 2; 15: nF8 ≤ 2; 16: rF8 ≤ 16; 17: t ≤ 118.

Identity: 18·Q21 + Q22 + 7·U2 + 18·U7 + 15·U9 + 21·U13 + 96·U14 + 18·U16 = −18.

L220. ρ; cap F/S=4/6.

φ=(−6, 6, 0, 0, 36, −9, 0, 0, 0, 0); threshold=(0, 1000000); full=(30, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: rS6 ≥ 12; 12: rS5 ≥ 2; 13: e ≥ 58; 14: nF8 ≥ 2; 15: nF8 ≤ 2; 16: rF8 ≤ 16; 17: t ≤ 118.

Identity: 18·Q21 + Q22 + U2 + 18·U7 + 6·U9 + 24·U13 + 30·U15 = −12.

L221. ρ; cap F/S=0/2.

φ=(−331, 136, 15, 0, 207, −72, 0, 0, 0, 0); threshold=(0, 2); full=(47, 0); μ=0.

ψ=(0, −28, 28, 0, 0, −7, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 7·Q11 + 24·Q21 + Q22 + 130·U7 + 8·U10 + 47·U12 = −24.

L222. ρ; cap F/S=2/2.

φ=(−124, 60, 0, 2, 91, −30, 0, 0, 0, 0); threshold=(20, 0); full=(33, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 2; 14:

nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 20·U4 + 60·U9 + 2·U12 + 33·U14 = −6.

L223. ρ; cap F/S=0/4.

φ=(−343, 144, 15, 0, 219, −76, 0, 0, 0, 0); threshold=(0, 2); full=(51, 0); μ=0.

ψ=(0, 0, 0, 28, −28, −7, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 7·Q11 + 24·Q21 + Q22 + 138·U7 + 8·U10 + 51·U12 = −24.

L224. ρ; cap F/S=2/4.

φ=(−130, 64, 0, 2, 97, −32, 0, 0, 0, 0); threshold=(22, 0); full=(35, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 22·U4 + 64·U9 + 2·U12 + 35·U14 = −6.

L225. ρ; cap F/S=0/6.

φ=(−355, 152, 15, 0, 231, −80, 0, 0, 0, 0); threshold=(0, 2); full=(55, 0); μ=0.

ψ=(0, 0, 0, 28, −28, −7, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 7·Q11 + 24·Q21 + Q22 + 146·U7 + 8·U10 + 55·U12 = −24.

L226. ρ; cap F/S=2/6.

φ=(−136, 68, 0, 2, 103, −34, 0, 0, 0, 0); threshold=(24, 0); full=(37, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 24·U4 + 68·U9 + 2·U12 + 37·U14 = −6.

L227. ρ; cap F/S=4/0.

φ=(−79, 29, 4, 0, 66, −20, 0, 0, 0, 0); threshold=(0, 1000000); full=(47, 0); μ=0.

ψ=(−70, 21, 0, 0, 21, −7, 0, 0, 0, 0); threshold=(0, 2); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + Q23 + 40·U7 + 10·U9 + 47·U12 = −12.

L228. ρ; cap F/S=4/2.

φ=(−15, 11, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(0, 1000000); full=(27, 0); μ=0.

ψ=(−70, 21, 0, 0, 21, −7, 0, 0, 0, 0); threshold=(7, 0); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: kS ≥ 0; 6: kS ≤ 2; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 4; 10: G ≤ 4; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 7·U6 + 20·U9 + 12·U11 + 27·U14 = −6.

L229. ρ; cap F/S=0/4.

φ=(−479, 150, 30, 0, 231, −83, 0, 0, 0, 0); threshold=(0, 4); full=(33, 0); μ=0.

ψ=(0, 0, 0, 28, −28, −7, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 14·Q11 + 48·Q21 + Q22 + 138·U7 + 16·U10 + 33·U12 = −48.

L230. ρ; cap F/S=2/4.

φ=(−54, 16, 2, 4, 47, −15, 0, 0, 0, 0); threshold=(0, 2); full=(18, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 2; 12: nF8 ≤ 2; 13: rF8 ≤ 16; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + U2 + 30·U7 + 18·U12 = −12.

L231. ρ; cap F/S=4/4.

φ=(−155, 54, 4, 0, 93, −29, 0, 0, 0, 0); threshold=(12, 0); full=(55, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 4; 10: G ≤ 4; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 2; 14: nF8 ≤ 2; 15: rF8 ≤ 16; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + 12·U4 + 58·U9 + 4·U12 + 55·U14 = −12.

L232. ρ; cap F/S=0/10.

φ=(−341, 120, 18, 0, 186, −65, 0, 0, 0, 0); threshold=(0, 4); full=(30, 0); μ=0.

ψ=(0, −8, 8, 0, 0, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 10; 10: S ≤ 10; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 8·Q11 + 30·Q21 + Q22 + 114·U7 + 10·U10 + 30·U12 = −30.

L233. ρ; cap F/S=2/10.

φ=(6, −24, 30, 0, 159, −45, 0, 0, 0, 0); threshold=(0, 2); full=(48, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 10; 10: S ≤ 10; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 48·Q21 + Q22 + 7·U2 + 90·U7 + 12·U9 + 48·U12 = −48.

L234. ρ; cap F/S=4/10.

φ=(−178, 94, 2, 0, 147, −48, 0, 0, 0, 0); threshold=(73, 0); full=(61, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 4; 10: G ≤ 4; 11: S ≥ 10; 12: S ≤ 10; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 73·U4 + 96·U9 + 2·U12 + 61·U14 = −6.

L235. ρ; cap F/S=0/2.

φ=(−390, 59, 16, 0, 135, −37, 0, 0, 0, 0); threshold=(0, 6); full=(−20, 0); μ=0.

ψ=(10, −3, 0, 0, −3, 1, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 48·Q21 + Q22 + Q23 + 74·U7 + 18·U10 + 20·U11 = −48.

L236. ρ; cap F/S=2/2.

φ=(−410, 94, 8, 0, 132, −40, 0, 0, 0, 0); threshold=(0, 4); full=(33, 0); μ=0.

ψ=(110, −33, 0, 0, −33, 11, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 24·Q21 + Q22 + 2·Q23 + 80·U7 + 52·U10 + 33·U12 = −24.

L237. ρ; cap F/S=4/2.

φ=(−64, 18, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(7, 2); full=(13, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 7·U4 + 20·U9 + 2·U12 + 13·U14 = −6.

L238. ρ; cap F/S=6/2.

φ=(−61, 16, 2, 0, 30, −9, 0, 0, 0, 0); threshold=(22, 0); full=(2, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 22·U4 + 18·U9 + 2·U12 + 2·U14 = −6.

L239. ρ; cap F/S=0/4.

φ=(−2010, 518, 85, 0, 654, −235, 0, 0, 0, 0); threshold=(0, 6); full=(0, 0); μ=0.

ψ=(1440, −532, 172, 0, −240, 47, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 312·Q21 + 2·Q22 + Q23 + 768·U7 + 264·U10 = −312.

L240. ρ; cap F/S=2/4.

φ=(−227, 59, 6, 0, 99, −30, 0, 0, 0, 0); threshold=(0, 4); full=(34, 0); μ=0.

ψ=(50, −15, 0, 0, −15, 5, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 18·Q21 + Q22 + Q23 + 60·U7 + 16·U10 + 34·U12 = −18.

L241. ρ; cap F/S=4/4.

φ=(−246, 66, 12, 0, 150, −45, 0, 0, 0, 0); threshold=(27, 2); full=(57, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 30·Q21 + Q22 + U2 + 27·U4 + 90·U9 + 6·U12 + 57·U14 = −30.

L242. ρ; cap F/S=6/4.

φ=(−64, 18, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(25, 0); full=(3, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 25·U4 + 20·U9 + 2·U12 + 3·U14 = −6.

L243. ρ; cap F/S=0/6.

φ=(−292, 78, 10, 0, 97, −34, 0, 0, 0, 0); threshold=(0, 6); full=(−1, 0); μ=0.

ψ=(24, −8, 2, 0, −4, 1, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 20·Q21 + Q22 + 5·Q23 + 58·U7 + 20·U10 + U11 = −20.

L244. ρ; cap F/S=2/6.

φ=(−262, 78, 8, 0, 141, −43, 0, 0, 0, 0); threshold=(0, 4); full=(55, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 24·Q21 + Q22 + 86·U7 + 8·U10 + 55·U12 = −24.

L245. ρ; cap F/S=4/6.

φ=(−300, 78, 18, 0, 201, −60, 0, 0, 0, 0); threshold=(33, 2); full=(75, 0); μ=2.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 42·Q21 + Q22 + 2·U2 + 33·U4 + 120·U9 + 6·U12 + 75·U14 = −42.

L246. ρ; cap F/S=6/6.

φ=(−67, 20, 2, 0, 36, −11, 0, 0, 0, 0); threshold=(28, 0); full=(4, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 28·U4 + 22·U9 + 2·U12 + 4·U14 = −6.

L247. ρ; cap F/S=0/8.

φ=(−257, 48, 10, 0, 102, −29, 0, 0, 0, 0); threshold=(0, 6); full=(−6, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 30·Q21 + Q22 + 58·U7 + 10·U10 + 6·U11 = −30.

L248. ρ; cap F/S=2/8.

φ=(−5568, 1566, 174, 0, 2871, −870, 0, 0, 0, 0); threshold=(0, 4); full=(1131, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 6; 8: G ≤ 6; 9: S ≥ 8; 10: S ≤ 8; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 522·Q21 + Q22 + 1740·U7 + 174·U10 + 1131·U12 = −696.

L249. ρ; cap F/S=4/8.

φ=(−54, 12, 6, 0, 51, −15, 0, 0, 0, 0); threshold=(12, 2); full=(18, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 12·Q21 + Q22 + U2 + 12·U4 + 30·U9 + 18·U14 = −12.

L250. ρ; cap F/S=6/8.

φ=(−70, 22, 2, 0, 39, −12, 0, 0, 0, 0); threshold=(31, 0); full=(5, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 6; 10: G ≤ 6; 11: S ≥ 8; 12: S ≤ 8; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 31·U4 + 24·U9 + 2·U12 + 5·U14 = −6.

L251. ρ; cap F/S=0/2.

φ=(−41325, 8874, 3132, 0, 14355, −5307, 0, 0, 0, 0); threshold=(0, 8); full=(−783, 0); μ=0.

ψ=(0, −1392, 1392, 0, 0, −348, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 1392·Q11 + 5220·Q21 + Q22 + 7830·U7 + 1740·U10 + 783·U11 = −14442.

L252. ρ; cap F/S=2/2.

φ=(−1255, 315, 14, 0, 360, −113, 0, 0, 0, 0); threshold=(0, 6); full=(31, 0); μ=0.

ψ=(1030, −309, 0, 0, −309, 103, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 42·Q21 + Q22 + Q23 + 226·U7 + 220·U10 + 31·U12 = −42.

L253. ρ; cap F/S=4/2.

φ=(−410, 94, 8, 0, 132, −40, 0, 0, 0, 0); threshold=(0, 4); full=(73, 0); μ=0.

ψ=(110, −33, 0, 0, −33, 11, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 24·Q21 + Q22 + 2·Q23 + 80·U7 + 52·U10 + 73·U12 = −24.

L254. ρ; cap F/S=6/2.

φ=(−64, 18, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(3, 2); full=(25, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 3·U4 + 20·U9 + 2·U12 + 25·U14 = −6.

L255. ρ; cap F/S=8/2.

φ=(−52, 10, 2, 0, 21, −6, 0, 0, 0, 0); threshold=(19, 0); full=(−1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 2; 12: S ≤ 2; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 19·U4 + 12·U9 + 2·U12 + U13 = −6.

L256. ρ; cap F/S=0/4.

φ=(−13775, 2958, 1044, 0, 4785, −1769, 0, 0, 0, 0); threshold=(0, 8); full=(−261, 0); μ=0.

ψ=(0, −464, 464, 0, 0, −116, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 464·Q11 + 1740·Q21 + Q22 + 2610·U7 + 580·U10 + 261·U11 = −3654.

L257. ρ; cap F/S=2/4.

φ=(−5568, 1566, 174, 0, 2871, −870, 0, 0, 0, 0); threshold=(0, 6); full=(1131, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 522·Q21 + Q22 + 1740·U7 + 174·U10 + 1131·U12 = −4872.

L258. ρ; cap F/S=4/4.

φ=(−5568, 1566, 174, 0, 2871, −870, 0, 0, 0, 0); threshold=(0, 4); full=(1740, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 4; 10: S ≤ 4; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 522·Q21 + Q22 + 1740·U7 + 174·U10 + 1740·U12 = −2436.

L259. ρ; cap F/S=6/4.

φ=(−64, 18, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(5, 2); full=(23, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 5·U4 + 20·U9 + 2·U12 + 23·U14 = −6.

L260. ρ; cap F/S=8/4.

φ=(−162, 34, 6, 0, 69, −20, 0, 0, 0, 0); threshold=(65, 0); full=(−1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 4; 12: S ≤ 4; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 18·Q21 + Q22 + 65·U4 + 40·U9 + 6·U12 + U13 = −18.

L261. ρ; cap F/S=0/6.

φ=(−34070, 7305, 2592, 0, 11841, −4379, 0, 0, 0, 0); threshold=(0, 8); full=(−648, 0); μ=0.

ψ=(0, −8, 8, 0, 0, −2, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 13·Q9 + 2304·Q11 + 8640·Q21 + 2·Q22 + 12960·U7 + 39·U8 + 2867·U10 + 1296·U11 = −12020.

L262. ρ; cap F/S=2/6.

φ=(−5568, 1566, 174, 0, 2871, −870, 0, 0, 0, 0); threshold=(0, 6); full=(1131, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 522·Q21 + Q22 + 1740·U7 + 174·U10 + 1131·U12 = −4524.

L263. ρ; cap F/S=4/6.

φ=(−128, 36, 4, 0, 66, −20, 0, 0, 0, 0); threshold=(0, 4); full=(49, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 8; 8: G ≤ 8; 9: S ≥ 6; 10: S ≤ 6; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 12·Q21 + Q22 + 40·U7 + 4·U10 + 49·U12 = −12.

L264. ρ; cap F/S=6/6.

φ=(−64, 18, 2, 0, 33, −10, 0, 0, 0, 0); threshold=(7, 2); full=(21, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 6·Q21 + Q22 + 7·U4 + 20·U9 + 2·U12 + 21·U14 = −6.

L265. ρ; cap F/S=8/6.

φ=(−168, 38, 6, 0, 75, −22, 0, 0, 0, 0); threshold=(73, 0); full=(1, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 8; 10: G ≤ 8; 11: S ≥ 6; 12: S ≤ 6; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 18·Q21 + Q22 + 73·U4 + 44·U9 + 6·U12 + U14 = −18.

L266. ρ; cap F/S=0/0.

φ=(−59190, 17657, 100, 0, 17817, −5959, 0, 0, 0, 0); threshold=(0, 10); full=(0, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 57390·Q6 + 108960·Q9 + 190990·Q11 + 786000·Q21 + 3956·Q22 + 35628061·U7 + 316480·U8 + 11716047·U10 + 11563007·U12 = −445759440.

L267. ρ; cap F/S=2/0.

φ=(−1190, 217, 95, 0, 423, −148, 0, 0, 0, 0); threshold=(0, 8); full=(111, 0); μ=0.

ψ=(0, −436, 436, 0, 0, −109, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 10736·Q9 + 38368·Q11 + 185856·Q21 + 1056·Q22 + 297792·U7 + 51216·U8 + 51216·U10 + 117216·U14 = −185856.

L268. ρ; cap F/S=4/0.

φ=(−1565, 394, 14, 0, 411, −130, 0, 0, 0, 0); threshold=(0, 6); full=(73, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 186·Q9 + 42·Q21 + Q22 + 186·U7 + 88·U10 + 260·U12 + 73·U14 = −42.

L269. ρ; cap F/S=6/0.

φ=(−280, 68, 4, 0, 84, −26, 0, 0, 0, 0); threshold=(0, 4); full=(71, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 42·Q9 + 12·Q21 + Q22 + 42·U7 + 14·U10 + 52·U12 + 71·U14 = −12.

L270. ρ; cap F/S=8/0.

φ=(−115, 27, 2, 0, 36, −11, 0, 0, 0, 0); threshold=(0, 2); full=(35, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 0; 10: S ≤ 0; 11: A ≥ 0; 12: A ≤ 0; 13: nF8 ≥ 4; 14: nF8 ≤ 4; 15: rF8 ≤ 36; 16: t ≤ 118.

Identity: 37·Q9 + 12·Q21 + 2·Q22 + 37·U7 + 11·U10 + 44·U12 + 70·U14 = −12.

L271. ρ; cap F/S=10/0.

φ=(−1045, 270, 12, 0, 327, −103, 0, 0, 0, 0); threshold=(93, 0); full=(377, 0); μ=0.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 10; 10: G ≤ 10; 11: S ≥ 0; 12: S ≤ 0; 13: A ≥ 0; 14: A ≤ 0; 15: nF8 ≥ 4; 16: nF8 ≤ 4; 17: rF8 ≤ 36; 18: t ≤ 118.

Identity: 168·Q9 + 36·Q21 + Q22 + 93·U4 + 168·U9 + 50·U12 + 206·U14 + 377·U16 = −36.

L272. ρ; cap F/S=0/2.

φ=(−41325, 8874, 3132, 0, 14355, −5307, 0, 0, 0, 0); threshold=(0, 10); full=(−783, 0); μ=0.

ψ=(0, −1392, 1392, 0, 0, −348, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 1392·Q11 + 5220·Q21 + Q22 + 7830·U7 + 1740·U10 + 783·U11 = −30102.

L273. ρ; cap F/S=2/2.

φ=(−17835, 4089, 348, 0, 5742, −1740, 0, 0, 0, 0); threshold=(0, 8); full=(1305, 0); μ=0.

ψ=(870, −261, 0, 0, −261, 87, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 10; 8: G ≤ 10; 9: S ≥ 2; 10: S ≤ 2; 11: nF8 ≥ 4; 12: nF8 ≤ 4; 13: rF8 ≤ 36; 14: t ≤ 118.

Identity: 957·Q20 + 2088·Q21 + 2·Q22 + 6960·U7 + 696·U10 + 2610·U12 = −8004.

L274. ρ′; cap F/S=2/14.

φ=(96, −42, 12, 0, 13, −3, 0, 0, 0, 0); threshold=(0, 1000000); full=(10, 0); μ=3.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 14; 10: S ≤ 14; 11: rS5 ≤ 2; 12: e ≤ 56; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 8·Q21 + Q22 + 3·U2 + 6·U7 + 6·U9 + 10·U12 + 10·U14 = −8.

L275. ρ′; cap F/S=2/14.

φ=(30, −18, 0, 0, 0, 3, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 14; 10: S ≤ 14; 11: rS5 ≤ 2; 12: e ≥ 58; 13: nF8 ≥ 6; 14: nF8 ≤ 6; 15: rF8 ≤ 60; 16: t ≤ 118.

Identity: 18·Q21 + Q22 + U2 + 6·U8 + 6·U9 + 24·U12 = −12.

L276. ρ′; cap F/S=2/14.

φ=(64, −61, 26, 0, 72, −18, 0, 0, 0, 0); threshold=(0, 1000000); full=(12, 0); μ=7.

ψ=(387, −43, 0, 0, 0, 0, 0, 0, −172, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 14; 10: S ≤ 14; 11: rS5 ≥ 4; 12: nF8 ≥ 6; 13: nF8 ≤ 6; 14: rF8 ≤ 60; 15: t ≤ 118.

Identity: 43·Q9 + 2·Q23 + 43·U7 + 43·U9 + 344·U11 = −258.

43 The one-full-pair residue: four tetrahedra instead of two

Proposition 43.1. *The (16,18) endpoint with one full facet pair is excluded.* By Section 42 a counterexample has G=0,S=10. All local g vanish by stress localization. The local stress sum of Section 12, Σ_{pairs}g=G+64−I/2, therefore gives I=128. The one full pair contributes sixteen vertex–facet incidences and the other sixteen facets have at most seven vertices. Equality 128=16+16·7 forces all of them to have seven vertices.

The full row. At G=0 every quadrilateral has exactly three facets in its upper link, by Section 13. In a full eight-vertex facet, two ridges of at least six vertices through a quadrilateral cover its eight vertices. Section 36 proves their link has at least four facets; thus this cannot occur. Each quadrilateral of a triangular-prism ridge must consequently have a different square-pyramidal ridge as partner: any partner with at least six vertices would give that full cover. This requires at least three square pyramids.

The complete small dual catalogue shows that the only eight-vertex zero-g rows with fewer than eight ridges are the prism, (8,18,17,7;35) and (8,19,18,7;35). The prism has already been excluded. In the middle row the catalogue gives either a prism with fewer than three square pyramids or a seven-vertex ridge and a square pyramid. The latter facet is a pyramid over that seven-vertex ridge; the square pyramid’s quadrilateral lies in the base. Their seven plus five vertices minus the shared four again cover the full facet. In the last row the catalogue gives a quadrilateral whose two containing ridges cover all eight vertices directly. These are forbidden full covers. All statements are obtained by the signed-order face criterion of Section 14, on the dual seven-vertex rows; the independent catalogue check includes every such code and the indicated cover or ridge-count witnesses.

The full-ridge bound $R_8\le 16$ forces each of the two full facets to have exactly eight ridges. Local g=0 gives J=38. Euler gives p=e; J+4n≤4e and η≥0 give 18≤e≤22. Section 36 excludes the e=18 fake row. The full row is therefore H_k=(8,k,k,8;38), 19≤k≤22.

The numerical profile. The nine zero-g seven-vertex rows are Y=(7,15,14,6;29), E=(7,16,16,7;32), J=(7,17,17,7;32), followed by (7,17,18,8;35), (7,17,19,9;38), (7,18,19,8;35), (7,18,20,9;38), (7,18,21,10;41), (7,18,22,11;44). Write their pair counts as y,q,j,a,b,c,d,u,v. They sum to eight. The flag identities at G=0,S=10,I=128 give f_{14}=3e+122. Summing their edges, ridges and Section 41 weights yields

$$\begin{gathered}3e+122=2k+30y+32q+34j+34a+34b+36c+36d+36u+36v,\\ r=8+6y+7q+7j+8a+9b+8c+9d+10u+11v,\\ 9y+4q+3[k=19]+3e\le 210.\end{gathered}\tag{43.1}$$

Together with e+r≤118 and the count sum, the first two lines give 2k+5q+7j+10a+13b+12c+15d+18u+21v≤68. The weight line gives 2k+3[k=19]≤20+3q+5j+5a+5b+3c+3d+3u+3v. Five times the first and seven times the second combine to

$$24k+7\cdot 3[k=19]+4q+15a+30b+39c+54d+69u+84v\le 480.\tag{43.2}$$

For k≥21 this is impossible. For k=19 its constant is 477, and for k=20 it is 480. Each nonzero variable except j,y contributes at least four, so q=a=b=c=d=u=v=0. For k=19 the two preceding inequalities force j≤4 and j≥5, impossible. For k=20 they force j=4,y=4. Equation (43.1) then gives e=58,r=60. Thus the actual facet multiset is

$$Y^8\,J^8\,H^2,\quad H=(8,20,20,8;38). \tag{43.3}$$

The exponents count actual facets. The independent nine-count enumeration visits 51,480 integer profiles and leaves exactly (43.3), but the displayed linear combination already proves the reduction.

Local equality data. The global quadrilateral budget is 72, equal to the weight 8·9 of the Y facets. Since each local weight is at most q−3b+2η, all facets attain their local lower bound. In H, η=2 and q−3b+4=0 imply q=b=2: use b≤q≤2 and η−q≠1. Both quadrilaterals of H are between square-pyramidal ridges, giving four square pyramids. The reserved-quadrilateral rules exclude prisms and every other non-five-vertex ridge with positive polygon excess. The twelve count equations leave either two tetrahedra, four square pyramids and two triangular bipyramids, or three tetrahedra, four square pyramids and one octahedron.

Y has two tetrahedra and J has three, as checked by the small catalogue. In (43.3) the number of global tetrahedral ridges equals the sum of local tetrahedron counts over facet-pair representatives. It is 4·2+4·3 plus H's tetrahedron count and must be even by symmetry. Hence H has exactly two tetrahedral ridges, four square pyramids and two bipyramids. Each ridge of H contains at most one quadrilateral. The small catalogue also gives J exactly one quadrilateral and no six-vertex ridge.

Two Y facets around each quadrilateral. Y has three quadrilaterals, each between its prism base and one square pyramid; its local q−3b contribution at each is one. J and H contribute minus two at each of their quadrilaterals. All upper links are triangles. The Section 36 local link inequality gives a nonpositive sum around each link. Its global sum is 8·3−8·2−2·4=0. Therefore each link sum is zero, forcing exactly two Y facets and one other facet in every quadrilateral link.

Each Y prism base K belongs to a second Y facet Z: neither J nor H has a six-vertex ridge. This partner is unique, and all three quadrilaterals of K lie in Y and Z. Choose one full facet H and its two quadrilaterals Q_1,Q_2. Each Q_i is therefore accompanied by a Y,Z pair sharing a prism K_i. Its other two quadrilaterals will be called mixed quadrilaterals. H shares at most one quadrilateral with any other facet, since their intersection is contained in a ridge of H. Consequently none of the mixed quadrilaterals is contained in H. The two sets of mixed quadrilaterals are disjoint: a shared one would force its two Y facets to be both pairs, and one of those Y facets would share Q_1,Q_2 with H. Thus there are four distinct mixed quadrilaterals.

Each mixed quadrilateral forces a tetrahedron. Fix Q⊂H, its two Y facets Y,Z and their prism K. The intersections H∩Y and H∩Z are square pyramids over Q, because the upper link is a triangle. Their fifth vertices are precisely the apices of Y and Z outside K. In particular both apices belong to H. For a mixed quad R of K, the third facet J′ is of type J. It is not H by the preceding paragraph, nor −H, since R meets Q in an edge and therefore contains H vertices. The intersections J′∩Y,J′∩Z are square pyramids over R; each contains the corresponding apex.

Here is the exact signed-vertex count, to justify the intersection rather than infer it from a drawing. $U=V(Y)\cup V(Z)$ has $7+7-6=8$ vertices and contains no antipodal pair: a potential cross pair either has a member in K, contradicting Y or Z, or consists of the two apices in H. It therefore selects one vertex of each antipodal pair of P. Exactly two members of U are outside H, namely $V(K)\setminus V(Q)$. Since two distinct prism quadrilaterals cover its six vertices, these two belong to R and hence to J′. Also $U\cap V(J')$ has $5+5-4=6$ vertices. Any $H\cap J'$ vertex outside U would be the antipode of a member of U. If that member belongs to H it is forbidden by H; otherwise it is one of the two members just shown to belong to J′ and is forbidden by J′. Thus $H\cap J'$ has exactly $6-2=4$ vertices.

This intersection cannot be a polygon: it would be a second quadrilateral of J′, different from R, which is not contained in H. Nor can a face with four vertices have dimension at most one. It is therefore a tetrahedral ridge of H. Different mixed quadrilaterals give different J′ facets, since a J facet has only one quadrilateral. Different neighboring facets give different ridges of H, since a ridge belongs to exactly two facets. The four mixed quadrilaterals therefore require four tetrahedral ridges of H, contrary to its exactly two. Proposition 43.1 follows.

44 A saturated eight-vertex facet and the two-full-pair zero-stress case

Lemma 44.1. *For the row B=(8,19,19,8;38), the local weight $q-3b+2\eta$ is at least six.* Section 41 already gives four, and $\eta=3$. Weight five would require $q=2,b=1$, contrary to $\eta-q\neq1$. It suffices to exclude weight four. That equality forces $q=b=1$ and exactly one pentagonal two-face S: its contribution $\eta-q=2$ leaves no other long polygon. Denote the quadrilateral by Q. The two ridges A,B over Q are square pyramids. The two ridges C,D over S have six vertices and are pentagonal pyramids. Here A,B are ridge names; the row symbol B is used only in the lemma statement.

For completeness, the twelve ridge-count equations give four tetrahedra, two square pyramids and two ridges of triple (6,10,6). A seven-vertex ridge would put Q in that base, contradicting $b=1$. A prism reserves three quadrilaterals and is impossible. A (6,11,7) ridge has polygon excess one; it cannot contain S and would reserve a quadrilateral, again impossible. Substitute these zeros and two square pyramids in $\Sigma c=8$, $\Sigma nc=38$, $\Sigma fc=38$ to obtain the stated counts. Thus $A\cap B=Q$ and $C\cap D=S$; the other four ridges are tetrahedra.

The intersection of a square-pyramidal ridge with C or D has at most three vertices. Indeed, it is a proper face of a pentagonal pyramid. It cannot be the pentagonal base, since the square pyramid has only five vertices; every other polygonal face is a triangle. Consequently $|A\cap C|,|A\cap D|,|B\cap C|,|B\cap D|\leq3$, where cardinalities count vertices.

An elementary four-set identity now supplies the missing geometric information. For each of the eight vertices v, let $\alpha(v)=1_A(v)+1_B(v)$ and $\beta(v)=1_C(v)+1_D(v)$. Then $(2-\alpha)(2-\beta)+2\alpha+2\beta=4+\alpha\beta$. Summing and using $|A|=|B|=5$, $|C|=|D|=6$ gives

$$\Sigma_v(2-\alpha(v))(2-\beta(v)) = |A\cap C|+|A\cap D|+|B\cap C|+|B\cap D|-12 \leq 0. \tag{44.1}$$

Each summand is nonnegative, so all are zero. Every vertex is in $A\cap B$ or $C\cap D$: $Q\cup S$ is the whole eight-vertex set. Since $|Q|+|S|=9$, their intersection is one vertex x. Write $A=Q\cup\{a\}$, $B=Q\cup\{b\}$, $C=S\cup\{c\}$, $D=S\cup\{d\}$ at vertex-set level. Then a,b are distinct in $S\setminus Q$, and c,d are distinct in $Q\setminus S$.

The apex c belongs to at least six ridges of the four-dimensional facet: C itself and the five distinct other ridges meeting C in its five side triangles. Distinct side triangles give distinct partners because two ridges have a single intersection face. The same applies to d and D. With only eight ridges, at least four contain both c and d. Only A,B can contain Q, since a two-face has two ridge neighbours inside a four-polytope. Choose a common ridge T other than A,B. The proper face $Q\cap T$ contains c,d but is not Q; it is therefore the edge cd. Likewise $Q\cap C$ and $Q\cap D$ are the edges xc and xd. This gives a triangle of edges in the quadrilateral Q, impossible. The equality case is excluded and the lemma follows. □

Proposition 44.2. *The (16,18) endpoint with two full facet pairs and G=0 is excluded.* Section 42 gives S=10; all local g vanish, and their sum gives I=128. Four full facets contribute 32 incidences. The fourteen remaining facets contribute 96 rather than their maximum 98. Antipodal parity therefore forces one six-vertex pair and six seven-vertex pairs. The Section 43 full-cover argument gives at least eight ridges per full facet. $R_8\leq36$ and parity then give at most ten.

There are three six-vertex zero-g rows: (6,13,13,6;26), (6,14,15,7;29), (6,14,16,8;32). The nine seven-vertex rows are those of (43.1). The full row is (8,k,k+r−8,r;14+3r), with (k,r) in the eleven-element list (19,8),(19,9),(20,8),(20,9),(20,10), (21,8),(21,9),(21,10),(22,8),(22,9),(22,10). This is the Euler/excess arithmetic superset with the Section 36 fake rows removed; it assumes no catalogue of all eight-vertex four-polytopes.

Let s_0,s_1,s_2 count the six-vertex pairs, y,q,j,a,b,c,d,u,v the seven-vertex pairs, and $f_0,\dots,f_{10}$ the full pairs in the displayed orders. They sum to 1,6,2 respectively. Use the edge and ridge sums, $3e+122=\Sigma_{\text{actual facets}}e(F)$, $r=\Sigma_{\text{pairs}}r(F)$, $e+r\le 118$, even e,r, and the Section 41 bound $9y+4q+4f_0+8f_1+3e\le 210$. Eliminating y with the count sum, five times the face bound plus seven times the weight bound yields

$$\begin{aligned}&39s_1+54s_2+4f_0+47f_1+15f_3+30f_4+24f_5+39f_6+54f_7\\&+48f_8+63f_9+78f_{10}+4q+15a+30b+39c+54d+69u+84v\le 24.\end{aligned}\tag{44.2}$$

All coefficients are nonnegative. The remaining finite calculation consists only of three count compositions with sums 1,6,2 and the displayed integer equations. Its complete output is two profiles. Write D=(6,13,13,6;26), H_k=(8,k,k,8;38):

Actual facet multiset	**(e,r)**
$D^2Y^4E^8H_{19}^4$	(56,62)
$D^2Y^8J^4H_{20}^2H_{21}^2$	(58,60)

The independent count recurrence enumerates every nonnegative composition, with no geometric filtering. Alternatively, (44.2) sets every variable whose coefficient exceeds 24 to zero, bounds the remaining variables by 24 divided by their coefficients, and reduces the check to substitution in the three original sums. No unknown realization is deleted. The first profile saturates the weight-four lower bound on every H_{19}. Lemma 44.1 excludes it.

In the second profile the weight budget is 72 and is attained. The H_{20} facets have q=b=2 and four square pyramids, with two or three tetrahedral ridges by the twelve count equations. Every ridge of H_{20} contains at most one quadrilateral. H_{21} has q=b=1; D has no quadrilateral, and J has one. The prism base of a Y facet therefore has another Y facet as its unique partner. Saturation of the Section 35 link budget gives exactly two Y facets and one other facet over every quadrilateral, exactly as in Section 43.

Choose an H_{20} facet H. Its two quadrilaterals give four distinct mixed quadrilaterals of the paired Y prisms, by the disjointness argument of Section 43. The third facet over a mixed quadrilateral cannot be H or −H. There is only one H_{20} pair, so that third facet is D,J or H_{21}, each with at most one quadrilateral. The signed-vertex count of Section 43 still gives $|H\cap J'|=4$: it uses only the two Y apices, the prism quad overlap and the absence of antipodal vertices in a facet. Its four-vertex intersection cannot be a quadrilateral, since J′ already has its distinct mixed quadrilateral; it is a tetrahedral ridge. Distinct mixed quadrilaterals have distinct third facets and distinct ridge intersections with H. Thus H would have at least four tetrahedral ridges, exceeding three. The second profile is impossible as well. □

45 The global counts in the positive two-pair cells

There are four full facets. Hence I≤4·8+14·7=130. The local stress sum is Σg(F)=2G+128−I. At G=2 each local stress is at most one, and Section 34 bounds the number of positive-stress facets by two. Thus I=130. At G=4 a facet with g=2 would, by the same two-facet bound with threshold two, allow at most two positive facets, each with g≤2. This contradicts Σg(F)≥6. Consequently g(F)≤1 in all eight positive cells.

We claim e=58,r=60,I=130. The following 27 rational identities prove the claim without assuming those counts. Their row domain is Section 22 with n≤8 on P and n≤9 on the polar, intersected with J≥4r. Both domains omit the impossible rows of Section 36 and the product T; the polar also omits Q and U by Sections 20,38,39,42. The local stress cap on P is one, that on the polar is S/2. The costs use the Section 41 weight with the B entry raised to five, valid by Lemma 44.1. The full count is N_8=4 and R_8≤36, and I≤130. For G=2 the threshold-zero count is at most two; for G=4 the threshold-two count is zero.

As before, $Q_1,\dots,Q_{23}$ are the base inequalities and U lists the additional quadrilateral, threshold, full-count and cell bounds in order. The coefficient vectors φ,ψ use the Section 34 order. Every multiplier in the identity is nonnegative, and its sum is the stated negative integer. These identities therefore exclude the stated branches for every admissible real polytope.

P1. E18positive g2s10 eLow; cap F/S=2/10.

φ=(228, −93, 27, 0, 36, −9, 0, 0, 0, 0); threshold=(0, 0); full=(30, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 2; 8: G ≤ 2; 9: S ≥ 10; 10: S ≤ 10; 11: e ≤ 56; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 18·Q21 + Q22 + 7·U2 + 18·U7 + 15·U9 + 21·U11 + 30·U13 = −18.

P2. E18positive g2s10 eHigh; cap F/S=2/10.

φ=(−1450, 435, 0, 0, 435, −145, 0, 0, 0, 0); threshold=(145, 0); full=(0, 0); μ=0.

ψ=(115, −227, 98, 0, 40, −8, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 10; 12: S ≤ 10; 13: e ≥ 60; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 294·Q21 + Q22 + Q23 + 145·U4 + 290·U9 + 98·U10 + 16·U11 + 262·U13 = −294.

P3. E18positive g2s10 tLow; cap F/S=2/10.

φ=(−90, 27, 0, 0, 27, −9, 0, 0, 0, 0); threshold=(9, 0); full=(0, 0); μ=0.

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 10; 12: S ≤ 10; 13: e ≥ 58; 14: e ≤ 58; 15: t ≤ 116; 16: nF8 ≥ 4; 17: nF8 ≤ 4; 18: rF8 ≤ 36; 19: t ≤ 118.

Identity: 36·Q9 + Q22 + 9·Q23 + 9·U4 + 36·U11 + 162·U13 + 162·U15 = −198.

P4. E18positive g2s2 eLow; cap F/S=2/2.

φ=(−4140, 927, 300, 0, 1497, −549, 0, 0, 0, 0); threshold=(479, 0); full=(−20, 0); μ=10.

ψ=(415, −415, 166, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 2; 12: S ≤ 2; 13: e ≤ 56; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 198·Q21 + Q22 + Q23 + 10·U2 + 479·U4 + 932·U9 + 240·U12 + 642·U13 + 20·U14 = −198.

P5. E18positive g2s2 eHigh; cap F/S=2/2.

φ=(−890, 257, 10, 0, 273, −93, 0, 0, 0, 0); threshold=(91, 0); full=(0, 0); μ=0.

ψ=(45, −45, 18, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 2; 12: S ≤ 2; 13: e ≥ 60; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 150·Q21 + Q22 + 3·Q23 + 91·U4 + 132·U9 + 10·U12 + 120·U13 = −150.

P6. E18positive g2s2 tLow; cap F/S=2/2.

φ=(−90, 27, 0, 0, 27, −9, 0, 0, 0, 0); threshold=(9, 0); full=(0, 0); μ=0.

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 2; 12: S ≤ 2; 13: e ≥ 58; 14: e ≤ 58; 15: t ≤ 116; 16: nF8 ≥ 4; 17: nF8 ≤ 4; 18: rF8 ≤ 36; 19: t ≤ 118.

Identity: Q22 + Q23 + 9·U4 + 12·U9 + 18·U13 + 18·U15 = −30.

P7. E18positive g2s4 eLow; cap F/S=2/4.

φ=(−218, 37, 28, 0, 99, −37, 0, 0, 0, 0); threshold=(37, 0); full=(0, 0); μ=2.

ψ=(25, −25, 10, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 4; 12: S ≤ 4; 13: e ≤ 56; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 2·U2 + 37·U4 + 64·U9 + 16·U12 + 66·U13 = −6.

P8. E18positive g2s4 eHigh; cap F/S=2/4.

φ=(−520, 151, 5, 0, 159, −54, 0, 0, 0, 0); threshold=(53, 0); full=(0, 0); μ=0.

ψ=(40, −40, 16, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 4; 12: S ≤ 4; 13: e ≥ 60; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 90·Q21 + Q22 + 2·Q23 + 53·U4 + 76·U9 + 5·U12 + 75·U13 = −90.

P9. E18positive g2s4 tLow; cap F/S=2/4.

φ=(−90, 27, 0, 0, 27, −9, 0, 0, 0, 0); threshold=(9, 0); full=(0, 0); μ=0.

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 4; 12: S ≤ 4; 13: e ≥ 58; 14: e ≤ 58; 15: t ≤ 116; 16: nF8 ≥ 4; 17: nF8 ≤ 4; 18: rF8 ≤ 36; 19: t ≤ 118.

Identity: Q22 + Q23 + 9·U4 + 12·U9 + 18·U13 + 18·U15 = −30.

P10. E18positive g2s6 eLow; cap F/S=2/6.

φ=(115, −56, 25, 0, 39, −13, 0, 0, 0, 0); threshold=(13, 0); full=(15, 0); μ=5.

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 6; 12: S ≤ 6; 13: e ≤ 56; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 6·Q21 + Q22 + Q23 + 5·U2 + 13·U4 + 26·U9 + 2·U10 + 5·U11 + 39·U13 + 15·U15 = −6.

P11. E18positive g2s6 eHigh; cap F/S=2/6.

φ=(−1190, 347, 10, 0, 363, −123, 0, 0, 0, 0); threshold=(121, 0); full=(0, 0); μ=0.

ψ=(185, −185, 74, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 6; 12: S ≤ 6; 13: e ≥ 60; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 210·Q21 + Q22 + Q23 + 121·U4 + 172·U9 + 10·U12 + 180·U13 = −210.

P12. E18positive g2s6 tLow; cap F/S=2/6.

φ=(−90, 27, 0, 0, 27, −9, 0, 0, 0, 0); threshold=(9, 0); full=(0, 0); μ=0.

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 6; 12: S ≤ 6; 13: e ≥ 58; 14: e ≤ 58; 15: t ≤ 116; 16: nF8 ≥ 4; 17: nF8 ≤ 4; 18: rF8 ≤ 36; 19: t ≤ 118.

Identity: Q22 + Q23 + 9·U4 + 12·U9 + 18·U13 + 18·U15 = −30.

P13. E18positive g2s8 eLow; cap F/S=2/8.

φ=(200, −103, 50, 0, 87, −29, 0, 0, 0, 0); threshold=(29, 0); full=(30, 0); μ=10.

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 8; 12: S ≤ 8; 13: e ≤ 56; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 18·Q21 + Q22 + 3·Q23 + 10·U2 + 29·U4 + 58·U9 + 6·U10 + 10·U11 + 72·U13 + 30·U15 = −18.

P14. E18positive g2s8 eHigh; cap F/S=2/8.

φ=(−670, 196, 5, 0, 204, −69, 0, 0, 0, 0); threshold=(68, 0); full=(0, 0); μ=0.

ψ=(35, −35, 14, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 8; 12: S ≤ 8; 13: e ≥ 60; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: 120·Q21 + Q22 + 3·Q23 + 68·U4 + 96·U9 + 5·U12 + 105·U13 = −120.

P15. E18positive g2s8 tLow; cap F/S=2/8.

φ=(−90, 27, 0, 0, 27, −9, 0, 0, 0, 0); threshold=(9, 0); full=(0, 0); μ=0.

ψ=(15, −15, 6, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: kF ≥ 0; 4: kF ≤ 2; 5: nF8 ≥ 0; 6: rF8 ≥ 0; 7: nS8 ≥ 0; 8: rS8 ≥ 0; 9: G ≥ 2; 10: G ≤ 2; 11: S ≥ 8; 12: S ≤ 8; 13: e ≥ 58; 14: e ≤ 58; 15: t ≤ 116; 16: nF8 ≥ 4; 17: nF8 ≤ 4; 18: rF8 ≤ 36; 19: t ≤ 118.

Identity: 36·Q9 + Q22 + 9·Q23 + 9·U4 + 36·U11 + 162·U13 + 162·U15 = −126.

P16. E18positive g4s2 eLow; cap F/S=2/2.

φ=(−69, −18, 27, 0, 75, −24, 0, 0, 0, 0); threshold=(0, 2); full=(6, 0); μ=4.

ψ=(5, −5, 2, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 2; 10: S ≤ 2; 11: e ≤ 56; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 408·Q21 + 17·Q22 + 51·Q23 + 68·U2 + 816·U7 + 102·U8 + 51·U10 + 561·U11 + 102·U13 = −408.

P17. E18positive g4s2 eHigh; cap F/S=2/2.

φ=(−3495, 819, −6, 0, 1164, −318, 0, 0, 0, 0); threshold=(0, 2); full=(201, 0); μ=10.

ψ=(165, −165, 66, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 2; 10: S ≤ 2; 11: e ≥ 60; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 1014·Q21 + Q22 + 3·Q23 + 10·U2 + 636·U7 + 198·U8 + 66·U9 + 1092·U11 + 201·U13 = −1014.

P18. E18positive g4s2 tLow; cap F/S=2/2.

φ=(−225, 51, 0, 0, 78, −21, 0, 0, 0, 0); threshold=(0, 2); full=(12, 0); μ=1.

ψ=(35, −35, 14, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 2; 10: S ≤ 2; 11: e ≥ 58; 12: e ≤ 58; 13: t ≤ 116; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: Q22 + Q23 + U2 + 42·U7 + 14·U8 + 6·U9 + 78·U11 + 72·U13 + 12·U15 = −66.

P19. E18positive g4s2 iLow; cap F/S=2/2.

φ=(−246, 12, 30, 0, 159, −45, 0, 0, 0, 0); threshold=(0, 2); full=(12, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 2; 10: S ≤ 2; 11: I ≤ 128; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 48·Q21 + Q22 + 7·U2 + 90·U7 + 12·U9 + 36·U11 + 12·U13 = −24.

P20. E18positive g4s4 eLow; cap F/S=2/4.

φ=(18, −51, 51, 0, 204, −60, 0, 0, 0, 0); threshold=(0, 2); full=(54, 0); μ=10.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 4; 10: S ≤ 4; 11: e ≤ 56; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 48·Q21 + Q22 + 10·U2 + 120·U7 + 9·U9 + 45·U11 + 54·U13 = −48.

P21. E18positive g4s4 eHigh; cap F/S=2/4.

φ=(−5535, 1335, −18, 0, 1776, −492, 0, 0, 0, 0); threshold=(0, 2); full=(339, 0); μ=8.

ψ=(715, −715, 286, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 4; 10: S ≤ 4; 11: e ≥ 60; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 1458·Q21 + Q22 + Q23 + 8·U2 + 984·U7 + 286·U8 + 66·U9 + 1560·U11 + 339·U13 = −1458.

P22. E18positive g4s4 tLow; cap F/S=2/4.

φ=(−225, 51, 0, 0, 78, −21, 0, 0, 0, 0); threshold=(0, 2); full=(12, 0); μ=1.

ψ=(35, −35, 14, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 4; 10: S ≤ 4; 11: e ≥ 58; 12: e ≤ 58; 13: t ≤ 116; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: Q22 + Q23 + U2 + 42·U7 + 14·U8 + 6·U9 + 78·U11 + 72·U13 + 12·U15 = −78.

P23. E18positive g4s4 iLow; cap F/S=2/4.

φ=(−246, 12, 30, 0, 159, −45, 0, 0, 0, 0); threshold=(0, 2); full=(12, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 4; 10: S ≤ 4; 11: I ≤ 128; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 48·Q21 + Q22 + 7·U2 + 90·U7 + 12·U9 + 36·U11 + 12·U13 = −48.

P24. E18positive g4s6 eLow; cap F/S=2/6.

φ=(18, −39, 39, 0, 165, −48, 0, 0, 0, 0); threshold=(0, 2); full=(45, 0); μ=8.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: e ≤ 56; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 42·Q21 + Q22 + 8·U2 + 96·U7 + 9·U9 + 27·U11 + 45·U13 = −42.

P25. E18positive g4s6 eHigh; cap F/S=2/6.

φ=(−2525, 617, −10, 0, 796, −222, 0, 0, 0, 0); threshold=(0, 2); full=(159, 0); μ=2.

ψ=(935, −935, 374, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: e ≥ 60; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 1902·Q21 + 3·Q22 + Q23 + 6·U2 + 1332·U7 + 374·U8 + 66·U9 + 2028·U11 + 477·U13 = −1902.

P26. E18positive g4s6 tLow; cap F/S=2/6.

φ=(−78, 30, 0, 0, 78, −21, 0, 0, 0, 0); threshold=(0, 2); full=(33, 0); μ=1.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: e ≥ 58; 12: e ≤ 58; 13: t ≤ 116; 14: nF8 ≥ 4; 15: nF8 ≤ 4; 16: rF8 ≤ 36; 17: t ≤ 118.

Identity: Q22 + U2 + 42·U7 + 6·U9 + 36·U11 + 30·U13 + 33·U15 = −48.

P27. E18positive g4s6 iLow; cap F/S=2/6.

φ=(−162, 0, 30, 0, 159, −45, 0, 0, 0, 0); threshold=(0, 2); full=(24, 0); μ=7.

ψ=(0, 0, 0, 0, 0, 0, 0, 0, 0, 0); threshold=(0, 1000000); full=(0, 0); μ=0.

U: 1: QW ≥ 0; 2: quadrilateral budget; 3: nF8 ≥ 0; 4: rF8 ≥ 0; 5: nS8 ≥ 0; 6: rS8 ≥ 0; 7: G ≥ 4; 8: G ≤ 4; 9: S ≥ 6; 10: S ≤ 6; 11: I ≤ 128; 12: nF8 ≥ 4; 13: nF8 ≤ 4; 14: rF8 ≤ 36; 15: t ≤ 118.

Identity: 48·Q21 + Q22 + 7·U2 + 90·U7 + 12·U9 + 24·U11 + 24·U13 = −48.

Coverage is particularly short here. Since e is even, the alternatives are e≤56, e=58 and e≥60; the first and third are excluded in each cell. At e=58, even t≤118 is either at most 116 or equal to 118; the former is excluded. Thus r=60. For G=2,

I=130 was already proved. For G=4, even I≤130 is either at most 128 or equal to 130; the former is excluded. Euler then gives p=86. We have proved

$$(v,e,p,r,h)=(16,58,86,60,18),\ I=130,\quad \Sigma_F g(F)=2G-2,\ \Sigma_F w(F)\le 156-6S. \tag{45.1}$$

Here $w=q-3b+2\eta$ is the actual geometric weight. The last inequality is Section 36 with these global counts substituted. The equality I=130 also forces all fourteen nonfull facets to have seven vertices.

46 Nine-pair enumeration without an eight-vertex catalogue

Central symmetry lets us choose one facet from each of the nine antipodal pairs. Seven choices have seven vertices; two choices are full. Number the possible rows by the following table. Rows 1–15 come from the planar Gale enumeration. Rows 16–47 are an arithmetic superset: Euler, $0 \le g \le 1$, $\eta \ge 0$, $J \ge 4r$, $J + 4n \le 4e$ and the excluded rows give precisely the listed possibilities. A full row with seven ridges is the dual of a classified seven-vertex row; the other full rows need no realization catalogue. The last column is a proved lower bound for w, including Lemma 44.1.

ID	Row (n,e,p,r;J)	g	r−n+1	Weight
1	(7, 15, 14, 6, 29)	0	0	9
2	(7, 16, 16, 7, 32)	0	1	4
3	(7, 17, 17, 7, 32)	0	1	0
4	(7, 17, 18, 8, 35)	0	2	0
5	(7, 17, 18, 8, 36)	1	2	0
6	(7, 17, 19, 9, 38)	0	3	0
7	(7, 18, 19, 8, 35)	0	2	0
8	(7, 18, 20, 9, 38)	0	3	0
9	(7, 18, 20, 9, 39)	1	3	0
10	(7, 18, 21, 10, 41)	0	4	0
11	(7, 18, 21, 10, 42)	1	4	0
12	(7, 18, 22, 11, 44)	0	5	0
13	(7, 19, 22, 10, 42)	1	4	0
14	(7, 19, 23, 11, 45)	1	5	0
15	(7, 19, 24, 12, 48)	1	6	0
16	(8, 18, 17, 7, 35)	0	0	10
17	(8, 18, 17, 7, 36)	1	0	15
18	(8, 19, 18, 7, 35)	0	0	9
19	(8, 19, 19, 8, 38)	0	1	6
20	(8, 19, 19, 8, 39)	1	1	8
21	(8, 19, 20, 9, 41)	0	2	8

ID	Row (n,e,p,r;J)	g	r−n+1	Weight
22	(8, 20, 20, 8, 38)	0	1	0
23	(8, 20, 20, 8, 39)	1	1	3
24	(8, 20, 21, 9, 41)	0	2	0
25	(8, 20, 21, 9, 42)	1	2	0
26	(8, 20, 22, 10, 44)	0	3	0
27	(8, 20, 22, 10, 45)	1	3	0
28	(8, 21, 21, 8, 38)	0	1	0
29	(8, 21, 21, 8, 39)	1	1	0
30	(8, 21, 22, 9, 41)	0	2	0
31	(8, 21, 22, 9, 42)	1	2	0
32	(8, 21, 23, 10, 44)	0	3	0
33	(8, 21, 23, 10, 45)	1	3	0
34	(8, 21, 24, 11, 47)	0	4	0
35	(8, 21, 24, 11, 48)	1	4	0
36	(8, 22, 22, 8, 38)	0	1	0
37	(8, 22, 22, 8, 39)	1	1	0
38	(8, 22, 23, 9, 41)	0	2	0
39	(8, 22, 23, 9, 42)	1	2	0
40	(8, 22, 24, 10, 44)	0	3	0
41	(8, 22, 24, 10, 45)	1	3	0
42	(8, 22, 25, 11, 47)	0	4	0
43	(8, 22, 25, 11, 48)	1	4	0
44	(8, 23, 23, 8, 39)	1	1	0
45	(8, 23, 24, 9, 42)	1	2	0
46	(8, 23, 25, 10, 45)	1	3	0
47	(8, 23, 26, 11, 48)	1	4	0

If m_i counts pair choices of row i, the entire search is the following system of nonnegative integer equations and one inequality:

$$\Sigma_{i\le 15} m_i=7,\ \Sigma_{i\ge 16} m_i=2,\ \Sigma m_i(r_i-n_i+1)=4,\\ 2\Sigma m_i e_i=288+S,\ \Sigma m_i g_i=G-1,\ \Sigma m_i \rho_i\le 78-3S. \tag{46.1}$$

The ridge-excess equality follows from $\Sigma n_i=65$ and $\Sigma r_i=60$. For a direct enumeration, choose two full row IDs with repetition. Subtract their ridge excesses from four. Visit the seven-vertex IDs in increasing order and choose a multiplicity between zero and the remaining pair count, stopping when its nonnegative ridge excess exceeds the remaining budget. Keep precisely the length-seven lists and test (46.1). This recurrence produces 1914 lists before the cell equations and the following 151 lists after them. Every nine-ID list records pair choices; double it to recover the actual eighteen facets. The table is a complete output, not a sample.

(G,S)	Lists
(2, 2)	42
(2, 4)	54
(2, 6)	34
(2, 8)	14
(2, 10)	1
(4, 2)	4
(4, 4)	1
(4, 6)	1

(G,S)=(2, 2). (1, 1, 1, 1, 2, 2, 5, 16, 16); (1, 1, 1, 1, 1, 4, 5, 16, 16); (1, 1, 1, 1, 1, 2, 9, 16, 16); (1, 1, 1, 2, 2, 2, 2, 16, 17); (1, 1, 1, 1, 2, 2, 4, 16, 17); (1, 1, 1, 1, 1, 4, 4, 16, 17); (1, 1, 1, 1, 1, 3, 6, 16, 17); (1, 1, 1, 1, 1, 2, 5, 16, 19); (1, 1, 1, 1, 1, 1, 9, 16, 19); (1, 1, 1, 1, 2, 2, 2, 16, 20); (1, 1, 1, 1, 1, 2, 4, 16, 20); (1, 1, 1, 1, 1, 1, 8, 16, 20); (1, 1, 1, 1, 1, 1, 6, 16, 23); (1, 1, 1, 1, 1, 1, 5, 16, 24); (1, 1, 1, 1, 1, 2, 2, 16, 25); (1, 1, 1, 1, 1, 1, 4, 16, 25); (1, 1, 1, 1, 1, 1, 3, 16, 27); (1, 1, 1, 1, 1, 1, 2, 16, 33); (1, 1, 1, 1, 2, 2, 2, 17, 19); (1, 1, 1, 1, 1, 2, 4, 17, 19); (1, 1, 1, 1, 1, 2, 3, 17, 21); (1, 1, 1, 1, 1, 1, 6, 17, 22); (1, 1, 1, 1, 1, 2, 2, 17, 24); (1, 1, 1, 1, 1, 1, 4, 17, 24); (1, 1, 1, 1, 1, 1, 3, 17, 26); (1, 1, 1, 1, 1, 1, 6, 18, 20); (1, 1, 1, 1, 1, 1, 5, 18, 21); (1, 1, 1, 1, 1, 1, 2, 18, 27); (1, 1, 1, 1, 1, 1, 1, 18, 35); (1, 1, 1, 1, 1, 1, 5, 19, 19); (1, 1, 1, 1, 1, 1, 2, 2, 19, 20); (1, 1, 1, 1, 1, 1, 4, 19, 20); (1, 1, 1, 1, 1, 1, 2, 19, 25); (1, 1, 1, 1, 1, 1, 1, 19, 33); (1, 1, 1, 1, 1, 1, 3, 20, 21); (1, 1, 1, 1, 1, 1, 2, 20, 24); (1, 1, 1, 1, 1, 1, 1, 20, 32); (1, 1, 1, 1, 1, 1, 2, 21, 23); (1, 1, 1, 1, 1, 1, 1, 21, 31); (1, 1, 1, 1, 1, 1, 1, 22, 27); (1, 1, 1, 1, 1, 1, 1, 23, 26); (1, 1, 1, 1, 1, 1, 1, 24, 25).

(G,S)=(2, 4). (1, 1, 1, 1, 2, 3, 5, 16, 16); (1, 1, 1, 1, 1, 5, 7, 16, 16); (1, 1, 1, 1, 1, 3, 9, 16, 16); (1, 1, 1, 2, 2, 2, 3, 16, 17); (1, 1, 1, 1, 2, 3, 4, 16, 17); (1, 1, 1, 1, 2, 2, 5, 16, 18); (1, 1, 1, 1, 1, 4, 5, 16, 18); (1, 1, 1, 1, 1, 3, 5, 16, 19); (1, 1, 1, 1, 2, 2, 3, 16, 20); (1, 1, 1, 1, 1, 3, 4, 16, 20); (1, 1, 1, 1, 1, 2, 5, 16, 22); (1, 1, 1, 1, 1, 1, 9, 16, 22);

(1, 1, 1, 1, 2, 2, 2, 16, 23); (1, 1, 1, 1, 1, 2, 4, 16, 23); (1, 1, 1, 1, 1, 2, 3, 16, 25); (1, 1, 1, 1, 1, 1, 7, 16, 25); (1, 1, 1, 1, 1, 1, 6, 16, 29); (1, 1, 1, 1, 1, 1, 5, 16, 30); (1, 1, 1, 1, 1, 2, 2, 16, 31); (1, 1, 1, 1, 1, 1, 4, 16, 31); (1, 1, 1, 1, 1, 1, 3, 16, 33); (1, 1, 1, 1, 2, 2, 3, 17, 19); (1, 1, 1, 1, 1, 3, 4, 17, 19); (1, 1, 1, 1, 2, 2, 2, 17, 22); (1, 1, 1, 1, 1, 2, 4, 17, 22); (1, 1, 1, 1, 1, 2, 3, 17, 24); (1, 1, 1, 1, 1, 2, 5, 18, 19); (1, 1, 1, 1, 2, 2, 2, 18, 20); (1, 1, 1, 1, 1, 2, 4, 18, 20); (1, 1, 1, 1, 1, 1, 6, 18, 23); (1, 1, 1, 1, 1, 1, 5, 18, 24); (1, 1, 1, 1, 1, 2, 2, 18, 25); (1, 1, 1, 1, 1, 1, 4, 18, 25); (1, 1, 1, 1, 1, 1, 3, 18, 27); (1, 1, 1, 1, 1, 2, 3, 19, 20); (1, 1, 1, 1, 1, 1, 5, 19, 22); (1, 1, 1, 1, 1, 2, 2, 19, 23); (1, 1, 1, 1, 1, 1, 4, 19, 23); (1, 1, 1, 1, 1, 1, 3, 19, 25); (1, 1, 1, 1, 1, 1, 2, 19, 31); (1, 1, 1, 1, 1, 2, 2, 20, 22); (1, 1, 1, 1, 1, 1, 4, 20, 22); (1, 1, 1, 1, 1, 1, 3, 20, 24); (1, 1, 1, 1, 1, 1, 2, 20, 30); (1, 1, 1, 1, 1, 1, 3, 21, 23); (1, 1, 1, 1, 1, 1, 2, 21, 29); (1, 1, 1, 1, 1, 1, 2, 22, 25); (1, 1, 1, 1, 1, 1, 1, 22, 33); (1, 1, 1, 1, 1, 1, 2, 23, 24); (1, 1, 1, 1, 1, 1, 1, 23, 32); (1, 1, 1, 1, 1, 1, 1, 24, 31); (1, 1, 1, 1, 1, 1, 1, 25, 30); (1, 1, 1, 1, 1, 1, 1, 26, 29); (1, 1, 1, 1, 1, 1, 1, 27, 28).

(G,S)=(2, 6). (1, 1, 1, 1, 3, 3, 5, 16, 16); (1, 1, 1, 2, 2, 3, 3, 16, 17); (1, 1, 1, 1, 2, 3, 5, 16, 18); (1, 1, 1, 1, 2, 3, 3, 16, 20); (1, 1, 1, 1, 1, 3, 5, 16, 22); (1, 1, 1, 1, 2, 2, 3, 16, 23); (1, 1, 1, 1, 1, 3, 4, 16, 23); (1, 1, 1, 1, 1, 3, 3, 16, 25); (1, 1, 1, 1, 1, 2, 5, 16, 28); (1, 1, 1, 1, 2, 2, 2, 16, 29); (1, 1, 1, 1, 1, 2, 4, 16, 29); (1, 1, 1, 1, 1, 2, 3, 16, 31); (1, 1, 1, 1, 2, 2, 3, 17, 22); (1, 1, 1, 1, 1, 3, 4, 17, 22); (1, 1, 1, 1, 1, 3, 3, 17, 24); (1, 1, 1, 1, 1, 3, 5, 18, 19); (1, 1, 1, 1, 1, 2, 5, 18, 22); (1, 1, 1, 1, 2, 2, 2, 18, 23); (1, 1, 1, 1, 1, 2, 3, 18, 25); (1, 1, 1, 1, 1, 3, 3, 19, 20); (1, 1, 1, 1, 1, 2, 3, 19, 23); (1, 1, 1, 1, 1, 1, 5, 19, 28); (1, 1, 1, 1, 1, 2, 2, 19, 29); (1, 1, 1, 1, 1, 1, 4, 19, 29); (1, 1, 1, 1, 1, 1, 3, 19, 31); (1, 1, 1, 1, 1, 2, 3, 20, 22); (1, 1, 1, 1, 1, 1, 5, 22, 22); (1, 1, 1, 1, 1, 2, 2, 22, 23); (1, 1, 1, 1, 1, 1, 4, 22, 23); (1, 1, 1, 1, 1, 1, 3, 22, 25); (1, 1, 1, 1, 1, 1, 2, 22, 31); (1, 1, 1, 1, 1, 1, 3, 23, 24); (1, 1, 1, 1, 1, 1, 2, 24, 29); (1, 1, 1, 1, 1, 1, 2, 25, 28).

(G,S)=(2, 8). (1, 1, 1, 1, 3, 3, 3, 16, 20); (1, 1, 1, 1, 2, 3, 3, 16, 23); (1, 1, 1, 1, 2, 2, 3, 16, 29); (1, 1, 1, 1, 1, 3, 5, 18, 22); (1, 1, 1, 1, 1, 3, 3, 18, 25); (1, 1, 1, 1, 1, 3, 3, 19, 23); (1, 1, 1, 1, 1, 3, 3, 20, 22); (1, 1, 1, 1, 1, 2, 3, 22, 23); (1, 1, 1, 1, 1, 1, 5, 22, 28); (1, 1, 1, 1, 1, 2, 2, 22, 29); (1, 1, 1, 1, 1, 1, 4, 22, 29); (1, 1, 1, 1, 1, 1, 3, 22, 31); (1, 1, 1, 1, 1, 1, 3, 24, 29); (1, 1, 1, 1, 1, 1, 3, 25, 28).

(G,S)=(2, 10). (1, 1, 1, 1, 1, 3, 3, 22, 23).

(G,S)=(4, 2). (1, 1, 1, 1, 1, 5, 5, 16, 17); (1, 1, 1, 1, 1, 2, 5, 17, 20); (1, 1, 1, 1, 1, 1, 5, 17, 25); (1, 1, 1, 1, 1, 1, 5, 20, 20).

(G,S)=(4, 4). (1, 1, 1, 1, 1, 1, 5, 20, 23).

(G,S)=(4, 6). (1, 1, 1, 1, 1, 1, 5, 23, 23).

47 Prisms, tetrahedral parity, and four positive-cell exclusions

Lemma 47.1. *If a four-polytope has at most eight vertices and a triangular-prismatic ridge, then b=0.* Suppose instead that a quadrilateral Q is shared by square-pyramidal ridges R,S. Let T be a prismatic ridge. Q is not in T, because its two ridge neighbours are already R,S. Each of T∩R,T∩S has at least 6+5−8=3 vertices. These intersections are polygonal faces of square pyramids different from Q, hence triangles. They are different: otherwise three ridges would meet in a two-face. A triangular prism has just two triangular faces, its disjoint bases. Thus T∩R∩S=T∩Q has no vertices. But T and Q would then use ten distinct vertices. This contradiction proves the lemma. □

Lemma 47.2. *For V=(8,19,19,8;39), the actual weight is twelve.* Count ridge vertices minus ridge polygons: $\Sigma(n(T)-f_2(T)) = 39-2\cdot 19=1$. Among the twelve possible ridge triples, only (6,9,5) and (7,11,6) have positive difference, namely one. The second is impossible. Indeed, an eight-vertex facet with a seven-vertex ridge is a pyramid, and has one more ridge than that base has polygons: here at most seven instead of eight. Therefore V has a prism. Lemma 47.1 gives b=0. Since η=4, its three quadrilaterals and η−q≠1 force q=4 and no larger polygons. Thus w=4+8=12. □

Lemma 47.3. *An eight-vertex four-polytope with three tetrahedral ridges and all other ridges square pyramids has at least 22 edges.* At a vertex x, let a(x) be the number of tetrahedra containing x and q(x) the number of quadrilaterals containing x. A tetrahedron contributes degree three at x. A square pyramid contributes four at its apex, three at a base vertex. If k(x) is the number of ridges through x, summing these degrees gives 4k(x)−a(x)−2q(x). It is even: it is the sum of the polygon lengths of the three-dimensional vertex figure, twice its edge count. Consequently a(x) is even.

There are three tetrahedra A,B,C, so each vertex belongs to zero or two. In particular the triple intersection is empty. Put α=|A∩B|, β=|A∩C|, γ=|B∩C|. Their four-vertex sets give α+β=α+γ=β+γ=4. Thus α=β=γ=2, and their union has six vertices. Its tetrahedral edges number at least 3·6−3=15, because each pair can share at most the one edge on its two common vertices. The other two vertices have degree at least four each, as every vertex of a four-polytope does. Their edge stars are outside the six-vertex core and overlap in at most their mutual edge. They add at least seven edges, giving 22. □

Use IDs from Section 46, and put D=(7,17,18,8;36), R_0=(8,18,17,7;36), R_{35}=(8,18,17,7;35), C_{20}=(8,20,20,8;39), Q_9=(8,20,21,9;42). These letters label facet rows only. The planar Gale recurrence, applied also to dual rows, gives: D has four tetrahedra and four square pyramids; every ridge of R_0 is a tetrahedron, square pyramid or triangular prism; R_{35} necessarily has a (6,10,6) or (7,11,6) ridge. These tests involve seven-vertex diagrams and are included in the independent catalogue calculation.

The (4,4) cell. Its only pair list is Y^6,D,V,C_{20}. The pair weight is at least 54+12+3=69, exceeding 78−3·4=66. Thus it is impossible.

The (4,2) cell. The four lists are Y^5,D^2,R_{35},R_0; Y^5,E,D,R_0,V; Y^6,D,R_0,Q_9; Y^6,D,V^2. The second and fourth have pair weights at least 45+4+15+12=76 and 54+24=78

respectively, exceeding the budget 72. In the first list only the one R_{35} pair could supply its special ridge type. Every ridge has a distinct second facet, which cannot be the antipode of the first: a proper face contains no antipodal vertices. Thus a ridge type absent from every other row cannot occur in a row represented by just one antipodal pair. This excludes the first list.

Apply that same transfer principle to Q_9 in the third list. Its ridges must be tetrahedra, square pyramids or prisms, since these are the only types supplied by Y,D,R_0. The difference $\Sigma(n(T)-f_2(T))=42-42=0$ then forces no prisms; the other two types have difference zero, prisms difference one. The count equations $\Sigma c=9$, $\Sigma nc=42$ give three tetrahedra and six square pyramids. Lemma 47.3 would require 22 edges, whereas Q_9 has 20. This excludes the third list and the cell.

The (2,10) cell. The only pair list is Y^5,J^2,H_{20},C_{20}, where $H_{20}=(8,20,20,8;38)$. Its lower weight is $45+3=48$, exactly the budget $78-30$. Thus C_{20} attains weight three. A prism in C_{20} would give $b=0$ by Lemma 47.1 and $q\geq 3$, hence $w\geq 9$ since $\eta=3$. There is no prism in C_{20}. There is none in H_{20} either: $\eta=2$ cannot contain three quadrilaterals. The J profile has no prism, whereas Y has one. Consequently the actual ten Y facets account for exactly ten prism incidences. Every ridge belongs to two facets and every ridge has a distinct antipode, so such a count is divisible by four. Ten is not. The cell is excluded.

The (4,6) cell. The only pair list is Y^6,D,C_{20}^2, and its pair weight $54+6=60$ equals the budget. Each C_{20} attains weight three. The twelve ridge-count equations and reserved quadrilateral inequalities of Section 41 now give $q=3,b=2$, no larger polygon, and exactly two tetrahedra, five square pyramids and one (6,11,7) ridge. This is the unique minimizing count vector for C_{20} in the printed Section 41 calculation.

Write F,H for full C_{20} facets sharing the (6,11,7) ridge T. Its second facet must be full, because Y,D lack that ridge type. Full facets choose exactly one vertex from each of the eight antipodal pairs. They agree on the six vertices of T and disagree on the remaining two: write $F\setminus T=\{a,b\}$, $H\setminus T=\{-a,-b\}$. The four full facets are $F,H,-F,-H$. Only H can be a full ridge neighbour of F: $-F$ is disjoint and $F\cap(-H)$ has only two vertices.

Each full facet has eight ridges, so each has seven nonfull neighbours. A nonfull facet cannot neighbour both members of an antipodal full pair: the disjoint intersections would be two ridges of at least four vertices each, exceeding its seven vertices. It has at most two full neighbours. The four full facets give 28 full/nonfull adjacencies and the fourteen nonfull facets can give at most 28. Therefore each nonfull facet has exactly two full neighbours.

In F choose a square-pyramidal ridge R whose intersection with T is a triangle. Such a choice exists: $5+6-8=3$ forces a polygonal intersection for each of the five square pyramids, whereas T has only one quadrilateral and that face has only one partner other than T. Thus R contains a,b and three vertices of T. Its second facet X is nonfull.

X cannot neighbour $-F$, since its ridge in $-F$ would be disjoint from the five vertices of R. It cannot neighbour H: it contains a,b and hence not $-a,-b$; a ridge in H would be contained in T and therefore equal T, forcing eight vertices in X. Finally, T has seven polygons and H has exactly seven other ridges. The other ridge over each polygon of T is distinct, so every ridge of H except T meets T in a

polygon. Every ridge U of −H except −T therefore has at least three vertices in −T. These are disjoint from R⊆F, so X cannot contain U. Nor can X contain −T itself. Hence X has F as its only full neighbour, contradicting the equality count above. This excludes the (4,6) cell. □

Of the eight positive two-pair cells, only (2,2),(2,4),(2,6),(2,8) remain for the further geometric argument. The arithmetic lists for these cells are retained in Section 46; no existence claim is made about any surviving list.

48 Local ridge patterns and antipodal partner capacity

Lemma 48.1. *An eight-vertex four-polytope with three tetrahedra, three square pyramids, one triangular bipyramid and one triangular prism has at least twenty edges.* Let T be its bipyramid and R its prism. They have at least $5+6-8=3$ common vertices, so intersect in a polygon. The bipyramid has only triangles, hence their intersection A is a triangular prism base and their vertex sets cover all eight vertices. Their edges give $9+9-3=15$ edges.

Each of the prism’s three quadrilaterals has a different other ridge, necessarily a square pyramid: neither a tetrahedron nor a bipyramid contains a quadrilateral, and the prism is unique. The apex of each such pyramid is outside R, so is one of the two vertices of T\R. Each square pyramid intersects T in the triangle consisting of that apex and the corresponding edge of A. These three triangles are distinct, because each two-face has only two ridge partners.

The three apices cannot all be the same vertex a. The other vertex of T\R would avoid those three triangles and A. It would belong to at most two of the bipyramid’s six polygons, whereas every vertex of a three-polytope belongs to at least three. Thus two pyramids use one apex a and the third the other apex b. The first two prism quadrilaterals cover all of R; their apex edges therefore join a to each of the three vertices of R\T. The third quadrilateral has two vertices outside T and supplies two further edges from b. These five edges are distinct and outside the fifteen-edge core. This proves the lemma. □

We can now perform a sharper necessary-pattern check. For each small row use its exact ridge-count vector c, quadrilateral count q and b from the Gale face lattice. For the full rows with seven ridges use the dual seven-vertex lattice. The following table prints these finite inputs in the twelve-type order of Section 36. A row may have several distinct patterns; none is chosen in advance.

ID	c	q	b	w
1	(2, 3, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0)	3	0	9
2	(4, 2, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0)	2	0	6
2	(5, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0)	0	0	4
3	(3, 2, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0)	1	1	0
4	(5, 2, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)	1	1	0
4	(6, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0)	1	0	3
5	(4, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)	2	2	0
6	(7, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)	1	1	0
7	(5, 0, 3, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0	0	0
8	(7, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0	0	0
8	(8, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)	0	0	0
9	(6, 2, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)	1	1	0
10	(9, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0	0	0
11	(8, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)	1	1	0
12	(11, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0	0	0

ID	c	q	b	w
13	(8, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0	0	0
14	(10, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0	0	0
15	(12, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)	0	0	0
16	(2, 4, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0)	4	0	12
16	(3, 1, 0, 1, 2, 0, 0, 0, 0, 0, 0, 0)	4	0	12
16	(3, 2, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0)	2	0	10
17	(1, 4, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0)	5	0	15
18	(2, 2, 1, 0, 2, 0, 0, 0, 0, 0, 0, 0)	3	0	9

For every other full row, generate all c from $\Sigma c=r$, $\Sigma nc=J$, $\Sigma f_2 c=2p$ using the twelve listed triples. Retain the Section 36 conditions P1–P3 and vertex-parity exclusions. Generate q,b by the explicit Section 41 rules, and impose b=0 whenever a prism occurs (Lemma 47.1). Retain only $w\geq\rho$ from Sections 41,44,47. Delete a ridge type with $f_2(T)+1>r$: each polygon of T needs a different other ridge in the facet. Delete a seven-vertex ridge with $r>f_2(T)+1$: the facet would be a pyramid and have exactly $f_2(T)+1$ ridges. Finally apply Lemmas 47.3 and 48.1 to delete their impossible edge counts. These rules define the full local option list; they need no list of all realizable eight-vertex polytopes.

Partner capacity. Distinct ridges of a facet have distinct other facets: the intersection of two facets is one face. If both members of an antipodal partner pair supply ridges, those ridges have disjoint vertex sets. Thus in a seven-vertex facet every partner pair supplies at most one ridge. In an eight-vertex facet, this remains true for every family of ridges having at least five vertices. For a set J of such ridge types and a fixed facet i, consequently

$$\Sigma_{j\in J} c_{ij} \leq |\{l\neq i : \text{some } c_{lj}>0 \text{ with } j\in J\}|. \tag{48.1}$$

For a single ridge type with n vertices, a partner pair can supply two only if both facets have at least 2n vertices. This gives the additional single-type bound obtained by assigning capacity one or two to each other pair. The incidence sum for every ridge type is divisible by four; equivalently $\Sigma_{\text{pairs}} c_{ij}$ is even.

A full facet has at most nine ridges. Its seven nonfull partner pairs supply at most one each, since two antipodal intersections would put eight vertices in a nonfull partner. The other full pair supplies at most two. If the two full pairs share a ridge of at least five vertices, each full facet has just one full neighbour: the antipodal alternative intersection has at most three vertices. The nonfull facets each have at most two full neighbours. Counting these adjacencies gives $r(F)+r(H)\leq 16$. To test whether such a full shared ridge is forced, compare a demand in (48.1) with just its nonfull carrier pairs; excess demand must use the other full pair.

A polygon with at least five vertices requires at least three facet pairs able to contain such a polygon. Its two-dimensional face figure is a polygon and has at least three facets. No proper face is in both members of an antipodal pair. For a local option, $\eta-q>0$ means that a larger polygon occurs. Therefore a choice of local options cannot have exactly one or two such carrier pairs.

The finite test is now specified completely: visit each list of Section 46 in the four G=2 cells; choose one local option at every position; require total pair weight at most 78−3S, every type-incidence sum even, the single-type capacities, all subset inequalities (48.1), the full-adjacency bound when forced, and the polygon carrier condition. All sums are over finite integer lists. The resulting complete residual table is below. The count is of local-option choices, not of actual polytopes.

(G,S)	**Nine row IDs**	**Local choices**
(2, 2)	(1, 1, 1, 1, 2, 2, 5, 16, 16)	3
(2, 2)	(1, 1, 1, 1, 1, 2, 5, 16, 19)	2
(2, 2)	(1, 1, 1, 1, 2, 2, 2, 16, 20)	1
(2, 2)	(1, 1, 1, 1, 1, 2, 2, 16, 25)	2
(2, 2)	(1, 1, 1, 1, 1, 1, 4, 17, 24)	2
(2, 2)	(1, 1, 1, 1, 1, 1, 5, 19, 19)	6
(2, 2)	(1, 1, 1, 1, 1, 2, 2, 19, 20)	4
(2, 4)	(1, 1, 1, 1, 1, 1, 4, 19, 23)	1
(2, 6)	(1, 1, 1, 1, 1, 1, 4, 22, 23)	1

No list survives at (2,8). Nine lists remain across the other cells. Their closure uses the full support functions themselves, as follows.

49 Signed face cuts close the two-full-pair endpoint

Let $F,-F,H,-H$ be the four full facets. Their normalized supporting functions take values ±1 on every vertex of P. Choose signs so that F and H agree on $k\geq4$ of the eight antipodal vertex pairs. Distinct pairs give $k\leq7$. Their actual common vertex set has k vertices; $F\cap(-H)$ has $8-k$.

The seven-common-vertex case. If $k=7$, the seven common vertices lie in the intersection of two distinct support hyperplanes, of affine dimension three. Their linear span L has dimension at most four. All vertices of P are these seven points, their negatives and one remaining antipodal pair $\pm a$. Full dimension forces $\dim L=4$ and $a\notin L$. Thus P is the free sum of the centrally symmetric four-polytope $Q=\operatorname{conv}(\pm B)$ in L and the segment $[-a,a]$. Polarity turns this into a product; the nonempty-face count is $3s(Q)$. Section 27 gives $s(Q)\geq81$, so $s(P)\geq243$. Hence $k=7$ cannot be a counterexample.

It remains to test $k=4,5,6$. For each seven-vertex nonfull facet X, label its abstract vertices $0,\dots,6$ and let M be their set. Let S be the vertices of X in F and U those in H. Fullness gives the complementary cuts $M\setminus S$ and $M\setminus U$ for $-F$ and $-H$. All four are faces of X, with the empty face allowed and X itself excluded. They can be read from the exact seven-vertex Gale lattices. Every label has one of the four sign combinations. Thus

$$\begin{aligned}|S\cap U|+|(M\setminus S)\cap(M\setminus U)|&\leq k,\\ |S\cap(M\setminus U)|+|(M\setminus S)\cap U|&\leq 8-k.\end{aligned}\tag{49.1}$$

These inequalities simply say that the seven vertices occupy distinct antipodal pairs; the one omitted pair can be in either sign class. An embedding of X in an actual P must satisfy them. Conversely we use them only as necessary conditions; any unrealizable cut retained by this test enlarges the search and cannot create a false exclusion.

For every allowed cut record the ridge type j_F when either S or $M\setminus S$ has dimension three, or zero when neither does. Both cannot have dimension three, since two disjoint ridges need at least eight vertices. Define j_H similarly. The type indices are $1,\dots,12$ in Section 36. Record the exact weight w of X and the parity mask $\pi=\Sigma 2^{j-1}(c_j \bmod 2)$.

Also record eight numbers z. For the positive full intersection $F\cap H$, count the two-faces $S\cap U$ and $(M\setminus S)\cap(M\setminus U)$ that are in a three-dimensional cut belonging to F; record their count and total vertex count. Do the same for three-dimensional cuts belonging to H. These give the first four entries of z. For the negative intersection $F\cap(-H)$, replace U by $M\setminus U$ and obtain the last four. Every dimension and vertex count here is determined by the small facet face lattice. Retain each distinct tuple $(j_F, j_H, w, \pi; z)$ once.

For transparency the complete cut-tuple inputs needed for the nine residual lists are printed below. They are generated by visiting all face masks S,U whose complements are proper faces, testing (49.1), and applying the preceding definitions. Facet row IDs are from Section 46. The symbol 0 denotes no ridge neighbour; π is written as an ordinary integer bit mask. The eight z entries are in the displayed order, including unused negative-intersection entries.

ID 1, k=4. (1, 2, 9, 10):(0, 0, 0, 0, 1, 3, 1, 3); (1, 2, 9, 10):(1, 3, 1, 3, 0, 0, 0, 0); (1, 4, 9, 10):(0, 0, 1, 3, 1, 3, 1, 3); (1, 4, 9, 10):(1, 3, 1, 3, 0, 0, 1, 3); (1, 0, 9, 10):(0, 0, 0, 0, 0, 0, 0, 0); (2, 1, 9, 10):(0, 0, 0, 0, 1, 3, 1, 3); (2, 1, 9, 10):(1, 3, 1, 3, 0, 0, 0, 0); (2, 2, 9, 10):(0, 0, 0, 0, 1, 3, 1, 3); (2, 2, 9, 10):(1, 3, 1, 3, 0, 0, 0, 0); (2, 4, 9, 10):(0, 0, 0, 0, 1, 4, 1, 4); (2, 4, 9, 10):(1, 4, 1, 4, 0, 0, 0, 0); (4, 1, 9, 10):(1, 3, 0, 0, 1, 3, 1, 3); (4, 1, 9, 10):(1, 3, 1, 3, 1, 3, 0, 0); (4, 2, 9, 10):(0, 0, 0, 0, 1, 4, 1, 4); (4, 2, 9, 10):(1, 4, 1, 4, 0, 0, 0, 0); (0, 1, 9, 10):(0, 0, 0, 0, 0, 0, 0, 0); (0, 0, 9, 10):(0, 0, 0, 0, 0, 0, 0, 0).

ID 1, k=5. (1, 2, 9, 10):(1, 3, 1, 3, 0, 0, 0, 0); (1, 4, 9, 10):(0, 0, 1, 3, 1, 3, 1, 3); (1, 0, 9, 10):(0, 0, 0, 0, 0, 0, 0, 0); (2, 1, 9, 10):(1, 3, 1, 3, 0, 0, 0, 0); (2, 2, 9, 10):(0, 0, 0, 0, 1, 3, 1, 3); (2, 4, 9, 10):(1, 4, 1, 4, 0, 0, 0, 0); (2, 0, 9, 10):(1, 3, 0, 0, 0, 0, 0, 0); (4, 1, 9, 10):(1, 3, 0, 0, 1, 3, 1, 3); (4, 2, 9, 10):(1, 4, 1, 4, 0, 0, 0, 0); (4, 0, 9, 10):(1, 4, 0, 0, 0, 0, 0, 0); (0, 1, 9, 10):(0, 0, 0, 0, 0, 0, 0, 0); (0, 2, 9, 10):(0, 0, 1, 3, 0, 0, 0, 0); (0, 4, 9, 10):(0, 0, 1, 4, 0, 0, 0, 0); (0, 0, 9, 10):(0, 0, 0, 0, 0, 0, 0, 0).

ID 1, k=6. (1, 1, 9, 10):(1, 3, 1, 3, 0, 0, 0, 0); (2, 0, 9, 10):(1, 3, 0, 0, 0, 0, 0, 0); (2, 0, 9, 10):(1, 4, 0, 0, 0, 0, 0, 0); (4, 0, 9, 10):(1, 4, 0, 0, 0, 0, 0, 0); (0, 2, 9, 10):(0, 0, 1, 3, 0, 0, 0, 0); (0, 2, 9, 10):(0, 0, 1, 4, 0, 0, 0, 0); (0, 4, 9, 10):(0, 0, 1, 4, 0, 0, 0, 0).

ID 2, k=4. (1, 2, 6, 16):(0, 0, 0, 0, 1, 3, 1, 3); (1, 2, 6, 16):(1, 3, 1, 3, 0, 0, 0, 0); (1, 5, 6, 16):(0, 0, 1, 3, 1, 3, 1, 3); (1, 5, 6, 16):(1, 3, 1, 3, 0, 0, 1, 3); (1, 0, 6, 16):(0, 0, 0, 0, 0, 0, 0, 0); (2, 1, 6, 16):(0, 0, 0, 0, 1, 3, 1, 3); (2, 1, 6, 16):(1, 3, 1, 3, 0, 0, 0, 0); (2, 2, 6, 16):(0, 0, 0, 0, 1, 3, 1, 3); (2, 2, 6, 16):(1, 3, 1, 3, 0, 0, 0, 0); (2, 5, 6, 16):(0, 0, 0, 0, 1, 4, 1, 4); (2, 5, 6, 16):(1, 4, 1, 4, 0, 0, 0, 0); (5, 1, 6, 16):(1, 3, 0, 0, 1, 3, 1, 3); (5, 1, 6, 16):(1, 3, 1, 3, 1, 3, 0, 0); (5, 2, 6, 16):(0, 0, 0, 0, 1, 4, 1, 4); (5, 2, 6, 16):(1, 4, 1, 4, 0, 0, 0, 0); (0, 1, 6, 16):(0, 0, 0, 0, 0, 0, 0, 0); (0, 0, 6, 16):(0, 0, 0, 0, 0, 0, 0, 0).

ID 2, k=5. (1, 2, 6, 16):(1, 3, 1, 3, 0, 0, 0, 0); (1, 5, 6, 16):(0, 0, 1, 3, 1, 3, 1, 3); (1, 0, 6, 16):(0, 0, 0, 0, 0, 0, 0, 0); (2, 1, 6, 16):(1, 3, 1, 3, 0, 0, 0, 0); (2, 2, 6, 16):(0, 0, 0, 0, 1, 3, 1, 3); (2, 5, 6, 16):(1, 4, 1, 4, 0, 0, 0, 0); (2, 0, 6, 16):(1, 3, 0, 0, 0, 0, 0, 0); (5, 1, 6, 16):(1, 3, 0, 0, 1, 3, 1, 3); (5, 2, 6, 16):(1, 4, 1, 4, 0, 0, 0, 0); (5, 5, 4, 1):(1, 5, 1, 5, 0, 0, 0, 0); (5, 0, 6, 16):(1, 4, 0, 0, 0, 0, 0, 0); (0, 1, 6, 16):(0, 0, 0, 0, 0, 0, 0, 0); (0, 2, 6, 16):(0, 0, 1, 3, 0, 0, 0, 0); (0, 5, 6, 16):(0, 0, 1, 4, 0, 0, 0, 0); (0, 0, 6, 16):(0, 0, 0, 0, 0, 0, 0, 0).

ID 2, k=6. (1, 1, 6, 16):(1, 3, 1, 3, 0, 0, 0, 0); (2, 0, 6, 16):(1, 3, 0, 0, 0, 0, 0, 0); (2, 0, 6, 16):(1, 4, 0, 0, 0, 0, 0, 0); (5, 5, 4, 1):(1, 5, 1, 5, 0, 0, 0, 0); (5, 0, 4, 1):(1, 5, 0, 0, 0, 0, 0, 0); (5, 0, 6, 16):(1, 4, 0, 0, 0, 0, 0, 0); (0, 2, 6, 16):(0, 0, 1, 3, 0, 0, 0, 0); (0, 2, 6, 16):(0, 0, 1, 4, 0, 0, 0, 0); (0, 5, 4, 1):(0, 0, 1, 5, 0, 0, 0, 0); (0, 5, 6, 16):(0, 0, 1, 4, 0, 0, 0, 0).

ID 4, k=4. (1, 1, 3, 34):(0, 0, 0, 0, 0, 0, 0, 0); (1, 2, 0, 5):(0, 0, 0, 0, 1, 3, 1, 3); (1, 2, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (1, 2, 3, 34):(0, 0, 0, 0, 1, 3, 1, 3); (1, 2, 3, 34):(1, 3, 1, 3, 0, 0, 0, 0); (1, 3, 0, 5):(0, 0, 0, 0, 1, 3, 1, 3); (1, 3, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (1, 6, 3, 34):(0, 0, 1, 3, 1, 3, 1, 3); (1, 6, 3, 34):(1, 3, 1, 3, 0, 0, 1, 3); (1, 0, 0, 5):(0, 0, 0, 0, 0, 0, 0, 0); (1, 0, 3, 34):(0, 0, 0, 0, 0, 0, 0, 0); (2, 1, 0, 5):(0, 0, 0, 0, 1, 3, 1, 3); (2, 1, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (2, 1, 3, 34):(0, 0, 0, 0, 1, 3, 1, 3); (2, 1, 3, 34):(1, 3, 1, 3, 0, 0, 0, 0); (2, 3, 0, 5):(0, 0, 0, 0, 1, 3, 1, 3); (2, 3, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (2, 6, 3, 34):(0, 0, 0, 0, 1, 4, 1, 4); (2, 6, 3, 34):(1, 4, 1, 4, 0, 0, 0, 0); (3, 1, 0, 5):(0, 0, 0, 0, 1, 3, 1, 3); (3, 1, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (3, 2, 0, 5):(0, 0, 0, 0, 1, 3, 1, 3); (3, 2, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (6, 1, 3, 34):(1, 3, 0, 0, 1, 3, 1, 3); (6, 1, 3, 34):(1, 3, 1, 3, 1, 3, 0, 0); (6, 2, 3, 34):(0, 0, 0, 0, 1, 4, 1, 4); (6, 2, 3, 34):(1, 4, 1, 4, 0, 0, 0, 0); (0, 1, 0,

5):(0, 0, 0, 0, 0, 0, 0, 0); (0, 1, 3, 34):(0, 0, 0, 0, 0, 0, 0, 0).

ID 4, k=5. (1, 1, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (1, 1, 3, 34):(0, 0, 0, 0, 0, 0, 0, 0); (1, 1, 3, 34):(1, 3, 1, 3, 0, 0, 0, 0); (1, 2, 0, 5):(0, 0, 1, 3, 0, 0, 0, 0); (1, 2, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (1, 2, 3, 34):(1, 3, 1, 3, 0, 0, 0, 0); (1, 3, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (1, 6, 3, 34):(0, 0, 1, 3, 1, 3, 1, 3); (1, 0, 0, 5):(0, 0, 0, 0, 0, 0, 0, 0); (1, 0, 3, 34):(0, 0, 0, 0, 0, 0, 0, 0); (2, 1, 0, 5):(1, 3, 0, 0, 0, 0, 0, 0); (2, 1, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (2, 1, 3, 34):(1, 3, 1, 3, 0, 0, 0, 0); (2, 2, 0, 5):(1, 4, 1, 4, 0, 0, 0, 0); (2, 3, 0, 5):(0, 0, 0, 0, 1, 3, 1, 3); (2, 6, 3, 34):(1, 4, 1, 4, 0, 0, 0, 0); (3, 1, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (3, 2, 0, 5):(0, 0, 0, 0, 1, 3, 1, 3); (3, 0, 0, 5):(1, 3, 0, 0, 0, 0, 0, 0); (6, 1, 3, 34):(1, 3, 0, 0, 1, 3, 1, 3); (6, 2, 3, 34):(1, 4, 1, 4, 0, 0, 0, 0); (6, 0, 3, 34):(1, 4, 0, 0, 0, 0, 0, 0); (0, 1, 0, 5):(0, 0, 0, 0, 0, 0, 0, 0); (0, 1, 3, 34):(0, 0, 0, 0, 0, 0, 0, 0); (0, 3, 0, 5):(0, 0, 1, 3, 0, 0, 0, 0); (0, 6, 3, 34):(0, 0, 1, 4, 0, 0, 0, 0).

ID 4, k=6. (1, 1, 0, 5):(1, 3, 1, 3, 0, 0, 0, 0); (1, 1, 3, 34):(1, 3, 1, 3, 0, 0, 0, 0); (1, 2, 0, 5):(0, 0, 1, 3, 0, 0, 0, 0); (2, 1, 0, 5):(1, 3, 0, 0, 0, 0, 0, 0); (2, 2, 0, 5):(1, 4, 1, 4, 0, 0, 0, 0); (2, 0, 0, 5):(1, 4, 0, 0, 0, 0, 0, 0); (2, 0, 3, 34):(1, 4, 0, 0, 0, 0, 0, 0); (3, 0, 0, 5):(1, 3, 0, 0, 0, 0, 0, 0); (6, 0, 3, 34):(1, 4, 0, 0, 0, 0, 0, 0); (0, 2, 0, 5):(0, 0, 1, 4, 0, 0, 0, 0); (0, 2, 3, 34):(0, 0, 1, 4, 0, 0, 0, 0); (0, 3, 0, 5):(0, 0, 1, 3, 0, 0, 0, 0); (0, 6, 3, 34):(0, 0, 1, 4, 0, 0, 0, 0).

ID 5, k=4. (1, 1, 0, 0):(0, 0, 0, 0, 0, 0, 0, 0); (1, 2, 0, 0):(0, 0, 0, 0, 1, 3, 1, 3); (1, 2, 0, 0):(1, 3, 1, 3, 0, 0, 0, 0); (2, 1, 0, 0):(0, 0, 0, 0, 1, 3, 1, 3); (2, 1, 0, 0):(1, 3, 1, 3, 0, 0, 0, 0); (2, 2, 0, 0):(0, 0, 0, 0, 1, 3, 1, 3); (2, 2, 0, 0):(1, 3, 1, 3, 0, 0, 0, 0).

ID 5, k=5. (1, 1, 0, 0):(0, 0, 0, 0, 0, 0, 0, 0); (1, 1, 0, 0):(1, 3, 1, 3, 0, 0, 0, 0); (1, 2, 0, 0):(0, 0, 1, 3, 0, 0, 0, 0); (1, 2, 0, 0):(1, 3, 1, 3, 0, 0, 0, 0); (2, 1, 0, 0):(1, 3, 0, 0, 0, 0, 0, 0); (2, 1, 0, 0):(1, 3, 1, 3, 0, 0, 0, 0); (2, 2, 0, 0):(0, 0, 0, 0, 1, 3, 1, 3); (2, 2, 0, 0):(1, 4, 1, 4, 0, 0, 0, 0).

ID 5, k=6. (1, 1, 0, 0):(1, 3, 1, 3, 0, 0, 0, 0); (1, 2, 0, 0):(0, 0, 1, 3, 0, 0, 0, 0); (2, 1, 0, 0):(1, 3, 0, 0, 0, 0, 0, 0); (2, 2, 0, 0):(1, 4, 1, 4, 0, 0, 0, 0).

The exact recurrence. Choose local options (c_F,q_F,b_F,w_F) and the corresponding H option from Section 48. The possible shared full ridges are: none; one ridge of a type with k vertices present in both options; and, only when k=4, two tetrahedra. The last case uses both F∩H and F∩(−H). For k≥5, no shared ridge would make F∩H a k-gon. Therefore it requires η(F)−q(F)≥k−3 and η(H)−q(H)≥k−3. Discard the no-ridge choice unless both inequalities hold.

Subtract each shared ridge from c_F and c_H to obtain target vectors a,d for the seven nonfull pairs. A shared ridge of type (n,e,r) has r polygons and total polygon vertex count 2e. Set its corresponding four z targets to (r,2e,r,2e). With one shared ridge use the positive-intersection entries; with two use both blocks. Unused blocks are ignored. Set the weight allowance to $L=78-3S-w_F-w_H$ and the final parity target to the exclusive-or of their masks.

Start with the state (a,d,z,π=0,L). Visit the seven nonfull row IDs in their printed order. For each tuple from that row's cut table, subtract one from a at j_F when nonzero, one from d at j_H when nonzero, subtract its used z entries and its weight, and replace π by its exclusive-or with the tuple's mask. Discard any state with a negative integer entry. At the end accept exactly zero target vectors and the required parity, with a nonnegative remaining weight allowance. Equal states may be combined; this is only a way of avoiding repeated additions and subtractions.

Why must a real polytope give an accepted state? Every ridge of F and H has its unique second facet, so the two ridge-count targets are exhausted exactly. Every polygon of a shared ridge T has, inside F, a unique other ridge R. Its second facet X is nonfull and R∩T=T∩X is that polygon. Thus the F-side polygon targets are exhausted exactly; the same holds on the H side. The weight and type-parity targets were proved in Sections 45,48. All possible sign cuts of each X have been retained. Hence rejecting every state excludes every actual realization.

For each of the nine residual lists the recurrence gives the following complete output. The supplied integer program recomputes the Gale cuts, table entries and state counts from these definitions; all calculations are exact.

(G,S)	**Nine row IDs**	**Accepted states**
(2, 2)	(1, 1, 1, 1, 2, 2, 5, 16, 16)	0
(2, 2)	(1, 1, 1, 1, 1, 2, 5, 16, 19)	0
(2, 2)	(1, 1, 1, 1, 2, 2, 2, 16, 20)	0
(2, 2)	(1, 1, 1, 1, 1, 2, 2, 16, 25)	0
(2, 2)	(1, 1, 1, 1, 1, 1, 4, 17, 24)	0
(2, 2)	(1, 1, 1, 1, 1, 1, 5, 19, 19)	0
(2, 2)	(1, 1, 1, 1, 1, 2, 2, 19, 20)	0
(2, 4)	(1, 1, 1, 1, 1, 1, 4, 19, 23)	0
(2, 6)	(1, 1, 1, 1, 1, 1, 4, 22, 23)	0

Every count is zero. Section 47 excluded the four other positive cells, and Section 44 excluded the zero-stress cell. Therefore the (16,18) endpoint with two full facet pairs is completely excluded. □

50 The three-full-pair normal form and its five-point fibre

Three full facet pairs have three independent normalized support functions, by the rank argument of Section 28. Project P by those functions to a subset of the three-cube. The six-corner image was handled in Sections 29,33: a point fibre splits off a segment, and otherwise its eight lifted triangular faces are full facets. The present branch has exactly six full facets, so that alternative is excluded. All eight cube corners are occupied.

Choose positive corners $c_0=(1,1,1)$, $c_1=(1,1,-1)$, $c_2=(1,-1,1)$, $c_3=(1,-1,-1)$. Their relation is $c_0+c_3=c_1+c_2$. After a linear choice of kernel coordinates, the polytope has the form

$$\begin{gathered} P=\operatorname{conv}\{\pm(c_j,z) : z\in K_j,\ 0\le j\le 3\}, \\ K_j\subset\mathbb{R}^2,\ n_j=|K_j|\ge 1,\ \Sigma n_j=8. \end{gathered} \tag{50.1}$$

Every K_j is in convex position: a nonextreme point in its fibre would be a nonextreme vertex of P. Hence $n_j\ge 3$ gives a genuine planar polygon. The possible unordered sizes are (5,1,1,1), (4,2,1,1), (3,3,1,1), (3,2,2,1), (2,2,2,2). An extra full support with a nonzero kernel part would increase the full normal rank to at least four, already treated in Sections 30–32. The remaining analysis can therefore assume exactly the three-coordinate full normal space.

Proposition 50.1. *The size pattern (5,1,1,1) has at least twenty facets.* Relabel so the large fibre is K_0. The other three cube corner vectors are linearly independent, so a linear shear makes their singleton heights zero. Write $p_1,\dots,p_5$ for the translated large-fibre vertices. In dual coordinates x_1,x_2,x_3,b the polar is precisely

$$\begin{gathered} |x_i|\le 1\ (i=1,2,3), \\ |x_1+x_2+x_3+\langle b,p_k\rangle|\le 1\ (k=1,\dots,5). \end{gathered} \tag{50.2}$$

Put $D=\{(s,b): |s+\langle b,p_k\rangle|\le 1 \text{ for all } k\}$. It is the polar of $\operatorname{conv}\{\pm(1,p_k)\}$ in dimension three. That convex hull is full dimensional and has exactly ten vertices: the top and bottom polygons lie in the exposing planes $s=\pm 1$. A three-polytope with ten vertices has at least seven facets, from $2e\ge 3v$ and Euler. Thus D has at least seven vertices. It is compact, centrally symmetric and has the origin in its interior.

Equation (50.2) identifies P^* with the fibre product of D and the cube $B=[-1,1]^3$ over $s=x_1+x_2+x_3$. We use two elementary vertex rules. A vertex y of D together with a vertex x of the corresponding cube section is a vertex of the fibre product: any nontrivial convex decomposition first fixes y, then fixes x. Conversely, a vertex x of B and a vertex y of its corresponding D section also give a vertex by the same argument in the reverse order.

If all vertices $y=(s,b)$ of D have $|s|<3$, each cube section $x_1+x_2+x_3=s$ is a polygon with at least three vertices. The first rule gives at least $3\cdot 7=21$ different vertices of P^*: different D vertices have different first coordinates in the product. In particular P has at least twenty facets.

Otherwise symmetry gives a point $y\in D$ with $s\ge 3$. Convexity and an interior ball at zero show that the section $s=1$ has nonempty planar interior: scale y by $1/s<1$ and retain a scaled interior ball. Its compact polygon has at least three vertices. The section $s=3$ is nonempty, by scaling y by $3/s\le 1$, and has at least one vertex. There

are three cube vertices with coordinate sum one and one cube vertex with coordinate sum three. The second rule therefore supplies nine distinct vertices at height one and one at height three. Their negatives are ten further vertices, distinguished by negative height. Thus P* has at least twenty vertices in this case too. All boundary equalities, including maximum height exactly three, are included. □

This leaves four fibre-size patterns in the three-full-pair branch. The proposition is an all-real argument and uses no sample realizations or parameter rounding.

51 Balanced box sections: exact vertices and their lifting

We complete the four remaining patterns of Section 50. Replace the planar fibres K_0,K_1,K_2,K_3 by their signed copies $L_0=K_0,L_1=-K_1,L_2=-K_2,L_3=K_3$. The offsets now satisfy $a_0+a_1+a_2+a_3=0$. Thus the actual polar, in five independent coordinates, is the set of (a,b) with

$$\Sigma a_j=0, \quad |a_j+\langle b,z\rangle|\le 1 \ (z\in L_j). \tag{51.1}$$

The change of coordinates is invertible: a vector giving zero on all sixteen original vertices is zero because they span $\mathbb{R}^5$. In particular (51.1) is compact. At $b=0$ it is the three-dimensional section $\Sigma a_j=0$ of the four-cube. Its six vertices have two coordinates $+1$ and two -1, and they are actual polar vertices: any convex decomposition preserves all their tight vertex inequalities. We count additional vertices in antipodal pairs.

Fix a nonzero planar direction n. For each L_j let h_j and l_j be its maximum and minimum projection on n, and put $c_j=(h_j+l_j)/2$, $r_j=(h_j-l_j)/2$. On the line $b=tn$ set $u_j=a_j+tc_j$ and $\delta=\Sigma c_j$. The positive half of the section is

$$t\ge 0, \quad \Sigma u_j=\delta t, \quad |u_j|+r_jt\le 1 \ (0\le j\le 3). \tag{51.2}$$

All r_j are nonnegative. A singleton has radius zero; a genuine polygon has positive radius. A segment has radius zero precisely when n is normal to it. The maximal feasible height is

$$T=\min\{1/\max r_j,\ 4/(|\delta|+\Sigma r_j)\}. \tag{51.3}$$

Here at least one radius is positive in the sections counted below. The first bound is the nonnegativity of every interval length. The second is necessary and sufficient for the sum δt to belong to the sum of the four intervals, whose radius is $4-t\Sigma r_j$.

Lemma 51.1 (Exact enumeration). *All positive-height vertices of (51.2) are given by the following two rules.* Below T, choose $\varepsilon\in\{-1,+1\}^4$ and use

$$t=(\Sigma\varepsilon_j)/(\delta+\Sigma\varepsilon_j r_j), \quad u_j=\varepsilon_j(1-r_jt), \quad 0<t<T. \tag{51.4}$$

Discard zero denominators and zero numerators. At $t=T$ let $b_j=1-r_jT$. Fix all but at most one of the coordinates with $b_j>0$ to $\pm b_j$, set coordinates with $b_j=0$ to zero, and solve $\Sigma u_j=\delta T$ for the remaining coordinate. Keep it if it lies in its interval. Remove duplicate vectors.

Proof. Below T none of the interval radii vanishes and the section has dimension four. A vertex needs four independent tight inequalities. Each class contributes at most one, so all four endpoints are tight, giving (51.4). When $\Sigma\varepsilon_j=0$, either $t=0$ or the resulting point is on an entire segment, rather than a positive interior-height vertex. At T, either the sum bound is tight and the cap is a single point, or a maximal-width interval vanishes. Its two endpoint inequalities force $t=T$ in any convex decomposition. Cap vertices are exactly vertices of a box cut by one sum equation; two coordinates strictly inside their intervals could be varied by opposite amounts. Thus the stated cap rule is exhaustive. □

Lemma 51.2 (Marking and lifting). *A section vertex is an actual polar vertex if one of its tight endpoint conditions forces b to remain on the chosen line.* In

particular we may mark $u_j=1-r_jt$ if the upper projection of L_j contains a polygon edge; mark $u_j=-1+r_jt$ if its lower projection contains an edge; and mark either $u_j=\pm1$ for a zero-radius segment. To prove this, the two endpoints of that edge or segment have equal tight signed values. In a convex decomposition their inequalities remain tight, so each summand's planar coordinate is orthogonal to their nonzero difference. It lies on the line n. The section vertex property then makes the decomposition trivial. Different marked points on one positive ray, their negatives, and points on different nonparallel lines are all distinct.

For two positive widths normalize their maximum to one and write $r=(a,1,0,0)$, $0<a\le1$, $d=\delta$ after the same scaling. For $0<a<1$ the only changes in (51.4) or the cap rule occur at $d=\pm(3-a),\pm(1+a),\pm(1-a)$. Indeed the endpoint equations at $t=1$ are $d=\Sigma\varepsilon_j(1-r_j)$; the sum-bound cap begins at $|d|=3-a$. The order of these six numbers is fixed throughout $0<a<1$. Substitution into the two rules gives the following tables. These are entire real parameter cells, including their boundary equalities, rather than sample realizations. A means either upper edge of classes 0,1; O means the upper edge of class 0 or the lower edge of class 1; Z means either zero segment of classes 2,3; M_j means the upper edge of class j or the zero segment of class 2; E_j means the upper edge of class j alone; B_j means either upper or lower edge of class j.

0<a<1	**A**	**O**	**Z**	**M_0**	**M_1**
$d<-3+a$	2	4	5	5	5
$d=-3+a$	2	4	4	4	4
$-3+a<d<-1-a$	4	6	6	5	6
$d=-1-a$	3	5	5	4	5
$-1-a<d<-1+a$	4	6	6	5	6
$d=-1+a$	4	4	4	4	4
$-1+a<d<1-a$	6	6	6	5	6
$d=1-a$	4	4	4	3	4
$1-a<d<1+a$	6	6	6	5	6
$d=1+a$	5	5	5	5	5
$1+a<d<3-a$	6	5	6	6	6
$d=3-a$	4	3	4	4	4
$d>3-a$	5	4	5	5	5

0<a<1	**E_0**	**E_1**	**B_0**	**B_1**
$d<-3+a$	1	1	5	5
$d=-3+a$	1	1	4	4

0<a<1	**E_0**	**E_1**	**B_0**	**B_1**
$-3+a<d<-1-a$	1	3	5	6
$d=-1-a$	1	3	5	5
$-1-a<d<-1+a$	2	4	6	6
$d=-1+a$	2	4	4	4
$-1+a<d<1-a$	2	6	4	6
$d=1-a$	2	4	4	4
$1-a<d<1+a$	4	6	6	6
$d=1+a$	4	5	5	5
$1+a<d<3-a$	4	6	5	6
$d=3-a$	3	4	4	4
$d>3-a$	4	4	5	5

a=1	**A**	**O**	**Z**	**M**	**E**	**B**
$d<-2$	2	4	5	5	1	5
$d=-2$	1	3	3	3	1	3
$-2<d<0$	2	4	4	4	2	4
$d=0$	2	2	2	2	2	2
$0<d<2$	4	4	4	4	4	4
$d=2$	3	3	3	3	3	3
$d>2$	5	4	5	5	4	5

At $a=1$ the values $d=-2,0,2$ produce an extra full support whenever both positive-width classes have just two projection levels: at $t=1$ their centred values are zero and their endpoints have values ±1, while the two zero classes can be assigned ±1 with sum d. Thus every original vertex has absolute value one and $b\neq0$. This is excluded by Section 50. The two-level hypothesis holds for segments, triangles at edge normals, and a quadrilateral at a pair of opposite parallel edges. It is not imposed on an arbitrary direction of a polygon.

For one positive width normalize $r=(1,0,0,0)$. The parameter cells have boundaries $d=-3,-1,1,3$. Let Z_1 mark one zero segment, Z_2 mark either of two zero segments, and M mark the upper polygon edge or that first zero segment. Direct substitution into Lemma 51.1 gives:

One width	All	Z_1	Z_2	M
$d<-3$	5	5	5	5
$d=-3$	4	4	4	4
$-3<d<-1$	6	5	6	6
$d=-1$	3	3	3	3
$-1<d<1$	6	4	6	6
$d=1$	3	3	3	3
$1<d<3$	6	5	6	6
$d=3$	4	4	4	4
$d>3$	5	5	5	5

If the nontrivial positive-width fibre has only two levels, $d=\pm1,\pm3$ gives an extra full support at $t=1$ by the same argument, now with three signs summing to d. Outside those four values, all positive section vertices number at least five and Z_2 also numbers at least five. Without the two-level hypothesis, Z_1 still numbers at least three; M numbers three only at $d=-1$ or $d=1$, and at least four elsewhere.

Support form of the table. In the two-width situation let $H=h_0+h_1+q\cdot n$, where q is the sum of the two singleton points. Then $H=\delta+r_0+r_1$. Each individual upper-edge count is at least two when $H>0$. For two aligned upper edges, their union has at least two; it has at least four when $H\geq 2\min(r_0,r_1)$, and at least three when $r_0\neq r_1$ and $H\geq 0$. These last aligned statements exclude the two-level exceptional full supports as specified above. Opposed upper/lower edges give at least three. An edge and a zero segment give at least three, two zero segments at least four, and both opposite edges of one class at least four. Each assertion is a column minimum on the displayed cells, with the inequality on H translated to one of their boundaries.

52 Four segments and a triangle with two segments

A zero-radius segment contributes at least two positive polar vertices on each of its normal rays. For completeness, the two active faces making the whole segment’s signed value +1 or −1 are nonempty for small positive t: set its value to that sign and distribute the required balancing offset among the other three classes. Their inequalities are initially strict. On each compact active face maximize n·b and choose a vertex in the maximizing face. The maximum is positive and the active segment pins b to its normal line. The two signs distinguish the two vertices. Similarly an upper polygon edge contributes at least one: set its upper value to +1 at small t, keep the other classes strict, and apply the same maximization. These observations also apply when the other fibres are arbitrary.

Proposition 52.1. *Four segment fibres give at least twenty facets unless there is an extra full support.* If their four directions are distinct modulo sign, their normal lines give $4\cdot2=8$ positive vertices, their negatives, and the six central vertices, hence twenty-two facets. If exactly two segments are parallel and no three are, their common normal has two zero segment radii and two positive radii. Column Z gives at least four positive vertices; the exceptional equal-width cells would be full supports. If the remaining two segments are parallel, their different common normal gives four more. Otherwise each has its own line and gives two more. Again the total is at least twenty-two.

If exactly three segments are parallel, their common normal has one positive width and three zero segment radii. Every positive section vertex is marked: at the cap, at least two zero-class coordinates are endpoints; below it all four endpoints are tight. The one-width table, with its four full-support cells excluded, gives at least five. The fourth segment has a different normal line and gives two, yielding $6+2(5+2)=20$. Finally, if all four segments are parallel, all radii vanish on their common normal. If $\delta=0$ the evaluation map has a nonzero kernel, contrary to full dimension. If $\delta\neq0$ choose all four u_j equal to one and $t=4/\delta$. This is a nonzero-kernel full support. Every possibility is therefore excluded in our branch. □

Proposition 52.2. *The pattern (3,2,2,1) has at least twenty facets in the absence of an extra full support.* Suppose first that the two segments are not parallel. Their normal lines are distinct. A triangle has three distinct nonopposite edge-normal lines, and a segment direction can coincide with at most one of them. Give each triangle line weight one; replace its weight by three when it coincides with a segment normal, using column M_0 or M_1. An unused segment line has weight two. If $c\in\{0,1,2\}$ triangle lines coincide, the total is $(3-c)+3c+2(2-c)=7$. All possible exceptional cells have two-level fibres and would give an extra full support. Thus six central vertices plus these seven pairs give twenty facets.

If the segments are parallel and their common normal is a triangle edge-normal line, the triangle has two projection levels. The four exceptional one-width cells give extra full supports, so $Z_2\geq5$ on that line. The other two triangle lines give two more, hence twenty facets.

Suppose their common normal is not a triangle edge-normal line. It gives at least three positive vertices by Z_2, and the three triangle edge lines give one each. We now produce a second vertex on one of those triangle rays, making seven in total.

Lemma 52.3 (A transverse edge bonus). *With one polygon upper edge, two positive segment widths r_1,r_2 and one singleton, the upper-edge ray gives at least two positive polar vertices if $H=\delta+r_0+r_1+r_2>2\min(r_1,r_2)$.* To prove this, intersect (51.2) with $u_0=1-r_0t$. This is a three-dimensional compact polytope whose $t=0$ base is the triangle $u_1+u_2+u_3=-1$ in the three-cube. If it had just one positive vertex, it would be a tetrahedron over that base. Each base edge is in one of the three lower endpoint planes $u_j=-1+r_jt$. Apart from $t=0$, that is the only defining plane containing the whole base edge, so it must be the corresponding lateral facet plane. The apex would therefore lie in all three planes, giving

$$\begin{aligned} t&=2/(r_1+r_2-r_0-\delta),\\ t\le 1/\max(r_1,r_2) &\Rightarrow \delta+r_0\le -|r_1-r_2|\\ &\Rightarrow H\le 2\min(r_1,r_2). \end{aligned} \tag{52.1}$$

This contradicts the strict hypothesis. There are at least two positive vertices of the active upper-edge face; its two edge endpoints pin the planar coordinate to the ray, so both are actual polar vertices. □

Write the two parallel signed segments as $v+[-a,a]e$ and $w+[-b,b]e$, where $a,b>0$ and e is a unit vector; let z be the singleton. Put $q=v+w+z$ and consider the genuine planar polygon $D=T+q+[-|a-b|,|a-b|]e$. In a triangle edge direction n_i its support is $H(n_i)-2\min(r_1,r_2)$. The three triangle normals positively span the plane. These supports cannot all be nonpositive, since that would force $D\subset\{0\}$. One is positive. Both segment widths in that direction are positive by the non-edge-normal assumption, so Lemma 52.3 gives the required second vertex. The count is now $3+2+1+1=7$ pairs, hence twenty facets. This completes every parallel and nonparallel alternative of $(3,2,2,1)$. □

53 Two triangles: matching, translation and the extra vertex

Write the signed triangles as T,U and their singleton points as v,w; put q=v+w. For a triangle label its vertex p_i opposite edge i, with outward unit normal n_i, support h_i, and altitude $w_i>0$. The barycentric coordinates of a point $-z$ are $(h_i+n_i\cdot z)/w_i$, giving the identities

$$\Sigma n_i/w_i=0,\quad \Sigma h_i/w_i=1,\qquad \Sigma(h_i+n_i\cdot z)/w_i=1. \tag{53.1}$$

Each triangle supplies at least four positive vertices among its three edge-normal rays. Indeed fix any point $u\in U$. By (53.1) some $h_i+n_i\cdot(u+q)>0$. Consequently $H(n_i)=h_T(n_i)+h_U(n_i)+n_i\cdot q>0$, and the individual upper-edge column in Section 51 gives at least two on that ray. The other two rays give one each. If the two triangles have no parallel edges, these two families are disjoint even up to sign, giving $6+2(4+4)=22$ facets.

A shared oriented normal is counted with weight two; an opposed normal with weight three, by columns A,O. Every unshared edge line has weight one. Because each triangle's own three normals are nonopposite, sharing is a matching between their two triples. The sum of the weights is six plus the number of opposed matches. If there are at most eighteen facets, there is no opposed match and there can be no additional vertex beyond these weights. In particular an unshared edge satisfies $H(n)\le 0$. For an aligned shared normal with altitudes w_T,w_U, the aligned four-vertex bound gives

$$H(n)<\min(w_T,w_U). \tag{53.2}$$

There is at least one aligned match, since otherwise the preceding four-vertex families give twenty-two. Suppose exactly one edge i of T matches edge j of U. Apply (53.1) with $z=p_j(U)+q$. Some $A_k=h_k(T)+n_k(T)\cdot z$ is positive. For an unshared edge $A_k\le H(n_k)\le 0$. For the shared edge, $A_i=H(n_i)-w_j(U)<0$ by (53.2). This is a contradiction, so there are at least two aligned matches.

If exactly two match, call the matched T edges a,b and the remaining edge c. Let the matching U edges be i,j and again use $z=p_i(U)+q$. Then $A_a=H(n_a)-w_i(U)<0$ and $A_b=H(n_b)<w_b(T)$. Equation (53.1) forces $A_c/w_c(T)=1-A_a/w_a(T)-A_b/w_b(T)>0$. But c is unshared and $A_c\le H(n_c)\le 0$. Thus all three outward normals match.

Two triangles with the same three outward normal directions are positively homothetic. To see this directly, translate corresponding vertices opposite two edges into coincidence. The third support line then fixes the common positive dilation factor α; all three corresponding altitudes scale by α. If $\alpha\ne 1$, the unequal-width aligned bound from Section 51 makes $H(n_i)<0$ for every i, since $H\ge 0$ would give a third vertex on a weight-two ray. Yet writing $U=\alpha T+d$ gives $\Sigma H(n_i)/w_i(T)=1+\alpha>0$ by (53.1). Therefore $\alpha=1$ and $U=T+d$ is a translate.

Put $a=-(d+q)/2$ and let β_i be its barycentric coordinates in T. Then $H(n_i)=2w_i\beta_i$; (53.2) gives $\beta_i<1/2$. Since $\Sigma\beta_i=1$, all three β_i are strictly positive. Set

$$f_0=1,\quad f_1=-1,\quad f_2=(\beta_1-\beta_0)/\beta_2;\qquad |f_2|<1,\quad \Sigma\beta_i f_i=0. \tag{53.3}$$

Interpolate the unique affine function $f(z)=A+b\cdot z$ on T with these three values. The last equality says $A+b\cdot a=0$. Assign signed class offsets $1-b\cdot v$, $-1-b\cdot w$, A, $A-b\cdot d$ to the two singletons, T and U respectively. Their sum is $2A-b\cdot(v+w+d)=0$. Both triangles then have the same matching values f_i and the singletons have values $+1,-1$. This constructs a point of the polar with six tight original vertex inequalities: the two singletons and the matching vertices with f_0,f_1. Its active face is nonempty and compact, so choose a vertex on it.

Its planar coordinate is nonzero, because two vertices in T have opposite values. If this coordinate were on any of T's three edge-normal lines, the two endpoints of that edge would have equal values. For the edge joining vertices 0,1 this is impossible. For either other edge it makes vertex 2 equal to $+1$ or -1. The difference between matching values on U and T is independent of the vertex, and is already zero at vertex 0. All eight positive representatives would consequently have absolute value one, an extra full support. Thus this vertex is off all three matched normal lines. The matched rays already give six pairs; this vertex and its negative give a seventh. Hence $6+2\cdot7=20$ facets, a contradiction. This excludes $(3,3,1,1)$. □

54 A quadrilateral and a segment: boundary drifts included

Let Q be the signed quadrilateral, S the signed segment, and v,w the singleton points. Put M=Q+S+v+w, a two-dimensional Minkowski sum. Its support in direction n is H(n)=h_Q(n)+h_S(n)+n·(v+w). A segment normal always supplies at least three positive vertices, by the one-width column Z_1. The other two zero classes here are singletons.

If Q has a pair of parallel opposite edges, use their common normal line. If it is transverse to S, the two positive widths have two levels, and column B_0 or B_1 gives at least four positive vertices after excluding the equal-width full-support cells. The different segment normal gives three more. If that line is normal to S, there is one positive two-level width, so the exceptional d=±1,±3 are full supports. All positive section vertices lift, since at the width cap both the upper and lower quadrilateral edges are tight; below the cap the zero segment is tight. The one-width table gives at least five. The other two edges of Q provide two different normal lines, or another opposed line transverse to S providing four. In either case there are at least seven pairs, hence twenty facets.

It remains that Q has four pairwise nonparallel edge-normal lines. If none is the segment normal, their four upper edges together with the segment's three vertices give seven pairs. Suppose one coincides. Orient that line as the outward normal n_0 of the corresponding Q edge, and normalize r_Q=1. Column M in the one-width table counts the union of its upper edge and the segment. If there are at most eighteen facets, the union can have only three positive vertices and the other three edge rays only one each. Thus d=±1. In unscaled coordinates this means

$$H(n_0)=0 \quad \text{or} \quad H(n_0)=w_Q(n_0), \tag{54.1}$$

where w_Q is the full projection width of Q. For every other edge normal n_i, both Q and S have positive widths and two singleton classes have width zero. The individual upper-edge bound says that $H(n_i)>0$ would give at least two vertices. Hence all three remaining supports $H(n_i)$ would have to be nonpositive.

The four outward normals of a bounded quadrilateral positively span the plane. Consequently $\{z : n_i \cdot z \le 0 \text{ for all } i\}=\{0\}$: a nonzero vector in that cone would give a recession direction of Q. If $H(n_0)=0$, all four supports of M are nonpositive, forcing the genuine two-dimensional polygon M to be {0}, impossible. If $H(n_0)=w_Q(n_0)$, the minimum of n_0 on M is zero, because S is parallel to Q's upper edge and has zero width in that direction. The opposite exposed face of M is the opposite exposed vertex of Q plus the nondegenerate segment S plus v+w. It is a nondegenerate segment in $n_0 \cdot z=0$. All its points also satisfy the other three $n_i \cdot z \le 0$, so the same recession argument forces the entire segment to be {0}. This too is impossible. Some transverse edge therefore supplies a second vertex, and the count is 3+2+1+1=7 pairs plus the six central vertices. This excludes (4,2,1,1), including both boundary drifts. □

55 Completion of the endpoint ledger

Every possibility in the fifteen-pair reduction of Section 5 has now been excluded under $t \le 118$. For clarity, the following ledger specifies where each complete exclusion is assembled. The references are to mathematical arguments and exact tables printed in this manuscript.

(v,h)	Complete exclusion
(14,16), (14,18), (14,20), (14,22), (14,24), (14,26)	§40.1: incidence bounds and row sums after the all-real full-normal analysis of §§28–32
(16,16)	§34: full-normal geometry, no-full certificates and complete even-cell coverage
(16,18)	No full pairs: §34; four or more: §§29–33; one: §43; two: §§44–49; three: §§50–54
(16,20)	§§35–37: quadrilateral localization, row weights, full-normal geometry and exact coverage
(16,22)	§19: forced ridge profile and antipodal incidence divisibility
(16,24)	§8: L_1 row sum, with the full prism excluded by §§16–17
(18,18)	§40.2: product and prism bounds of §§20,38–39, and the complete D certificates of §34
(18,20)	§40.3: product and prism alternatives, extra-pair geometry of §21, and complete C certificates of §34
(18,22)	§23: product bound and four exact slack alternatives
(20,20)	§24: product bound on either side of polarity and two remaining exact certificates

For (16,18), the few-full reduction of Section 42 covers every possible slack cell and local maximum. Its three-full-pair residues no longer need an additional numerical certificate: Section 50 puts their three independent full normals into the cube-fibre form. Its five-point fibre is excluded there, and all four other distributions are excluded in Sections 52–54. Each has at least twenty facets unless a nonzero-kernel full support occurs; that alternative has rank at least four and is excluded by Sections 30–32. This closes the final branch of the ledger.

The finite arithmetic has a direct meaning on paper: row domains, integer recurrences, parameter cells and exact negative linear combinations are displayed. Their accompanying fraction-arithmetic programs independently check the tables. They do not establish the geometric lemmas; those are proved in the preceding sections. As usual, the classical three-dimensional convex rigidity theorem stated in Section 10 is used as a standard mathematical theorem; the dimensional extension and every stress localization used here are supplied explicitly.

56 Conclusion

Suppose a centrally symmetric five-dimensional convex polytope had fewer than 243 nonempty faces. Euler’s relation and antipodal parity would give $s(P)=3+2t$ with even $t\le 118$. The cube-section argument and polarity exclude vertex or facet counts at most twelve. Orienting polarity so $v\le h$, the flag and rigidity inequalities place (v,h) in the fifteen-pair ledger of Section 5. Section 55 excludes every entry. Therefore no such polytope exists, and

$$\begin{aligned} f_1(P)+f_3(P)&\ge 120,\\ s(P)=3+2(f_1(P)+f_3(P))&\ge 243=3^5. \end{aligned} \tag{56.1}$$

The five-cube has face counts $(32,80,80,40,10)$. Including the cube itself gives exactly 243 nonempty faces. Thus the five-dimensional bound is sharp. □